

Complete Hospital ERP PostgreSQL Schema

📁 PART 1: FOUNDATION & MULTI-TENANCY

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-- HOSPITAL ERP - COMPLETE DATABASE SCHEMA
-- PostgreSQL 15+
-- =====

-- Enable required extensions
CREATE EXTENSION IF NOT EXISTS "uuid-ossf";
CREATE EXTENSION IF NOT EXISTS "pgcrypto";
CREATE EXTENSION IF NOT EXISTS "pg_trgm";
CREATE EXTENSION IF NOT EXISTS "btree_gist";

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-- SECTION 1: CORE ENUMS
-- =====

-- Gender
CREATE TYPE gender_type AS ENUM ('male', 'female', 'other', 'unknown');

-- Blood Groups
CREATE TYPE blood_group_type AS ENUM ('A+', 'A-', 'B+', 'B-', 'AB+', 'AB-', 'O+', 'O-', 'unknown');

-- Marital Status
CREATE TYPE marital_status_type AS ENUM ('single', 'married', 'divorced', 'widowed', 'separated', 'other');

-- Patient Types
CREATE TYPE patient_type AS ENUM ('opd', 'ipd', 'emergency', 'daycare', 'telemedicine');

-- Visit Status
CREATE TYPE visit_status_type AS ENUM ('scheduled', 'checked_in', 'in_progress', 'completed', 'cancelled', 'no_show');

-- Admission Status
CREATE TYPE admission_status_type AS ENUM ('pending', 'admitted', 'discharged', 'transferred', 'expired', 'lama', 'dor');

-- Bed Status
CREATE TYPE bed_status_type AS ENUM ('available', 'occupied', 'reserved', 'cleaning', 'maintenance', 'blocked');

-- Order Status
CREATE TYPE order_status_type AS ENUM ('pending', 'in_progress', 'completed', 'cancelled', 'on_hold');
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-- Priority Levels
CREATE TYPE priority_type AS ENUM ('routine', 'urgent', 'stat', 'emergency');

-- Payment Status
CREATE TYPE payment_status_type AS ENUM ('pending', 'partial', 'paid', 'refunded',
'cancelled', 'write_off');

-- Payment Methods
CREATE TYPE payment_method_type AS ENUM ('cash', 'card', 'upi', 'bank_transfer',
'cheque', 'mobile_banking', 'insurance', 'corporate', 'other');

-- Shift Types
CREATE TYPE shift_type AS ENUM ('morning', 'evening', 'night', 'general',
'split');

-- Employment Types
CREATE TYPE employment_type AS ENUM ('permanent', 'contract', 'probation',
'intern', 'consultant', 'visiting');

-- Leave Status
CREATE TYPE leave_status_type AS ENUM ('pending', 'approved', 'rejected',
'cancelled');

-- Maintenance Priority
CREATE TYPE maintenance_priority_type AS ENUM ('critical', 'high', 'normal',
'low');

-- Ticket Status
CREATE TYPE ticket_status_type AS ENUM ('open', 'assigned', 'in_progress',
'pending_parts', 'resolved', 'closed');

-- Consent Status
CREATE TYPE consent_status_type AS ENUM ('pending', 'obtained', 'refused',
'withdrawn', 'not_applicable');

-- Insurance Claim Status
CREATE TYPE claim_status_type AS ENUM ('draft', 'submitted', 'under_review',
'query_raised', 'approved', 'partial_approved', 'rejected', 'settled',
'appealed');

-- Specimen Status
CREATE TYPE specimen_status_type AS ENUM ('ordered', 'collected', 'in_transit',
'received', 'processing', 'completed', 'rejected');

-- Result Status
CREATE TYPE result_status_type AS ENUM ('pending', 'preliminary', 'final',
'amended', 'cancelled');

-- Document Types
CREATE TYPE document_type AS ENUM ('lab_report', 'radiology', 'prescription',
'consent', 'discharge_summary', 'insurance', 'id_proof', 'photo', 'other');

-- Notification Channels
CREATE TYPE notification_channel_type AS ENUM ('sms', 'email', 'whatsapp', 'push',
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'in_app');

-- =====
-- SECTION 2: MULTI-TENANCY FOUNDATION
-- =====

-- Organizations (Top-level tenants)
CREATE TABLE organizations (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  code VARCHAR(20) UNIQUE NOT NULL,
  name VARCHAR(255) NOT NULL,
  name_bangla VARCHAR(255),
  legal_name VARCHAR(255),
  organization_type VARCHAR(50) DEFAULT 'hospital',

  -- Contact Information
  email VARCHAR(255),
  phone VARCHAR(20),
  website VARCHAR(255),

  -- Address
  address_line1 VARCHAR(255),
  address_line2 VARCHAR(255),
  city VARCHAR(100),
  district VARCHAR(100),
  division VARCHAR(100),
  postal_code VARCHAR(20),
  country VARCHAR(100) DEFAULT 'Bangladesh',

  -- Legal & Registration
  trade_license VARCHAR(100),
  tin_number VARCHAR(50),
  bin_number VARCHAR(50),
  dghs_license VARCHAR(100),

  -- Branding
  logo_url TEXT,
  favicon_url TEXT,
  primary_color VARCHAR(10),
  secondary_color VARCHAR(10),

  -- Settings (JSONB for flexibility)
  settings JSONB DEFAULT '{}',
  features JSONB DEFAULT '{}',

  -- Subscription
  subscription_plan VARCHAR(50) DEFAULT 'basic',
  subscription_start DATE,
  subscription_end DATE,
  max_branches INTEGER DEFAULT 1,
  max_users INTEGER DEFAULT 50,

  -- Status
  is_active BOOLEAN DEFAULT TRUE,

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        created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
        updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
        created_by UUID,
        updated_by UUID
    );

-- Branches
CREATE TABLE branches (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),
    code VARCHAR(20) NOT NULL,
    name VARCHAR(255) NOT NULL,
    name_bangla VARCHAR(255),
    branch_type VARCHAR(50) DEFAULT 'hospital',

    -- Contact
    email VARCHAR(255),
    phone VARCHAR(20),
    emergency_phone VARCHAR(20),

    -- Address
    address_line1 VARCHAR(255),
    address_line2 VARCHAR(255),
    city VARCHAR(100),
    district VARCHAR(100),
    division VARCHAR(100),
    postal_code VARCHAR(20),
    latitude DECIMAL(10, 8),
    longitude DECIMAL(11, 8),

    -- Licenses
    dghs_license VARCHAR(100),

    -- Capacity
    total_beds INTEGER DEFAULT 0,
    total_ots INTEGER DEFAULT 0,

    -- Settings
    settings JSONB DEFAULT '{}',
    timezone VARCHAR(50) DEFAULT 'Asia/Dhaka',

    is_main_branch BOOLEAN DEFAULT FALSE,
    is_active BOOLEAN DEFAULT TRUE,
    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
    created_by UUID,
    updated_by UUID,

    UNIQUE(organization_id, code)
);

-- Departments
CREATE TABLE departments (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),

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organization_id UUID NOT NULL REFERENCES organizations(id),
branch_id UUID REFERENCES branches(id),
parent_department_id UUID REFERENCES departments(id),

code VARCHAR(20) NOT NULL,
name VARCHAR(255) NOT NULL,
name_bangla VARCHAR(255),
department_type VARCHAR(50), -- clinical, administrative, support

head_of_department_id UUID,

-- Contact
email VARCHAR(255),
phone VARCHAR(20),
extension VARCHAR(10),
location VARCHAR(255),

-- Settings
settings JSONB DEFAULT '{}',
color_code VARCHAR(10),
icon VARCHAR(50),

display_order INTEGER DEFAULT 0,
is_clinical BOOLEAN DEFAULT FALSE,
is_active BOOLEAN DEFAULT TRUE,
created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

UNIQUE(organization_id, code)
);

-- =====
-- SECTION 3: USER & ACCESS CONTROL
-- =====

-- Roles
CREATE TABLE roles (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID REFERENCES organizations(id), -- NULL for system roles

    code VARCHAR(50) NOT NULL,
    name VARCHAR(100) NOT NULL,
    description TEXT,

    role_type VARCHAR(50) DEFAULT 'custom', -- system, custom
    role_level INTEGER DEFAULT 0, -- hierarchy level

    parent_role_id UUID REFERENCES roles(id),

    is_system_role BOOLEAN DEFAULT FALSE,
    is_active BOOLEAN DEFAULT TRUE,
    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

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-- Permissions
CREATE TABLE permissions (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),

    module VARCHAR(100) NOT NULL,
    sub_module VARCHAR(100),
    action VARCHAR(50) NOT NULL, -- create, read, update, delete, export, approve,
    etc.

    code VARCHAR(150) NOT NULL UNIQUE,
    name VARCHAR(255) NOT NULL,
    description TEXT,

    is_active BOOLEAN DEFAULT TRUE,
    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Role Permissions Mapping
CREATE TABLE role_permissions (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    role_id UUID NOT NULL REFERENCES roles(id) ON DELETE CASCADE,
    permission_id UUID NOT NULL REFERENCES permissions(id) ON DELETE CASCADE,

    -- Scope restrictions
    branch_restricted BOOLEAN DEFAULT FALSE,
    department_restricted BOOLEAN DEFAULT FALSE,
    own_records_only BOOLEAN DEFAULT FALSE,

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
    created_by UUID,

    UNIQUE(role_id, permission_id)
);

-- Users
CREATE TABLE users (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),

    -- Authentication
    username VARCHAR(100) NOT NULL,
    email VARCHAR(255),
    phone VARCHAR(20),
    password_hash VARCHAR(255) NOT NULL,

    -- Profile
    first_name VARCHAR(100) NOT NULL,
    last_name VARCHAR(100),
    full_name VARCHAR(255) GENERATED ALWAYS AS (first_name || ' ' ||
COALESCE(last_name, '')) STORED,
    name_bangla VARCHAR(255),

    gender gender_type,

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    date_of_birth DATE,
    photo_url TEXT,

    -- Employment (if staff)
    employee_id VARCHAR(50),
    designation VARCHAR(100),
    department_id UUID REFERENCES departments(id),

    -- Security
    is_email_verified BOOLEAN DEFAULT FALSE,
    is_phone_verified BOOLEAN DEFAULT FALSE,
    two_factor_enabled BOOLEAN DEFAULT FALSE,
    two_factor_secret VARCHAR(255),

    password_changed_at TIMESTAMPTZ,
    must_change_password BOOLEAN DEFAULT FALSE,
    failed_login_attempts INTEGER DEFAULT 0,
    locked_until TIMESTAMPTZ,

    -- Status
    is_active BOOLEAN DEFAULT TRUE,
    last_login_at TIMESTAMPTZ,
    last_login_ip INET,

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
    created_by UUID,

    UNIQUE(organization_id, username),
    UNIQUE(organization_id, email)
);

-- User Roles Mapping
CREATE TABLE user_roles (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    user_id UUID NOT NULL REFERENCES users(id) ON DELETE CASCADE,
    role_id UUID NOT NULL REFERENCES roles(id) ON DELETE CASCADE,

    -- Scope
    branch_id UUID REFERENCES branches(id),
    department_id UUID REFERENCES departments(id),

    is_primary BOOLEAN DEFAULT FALSE,
    valid_from DATE,
    valid_until DATE,

    is_active BOOLEAN DEFAULT TRUE,
    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
    created_by UUID,

    UNIQUE(user_id, role_id, branch_id, department_id)
);

-- User Branch Access

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CREATE TABLE user_branch_access (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  user_id UUID NOT NULL REFERENCES users(id) ON DELETE CASCADE,
  branch_id UUID NOT NULL REFERENCES branches(id) ON DELETE CASCADE,

  is_default BOOLEAN DEFAULT FALSE,

  created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
  created_by UUID,

  UNIQUE(user_id, branch_id)
);
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-- User Sessions
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CREATE TABLE user_sessions (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  user_id UUID NOT NULL REFERENCES users(id) ON DELETE CASCADE,

  session_token VARCHAR(500) NOT NULL UNIQUE,
  refresh_token VARCHAR(500),

  device_type VARCHAR(50),
  device_name VARCHAR(255),
  browser VARCHAR(100),
  os VARCHAR(100),
  ip_address INET,
  user_agent TEXT,

  location_city VARCHAR(100),
  location_country VARCHAR(100),

  is_active BOOLEAN DEFAULT TRUE,
  expires_at TIMESTAMPTZ NOT NULL,
  last_activity_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

  created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);
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-- SECTION 4: AUDIT LOGGING
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-- Audit Logs
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CREATE TABLE audit_logs (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID REFERENCES organizations(id),
  branch_id UUID REFERENCES branches(id),

  -- User Info
  user_id UUID REFERENCES users(id),
  username VARCHAR(100),
  ip_address INET,
  user_agent TEXT,
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-- Action Details
action VARCHAR(50) NOT NULL, -- CREATE, UPDATE, DELETE, LOGIN, LOGOUT, VIEW,
EXPORT, etc.
module VARCHAR(100),
sub_module VARCHAR(100),

-- Entity Info
table_name VARCHAR(100),
record_id UUID,

-- Data Changes
old_values JSONB,
new_values JSONB,
changed_fields TEXT[],

-- Context
description TEXT,
metadata JSONB,

created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Create hypertable for audit_logs if using TimescaleDB
-- SELECT create_hypertable('audit_logs', 'created_at', if_not_exists => TRUE);

-- =====
-- SECTION 5: MASTER DATA
-- =====

-- Countries
CREATE TABLE countries (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  code VARCHAR(3) UNIQUE NOT NULL,
  name VARCHAR(100) NOT NULL,
  name_bangla VARCHAR(100),
  phone_code VARCHAR(10),
  currency_code VARCHAR(3),
  is_active BOOLEAN DEFAULT TRUE
);

-- Divisions (Bangladesh specific)
CREATE TABLE divisions (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  country_id UUID REFERENCES countries(id),
  code VARCHAR(10) NOT NULL,
  name VARCHAR(100) NOT NULL,
  name_bangla VARCHAR(100),
  is_active BOOLEAN DEFAULT TRUE
);

-- Districts
CREATE TABLE districts (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  division_id UUID REFERENCES divisions(id),

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        code VARCHAR(10) NOT NULL,
        name VARCHAR(100) NOT NULL,
        name_bangla VARCHAR(100),
        is_active BOOLEAN DEFAULT TRUE
    );

-- Upazilas/Thanas
CREATE TABLE upazilas (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    district_id UUID REFERENCES districts(id),
    code VARCHAR(10),
    name VARCHAR(100) NOT NULL,
    name_bangla VARCHAR(100),
    is_active BOOLEAN DEFAULT TRUE
);

-- ICD-10 Diagnosis Codes
CREATE TABLE icd_codes (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    code VARCHAR(20) UNIQUE NOT NULL,
    short_description VARCHAR(255) NOT NULL,
    long_description TEXT,
    category VARCHAR(100),
    chapter VARCHAR(100),
    is_billable BOOLEAN DEFAULT TRUE,
    is_active BOOLEAN DEFAULT TRUE
);

-- CPT Procedure Codes
CREATE TABLE cpt_codes (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    code VARCHAR(20) UNIQUE NOT NULL,
    description VARCHAR(500) NOT NULL,
    category VARCHAR(100),
    base_rate DECIMAL(12, 2),
    is_active BOOLEAN DEFAULT TRUE
);

-- Drug Database (DGDA)
CREATE TABLE drug_master (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),

    generic_name VARCHAR(255) NOT NULL,
    brand_name VARCHAR(255),
    manufacturer VARCHAR(255),

    drug_type VARCHAR(50), -- tablet, capsule, syrup, injection, etc.
    strength VARCHAR(100),
    unit VARCHAR(50),

    therapeutic_class VARCHAR(100),
    pharmacological_class VARCHAR(100),

    -- Regulatory

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    dgda_registration VARCHAR(100),
    is_controlled BOOLEAN DEFAULT FALSE,
    is_narcotic BOOLEAN DEFAULT FALSE,
    schedule_type VARCHAR(20),

    -- Pricing
    mrp DECIMAL(10, 2),

    -- Clinical
    route_of_administration VARCHAR(50),
    pregnancy_category VARCHAR(10),

    -- Flags
    is_generic BOOLEAN DEFAULT FALSE,
    requires_prescription BOOLEAN DEFAULT TRUE,

    is_active BOOLEAN DEFAULT TRUE,
    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Drug Interactions
CREATE TABLE drug_interactions (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    drug_id_1 UUID NOT NULL REFERENCES drug_master(id),
    drug_id_2 UUID NOT NULL REFERENCES drug_master(id),

    severity VARCHAR(20) NOT NULL, -- mild, moderate, severe, contraindicated
    description TEXT,
    clinical_effect TEXT,
    management TEXT,

    is_active BOOLEAN DEFAULT TRUE,

    UNIQUE(drug_id_1, drug_id_2)
);

-- Allergens
CREATE TABLE allergens (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    code VARCHAR(50),
    name VARCHAR(255) NOT NULL,
    category VARCHAR(50), -- drug, food, environment, other
    description TEXT,
    is_active BOOLEAN DEFAULT TRUE
);

-- =====
-- SECTION 6: INDEXES FOR FOUNDATION TABLES
-- =====

CREATE INDEX idx_organizations_code ON organizations(code);
CREATE INDEX idx_organizations_active ON organizations(is_active) WHERE is_active
= TRUE;

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CREATE INDEX idx_branches_org ON branches(organization_id);
CREATE INDEX idx_branches_code ON branches(organization_id, code);
CREATE INDEX idx_branches_active ON branches(is_active) WHERE is_active = TRUE;

CREATE INDEX idx_departments_org ON departments(organization_id);
CREATE INDEX idx_departments_branch ON departments(branch_id);
CREATE INDEX idx_departments_parent ON departments(parent_department_id);

CREATE INDEX idx_users_org ON users(organization_id);
CREATE INDEX idx_users_username ON users(organization_id, username);
CREATE INDEX idx_users_email ON users(organization_id, email);
CREATE INDEX idx_users_employee ON users(employee_id);
CREATE INDEX idx_users_department ON users(department_id);

CREATE INDEX idx_user_roles_user ON user_roles(user_id);
CREATE INDEX idx_user_roles_role ON user_roles(role_id);

CREATE INDEX idx_audit_logs_org ON audit_logs(organization_id);
CREATE INDEX idx_audit_logs_user ON audit_logs(user_id);
CREATE INDEX idx_audit_logs_table ON audit_logs(table_name, record_id);
CREATE INDEX idx_audit_logs_created ON audit_logs(created_at DESC);
CREATE INDEX idx_audit_logs_action ON audit_logs(action);

CREATE INDEX idx_drug_master_generic ON drug_master(generic_name);
CREATE INDEX idx_drug_master_brand ON drug_master(brand_name);
CREATE INDEX idx_drug_master_search ON drug_master USING gin(
    (generic_name || ' ' || COALESCE(brand_name, '')) gin_trgm_ops
);

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PART 2: PATIENT MANAGEMENT

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-- =====
-- SECTION 7: PATIENT MANAGEMENT
-- =====

-- Patients
CREATE TABLE patients (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),

    -- Unique Identifiers
    mrn VARCHAR(50) NOT NULL, -- Medical Record Number
    old_mrn VARCHAR(50), -- For migration

    -- Basic Info
    title VARCHAR(20), -- Mr, Mrs, Dr, etc.
    first_name VARCHAR(100) NOT NULL,
    middle_name VARCHAR(100),
    last_name VARCHAR(100),
    full_name VARCHAR(255) GENERATED ALWAYS AS (
        COALESCE(title || ' ', '') || first_name ||

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        COALESCE(' ' || middle_name, '') ||
        COALESCE(' ' || last_name, '')
    ) STORED,
    name_bangla VARCHAR(255),

-- Demographics
    gender gender_type NOT NULL,
    date_of_birth DATE,
    blood_group blood_group_type,
    marital_status marital_status_type,

-- Identity Documents
    nid VARCHAR(20),
    birth_certificate_no VARCHAR(50),
    passport_no VARCHAR(50),
    passport_expiry DATE,

-- Contact
    primary_phone VARCHAR(20) NOT NULL,
    secondary_phone VARCHAR(20),
    email VARCHAR(255),
    emergency_contact_name VARCHAR(255),
    emergency_contact_phone VARCHAR(20),
    emergency_contact_relation VARCHAR(50),

-- Address - Present
    present_address_line1 VARCHAR(255),
    present_address_line2 VARCHAR(255),
    present_city VARCHAR(100),
    present_district VARCHAR(100),
    present_division VARCHAR(100),
    present_postal_code VARCHAR(20),
    present_country VARCHAR(100) DEFAULT 'Bangladesh',

-- Address - Permanent
    permanent_address_line1 VARCHAR(255),
    permanent_address_line2 VARCHAR(255),
    permanent_city VARCHAR(100),
    permanent_district VARCHAR(100),
    permanent_division VARCHAR(100),
    permanent_postal_code VARCHAR(20),
    permanent_country VARCHAR(100) DEFAULT 'Bangladesh',

-- Occupation & Social
    occupation VARCHAR(100),
    employer VARCHAR(255),
    religion VARCHAR(50),
    nationality VARCHAR(100) DEFAULT 'Bangladeshi',

-- Photo
    photo_url TEXT,

-- Tags/Flags
    is_vip BOOLEAN DEFAULT FALSE,

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is_corporate BOOLEAN DEFAULT FALSE,
corporate_id UUID, -- Reference to corporate company
is_foreign BOOLEAN DEFAULT FALSE,
is_medico_legal BOOLEAN DEFAULT FALSE,

-- Family Linking
family_id UUID,
relationship_to_head VARCHAR(50),

-- Registration Info
registration_date DATE DEFAULT CURRENT_DATE,
registered_branch_id UUID REFERENCES branches(id),
referred_by VARCHAR(255),
referral_source VARCHAR(100),

-- Consent
consent_for_sms BOOLEAN DEFAULT TRUE,
consent_for_email BOOLEAN DEFAULT TRUE,
consent_for_data_sharing BOOLEAN DEFAULT FALSE,

-- Portal Access
portal_enabled BOOLEAN DEFAULT FALSE,
portal_username VARCHAR(100),
portal_password_hash VARCHAR(255),
portal_last_login TIMESTAMPTZ,

-- Status
is_active BOOLEAN DEFAULT TRUE,
is_deceased BOOLEAN DEFAULT FALSE,
deceased_date DATE,

-- Metadata
metadata JSONB DEFAULT '{}',

created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
created_by UUID REFERENCES users(id),
updated_by UUID REFERENCES users(id),

UNIQUE(organization_id, mrn)
);

-- Patient QR/Barcode
CREATE TABLE patient_identifiers (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  patient_id UUID NOT NULL REFERENCES patients(id) ON DELETE CASCADE,

  identifier_type VARCHAR(50) NOT NULL, -- qr_code, barcode, rfid, wristband
  identifier_value VARCHAR(255) NOT NULL,

  issued_date DATE DEFAULT CURRENT_DATE,
  expiry_date DATE,
  is_active BOOLEAN DEFAULT TRUE,

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        created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
    );

-- Patient Medical History
CREATE TABLE patient_medical_history (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    patient_id UUID NOT NULL REFERENCES patients(id) ON DELETE CASCADE,

    history_type VARCHAR(50) NOT NULL, -- pmh, psh, family, social

    -- For medical/surgical history
    condition_name VARCHAR(255),
    icd_code_id UUID REFERENCES icd_codes(id),
    diagnosis_date DATE,
    is_current BOOLEAN DEFAULT TRUE,
    notes TEXT,

    -- For surgical history
    surgery_name VARCHAR(255),
    surgery_date DATE,
    surgeon VARCHAR(255),
    hospital VARCHAR(255),

    -- For family history
    relative_relation VARCHAR(50),
    relative_condition VARCHAR(255),

    -- For social history
    smoking_status VARCHAR(50), -- never, former, current
    smoking_years INTEGER,
    packs_per_day DECIMAL(4,2),
    alcohol_status VARCHAR(50),
    alcohol_frequency VARCHAR(50),
    occupation_hazards TEXT,

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
    created_by UUID REFERENCES users(id)
);

-- Patient Allergies
CREATE TABLE patient_allergies (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    patient_id UUID NOT NULL REFERENCES patients(id) ON DELETE CASCADE,

    allergen_id UUID REFERENCES allergens(id),
    allergen_name VARCHAR(255) NOT NULL,
    allergy_type VARCHAR(50), -- drug, food, environment, other

    reaction TEXT,
    severity VARCHAR(20), -- mild, moderate, severe, life_threatening

    onset_date DATE,
    is_confirmed BOOLEAN DEFAULT FALSE,
    confirmed_by UUID REFERENCES users(id),

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confirmed_date DATE,

notes TEXT,

is_active BOOLEAN DEFAULT TRUE,
created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
created_by UUID REFERENCES users(id)
);

-- Patient Current Medications
CREATE TABLE patient_medications (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  patient_id UUID NOT NULL REFERENCES patients(id) ON DELETE CASCADE,

  drug_id UUID REFERENCES drug_master(id),
  medication_name VARCHAR(255) NOT NULL,
  dose VARCHAR(100),
  frequency VARCHAR(100),
  route VARCHAR(50),

  start_date DATE,
  end_date DATE,

  prescriber VARCHAR(255),
  prescriber_facility VARCHAR(255),

  purpose TEXT,
  notes TEXT,

  is_current BOOLEAN DEFAULT TRUE,
  created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
  created_by UUID REFERENCES users(id)
);

-- Patient Chronic Conditions
CREATE TABLE patient_chronic_conditions (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  patient_id UUID NOT NULL REFERENCES patients(id) ON DELETE CASCADE,

  condition_name VARCHAR(255) NOT NULL,
  icd_code_id UUID REFERENCES icd_codes(id),

  diagnosis_date DATE,
  diagnosed_by VARCHAR(255),

  severity VARCHAR(20),
  is_controlled BOOLEAN,

  treatment_plan TEXT,
  notes TEXT,

  is_active BOOLEAN DEFAULT TRUE,
  created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
  created_by UUID REFERENCES users(id)
```



```

);

-- Patient Immunizations
CREATE TABLE patient_immunizations (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  patient_id UUID NOT NULL REFERENCES patients(id) ON DELETE CASCADE,

  vaccine_name VARCHAR(255) NOT NULL,
  dose_number INTEGER,

  administration_date DATE NOT NULL,
  administered_by VARCHAR(255),
  administration_site VARCHAR(100),

  lot_number VARCHAR(100),
  manufacturer VARCHAR(255),
  expiry_date DATE,

  next_due_date DATE,

  notes TEXT,

  created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
  created_by UUID REFERENCES users(id)
);

-- Patient Documents
CREATE TABLE patient_documents (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  patient_id UUID NOT NULL REFERENCES patients(id) ON DELETE CASCADE,

  document_type document_type NOT NULL,
  document_category VARCHAR(50),

  title VARCHAR(255) NOT NULL,
  description TEXT,

  file_name VARCHAR(255) NOT NULL,
  file_path TEXT NOT NULL,
  file_size BIGINT,
  mime_type VARCHAR(100),

  upload_source VARCHAR(50), -- hospital, external, patient_portal
  source_hospital VARCHAR(255),

  document_date DATE,

  is_confidential BOOLEAN DEFAULT FALSE,

  created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
  created_by UUID REFERENCES users(id)
);

-- Patient Family Group

```

```

CREATE TABLE patient_families (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),

  family_code VARCHAR(50) NOT NULL,
  family_name VARCHAR(255),
  head_patient_id UUID REFERENCES patients(id),

  address TEXT,
  phone VARCHAR(20),

  created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

  UNIQUE(organization_id, family_code)
);

-- Patient Death Registry
CREATE TABLE patient_death_records (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  patient_id UUID NOT NULL REFERENCES patients(id),
  organization_id UUID NOT NULL REFERENCES organizations(id),
  branch_id UUID NOT NULL REFERENCES branches(id),

  death_datetime TIMESTAMPTZ NOT NULL,
  place_of_death VARCHAR(255),

  primary_cause VARCHAR(500),
  secondary_causes TEXT[],

  -- Death certificate
  certificate_number VARCHAR(100),
  certificate_issued_date DATE,
  certificate_issued_by UUID REFERENCES users(id),

  -- Certifying physician
  certifying_doctor_id UUID REFERENCES users(id),

  -- Autopsy
  autopsy_performed BOOLEAN DEFAULT FALSE,
  autopsy_findings TEXT,

  -- For MLC cases
  is_medico_legal BOOLEAN DEFAULT FALSE,
  police_station VARCHAR(255),
  police_case_no VARCHAR(100),

  notes TEXT,

  created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
  created_by UUID REFERENCES users(id)
);

-- Patient Satisfaction Surveys
CREATE TABLE patient_surveys (

```

```

id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
organization_id UUID NOT NULL REFERENCES organizations(id),
patient_id UUID NOT NULL REFERENCES patients(id),
visit_id UUID, -- Reference to OPD visit or admission

survey_type VARCHAR(50), -- discharge, opd_visit, post_procedure

-- Ratings (1-5)
overall_rating INTEGER CHECK (overall_rating BETWEEN 1 AND 5),
doctor_rating INTEGER CHECK (doctor_rating BETWEEN 1 AND 5),
nurse_rating INTEGER CHECK (nurse_rating BETWEEN 1 AND 5),
facility_rating INTEGER CHECK (facility_rating BETWEEN 1 AND 5),
cleanliness_rating INTEGER CHECK (cleanliness_rating BETWEEN 1 AND 5),
food_rating INTEGER CHECK (food_rating BETWEEN 1 AND 5),
billing_rating INTEGER CHECK (billing_rating BETWEEN 1 AND 5),
wait_time_rating INTEGER CHECK (wait_time_rating BETWEEN 1 AND 5),

-- NPS
nps_score INTEGER CHECK (nps_score BETWEEN 0 AND 10),

-- Free text
positive_feedback TEXT,
improvement_suggestions TEXT,

-- Metadata
submitted_via VARCHAR(50), -- app, sms, call, kiosk
submitted_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Indexes for Patient tables
CREATE INDEX idx_patients_org ON patients(organization_id);
CREATE INDEX idx_patients_mrn ON patients(organization_id, mrn);
CREATE INDEX idx_patients_phone ON patients(primary_phone);
CREATE INDEX idx_patients_nid ON patients(nid) WHERE nid IS NOT NULL;
CREATE INDEX idx_patients_name ON patients USING gin(full_name gin_trgm_ops);
CREATE INDEX idx_patients_name_bangla ON patients USING gin(name_bangla
gin_trgm_ops) WHERE name_bangla IS NOT NULL;
CREATE INDEX idx_patients_family ON patients(family_id) WHERE family_id IS NOT
NULL;
CREATE INDEX idx_patients_active ON patients(organization_id, is_active) WHERE
is_active = TRUE;

CREATE INDEX idx_patient_allergies_patient ON patient_allergies(patient_id);
CREATE INDEX idx_patient_allergies_active ON patient_allergies(patient_id,
is_active) WHERE is_active = TRUE;

CREATE INDEX idx_patient_medical_history_patient ON
patient_medical_history(patient_id);
CREATE INDEX idx_patient_medications_patient ON patient_medications(patient_id);
CREATE INDEX idx_patient_medications_current ON patient_medications(patient_id,
is_current) WHERE is_current = TRUE;

CREATE INDEX idx_patient_documents_patient ON patient_documents(patient_id);

```

```
CREATE INDEX idx_patient_documents_type ON patient_documents(patient_id,
document_type);
```

📁 PART 3: VISITS, APPOINTMENTS & ADMISSIONS

```
-- =====
-- SECTION 8: OPD VISITS & APPOINTMENTS
-- =====

-- Doctor Profiles
CREATE TABLE doctors (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  user_id UUID NOT NULL REFERENCES users(id),
  organization_id UUID NOT NULL REFERENCES organizations(id),

  -- Professional Info
  doctor_code VARCHAR(50),
  bmhc_registration VARCHAR(50),
  bmhc_expiry_date DATE,

  specialization VARCHAR(100),
  sub_specialization VARCHAR(100),

  qualification TEXT,
  experience_years INTEGER,

  -- Fees
  opd_fee DECIMAL(10, 2),
  followup_fee DECIMAL(10, 2),
  telemedicine_fee DECIMAL(10, 2),

  -- Settings
  consultation_duration INTEGER DEFAULT 15, -- minutes
  max_patients_per_slot INTEGER DEFAULT 1,

  -- Digital
  digital_signature_url TEXT,

  is_available_opd BOOLEAN DEFAULT TRUE,
  is_available_telemedicine BOOLEAN DEFAULT FALSE,
  is_visiting BOOLEAN DEFAULT FALSE,

  is_active BOOLEAN DEFAULT TRUE,
  created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
  updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

  UNIQUE(organization_id, doctor_code)
);

-- Doctor Schedule Templates
CREATE TABLE doctor_schedule_templates (
```

```

    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    doctor_id UUID NOT NULL REFERENCES doctors(id),
    branch_id UUID NOT NULL REFERENCES branches(id),

    day_of_week INTEGER NOT NULL CHECK (day_of_week BETWEEN 0 AND 6), -- 0=Sunday

    -- Session 1
    session1_start TIME,
    session1_end TIME,
    session1_max_patients INTEGER,
    session1_room VARCHAR(50),

    -- Session 2
    session2_start TIME,
    session2_end TIME,
    session2_max_patients INTEGER,
    session2_room VARCHAR(50),

    consultation_type VARCHAR(50) DEFAULT 'opd', -- opd, telemedicine

    is_active BOOLEAN DEFAULT TRUE,

    UNIQUE(doctor_id, branch_id, day_of_week, consultation_type)
);

-- Doctor Schedule Overrides (Leaves, Holidays)
CREATE TABLE doctor_schedule_overrides (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    doctor_id UUID NOT NULL REFERENCES doctors(id),
    branch_id UUID REFERENCES branches(id),

    override_date DATE NOT NULL,
    override_type VARCHAR(50) NOT NULL, -- leave, holiday, extra_session

    -- If extra session
    session_start TIME,
    session_end TIME,
    max_patients INTEGER,
    room VARCHAR(50),

    reason TEXT,

    is_active BOOLEAN DEFAULT TRUE,
    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
    created_by UUID REFERENCES users(id)
);

-- Appointments
CREATE TABLE appointments (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),
    branch_id UUID NOT NULL REFERENCES branches(id),

    -- Booking Info

```

```
appointment_number VARCHAR(50) NOT NULL,
token_number INTEGER,

-- Parties
patient_id UUID NOT NULL REFERENCES patients(id),
doctor_id UUID NOT NULL REFERENCES doctors(id),

-- Schedule
appointment_date DATE NOT NULL,
appointment_time TIME NOT NULL,
appointment_end_time TIME,
slot_duration INTEGER DEFAULT 15, -- minutes

-- Type
appointment_type VARCHAR(50) DEFAULT 'new', -- new, followup, review
consultation_type VARCHAR(50) DEFAULT 'opd', -- opd, telemedicine
visit_type patient_type DEFAULT 'opd',

-- Status
status visit_status_type DEFAULT 'scheduled',

-- Check-in/out
checked_in_at TIMESTAMPTZ,
consultation_start_at TIMESTAMPTZ,
consultation_end_at TIMESTAMPTZ,

-- Telemedicine specific
video_link TEXT,
meeting_password VARCHAR(50),

-- Referral
referred_by_doctor_id UUID REFERENCES doctors(id),
referral_notes TEXT,

-- Booking Details
booked_via VARCHAR(50), -- counter, app, web, call
booked_by UUID REFERENCES users(id),

-- Cancellation
cancelled_at TIMESTAMPTZ,
cancelled_by UUID REFERENCES users(id),
cancellation_reason TEXT,

-- Rescheduling
rescheduled_from UUID REFERENCES appointments(id),
rescheduled_to UUID REFERENCES appointments(id),

-- Payment
consultation_fee DECIMAL(10, 2),
payment_status payment_status_type DEFAULT 'pending',

notes TEXT,

created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
```

```

        updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
        created_by UUID REFERENCES users(id),

        UNIQUE(organization_id, appointment_number)
    );

-- OPD Visits (Consultation Records)
CREATE TABLE opd_visits (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),
    branch_id UUID NOT NULL REFERENCES branches(id),

    visit_number VARCHAR(50) NOT NULL,
    appointment_id UUID REFERENCES appointments(id),

    patient_id UUID NOT NULL REFERENCES patients(id),
    doctor_id UUID NOT NULL REFERENCES doctors(id),

    visit_date DATE NOT NULL,
    visit_time TIME,

    visit_type VARCHAR(50), -- new, followup, routine_checkup, emergency_opd

    -- Chief Complaint
    chief_complaint TEXT,
    complaint_duration VARCHAR(100),

    -- Vitals (can be recorded by nurse before consultation)
    vitals_recorded BOOLEAN DEFAULT FALSE,
    vitals_recorded_by UUID REFERENCES users(id),
    vitals_recorded_at TIMESTAMPTZ,

    -- Consultation
    consultation_start_at TIMESTAMPTZ,
    consultation_end_at TIMESTAMPTZ,
    consultation_duration INTEGER, -- minutes, auto-calculated

    -- Status
    status visit_status_type DEFAULT 'scheduled',

    -- Follow-up
    followup_required BOOLEAN DEFAULT FALSE,
    followup_date DATE,
    followup_notes TEXT,

    -- Admission Decision
    admission_advised BOOLEAN DEFAULT FALSE,

    -- MLC
    is_medico_legal BOOLEAN DEFAULT FALSE,
    mlc_number VARCHAR(50),

    notes TEXT,

```

```

        created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
        updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

        UNIQUE(organization_id, visit_number)
    );

-- =====
-- SECTION 9: IPD ADMISSIONS
-- =====

-- Admissions
CREATE TABLE admissions (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),
    branch_id UUID NOT NULL REFERENCES branches(id),

    -- Admission Info
    admission_number VARCHAR(50) NOT NULL,
    ip_number VARCHAR(50), -- Inpatient number

    patient_id UUID NOT NULL REFERENCES patients(id),

    -- Dates
    admission_date DATE NOT NULL,
    admission_time TIME NOT NULL,
    expected_discharge_date DATE,
    actual_discharge_date DATE,
    actual_discharge_time TIME,

    -- Type
    admission_type VARCHAR(50), -- planned, emergency, transfer_in, daycare
    admission_source VARCHAR(50), -- opd, emergency, direct, referral

    -- Status
    status admission_status_type DEFAULT 'admitted',

    -- Medical Team
    primary_doctor_id UUID REFERENCES doctors(id),
    admitting_doctor_id UUID REFERENCES doctors(id),
    consulting_doctors UUID[], -- Array of doctor IDs

    -- Department/Ward
    department_id UUID REFERENCES departments(id),
    admitting_diagnosis TEXT,

    -- Bed Info (current)
    current_ward_id UUID,
    current_bed_id UUID,

    -- Provisional Diagnosis
    provisional_diagnosis TEXT,
    icd_codes UUID[], -- Array of ICD code IDs

    -- Surgery Info

```



```

surgery_planned BOOLEAN DEFAULT FALSE,
surgery_date DATE,

-- MLC
is_medico_legal BOOLEAN DEFAULT FALSE,
mlc_number VARCHAR(50),
police_case_number VARCHAR(100),

-- Consent
general_consent_obtained BOOLEAN DEFAULT FALSE,

-- Attendant
attendant_name VARCHAR(255),
attendant_phone VARCHAR(20),
attendant_relation VARCHAR(50),
attendant_address TEXT,

-- Referral
referred_from VARCHAR(255),
referral_letter_url TEXT,

-- Package/Insurance
is_package BOOLEAN DEFAULT FALSE,
package_id UUID,
is_insurance BOOLEAN DEFAULT FALSE,
insurance_preauth_id UUID,

-- Discharge
discharge_type VARCHAR(50), -- normal, lama, dor, expired, transfer
discharge_summary_completed BOOLEAN DEFAULT FALSE,

notes TEXT,

created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
created_by UUID REFERENCES users(id),

UNIQUE(organization_id, admission_number)
);

-- Admission Deposits
CREATE TABLE admission_deposits (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  admission_id UUID NOT NULL REFERENCES admissions(id),

  deposit_number VARCHAR(50) NOT NULL,

  amount DECIMAL(12, 2) NOT NULL,
  payment_method payment_method_type NOT NULL,

  transaction_reference VARCHAR(100),

  deposit_date DATE DEFAULT CURRENT_DATE,
  deposited_by VARCHAR(255),

```

```

-- For refunds
is_refunded BOOLEAN DEFAULT FALSE,
refund_amount DECIMAL(12, 2),
refund_date DATE,
refund_reason TEXT,

notes TEXT,

created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
created_by UUID REFERENCES users(id)
);

-- Bed Allocations (History)
CREATE TABLE bed_allocations (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  admission_id UUID NOT NULL REFERENCES admissions(id),

  ward_id UUID NOT NULL,
  bed_id UUID NOT NULL,

  allocation_datetime TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
  release_datetime TIMESTAMPTZ,

  allocation_type VARCHAR(50), -- initial, transfer, upgrade, downgrade

  daily_rate DECIMAL(10, 2),

  allocated_by UUID REFERENCES users(id),
  released_by UUID REFERENCES users(id),

  notes TEXT
);

-- Discharge Summary
CREATE TABLE discharge_summaries (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  admission_id UUID NOT NULL REFERENCES admissions(id),

  -- Dates
  discharge_date DATE NOT NULL,
  discharge_time TIME,

  -- Diagnoses
  admission_diagnosis TEXT,
  final_diagnosis TEXT,
  icd_codes UUID[],

  -- Procedures
  procedures_performed TEXT,
  cpt_codes UUID[],

  -- Treatment Summary
  treatment_given TEXT,

```

```

course_in_hospital TEXT,

-- Condition at Discharge
condition_at_discharge VARCHAR(50), -- improved, unchanged, deteriorated,
expired

-- Instructions
discharge_advice TEXT,
diet_instructions TEXT,
activity_restrictions TEXT,
wound_care_instructions TEXT,

-- Medications
discharge_medications JSONB,

-- Follow-up
followup_date DATE,
followup_instructions TEXT,

-- References
investigations_pending TEXT,
referral_to VARCHAR(255),

-- Prepared By
prepared_by UUID REFERENCES users(id),
approved_by UUID REFERENCES doctors(id),
prepared_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

-- PDF
summary_pdf_url TEXT
);

-- Emergency Visits
CREATE TABLE emergency_visits (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),
  branch_id UUID NOT NULL REFERENCES branches(id),

  emergency_number VARCHAR(50) NOT NULL,

  patient_id UUID NOT NULL REFERENCES patients(id),

  arrival_datetime TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
  arrival_mode VARCHAR(50), -- walk_in, ambulance, police, referred

  -- Triage
  triage_level VARCHAR(20), -- resuscitation, emergency, urgent, standard,
non_urgent
  triage_color VARCHAR(20), -- red, orange, yellow, green, blue
  triage_time TIMESTAMPTZ,
  triaged_by UUID REFERENCES users(id),

  -- Chief Complaint
  chief_complaint TEXT,

```

```

presenting_symptoms TEXT,

-- Vitals at arrival
initial_vitals JSONB,

-- Attending
attending_doctor_id UUID REFERENCES doctors(id),

-- Status
status VARCHAR(50), -- waiting, under_treatment, observation, admitted,
discharged, expired

-- Disposition
disposition VARCHAR(50), -- discharged, admitted, referred, lama, expired
disposition_time TIMESTAMPTZ,

-- If admitted
admission_id UUID REFERENCES admissions(id),

-- MLC
is_medico_legal BOOLEAN DEFAULT FALSE,
mlc_number VARCHAR(50),
brought_by_police BOOLEAN DEFAULT FALSE,
police_officer_name VARCHAR(255),
police_station VARCHAR(255),

notes TEXT,

created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

UNIQUE(organization_id, emergency_number)
);

-- Indexes
CREATE INDEX idx_appointments_org ON appointments(organization_id);
CREATE INDEX idx_appointments_branch ON appointments(branch_id);
CREATE INDEX idx_appointments_patient ON appointments(patient_id);
CREATE INDEX idx_appointments_doctor ON appointments(doctor_id);
CREATE INDEX idx_appointments_date ON appointments(appointment_date);
CREATE INDEX idx_appointments_status ON appointments(status);
CREATE INDEX idx_appointments_doctor_date ON appointments(doctor_id,
appointment_date);

CREATE INDEX idx_opd_visits_org ON opd_visits(organization_id);
CREATE INDEX idx_opd_visits_patient ON opd_visits(patient_id);
CREATE INDEX idx_opd_visits_doctor ON opd_visits(doctor_id);
CREATE INDEX idx_opd_visits_date ON opd_visits(visit_date);

CREATE INDEX idx_admissions_org ON admissions(organization_id);
CREATE INDEX idx_admissions_patient ON admissions(patient_id);
CREATE INDEX idx_admissions_status ON admissions(status);
CREATE INDEX idx_admissions_date ON admissions(admission_date);
CREATE INDEX idx_admissions_doctor ON admissions(primary_doctor_id);

```

```
CREATE INDEX idx_admissions_active ON admissions(organization_id, status) WHERE
status = 'admitted';

CREATE INDEX idx_emergency_visits_org ON emergency_visits(organization_id);
CREATE INDEX idx_emergency_visits_patient ON emergency_visits(patient_id);
CREATE INDEX idx_emergency_visits_arrival ON emergency_visits(arrival_datetime);
```

PART 4: EMR & CLINICAL DOCUMENTATION

SQL

```
-- =====
-- SECTION 10: EMR - ELECTRONIC MEDICAL RECORDS
-- =====

-- Vitals
CREATE TABLE patient_vitals (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),

  patient_id UUID NOT NULL REFERENCES patients(id),

  -- Context
  visit_id UUID, -- OPD visit
  admission_id UUID REFERENCES admissions(id), -- IPD
  emergency_id UUID REFERENCES emergency_visits(id),

  recorded_datetime TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
  recorded_by UUID NOT NULL REFERENCES users(id),

  -- Vital Signs
  temperature DECIMAL(4, 1), -- Celsius
  temperature_site VARCHAR(20), -- oral, axillary, rectal, tympanic

  pulse_rate INTEGER, -- bpm
  pulse_rhythm VARCHAR(20), -- regular, irregular
  pulse_volume VARCHAR(20), -- normal, weak, bounding

  respiratory_rate INTEGER, -- per minute

  blood_pressure_systolic INTEGER, -- mmHg
  blood_pressure_diastolic INTEGER, -- mmHg
  blood_pressure_position VARCHAR(20), -- sitting, lying, standing

  spo2 INTEGER, -- percentage
  spo2_on_oxygen BOOLEAN DEFAULT FALSE,
  oxygen_flow_rate DECIMAL(4, 1), -- L/min

  -- Measurements
  height DECIMAL(5, 2), -- cm
```

```

weight DECIMAL(5, 2), -- kg
bmi DECIMAL(4, 1), -- auto-calculated
bsa DECIMAL(4, 2), -- body surface area

head_circumference DECIMAL(4, 1), -- cm (for pediatrics)
waist_circumference DECIMAL(5, 1),
hip_circumference DECIMAL(5, 1),

-- Blood Sugar
blood_sugar DECIMAL(5, 1), -- mg/dL
blood_sugar_type VARCHAR(20), -- fasting, random, pp

-- Pain
pain_score INTEGER CHECK (pain_score BETWEEN 0 AND 10),
pain_location VARCHAR(255),

-- Neurological
gcs_eye INTEGER CHECK (gcs_eye BETWEEN 1 AND 4),
gcs_verbal INTEGER CHECK (gcs_verbal BETWEEN 1 AND 5),
gcs_motor INTEGER CHECK (gcs_motor BETWEEN 1 AND 6),
gcs_total INTEGER GENERATED ALWAYS AS (gcs_eye + gcs_verbal + gcs_motor)
STORED,

pupil_left VARCHAR(50),
pupil_right VARCHAR(50),

-- Assessment Scores
fall_risk_score INTEGER,
pressure_ulcer_risk INTEGER, -- Braden scale

notes TEXT,

created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Clinical Notes
CREATE TABLE clinical_notes (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),

    patient_id UUID NOT NULL REFERENCES patients(id),

    -- Context
    visit_id UUID,
    admission_id UUID REFERENCES admissions(id),
    emergency_id UUID REFERENCES emergency_visits(id),

    note_type VARCHAR(50) NOT NULL,
    -- consultation_note, progress_note, nursing_note, procedure_note,
    -- handoff_note, discharge_note, telephone_note

    note_datetime TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
    author_id UUID NOT NULL REFERENCES users(id),

```

```

-- SOAP Format
subjective TEXT,
objective TEXT,
assessment TEXT,
plan TEXT,

-- Or free text
note_text TEXT,

-- Structured data
structured_data JSONB,

-- Signing
is_signed BOOLEAN DEFAULT FALSE,
signed_by UUID REFERENCES users(id),
signed_at TIMESTAMPTZ,

-- Co-sign (for residents)
requires_cosign BOOLEAN DEFAULT FALSE,
cosigned_by UUID REFERENCES users(id),
cosigned_at TIMESTAMPTZ,

-- Amendments
is_amended BOOLEAN DEFAULT FALSE,
amendment_reason TEXT,
amended_at TIMESTAMPTZ,

-- Confidential
is_confidential BOOLEAN DEFAULT FALSE,

created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Diagnoses (for visits/admissions)
CREATE TABLE clinical_diagnoses (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),

    patient_id UUID NOT NULL REFERENCES patients(id),

    visit_id UUID,
    admission_id UUID REFERENCES admissions(id),

    icd_code_id UUID REFERENCES icd_codes(id),
    diagnosis_code VARCHAR(20),
    diagnosis_description VARCHAR(500) NOT NULL,

    diagnosis_type VARCHAR(50), -- primary, secondary, admitting, discharge
    is_primary BOOLEAN DEFAULT FALSE,

    onset_date DATE,
    resolution_date DATE,

    diagnosed_by UUID REFERENCES users(id),

```

```

    diagnosed_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

    is_chronic BOOLEAN DEFAULT FALSE,
    is_resolved BOOLEAN DEFAULT FALSE,

    notes TEXT
);

-- Procedures Performed
CREATE TABLE clinical_procedures (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),

    patient_id UUID NOT NULL REFERENCES patients(id),

    visit_id UUID,
    admission_id UUID REFERENCES admissions(id),
    surgery_id UUID, -- Reference to OT surgery

    cpt_code_id UUID REFERENCES cpt_codes(id),
    procedure_code VARCHAR(20),
    procedure_name VARCHAR(500) NOT NULL,

    procedure_datetime TIMESTAMPTZ NOT NULL,
    performed_by UUID REFERENCES users(id),
    assistant_ids UUID[],

    -- Location
    location VARCHAR(100),
    laterality VARCHAR(20), -- left, right, bilateral

    -- Anesthesia
    anesthesia_type VARCHAR(50),
    anesthesia_by UUID REFERENCES users(id),

    -- Details
    indications TEXT,
    procedure_description TEXT,
    findings TEXT,
    complications TEXT,

    -- Duration
    start_time TIMESTAMPTZ,
    end_time TIMESTAMPTZ,
    duration_minutes INTEGER,

    -- Specimens
    specimen_sent BOOLEAN DEFAULT FALSE,
    specimen_details TEXT,

    notes TEXT,

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
    created_by UUID REFERENCES users(id)

```



```

);

-- Doctor Templates
CREATE TABLE doctor_templates (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),
  doctor_id UUID REFERENCES doctors(id), -- NULL for organization-wide

  template_type VARCHAR(50) NOT NULL, -- chief_complaint, examination,
  diagnosis, prescription, advice
  category VARCHAR(100),

  name VARCHAR(255) NOT NULL,
  shortcut VARCHAR(50),

  content TEXT NOT NULL,
  structured_content JSONB,

  is_active BOOLEAN DEFAULT TRUE,
  usage_count INTEGER DEFAULT 0,

  created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
  updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
  created_by UUID REFERENCES users(id)
);

-- Indexes for EMR
CREATE INDEX idx_vitals_patient ON patient_vitals(patient_id);
CREATE INDEX idx_vitals_admission ON patient_vitals(admission_id);
CREATE INDEX idx_vitals_datetime ON patient_vitals(recorded_datetime DESC);

CREATE INDEX idx_clinical_notes_patient ON clinical_notes(patient_id);
CREATE INDEX idx_clinical_notes_admission ON clinical_notes(admission_id);
CREATE INDEX idx_clinical_notes_type ON clinical_notes(note_type);
CREATE INDEX idx_clinical_notes_datetime ON clinical_notes(note_datetime DESC);

CREATE INDEX idx_diagnoses_patient ON clinical_diagnoses(patient_id);
CREATE INDEX idx_diagnoses_admission ON clinical_diagnoses(admission_id);
CREATE INDEX idx_diagnoses_icd ON clinical_diagnoses(icd_code_id);

CREATE INDEX idx_procedures_patient ON clinical_procedures(patient_id);
CREATE INDEX idx_procedures_admission ON clinical_procedures(admission_id);

```

PART 5: PRESCRIPTION (Rx) MANAGEMENT

SQL

```

-- =====
-- SECTION 11: PRESCRIPTION MANAGEMENT
-- =====

```

```

-- Prescriptions
CREATE TABLE prescriptions (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),
  branch_id UUID NOT NULL REFERENCES branches(id),

  prescription_number VARCHAR(50) NOT NULL,

  patient_id UUID NOT NULL REFERENCES patients(id),
  doctor_id UUID NOT NULL REFERENCES doctors(id),

  -- Context
  visit_id UUID,
  admission_id UUID REFERENCES admissions(id),

  prescription_date DATE NOT NULL DEFAULT CURRENT_DATE,
  prescription_datetime TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

  prescription_type VARCHAR(50) DEFAULT 'opd', -- opd, ipd, discharge,
  telemedicine

  -- Clinical Context
  diagnosis TEXT,
  icd_codes UUID[],

  -- Patient Info Snapshot
  patient_age_at_prescription VARCHAR(50),
  patient_weight DECIMAL(5, 2),

  -- Allergy Warning Acknowledged
  allergy_warning_shown BOOLEAN DEFAULT FALSE,
  allergy_warning_acknowledged BOOLEAN DEFAULT FALSE,

  -- Interaction Warning
  interaction_warning_shown BOOLEAN DEFAULT FALSE,
  interaction_warning_acknowledged BOOLEAN DEFAULT FALSE,

  -- Advice & Instructions
  general_advice TEXT,
  diet_advice TEXT,
  exercise_advice TEXT,

  -- Follow-up
  followup_date DATE,
  followup_instructions TEXT,

  -- Signature
  is_signed BOOLEAN DEFAULT FALSE,
  signed_at TIMESTAMPTZ,
  digital_signature_url TEXT,

  -- Dispensing
  is_dispensed BOOLEAN DEFAULT FALSE,

```

```

dispensed_at TIMESTAMPTZ,
dispensed_by UUID REFERENCES users(id),

-- PDF
prescription_pdf_url TEXT,
qr_code VARCHAR(255),

notes TEXT,

created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
created_by UUID REFERENCES users(id),

UNIQUE(organization_id, prescription_number)
);

-- Prescription Items (Medications)
CREATE TABLE prescription_items (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  prescription_id UUID NOT NULL REFERENCES prescriptions(id) ON DELETE CASCADE,

  item_order INTEGER NOT NULL,

  -- Drug Info
  drug_id UUID REFERENCES drug_master(id),
  drug_name VARCHAR(255) NOT NULL,
  generic_name VARCHAR(255),
  is_generic BOOLEAN DEFAULT FALSE,

  -- Dosage
  strength VARCHAR(100),
  dose VARCHAR(100) NOT NULL, -- e.g., "1 tablet", "5ml"
  dose_unit VARCHAR(50),

  -- Frequency
  frequency VARCHAR(100) NOT NULL, -- e.g., "1+0+1", "TID", "QID"
  frequency_times INTEGER, -- times per day
  specific_times TIME[], -- specific times if needed

  -- Duration
  duration INTEGER, -- number of days
  duration_unit VARCHAR(20) DEFAULT 'days', -- days, weeks, months

  -- Route
  route VARCHAR(50), -- oral, iv, im, topical, etc.

  -- Instructions
  instructions TEXT, -- Before food, After food, etc.
  instructions_bangla TEXT,

  -- Quantity
  quantity_prescribed INTEGER,
  quantity_unit VARCHAR(50),

```

```

-- Refill
refills_allowed INTEGER DEFAULT 0,
refills_remaining INTEGER DEFAULT 0,

-- Dispensing
quantity_dispensed INTEGER,
is_dispensed BOOLEAN DEFAULT FALSE,

-- Substitution
allow_substitution BOOLEAN DEFAULT TRUE,
substitution_reason TEXT,

-- PRN (As needed)
is_prn BOOLEAN DEFAULT FALSE,
prn_reason TEXT,
max_daily_dose VARCHAR(100),

-- Controlled
is_controlled BOOLEAN DEFAULT FALSE,

-- Warnings
has_interaction BOOLEAN DEFAULT FALSE,
interaction_details TEXT,
has_allergy_warning BOOLEAN DEFAULT FALSE,
allergy_details TEXT,

notes TEXT,

created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Prescription Refill Log
CREATE TABLE prescription_refills (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  prescription_id UUID NOT NULL REFERENCES prescriptions(id),
  prescription_item_id UUID NOT NULL REFERENCES prescription_items(id),

  refill_number INTEGER NOT NULL,
  refill_date DATE NOT NULL,

  quantity_dispensed INTEGER,

  dispensed_by UUID REFERENCES users(id),
  pharmacy_location VARCHAR(255),

  notes TEXT,

  created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Prescription Templates
CREATE TABLE prescription_templates (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),

```

```

    doctor_id UUID REFERENCES doctors(id),

    name VARCHAR(255) NOT NULL,
    category VARCHAR(100), -- e.g., "Hypertension", "Diabetes"
    description TEXT,

    -- Template items stored as JSONB
    medications JSONB NOT NULL,
    advice TEXT,

    is_active BOOLEAN DEFAULT TRUE,
    usage_count INTEGER DEFAULT 0,

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
    created_by UUID REFERENCES users(id)
);

-- Narcotic Drug Log
CREATE TABLE narcotic_prescriptions (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),

    prescription_id UUID NOT NULL REFERENCES prescriptions(id),
    prescription_item_id UUID NOT NULL REFERENCES prescription_items(id),

    drug_id UUID REFERENCES drug_master(id),
    drug_name VARCHAR(255) NOT NULL,

    quantity_prescribed INTEGER NOT NULL,

    -- Patient Info
    patient_id UUID NOT NULL REFERENCES patients(id),
    patient_name VARCHAR(255),
    patient_nid VARCHAR(20),

    -- Doctor Info
    doctor_id UUID NOT NULL REFERENCES doctors(id),
    doctor_bmdc VARCHAR(50),

    -- Register Entry
    register_number VARCHAR(50),

    dispensed_quantity INTEGER,
    dispensed_at TIMESTAMPTZ,
    dispensed_by UUID REFERENCES users(id),

    witness_name VARCHAR(255),
    witness_id VARCHAR(50),

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Indexes

```

```

CREATE INDEX idx_prescriptions_org ON prescriptions(organization_id);
CREATE INDEX idx_prescriptions_patient ON prescriptions(patient_id);
CREATE INDEX idx_prescriptions_doctor ON prescriptions(doctor_id);
CREATE INDEX idx_prescriptions_date ON prescriptions(prescription_date);
CREATE INDEX idx_prescriptions_admission ON prescriptions(admission_id);

CREATE INDEX idx_prescription_items_rx ON prescription_items(prescription_id);
CREATE INDEX idx_prescription_items_drug ON prescription_items(drug_id);

```

📁 PART 6: NURSE STATION MANAGEMENT

SQL

```

-- =====
-- SECTION 12: NURSE STATION MANAGEMENT
-- =====

-- Nurse Profiles
CREATE TABLE nurses (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    user_id UUID NOT NULL REFERENCES users(id),
    organization_id UUID NOT NULL REFERENCES organizations(id),

    nurse_code VARCHAR(50),
    nursing_council_reg VARCHAR(50),
    registration_expiry DATE,

    qualification VARCHAR(255),
    specialization VARCHAR(100),

    experience_years INTEGER,

    current_ward_id UUID,
    current_shift shift_type,

    is_head_nurse BOOLEAN DEFAULT FALSE,
    is_active BOOLEAN DEFAULT TRUE,

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

    UNIQUE(organization_id, nurse_code)
);

-- Nurse Duty Roster
CREATE TABLE nurse_duty_roster (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),
    branch_id UUID NOT NULL REFERENCES branches(id),

    nurse_id UUID NOT NULL REFERENCES nurses(id),
    ward_id UUID NOT NULL,

```

```

    duty_date DATE NOT NULL,
    shift shift_type NOT NULL,

    shift_start_time TIME NOT NULL,
    shift_end_time TIME NOT NULL,

    actual_start_time TIME,
    actual_end_time TIME,

    is_on_call BOOLEAN DEFAULT FALSE,

    status VARCHAR(20) DEFAULT 'scheduled', -- scheduled, completed, absent,
swapped

    swapped_with_nurse_id UUID REFERENCES nurses(id),
    swap_approved_by UUID REFERENCES users(id),

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
    created_by UUID REFERENCES users(id),

    UNIQUE(nurse_id, duty_date, shift)
);

-- Nurse Handover Notes
CREATE TABLE nurse_handover_notes (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),
    branch_id UUID NOT NULL REFERENCES branches(id),
    ward_id UUID NOT NULL,

    handover_datetime TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,

    outgoing_nurse_id UUID NOT NULL REFERENCES nurses(id),
    incoming_nurse_id UUID NOT NULL REFERENCES nurses(id),

    outgoing_shift shift_type NOT NULL,
    incoming_shift shift_type NOT NULL,

    -- Summary
    total_patients INTEGER,
    critical_patients INTEGER,
    new_admissions INTEGER,
    expected_discharges INTEGER,

    -- Notes
    general_notes TEXT,
    critical_patient_notes TEXT,
    pending_tasks TEXT,
    special_instructions TEXT,

    -- Acknowledgment
    acknowledged_by_incoming BOOLEAN DEFAULT FALSE,
    acknowledged_at TIMESTAMPTZ,

```

```

        created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
    );

-- Patient Handover Details
CREATE TABLE nurse_handover_patients (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    handover_id UUID NOT NULL REFERENCES nurse_handover_notes(id) ON DELETE
    CASCADE,

    admission_id UUID NOT NULL REFERENCES admissions(id),
    patient_id UUID NOT NULL REFERENCES patients(id),
    bed_number VARCHAR(20),

    diagnosis TEXT,
    current_status TEXT,

    -- Concerns/Alerts
    critical_alerts TEXT,
    pending_orders TEXT,
    special_care TEXT,

    notes TEXT
);

-- Doctor Orders (for Nurse execution)
CREATE TABLE doctor_orders (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),

    order_number VARCHAR(50) NOT NULL,

    patient_id UUID NOT NULL REFERENCES patients(id),
    admission_id UUID NOT NULL REFERENCES admissions(id),

    ordered_by UUID NOT NULL REFERENCES doctors(id),
    order_datetime TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,

    order_type VARCHAR(50) NOT NULL,
    -- medication, lab_test, radiology, diet, nursing_care,
    -- procedure, consultation, monitoring, iv_fluid, blood_transfusion

    priority priority_type DEFAULT 'routine',

    -- Order Details
    order_text TEXT NOT NULL,
    structured_order JSONB,

    -- Frequency (for recurring orders)
    frequency VARCHAR(50), -- once, daily, bid, tid, qid, prn
    start_datetime TIMESTAMPTZ,
    end_datetime TIMESTAMPTZ,

    -- Status

```



```

    status order_status_type DEFAULT 'pending',

    -- Execution
    executed_by UUID REFERENCES users(id),
    executed_at TIMESTAMPTZ,
    execution_notes TEXT,

    -- Discontinuation
    discontinued_by UUID REFERENCES users(id),
    discontinued_at TIMESTAMPTZ,
    discontinuation_reason TEXT,

    -- Clarification
    clarification_requested BOOLEAN DEFAULT FALSE,
    clarification_request_text TEXT,
    clarification_response TEXT,

    notes TEXT,

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

    UNIQUE(organization_id, order_number)
);

-- Order Execution Log
CREATE TABLE order_execution_log (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    order_id UUID NOT NULL REFERENCES doctor_orders(id),

    execution_datetime TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
    executed_by UUID NOT NULL REFERENCES users(id),

    action VARCHAR(50), -- executed, held, refused, missed

    notes TEXT,

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Medication Administration Record (MAR)
CREATE TABLE medication_administration_records (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),

    patient_id UUID NOT NULL REFERENCES patients(id),
    admission_id UUID NOT NULL REFERENCES admissions(id),
    order_id UUID REFERENCES doctor_orders(id),

    drug_id UUID REFERENCES drug_master(id),
    medication_name VARCHAR(255) NOT NULL,

    dose VARCHAR(100) NOT NULL,
    route VARCHAR(50) NOT NULL,

```

```

-- Scheduled
scheduled_datetime TIMESTAMPTZ NOT NULL,

-- Actual Administration
administered_datetime TIMESTAMPTZ,
administered_by UUID REFERENCES nurses(id),

-- Five Rights Verified
right_patient BOOLEAN DEFAULT FALSE,
right_drug BOOLEAN DEFAULT FALSE,
right_dose BOOLEAN DEFAULT FALSE,
right_route BOOLEAN DEFAULT FALSE,
right_time BOOLEAN DEFAULT FALSE,

-- Barcode Verification
patient_barcode_scanned BOOLEAN DEFAULT FALSE,
medication_barcode_scanned BOOLEAN DEFAULT FALSE,

-- Status
status VARCHAR(50), -- administered, held, refused, missed, delayed

-- If not administered
reason_not_given TEXT,

-- Witness (for narcotics)
witness_id UUID REFERENCES users(id),

notes TEXT,

created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- IV Fluid Chart
CREATE TABLE iv_fluid_charts (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  admission_id UUID NOT NULL REFERENCES admissions(id),
  patient_id UUID NOT NULL REFERENCES patients(id),

  order_id UUID REFERENCES doctor_orders(id),

  fluid_name VARCHAR(255) NOT NULL,
  volume_ml INTEGER NOT NULL,
  additives TEXT,

  start_datetime TIMESTAMPTZ NOT NULL,
  end_datetime TIMESTAMPTZ,

  rate_ml_per_hour INTEGER,

  site VARCHAR(50), -- right_hand, left_hand, etc.
  cannula_size VARCHAR(20),

  started_by UUID REFERENCES nurses(id),

```

```

    completed_by UUID REFERENCES nurses(id),

    status VARCHAR(50), -- running, completed, stopped
    stop_reason TEXT,

    notes TEXT,

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Intake Output Chart
CREATE TABLE intake_output_charts (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    admission_id UUID NOT NULL REFERENCES admissions(id),
    patient_id UUID NOT NULL REFERENCES patients(id),

    recorded_datetime TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
    recorded_by UUID NOT NULL REFERENCES nurses(id),

    chart_date DATE NOT NULL,
    shift shift_type,
    hour INTEGER CHECK (hour BETWEEN 0 AND 23),

    -- Intake (ml)
    oral_intake INTEGER DEFAULT 0,
    iv_fluid_intake INTEGER DEFAULT 0,
    blood_intake INTEGER DEFAULT 0,
    ng_tube_intake INTEGER DEFAULT 0,
    other_intake INTEGER DEFAULT 0,
    total_intake INTEGER GENERATED ALWAYS AS (
        COALESCE(oral_intake, 0) + COALESCE(iv_fluid_intake, 0) +
        COALESCE(blood_intake, 0) + COALESCE(ng_tube_intake, 0) +
        COALESCE(other_intake, 0)
    ) STORED,

    -- Output (ml)
    urine_output INTEGER DEFAULT 0,
    vomit_output INTEGER DEFAULT 0,
    drain_output INTEGER DEFAULT 0,
    ng_aspiration INTEGER DEFAULT 0,
    stool_output INTEGER DEFAULT 0,
    other_output INTEGER DEFAULT 0,
    total_output INTEGER GENERATED ALWAYS AS (
        COALESCE(urine_output, 0) + COALESCE(vomit_output, 0) +
        COALESCE(drain_output, 0) + COALESCE(ng_aspiration, 0) +
        COALESCE(stool_output, 0) + COALESCE(other_output, 0)
    ) STORED,

    notes TEXT,

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Nursing Assessment

```

```

CREATE TABLE nursing_assessments (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),

  patient_id UUID NOT NULL REFERENCES patients(id),
  admission_id UUID NOT NULL REFERENCES admissions(id),

  assessment_type VARCHAR(50), -- initial, daily, reassessment
  assessment_datetime TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
  assessed_by UUID NOT NULL REFERENCES nurses(id),

  -- General Appearance
  general_appearance TEXT,
  consciousness_level VARCHAR(50),
  orientation TEXT,

  -- System Review
  neurological_assessment JSONB,
  cardiovascular_assessment JSONB,
  respiratory_assessment JSONB,
  gastrointestinal_assessment JSONB,
  genitourinary_assessment JSONB,
  musculoskeletal_assessment JSONB,
  skin_assessment JSONB,

  -- Risk Assessments
  fall_risk_score INTEGER,
  fall_risk_level VARCHAR(20),
  pressure_ulcer_score INTEGER, -- Braden
  pressure_ulcer_risk VARCHAR(20),
  nutritional_score INTEGER,
  nutritional_risk VARCHAR(20),

  -- ADL Assessment
  mobility_level VARCHAR(50),
  self_care_ability VARCHAR(50),
  communication_ability VARCHAR(50),

  -- Psychosocial
  emotional_status VARCHAR(100),
  family_support TEXT,

  -- Care Plan
  nursing_problems TEXT[],
  nursing_goals TEXT,
  nursing_interventions TEXT,

  notes TEXT,

  created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
  updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

```

```

-- Nursing Care Log

```

```

CREATE TABLE nursing_care_log (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  admission_id UUID NOT NULL REFERENCES admissions(id),
  patient_id UUID NOT NULL REFERENCES patients(id),

  care_datetime TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
  performed_by UUID NOT NULL REFERENCES nurses(id),

  care_type VARCHAR(100) NOT NULL,
  -- bath, oral_care, position_change, wound_dressing,
  -- catheter_care, ng_care, tracheostomy_care, etc.

  details TEXT,

  created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Wound Care Documentation
CREATE TABLE wound_care_records (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  admission_id UUID NOT NULL REFERENCES admissions(id),
  patient_id UUID NOT NULL REFERENCES patients(id),

  wound_id VARCHAR(50), -- For tracking multiple wounds
  wound_location VARCHAR(100) NOT NULL,
  wound_type VARCHAR(50), -- surgical, pressure, traumatic, etc.

  assessment_datetime TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
  assessed_by UUID NOT NULL REFERENCES nurses(id),

  -- Wound Assessment
  length_cm DECIMAL(5,2),
  width_cm DECIMAL(5,2),
  depth_cm DECIMAL(5,2),

  wound_bed_description TEXT,
  exudate_type VARCHAR(50),
  exudate_amount VARCHAR(50),
  odor VARCHAR(50),
  surrounding_skin TEXT,
  signs_of_infection BOOLEAN DEFAULT FALSE,
  infection_signs TEXT,

  -- Dressing
  dressing_type VARCHAR(255),
  dressing_changed BOOLEAN DEFAULT FALSE,

  photo_url TEXT,

  healing_progress VARCHAR(50),

  notes TEXT,

  created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP

```

```

);

-- Specimen Collection
CREATE TABLE specimen_collections (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),

  patient_id UUID NOT NULL REFERENCES patients(id),
  admission_id UUID REFERENCES admissions(id),

  order_id UUID REFERENCES doctor_orders(id),
  lab_order_id UUID, -- Reference to lab orders

  specimen_number VARCHAR(50) NOT NULL,

  specimen_type VARCHAR(100) NOT NULL, -- blood, urine, stool, sputum, etc.
  collection_container VARCHAR(100),

  collected_datetime TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
  collected_by UUID NOT NULL REFERENCES users(id),

  collection_site VARCHAR(100),
  quantity VARCHAR(50),

  -- Verification
  patient_verified BOOLEAN DEFAULT TRUE,
  barcode_label_printed BOOLEAN DEFAULT FALSE,

  -- Transport
  sent_to_lab_at TIMESTAMPTZ,
  sent_by UUID REFERENCES users(id),
  received_in_lab_at TIMESTAMPTZ,
  received_by UUID REFERENCES users(id),

  status specimen_status_type DEFAULT 'collected',

  -- If rejected
  rejection_reason TEXT,
  recollection_required BOOLEAN DEFAULT FALSE,

  notes TEXT,

  created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

  UNIQUE(organization_id, specimen_number)
);

-- Patient Call Bell Log
CREATE TABLE call_bell_log (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),
  branch_id UUID NOT NULL REFERENCES branches(id),

  ward_id UUID NOT NULL,

```

```

    bed_id UUID NOT NULL,
    patient_id UUID NOT NULL REFERENCES patients(id),

    call_datetime TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,

    urgency VARCHAR(20), -- normal, urgent, emergency
    reason VARCHAR(255),

    responded_at TIMESTAMPTZ,
    response_time_seconds INTEGER,
    responded_by UUID REFERENCES nurses(id),

    resolution TEXT,
    resolved_at TIMESTAMPTZ,

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Indexes
CREATE INDEX idx_nurse_roster_nurse ON nurse_duty_roster(nurse_id);
CREATE INDEX idx_nurse_roster_date ON nurse_duty_roster(duty_date);
CREATE INDEX idx_nurse_roster_ward ON nurse_duty_roster(ward_id, duty_date);

CREATE INDEX idx_doctor_orders_patient ON doctor_orders(patient_id);
CREATE INDEX idx_doctor_orders_admission ON doctor_orders(admission_id);
CREATE INDEX idx_doctor_orders_status ON doctor_orders(status);
CREATE INDEX idx_doctor_orders_pending ON doctor_orders(admission_id, status)
WHERE status = 'pending';

CREATE INDEX idx_mar_admission ON medication_administration_records(admission_id);
CREATE INDEX idx_mar_scheduled ON
medication_administration_records(scheduled_datetime);
CREATE INDEX idx_mar_patient_date ON medication_administration_records(patient_id,
scheduled_datetime);

CREATE INDEX idx_io_chart_admission ON intake_output_charts(admission_id);
CREATE INDEX idx_io_chart_date ON intake_output_charts(chart_date);

CREATE INDEX idx_specimen_patient ON specimen_collections(patient_id);
CREATE INDEX idx_specimen_status ON specimen_collections(status);

```

PART 7: BED & WARD MANAGEMENT

SQL

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-- =====
-- SECTION 13: BED & WARD MANAGEMENT
-- =====

-- Wards

```

```

CREATE TABLE wards (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),
  branch_id UUID NOT NULL REFERENCES branches(id),

  code VARCHAR(20) NOT NULL,
  name VARCHAR(100) NOT NULL,
  name_bangla VARCHAR(100),

  ward_type VARCHAR(50), -- general, private, icu, nicu, picu, ccu, hdu,
  maternity, pediatric, psychiatric

  building VARCHAR(50),
  floor VARCHAR(20),

  -- Capacity
  total_beds INTEGER DEFAULT 0,

  -- Gender Policy
  gender_restriction gender_type, -- NULL means both allowed

  -- Nursing Station
  nursing_station_phone VARCHAR(20),

  -- Head Nurse
  head_nurse_id UUID REFERENCES nurses(id),

  -- Settings
  visiting_hours TEXT,
  max_attendants INTEGER DEFAULT 1,

  color_code VARCHAR(10),

  is_active BOOLEAN DEFAULT TRUE,
  created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
  updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

  UNIQUE(branch_id, code)
);

-- Rooms
CREATE TABLE rooms (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  ward_id UUID NOT NULL REFERENCES wards(id),

  room_number VARCHAR(20) NOT NULL,

  room_type VARCHAR(50), -- general, private, semi_private, suite, isolation

  -- Capacity
  total_beds INTEGER DEFAULT 1,

  -- Amenities
  has_attached_bathroom BOOLEAN DEFAULT FALSE,

```



```

has_ac BOOLEAN DEFAULT FALSE,
has_tv BOOLEAN DEFAULT FALSE,
has_refrigerator BOOLEAN DEFAULT FALSE,

floor_area_sqft DECIMAL(8,2),

-- Rates
daily_rate DECIMAL(10, 2),

is_active BOOLEAN DEFAULT TRUE,
created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

UNIQUE(ward_id, room_number)
);

-- Beds
CREATE TABLE beds (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),
  branch_id UUID NOT NULL REFERENCES branches(id),
  ward_id UUID NOT NULL REFERENCES wards(id),
  room_id UUID REFERENCES rooms(id),

  bed_number VARCHAR(20) NOT NULL,

  bed_type VARCHAR(50), -- standard, icu, nicu, bariatric, pediatric, electric,
manual

-- Features
has_oxygen BOOLEAN DEFAULT FALSE,
has_suction BOOLEAN DEFAULT FALSE,
has_monitor BOOLEAN DEFAULT FALSE,
has_ventilator_support BOOLEAN DEFAULT FALSE,
has_side_rails BOOLEAN DEFAULT FALSE,

-- Status
status bed_status_type DEFAULT 'available',
current_patient_id UUID REFERENCES patients(id),
current_admission_id UUID REFERENCES admissions(id),

-- Housekeeping
last_cleaned_at TIMESTAMPTZ,
last_cleaned_by UUID REFERENCES users(id),

-- Maintenance
last_maintenance_at TIMESTAMPTZ,
next_maintenance_due DATE,

-- Rates
daily_rate DECIMAL(10, 2),

-- Display
display_order INTEGER,

```

```

    is_active BOOLEAN DEFAULT TRUE,
    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

    UNIQUE(ward_id, bed_number)
);

-- Bed Status History
CREATE TABLE bed_status_history (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    bed_id UUID NOT NULL REFERENCES beds(id),

    previous_status bed_status_type,
    new_status bed_status_type NOT NULL,

    patient_id UUID REFERENCES patients(id),
    admission_id UUID REFERENCES admissions(id),

    changed_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
    changed_by UUID REFERENCES users(id),

    reason TEXT
);

-- Bed Reservations
CREATE TABLE bed_reservations (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),
    branch_id UUID NOT NULL REFERENCES branches(id),

    bed_id UUID REFERENCES beds(id), -- Specific bed
    ward_id UUID REFERENCES wards(id), -- Or just ward preference
    bed_type VARCHAR(50), -- Or just bed type preference

    patient_id UUID NOT NULL REFERENCES patients(id),

    reserved_from TIMESTAMPTZ NOT NULL,
    reserved_until TIMESTAMPTZ NOT NULL,

    expected_admission_date DATE,
    surgery_date DATE,

    reservation_reason TEXT,

    status VARCHAR(20) DEFAULT 'active', -- active, confirmed, cancelled, expired

    reserved_by UUID REFERENCES users(id),
    reserved_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

    confirmed_by UUID REFERENCES users(id),
    confirmed_at TIMESTAMPTZ,

    cancelled_by UUID REFERENCES users(id),
    cancelled_at TIMESTAMPTZ,

```

```

        cancellation_reason TEXT,

        created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
    );

-- Bed Transfer Requests
CREATE TABLE bed_transfer_requests (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),

    admission_id UUID NOT NULL REFERENCES admissions(id),
    patient_id UUID NOT NULL REFERENCES patients(id),

    -- From
    from_ward_id UUID NOT NULL REFERENCES wards(id),
    from_bed_id UUID NOT NULL REFERENCES beds(id),

    -- To
    to_ward_id UUID REFERENCES wards(id),
    to_bed_id UUID REFERENCES beds(id),

    transfer_type VARCHAR(50), -- upgrade, downgrade, icu_escalation, step_down,
    same_ward, patient_request

    reason TEXT NOT NULL,

    requested_by UUID NOT NULL REFERENCES users(id),
    requested_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

    -- Approval
    status VARCHAR(20) DEFAULT 'pending', -- pending, approved, rejected,
    completed, cancelled

    approved_by UUID REFERENCES users(id),
    approved_at TIMESTAMPTZ,

    rejection_reason TEXT,

    -- Execution
    transferred_at TIMESTAMPTZ,
    transferred_by UUID REFERENCES users(id),

    -- Rate change
    old_daily_rate DECIMAL(10, 2),
    new_daily_rate DECIMAL(10, 2),

    notes TEXT,

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Bed Cleaning Tasks
CREATE TABLE bed_cleaning_tasks (

```

```

    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),
    branch_id UUID NOT NULL REFERENCES branches(id),

    bed_id UUID NOT NULL REFERENCES beds(id),
    ward_id UUID NOT NULL REFERENCES wards(id),

    cleaning_type VARCHAR(50), -- routine, discharge, terminal, isolation

    -- Trigger
    triggered_by VARCHAR(50), -- discharge, transfer, manual, scheduled
    previous_patient_id UUID REFERENCES patients(id),

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

    -- Assignment
    assigned_to UUID REFERENCES users(id),
    assigned_at TIMESTAMPTZ,

    -- Progress
    started_at TIMESTAMPTZ,
    completed_at TIMESTAMPTZ,

    -- Checklist
    checklist JSONB,

    -- QA
    qa_checked BOOLEAN DEFAULT FALSE,
    qa_checked_by UUID REFERENCES users(id),
    qa_checked_at TIMESTAMPTZ,
    qa_passed BOOLEAN,
    qa_remarks TEXT,

    -- Photos
    before_photos TEXT[],
    after_photos TEXT[],

    status VARCHAR(20) DEFAULT 'pending', -- pending, assigned, in_progress,
    completed, qa_pending, approved

    notes TEXT
);

-- Ward Census (Daily snapshot)
CREATE TABLE ward_census (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),
    branch_id UUID NOT NULL REFERENCES branches(id),
    ward_id UUID NOT NULL REFERENCES wards(id),

    census_date DATE NOT NULL,
    census_time TIME DEFAULT '00:00',

    total_beds INTEGER,

```

```

    occupied_beds INTEGER,
    available_beds INTEGER,
    reserved_beds INTEGER,
    cleaning_beds INTEGER,
    maintenance_beds INTEGER,

    occupancy_percentage DECIMAL(5,2),

    -- Patient movements
    admissions_today INTEGER DEFAULT 0,
    discharges_today INTEGER DEFAULT 0,
    transfers_in INTEGER DEFAULT 0,
    transfers_out INTEGER DEFAULT 0,
    deaths_today INTEGER DEFAULT 0,

    recorded_by UUID REFERENCES users(id),

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

    UNIQUE(ward_id, census_date)
);

-- Indexes
CREATE INDEX idx_wards_branch ON wards(branch_id);
CREATE INDEX idx_wards_type ON wards(ward_type);

CREATE INDEX idx_beds_ward ON beds(ward_id);
CREATE INDEX idx_beds_status ON beds(status);
CREATE INDEX idx_beds_available ON beds(ward_id, status) WHERE status =
'available';

CREATE INDEX idx_bed_reservations_bed ON bed_reservations.bed_id);
CREATE INDEX idx_bed_reservations_patient ON bed_reservations(patient_id);
CREATE INDEX idx_bed_reservations_dates ON bed_reservations(reserved_from,
reserved_until);

CREATE INDEX idx_bed_cleaning_bed ON bed_cleaning_tasks.bed_id);
CREATE INDEX idx_bed_cleaning_status ON bed_cleaning_tasks(status);

```

PART 8: OT (OPERATION THEATRE) MANAGEMENT

SQL

```

-- =====
-- SECTION 14: OPERATION THEATRE MANAGEMENT
-- =====

-- OT Rooms
CREATE TABLE ot_rooms (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),

```

```

organization_id UUID NOT NULL REFERENCES organizations(id),
branch_id UUID NOT NULL REFERENCES branches(id),

room_number VARCHAR(20) NOT NULL,
name VARCHAR(100),

ot_type VARCHAR(50), -- major, minor, emergency, cardiac, neuro, ortho, eye,
ent

-- Specifications
floor_area_sqft DECIMAL(8,2),
laminar_flow BOOLEAN DEFAULT FALSE,

-- Equipment List
equipment JSONB,

-- Rates
hourly_rate DECIMAL(10, 2),

-- Status
status VARCHAR(20) DEFAULT 'available', -- available, in_use, cleaning,
maintenance

is_active BOOLEAN DEFAULT TRUE,
created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

UNIQUE(branch_id, room_number)
);

-- Surgery Types/Procedures
CREATE TABLE surgery_types (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),

  code VARCHAR(20) NOT NULL,
  name VARCHAR(255) NOT NULL,

  category VARCHAR(100), -- general, cardiac, neuro, ortho, etc.
  sub_category VARCHAR(100),

  cpt_code_id UUID REFERENCES cpt_codes(id),

  typical_duration_minutes INTEGER,

  ot_type_required VARCHAR(50),
  anesthesia_type_typical VARCHAR(50),

  -- Charges
  surgeon_fee DECIMAL(10, 2),
  ot_charge DECIMAL(10, 2),
  anesthesia_charge DECIMAL(10, 2),

  requires_blood_reservation BOOLEAN DEFAULT FALSE,
  requires_icu_booking BOOLEAN DEFAULT FALSE,

```

```

    pre_op_instructions TEXT,

    is_active BOOLEAN DEFAULT TRUE,

    UNIQUE(organization_id, code)
);

-- Surgery Scheduling
CREATE TABLE surgeries (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),
    branch_id UUID NOT NULL REFERENCES branches(id),

    surgery_number VARCHAR(50) NOT NULL,

    patient_id UUID NOT NULL REFERENCES patients(id),
    admission_id UUID REFERENCES admissions(id),

    -- Scheduling
    scheduled_date DATE NOT NULL,
    scheduled_start_time TIME NOT NULL,
    scheduled_end_time TIME,
    estimated_duration_minutes INTEGER,

    ot_room_id UUID NOT NULL REFERENCES ot_rooms(id),

    surgery_type_id UUID REFERENCES surgery_types(id),
    surgery_name VARCHAR(255) NOT NULL,

    surgery_category VARCHAR(50), -- elective, emergency, daycare

    -- Laterality
    laterality VARCHAR(20), -- left, right, bilateral, not_applicable

    -- Team
    primary_surgeon_id UUID NOT NULL REFERENCES doctors(id),
    assistant_surgeon_ids UUID[],
    anesthesiologist_id UUID REFERENCES doctors(id),
    scrub_nurse_id UUID REFERENCES nurses(id),
    circulating_nurse_id UUID REFERENCES nurses(id),

    -- Anesthesia Plan
    anesthesia_type VARCHAR(50), -- general, spinal, epidural, local, regional

    -- Status
    status VARCHAR(50) DEFAULT 'scheduled',
    -- scheduled, pre_op_started, in_ot, surgery_started, surgery_completed,
    in_pacu, completed, cancelled

    -- Actual Times
    actual_start_time TIMESTAMPTZ,
    actual_end_time TIMESTAMPTZ,
    actual_duration_minutes INTEGER,

```

```

-- Cancellation
cancelled_at TIMESTAMPTZ,
cancelled_by UUID REFERENCES users(id),
cancellation_reason TEXT,

-- Pre-Auth
insurance_preauth_id UUID,
preauth_approved BOOLEAN,

notes TEXT,

created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
created_by UUID REFERENCES users(id),

UNIQUE(organization_id, surgery_number)
);

-- Pre-Operative Checklist
CREATE TABLE surgery_preop_checklist (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  surgery_id UUID NOT NULL REFERENCES surgeries(id),

  -- Identity
  patient_identity_verified BOOLEAN DEFAULT FALSE,
  consent_signed BOOLEAN DEFAULT FALSE,
  site_marked BOOLEAN DEFAULT FALSE,

  -- Clinical
  allergies_verified BOOLEAN DEFAULT FALSE,
  fasting_verified BOOLEAN DEFAULT FALSE,
  fasting_hours INTEGER,

  -- Preparations
  preop_bath_done BOOLEAN DEFAULT FALSE,
  jewelry_removed BOOLEAN DEFAULT FALSE,
  dentures_removed BOOLEAN DEFAULT FALSE,
  nail_polish_removed BOOLEAN DEFAULT FALSE,
  hospital_gown_worn BOOLEAN DEFAULT FALSE,
  id_band_on BOOLEAN DEFAULT FALSE,

  -- Investigations
  blood_group_confirmed BOOLEAN DEFAULT FALSE,
  cross_match_done BOOLEAN DEFAULT FALSE,
  blood_units_reserved INTEGER,

  investigations_reviewed BOOLEAN DEFAULT FALSE,
  ecg_done BOOLEAN DEFAULT FALSE,
  chest_xray_done BOOLEAN DEFAULT FALSE,

  -- Medications
  preop_medications_given BOOLEAN DEFAULT FALSE,
  antibiotic_prophylaxis BOOLEAN DEFAULT FALSE,

```



```

    current_medications_held TEXT,

    -- Anesthesia
    anesthesia_clearance BOOLEAN DEFAULT FALSE,
    asa_grade VARCHAR(10),
    mallampati_score INTEGER,

    -- WHO Checklist Sign In
    who_sign_in_completed BOOLEAN DEFAULT FALSE,
    who_sign_in_time TIMESTAMPTZ,
    who_sign_in_by UUID REFERENCES users(id),

    completed_by UUID REFERENCES users(id),
    completed_at TIMESTAMPTZ,

    notes TEXT,

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Intra-Operative Record
CREATE TABLE surgery_intraop_records (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    surgery_id UUID NOT NULL REFERENCES surgeries(id),

    -- Times
    patient_in_ot_at TIMESTAMPTZ,
    anesthesia_start_at TIMESTAMPTZ,
    incision_time TIMESTAMPTZ,
    closure_time TIMESTAMPTZ,
    anesthesia_end_at TIMESTAMPTZ,
    patient_out_ot_at TIMESTAMPTZ,

    -- Positioning
    patient_position VARCHAR(100),

    -- Anesthesia Details
    anesthesia_type VARCHAR(50),
    anesthesia_drugs JSONB,
    airway_management VARCHAR(100),
    ett_size VARCHAR(20),

    -- Vitals during surgery (array of readings)
    intraop_vitals JSONB,

    -- IV Access
    iv_access_site VARCHAR(100),
    iv_cannula_size VARCHAR(20),
    central_line BOOLEAN DEFAULT FALSE,
    central_line_site VARCHAR(50),
    arterial_line BOOLEAN DEFAULT FALSE,

    -- Fluids

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```

iv_fluids_given JSONB,
estimated_blood_loss_ml INTEGER,
blood_transfused JSONB,
urine_output_ml INTEGER,

-- Counts
instrument_count_before INTEGER,
instrument_count_after INTEGER,
instrument_count_correct BOOLEAN,
swab_count_before INTEGER,
swab_count_after INTEGER,
swab_count_correct BOOLEAN,
needle_count_correct BOOLEAN,

-- Specimens
specimen_sent BOOLEAN DEFAULT FALSE,
specimen_details TEXT,
frozen_section BOOLEAN DEFAULT FALSE,
frozen_section_result TEXT,

-- Implants
implants_used JSONB, -- [{name, manufacturer, lot_number, size, expiry}]

-- WHO Checklist
who_time_out_completed BOOLEAN DEFAULT FALSE,
who_time_out_time TIMESTAMPTZ,
who_sign_out_completed BOOLEAN DEFAULT FALSE,
who_sign_out_time TIMESTAMPTZ,

-- Surgical Notes
procedure_performed TEXT,
findings TEXT,
complications TEXT,

-- Photos/Videos
surgical_photos TEXT[],
surgical_video_url TEXT,

recorded_by UUID REFERENCES users(id),

created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Anesthesia Record
CREATE TABLE anesthesia_records (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  surgery_id UUID NOT NULL REFERENCES surgeries(id),

  anesthesiologist_id UUID NOT NULL REFERENCES doctors(id),

  -- Pre-Anesthesia Assessment
  asa_classification VARCHAR(10),
  airway_assessment JSONB,

```

```

previous_anesthesia_history TEXT,
anticipated_difficulty TEXT,

-- Anesthesia Plan
anesthesia_type VARCHAR(50),
anesthesia_technique TEXT,

-- Pre-Medication
premedication JSONB,

-- Induction
induction_agents JSONB,
airway_device VARCHAR(100),
ETT_size VARCHAR(20),
cuff_pressure DECIMAL(4,1),

-- Maintenance
maintenance_agents JSONB,
muscle_relaxants JSONB,

-- Monitoring
monitoring_devices TEXT[],

-- Vital Signs (time series)
vital_signs_chart JSONB,

-- Drugs Administered
drugs_administered JSONB,

-- Fluids & Blood
fluid_balance JSONB,

-- Events
intraop_events TEXT,

-- Recovery
reversal_agents JSONB,
extubation_time TIMESTAMPTZ,
extubation_status VARCHAR(50),

-- Post-Anesthesia Instructions
post_op_orders TEXT,
pain_management_plan TEXT,

notes TEXT,

created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Post-Operative Record (PACU)
CREATE TABLE surgery_postop_records (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  surgery_id UUID NOT NULL REFERENCES surgeries(id),

```

```

-- PACU Arrival
pacu_arrival_time TIMESTAMPTZ,
received_by UUID REFERENCES nurses(id),

-- Initial Assessment
initial_vitals JSONB,
consciousness_level VARCHAR(50),
airway_status VARCHAR(50),

-- Aldrete Score (every 15 mins)
aldrete_scores JSONB, -- [{time, activity, respiration, circulation,
consciousness, oxygen_saturation, total}]

-- Pain Management
pain_scores JSONB,
analgesics_given JSONB,

-- Complications
nausea_vomiting BOOLEAN DEFAULT FALSE,
antiemetic_given JSONB,

shivering BOOLEAN DEFAULT FALSE,
hypothermia BOOLEAN DEFAULT FALSE,

complications TEXT,

-- Discharge from PACU
pacu_discharge_time TIMESTAMPTZ,
discharge_aldrete_score INTEGER,
discharge_destination VARCHAR(50), -- ward, icu, hdu

discharged_by UUID REFERENCES users(id),

notes TEXT,

created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- OT Sterilization Log
CREATE TABLE ot_sterilization_log (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),
  branch_id UUID NOT NULL REFERENCES branches(id),

  ot_room_id UUID REFERENCES ot_rooms(id),

  sterilization_type VARCHAR(50), -- fumigation, uv, terminal_cleaning

  started_at TIMESTAMPTZ NOT NULL,
  completed_at TIMESTAMPTZ,

  performed_by UUID REFERENCES users(id),

```

```

    chemicals_used TEXT,
    duration_minutes INTEGER,

    verification_done BOOLEAN DEFAULT FALSE,
    verified_by UUID REFERENCES users(id),
    verified_at TIMESTAMPTZ,

    next_sterilization_due TIMESTAMPTZ,

    notes TEXT,

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- OT Equipment Usage
CREATE TABLE ot_equipment_usage (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    surgery_id UUID NOT NULL REFERENCES surgeries(id),

    equipment_id UUID, -- Reference to equipment registry
    equipment_name VARCHAR(255) NOT NULL,

    usage_start_time TIMESTAMPTZ,
    usage_end_time TIMESTAMPTZ,

    condition_before VARCHAR(50),
    condition_after VARCHAR(50),

    issues_reported TEXT,

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Indexes
CREATE INDEX idx_surgeries_org ON surgeries(organization_id);
CREATE INDEX idx_surgeries_patient ON surgeries(patient_id);
CREATE INDEX idx_surgeries_date ON surgeries(scheduled_date);
CREATE INDEX idx_surgeries_ot ON surgeries(ot_room_id, scheduled_date);
CREATE INDEX idx_surgeries_surgeon ON surgeries(primary_surgeon_id);
CREATE INDEX idx_surgeries_status ON surgeries(status);
CREATE INDEX idx_surgeries_admission ON surgeries(admission_id);

```

PART 9: DIAGNOSTIC MANAGEMENT (LAB & RADIOLOGY)

SQL

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-- =====
-- SECTION 15: DIAGNOSTIC MANAGEMENT
-- =====

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-- Test Catalog
CREATE TABLE test_catalog (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),

  code VARCHAR(20) NOT NULL,
  name VARCHAR(255) NOT NULL,
  name_bangla VARCHAR(255),

  department VARCHAR(50) NOT NULL, -- biochemistry, hematology, microbiology,
  pathology, radiology
  sub_department VARCHAR(50),
  category VARCHAR(100),

  -- Test Type
  test_type VARCHAR(50), -- single, panel, profile

  -- For panels/profiles
  is_panel BOOLEAN DEFAULT FALSE,
  panel_tests UUID[], -- Array of test IDs

  -- Specimen
  specimen_type VARCHAR(100),
  specimen_container VARCHAR(100),
  specimen_volume VARCHAR(50),
  collection_instructions TEXT,

  -- Processing
  processing_department VARCHAR(50),
  method VARCHAR(255),
  equipment VARCHAR(255),

  -- Turnaround Time
  tat_routine_hours INTEGER,
  tat_urgent_hours INTEGER,
  tat_stat_minutes INTEGER,

  -- Reporting
  report_format VARCHAR(50), -- numeric, text, image
  result_unit VARCHAR(50),

  -- Pricing
  price DECIMAL(10, 2),
  cost DECIMAL(10, 2),

  -- Clinical
  loinc_code VARCHAR(20),
  cpt_code_id UUID REFERENCES cpt_codes(id),

  -- Flags
  requires_fasting BOOLEAN DEFAULT FALSE,
  is_outsourced BOOLEAN DEFAULT FALSE,
  outsource_lab VARCHAR(255),

```

```

    is_active BOOLEAN DEFAULT TRUE,
    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

    UNIQUE(organization_id, code)
);

-- Test Reference Ranges
CREATE TABLE test_reference_ranges (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    test_id UUID NOT NULL REFERENCES test_catalog(id),

    -- Demographics
    gender gender_type, -- NULL for all
    age_min_years DECIMAL(5,2),
    age_max_years DECIMAL(5,2),
    age_min_days INTEGER, -- For neonates
    age_max_days INTEGER,

    -- Normal Range
    normal_min DECIMAL(15, 5),
    normal_max DECIMAL(15, 5),
    normal_text VARCHAR(255), -- For non-numeric results

    -- Critical Values
    critical_low DECIMAL(15, 5),
    critical_high DECIMAL(15, 5),

    -- Panic Values
    panic_low DECIMAL(15, 5),
    panic_high DECIMAL(15, 5),

    unit VARCHAR(50),

    interpretation_guide TEXT,

    is_active BOOLEAN DEFAULT TRUE,
    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Lab Orders
CREATE TABLE lab_orders (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),
    branch_id UUID NOT NULL REFERENCES branches(id),

    order_number VARCHAR(50) NOT NULL,

    patient_id UUID NOT NULL REFERENCES patients(id),
    visit_id UUID, -- OPD
    admission_id UUID REFERENCES admissions(id), -- IPD

    ordered_by UUID NOT NULL REFERENCES users(id),
    order_datetime TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,

```

```

    priority priority_type DEFAULT 'routine',

    -- Clinical Info
    clinical_history TEXT,
    provisional_diagnosis TEXT,

    -- Fasting Status
    is_fasting BOOLEAN,
    last_meal_time TIMESTAMPTZ,

    -- Status
    status VARCHAR(50) DEFAULT 'ordered',
    -- ordered, collected, received, processing, partially_completed, completed,
    cancelled

    -- Billing
    billing_status payment_status_type DEFAULT 'pending',
    bill_id UUID,

    notes TEXT,

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

    UNIQUE(organization_id, order_number)
);

-- Lab Order Items
CREATE TABLE lab_order_items (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    order_id UUID NOT NULL REFERENCES lab_orders(id) ON DELETE CASCADE,

    test_id UUID NOT NULL REFERENCES test_catalog(id),
    test_code VARCHAR(20),
    test_name VARCHAR(255),

    priority priority_type,

    -- Specimen
    specimen_id UUID, -- Reference to specimen_collections
    specimen_collected BOOLEAN DEFAULT FALSE,
    specimen_received BOOLEAN DEFAULT FALSE,

    -- Processing
    processing_started_at TIMESTAMPTZ,
    processing_by UUID REFERENCES users(id),

    -- Status
    status result_status_type DEFAULT 'pending',

    -- If cancelled
    cancelled_at TIMESTAMPTZ,
    cancelled_by UUID REFERENCES users(id),

```



```

    cancellation_reason TEXT,

    -- Outsourcing
    is_outsourced BOOLEAN DEFAULT FALSE,
    outsource_lab VARCHAR(255),
    outsource_sent_at TIMESTAMPTZ,
    outsource_received_at TIMESTAMPTZ,

    price DECIMAL(10, 2),

    notes TEXT,

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Lab Results
CREATE TABLE lab_results (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),

    order_item_id UUID NOT NULL REFERENCES lab_order_items(id),
    order_id UUID NOT NULL REFERENCES lab_orders(id),
    patient_id UUID NOT NULL REFERENCES patients(id),
    test_id UUID NOT NULL REFERENCES test_catalog(id),

    result_datetime TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,

    -- For panels, this might be a sub-test
    parameter_name VARCHAR(255),
    parameter_code VARCHAR(50),

    -- Result
    result_value VARCHAR(500),
    result_numeric DECIMAL(15, 5),
    result_unit VARCHAR(50),

    -- Reference Range (snapshot at time of result)
    reference_min DECIMAL(15, 5),
    reference_max DECIMAL(15, 5),
    reference_text VARCHAR(255),

    -- Flags
    abnormal_flag VARCHAR(10), -- N, L, H, LL, HH, A
    is_critical BOOLEAN DEFAULT FALSE,
    critical_notified BOOLEAN DEFAULT FALSE,
    critical_notified_to UUID,
    critical_notified_at TIMESTAMPTZ,

    -- Delta Check
    previous_value DECIMAL(15, 5),
    previous_date DATE,
    delta_flag BOOLEAN DEFAULT FALSE,

```

```

-- Comments
result_comment TEXT,
technical_comment TEXT,

-- Verification
entered_by UUID REFERENCES users(id),
entered_at TIMESTAMPTZ,

verified_by UUID REFERENCES users(id),
verified_at TIMESTAMPTZ,

status result_status_type DEFAULT 'preliminary',

-- Machine Interface
analyzer_name VARCHAR(100),
analyzer_result_id VARCHAR(100),

created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Lab Result Amendments
CREATE TABLE lab_result_amendments (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  result_id UUID NOT NULL REFERENCES lab_results(id),

  previous_value VARCHAR(500),
  new_value VARCHAR(500),

  amendment_reason TEXT NOT NULL,

  amended_by UUID NOT NULL REFERENCES users(id),
  amended_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

  approved_by UUID REFERENCES users(id),
  approved_at TIMESTAMPTZ
);

-- Microbiology Culture Results
CREATE TABLE microbiology_results (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  order_item_id UUID NOT NULL REFERENCES lab_order_items(id),
  patient_id UUID NOT NULL REFERENCES patients(id),

  specimen_type VARCHAR(100),

  -- Gram Stain
  gram_stain_result TEXT,

  -- Culture
  culture_result VARCHAR(50), -- positive, negative, no_growth
  growth_description TEXT,
  incubation_hours INTEGER,

```

```

-- Organisms
organisms JSONB, -- [{name, colony_count, significance}]

-- Sensitivity (Antibiogram)
sensitivity_results JSONB,
-- [{antibiotic, mic, interpretation (S/I/R)}]

-- Comments
clinical_significance TEXT,
recommendations TEXT,

reported_by UUID REFERENCES users(id),
reported_at TIMESTAMPTZ,

verified_by UUID REFERENCES users(id),
verified_at TIMESTAMPTZ,

created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Histopathology Reports
CREATE TABLE histopathology_reports (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  order_item_id UUID NOT NULL REFERENCES lab_order_items(id),
  patient_id UUID NOT NULL REFERENCES patients(id),

  -- Specimen Info
  specimen_type VARCHAR(255),
  specimen_received_date DATE,
  specimen_site VARCHAR(255),

  -- Gross Examination
  gross_description TEXT,

  -- Microscopic Examination
  microscopic_description TEXT,

  -- Diagnosis
  diagnosis TEXT,

  -- IHC (Immunohistochemistry)
  ihc_performed BOOLEAN DEFAULT FALSE,
  ihc_markers JSONB,

  -- Staging/Grading
  staging TEXT,
  grading TEXT,

  -- Margins
  margin_status VARCHAR(50),

  -- Additional Studies
  special_stains JSONB,
  molecular_studies JSONB,

```

```

-- Synoptic Report
synoptic_report JSONB,

-- Comments
pathologist_comments TEXT,

-- Sign-off
reported_by UUID REFERENCES users(id),
reported_at TIMESTAMPTZ,

verified_by UUID REFERENCES users(id),
verified_at TIMESTAMPTZ,

-- Images
microscopy_images TEXT[],

created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- =====
-- RADIOLOGY
-- =====

-- Radiology Modalities
CREATE TABLE radiology_modalities (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),
    branch_id UUID NOT NULL REFERENCES branches(id),

    code VARCHAR(20) NOT NULL,
    name VARCHAR(100) NOT NULL,
    modality_type VARCHAR(50), -- xray, ct, mri, usg, mammo, fluoroscopy, nuclear

    room_number VARCHAR(20),

    ae_title VARCHAR(50), -- DICOM AE Title

    is_active BOOLEAN DEFAULT TRUE,

    UNIQUE(branch_id, code)
);

-- Radiology Orders
CREATE TABLE radiology_orders (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),
    branch_id UUID NOT NULL REFERENCES branches(id),

    order_number VARCHAR(50) NOT NULL,
    accession_number VARCHAR(50), -- DICOM

    patient_id UUID NOT NULL REFERENCES patients(id),
    visit_id UUID,

```

```

admission_id UUID REFERENCES admissions(id),

ordered_by UUID NOT NULL REFERENCES users(id),
order_datetime TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,

-- Test Info
test_id UUID NOT NULL REFERENCES test_catalog(id),
procedure_name VARCHAR(255) NOT NULL,
modality_type VARCHAR(50),
body_part VARCHAR(100),
laterality VARCHAR(20),

-- Clinical
clinical_history TEXT,
indication TEXT,

priority priority_type DEFAULT 'routine',

-- Scheduling
scheduled_datetime TIMESTAMPTZ,
modality_id UUID REFERENCES radiology_modalities(id),

-- Status
status VARCHAR(50) DEFAULT 'ordered',
-- ordered, scheduled, patient_arrived, in_progress, completed, reported,
cancelled

-- Execution
performed_datetime TIMESTAMPTZ,
performed_by UUID REFERENCES users(id), -- Technician

-- Contrast
contrast_used BOOLEAN DEFAULT FALSE,
contrast_type VARCHAR(100),
contrast_volume VARCHAR(50),

-- For special preps
patient_prep TEXT,

-- Billing
billing_status payment_status_type DEFAULT 'pending',
price DECIMAL(10, 2),

notes TEXT,

created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

UNIQUE(organization_id, order_number)
);

-- Radiology Images (DICOM Study reference)
CREATE TABLE radiology_studies (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),

```

```

    order_id UUID NOT NULL REFERENCES radiology_orders(id),
    patient_id UUID NOT NULL REFERENCES patients(id),

    -- DICOM UUIDs
    study_instance_uid VARCHAR(100) UNIQUE,
    study_date DATE,
    study_time TIME,

    modality VARCHAR(20),

    -- Study Info
    study_description VARCHAR(255),
    number_of_series INTEGER,
    number_of_images INTEGER,

    -- Storage
    pacs_location TEXT,
    storage_size_mb DECIMAL(10, 2),

    -- Viewing
    dicom_viewer_url TEXT,

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Radiology Reports
CREATE TABLE radiology_reports (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    order_id UUID NOT NULL REFERENCES radiology_orders(id),
    study_id UUID REFERENCES radiology_studies(id),
    patient_id UUID NOT NULL REFERENCES patients(id),

    -- Report Content
    findings TEXT,
    impression TEXT,
    recommendations TEXT,

    -- Structured Report (for some modalities)
    structured_report JSONB,

    -- Comparison
    comparison_study_id UUID REFERENCES radiology_studies(id),
    comparison_notes TEXT,

    -- Critical Finding
    is_critical BOOLEAN DEFAULT FALSE,
    critical_finding TEXT,
    critical_notified BOOLEAN DEFAULT FALSE,
    critical_notified_to UUID,
    critical_notified_at TIMESTAMPTZ,

    -- Sign-off
    reported_by UUID REFERENCES users(id), -- Radiologist
    reported_at TIMESTAMPTZ,

```

```

verified_by UUID REFERENCES users(id),
verified_at TIMESTAMPTZ,

status result_status_type DEFAULT 'preliminary',

-- Addendum
addendum TEXT,
addendum_by UUID REFERENCES users(id),
addendum_at TIMESTAMPTZ,

-- PDF
report_pdf_url TEXT,

created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- QC (Quality Control) Records
CREATE TABLE lab_qc_records (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),
  branch_id UUID NOT NULL REFERENCES branches(id),

  test_id UUID NOT NULL REFERENCES test_catalog(id),
  analyzer_name VARCHAR(100),

  qc_date DATE NOT NULL,
  qc_level VARCHAR(20), -- level1, level2, level3
  qc_material_lot VARCHAR(50),

  expected_mean DECIMAL(15, 5),
  expected_sd DECIMAL(15, 5),

  observed_value DECIMAL(15, 5),

  -- Westgard Rules
  qc_status VARCHAR(20), -- pass, fail, warning
  rules_violated TEXT[],

  corrective_action TEXT,

  performed_by UUID REFERENCES users(id),
  reviewed_by UUID REFERENCES users(id),

  created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Indexes
CREATE INDEX idx_test_catalog_org ON test_catalog(organization_id);
CREATE INDEX idx_test_catalog_dept ON test_catalog(department);
CREATE INDEX idx_test_catalog_code ON test_catalog(organization_id, code);

CREATE INDEX idx_lab_orders_org ON lab_orders(organization_id);

```

```

CREATE INDEX idx_lab_orders_patient ON lab_orders(patient_id);
CREATE INDEX idx_lab_orders_date ON lab_orders(order_datetime);
CREATE INDEX idx_lab_orders_status ON lab_orders(status);

CREATE INDEX idx_lab_results_patient ON lab_results(patient_id);
CREATE INDEX idx_lab_results_order ON lab_results(order_id);
CREATE INDEX idx_lab_results_critical ON lab_results(is_critical) WHERE
is_critical = TRUE;

CREATE INDEX idx_radiology_orders_patient ON radiology_orders(patient_id);
CREATE INDEX idx_radiology_orders_date ON radiology_orders(order_datetime);
CREATE INDEX idx_radiology_orders_status ON radiology_orders(status);

```

PART 10: PHARMACY MANAGEMENT

SQL

```

-- =====
-- SECTION 16: PHARMACY MANAGEMENT
-- =====

-- Pharmacy Locations
CREATE TABLE pharmacy_locations (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),
    branch_id UUID NOT NULL REFERENCES branches(id),

    code VARCHAR(20) NOT NULL,
    name VARCHAR(100) NOT NULL,

    location_type VARCHAR(50), -- main, satellite, ward, emergency, ot

    floor VARCHAR(20),
    building VARCHAR(50),

    phone VARCHAR(20),

    operating_hours TEXT,

    -- For ward pharmacies
    ward_id UUID REFERENCES wards(id),

    is_active BOOLEAN DEFAULT TRUE,
    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

    UNIQUE(branch_id, code)
);

-- Medicine Inventory
CREATE TABLE medicine_inventory (

```



```

    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),
    branch_id UUID NOT NULL REFERENCES branches(id),
    pharmacy_id UUID NOT NULL REFERENCES pharmacy_locations(id),

    drug_id UUID NOT NULL REFERENCES drug_master(id),

    -- Current Stock
    quantity_on_hand INTEGER NOT NULL DEFAULT 0,
    quantity_reserved INTEGER DEFAULT 0, -- Reserved for orders
    quantity_available INTEGER GENERATED ALWAYS AS (quantity_on_hand -
quantity_reserved) STORED,

    -- Reorder Levels
    reorder_level INTEGER,
    reorder_quantity INTEGER,
    max_stock INTEGER,

    -- Unit
    stock_unit VARCHAR(50),

    -- Cost
    last_purchase_price DECIMAL(10, 2),
    average_cost DECIMAL(10, 2),
    selling_price DECIMAL(10, 2),

    -- Location
    bin_location VARCHAR(50),
    shelf_number VARCHAR(20),

    is_active BOOLEAN DEFAULT TRUE,
    last_stock_date TIMESTAMPTZ,

    updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

    UNIQUE(pharmacy_id, drug_id)
);

-- Medicine Batches
CREATE TABLE medicine_batches (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    inventory_id UUID NOT NULL REFERENCES medicine_inventory(id),
    drug_id UUID NOT NULL REFERENCES drug_master(id),
    pharmacy_id UUID NOT NULL REFERENCES pharmacy_locations(id),

    batch_number VARCHAR(50) NOT NULL,
    lot_number VARCHAR(50),

    manufacturing_date DATE,
    expiry_date DATE NOT NULL,

    -- Quantity
    received_quantity INTEGER NOT NULL,
    current_quantity INTEGER NOT NULL,

```

```

-- Cost
purchase_price DECIMAL(10, 2),
selling_price DECIMAL(10, 2),

-- Supplier
supplier_id UUID,
grn_number VARCHAR(50),

-- Status
is_active BOOLEAN DEFAULT TRUE,
is_quarantine BOOLEAN DEFAULT FALSE,

created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

UNIQUE(pharmacy_id, drug_id, batch_number)
);

-- Medicine Stock Transactions
CREATE TABLE medicine_stock_transactions (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),
    pharmacy_id UUID NOT NULL REFERENCES pharmacy_locations(id),

    drug_id UUID NOT NULL REFERENCES drug_master(id),
    batch_id UUID REFERENCES medicine_batches(id),

    transaction_type VARCHAR(50) NOT NULL,
    -- purchase, sale, return_supplier, return_patient, transfer_in, transfer_out,
    -- adjustment_in, adjustment_out, expired, damaged, consumption

    transaction_date TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,

    quantity INTEGER NOT NULL,
    unit_price DECIMAL(10, 2),
    total_amount DECIMAL(12, 2),

    -- Reference
    reference_type VARCHAR(50), -- purchase_order, prescription, return, transfer,
adjustment
    reference_id UUID,
    reference_number VARCHAR(50),

    -- Balance
    balance_before INTEGER,
    balance_after INTEGER,

    remarks TEXT,

    created_by UUID REFERENCES users(id),
    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

```

```

-- Medicine Dispensing
CREATE TABLE medicine_dispensing (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),
  branch_id UUID NOT NULL REFERENCES branches(id),
  pharmacy_id UUID NOT NULL REFERENCES pharmacy_locations(id),

  dispensing_number VARCHAR(50) NOT NULL,

  patient_id UUID NOT NULL REFERENCES patients(id),
  prescription_id UUID REFERENCES prescriptions(id),
  admission_id UUID REFERENCES admissions(id),

  dispensing_type VARCHAR(50), -- opd, ipd, discharge, emergency, etc

  dispensing_datetime TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
  dispensed_by UUID NOT NULL REFERENCES users(id),

  -- Billing
  total_amount DECIMAL(12, 2),
  discount_amount DECIMAL(10, 2),
  net_amount DECIMAL(12, 2),

  payment_status payment_status_type DEFAULT 'pending',
  bill_id UUID,

  -- Counseling
  counseling_done BOOLEAN DEFAULT FALSE,
  counseling_notes TEXT,

  status VARCHAR(50) DEFAULT 'completed', -- pending, partial, completed,
cancelled

  notes TEXT,

  created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

  UNIQUE(organization_id, dispensing_number)
);

-- Medicine Dispensing Items
CREATE TABLE medicine_dispensing_items (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  dispensing_id UUID NOT NULL REFERENCES medicine_dispensing(id) ON DELETE
CASCADE,

  drug_id UUID NOT NULL REFERENCES drug_master(id),
  batch_id UUID REFERENCES medicine_batches(id),

  prescription_item_id UUID REFERENCES prescription_items(id),

  drug_name VARCHAR(255) NOT NULL,

  quantity_prescribed INTEGER,

```

```

quantity_dispensed INTEGER NOT NULL,

-- Substitution
is_substituted BOOLEAN DEFAULT FALSE,
original_drug_id UUID REFERENCES drug_master(id),
substitution_reason TEXT,

unit_price DECIMAL(10, 2),
discount_percent DECIMAL(5, 2),
total_price DECIMAL(10, 2),

-- Dosage Instructions
dose VARCHAR(100),
frequency VARCHAR(100),
duration VARCHAR(50),
instructions TEXT,

-- Barcode Verification
barcode_verified BOOLEAN DEFAULT FALSE,

created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Medicine Returns
CREATE TABLE medicine_returns (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),
  pharmacy_id UUID NOT NULL REFERENCES pharmacy_locations(id),

  return_number VARCHAR(50) NOT NULL,

  return_type VARCHAR(50), -- patient_return, ward_return, supplier_return

  -- Source
  patient_id UUID REFERENCES patients(id),
  dispensing_id UUID REFERENCES medicine_dispensing(id),
  ward_id UUID REFERENCES wards(id),
  supplier_id UUID,

  return_date TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
  returned_by UUID REFERENCES users(id),

  reason TEXT,

  total_amount DECIMAL(12, 2),

  -- For patient returns - refund
  refund_amount DECIMAL(12, 2),
  refund_processed BOOLEAN DEFAULT FALSE,

  status VARCHAR(50) DEFAULT 'completed',

  created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

```

```

        UNIQUE(organization_id, return_number)
    );

-- Medicine Return Items
CREATE TABLE medicine_return_items (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    return_id UUID NOT NULL REFERENCES medicine_returns(id) ON DELETE CASCADE,

    drug_id UUID NOT NULL REFERENCES drug_master(id),
    batch_id UUID REFERENCES medicine_batches(id),

    quantity INTEGER NOT NULL,
    unit_price DECIMAL(10, 2),
    total_price DECIMAL(10, 2),

    condition VARCHAR(50), -- good, damaged, expired
    restock BOOLEAN DEFAULT FALSE,

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Ward Stock (Floor Stock)
CREATE TABLE ward_stock (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),
    ward_id UUID NOT NULL REFERENCES wards(id),

    drug_id UUID NOT NULL REFERENCES drug_master(id),

    current_quantity INTEGER DEFAULT 0,
    max_quantity INTEGER,
    reorder_level INTEGER,

    last_replenished_at TIMESTAMPTZ,
    last_replenished_by UUID REFERENCES users(id),

    updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

    UNIQUE(ward_id, drug_id)
);

-- Crash Cart Inventory
CREATE TABLE crash_cart_inventory (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),

    cart_code VARCHAR(20) NOT NULL,
    ward_id UUID REFERENCES wards(id),
    location VARCHAR(255),

    drug_id UUID NOT NULL REFERENCES drug_master(id),

    required_quantity INTEGER NOT NULL,
    current_quantity INTEGER DEFAULT 0,

```

```

    last_checked_at TIMESTAMPTZ,
    last_checked_by UUID REFERENCES users(id),

    expiry_date DATE, -- Earliest expiry in cart

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Narcotic Drug Register
CREATE TABLE narcotic_register (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),
    pharmacy_id UUID NOT NULL REFERENCES pharmacy_locations(id),

    drug_id UUID NOT NULL REFERENCES drug_master(id),

    -- Register Entry
    register_number VARCHAR(50) NOT NULL,
    entry_date DATE NOT NULL,

    transaction_type VARCHAR(50), -- receipt, issue, adjustment, wastage

    -- Quantities
    quantity INTEGER NOT NULL,
    balance_before INTEGER,
    balance_after INTEGER,

    -- For Issues
    patient_id UUID REFERENCES patients(id),
    patient_name VARCHAR(255),
    prescription_id UUID REFERENCES prescriptions(id),

    -- Witness
    witness_name VARCHAR(255),
    witness_id_number VARCHAR(50),

    -- Signatures (digital reference)
    pharmacist_id UUID REFERENCES users(id),
    witness_signature_url TEXT,

    remarks TEXT,

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

    UNIQUE(organization_id, register_number)
);

-- Cold Chain Log
CREATE TABLE cold_chain_log (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),
    pharmacy_id UUID NOT NULL REFERENCES pharmacy_locations(id),

```

```

refrigerator_id VARCHAR(50) NOT NULL,
refrigerator_name VARCHAR(100),

recorded_datetime TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,

temperature_celsius DECIMAL(4, 1) NOT NULL,
humidity_percent DECIMAL(4, 1),

is_within_range BOOLEAN,
min_temp DECIMAL(4, 1),
max_temp DECIMAL(4, 1),

excursion_detected BOOLEAN DEFAULT FALSE,
excursion_action TEXT,

recorded_by UUID REFERENCES users(id),

created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Indexes
CREATE INDEX idx_medicine_inventory_pharmacy ON medicine_inventory(pharmacy_id);
CREATE INDEX idx_medicine_inventory_drug ON medicine_inventory(drug_id);
CREATE INDEX idx_medicine_inventory_low_stock ON medicine_inventory(pharmacy_id)
    WHERE quantity_available <= reorder_level;

CREATE INDEX idx_medicine_batches_inventory ON medicine_batches(inventory_id);
CREATE INDEX idx_medicine_batches_expiry ON medicine_batches(expiry_date);
CREATE INDEX idx_medicine_batches_expiring ON medicine_batches(expiry_date)
    WHERE expiry_date <= CURRENT_DATE + INTERVAL '90 days';

CREATE INDEX idx_stock_transactions_pharmacy ON
medicine_stock_transactions(pharmacy_id);
CREATE INDEX idx_stock_transactions_drug ON medicine_stock_transactions(drug_id);
CREATE INDEX idx_stock_transactions_date ON
medicine_stock_transactions(transaction_date);

CREATE INDEX idx_dispensing_patient ON medicine_dispensing(patient_id);
CREATE INDEX idx_dispensing_prescription ON medicine_dispensing(prescription_id);
CREATE INDEX idx_dispensing_date ON medicine_dispensing(dispensing_datetime);

```

PART 11: INVENTORY & SUPPLY CHAIN

SQL

```

-- =====
-- SECTION 17: INVENTORY & SUPPLY CHAIN
-- =====

```

```
-- Item Categories
CREATE TABLE item_categories (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),

  code VARCHAR(20) NOT NULL,
  name VARCHAR(100) NOT NULL,

  parent_category_id UUID REFERENCES item_categories(id
```

```
-- =====
-- SECTION 17: INVENTORY & SUPPLY CHAIN (Continued)
-- =====

-- Item Categories (continued)
CREATE TABLE item_categories (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),

  code VARCHAR(20) NOT NULL,
  name VARCHAR(100) NOT NULL,

  parent_category_id UUID REFERENCES item_categories(id),

  category_type VARCHAR(50), -- medical_supplies, surgical, linen, housekeeping,
  office, food, it, ppe

  is_active BOOLEAN DEFAULT TRUE,
  created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

  UNIQUE(organization_id, code)
);

-- Item Master
CREATE TABLE item_master (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),

  item_code VARCHAR(50) NOT NULL,
  name VARCHAR(255) NOT NULL,
  name_bangla VARCHAR(255),
  description TEXT,

  category_id UUID REFERENCES item_categories(id),
  sub_category VARCHAR(100),

  -- Units
  base_unit VARCHAR(50) NOT NULL, -- piece, box, pack, kg, liter
  purchase_unit VARCHAR(50),
  issue_unit VARCHAR(50),
  conversion_factor DECIMAL(10, 4) DEFAULT 1,
```



```

-- Specifications
brand VARCHAR(100),
manufacturer VARCHAR(255),
model VARCHAR(100),
specifications JSONB,

-- Tracking
is_serialized BOOLEAN DEFAULT FALSE,
is_batch_tracked BOOLEAN DEFAULT FALSE,
has_expiry BOOLEAN DEFAULT FALSE,

-- Pricing
standard_cost DECIMAL(12, 2),
last_purchase_price DECIMAL(12, 2),

-- Stock Control
reorder_level INTEGER,
reorder_quantity INTEGER,
min_stock INTEGER,
max_stock INTEGER,
lead_time_days INTEGER,

-- Classification
abc_class VARCHAR(1), -- A, B, C
ved_class VARCHAR(1), -- V(ital), E(ssential), D(esirable)

-- For medical items
is_sterile BOOLEAN DEFAULT FALSE,
is_implant BOOLEAN DEFAULT FALSE,
is_consignment BOOLEAN DEFAULT FALSE,

-- Images
image_url TEXT,

is_active BOOLEAN DEFAULT TRUE,
created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

UNIQUE(organization_id, item_code)
);

-- Suppliers
CREATE TABLE suppliers (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),

  supplier_code VARCHAR(20) NOT NULL,
  name VARCHAR(255) NOT NULL,

  supplier_type VARCHAR(50), -- manufacturer, distributor, dealer, importer

-- Contact
contact_person VARCHAR(255),

```

```

phone VARCHAR(20),
email VARCHAR(255),
website VARCHAR(255),

-- Address
address_line1 VARCHAR(255),
address_line2 VARCHAR(255),
city VARCHAR(100),
district VARCHAR(100),
postal_code VARCHAR(20),
country VARCHAR(100) DEFAULT 'Bangladesh',

-- Legal/Tax
trade_license VARCHAR(100),
tin_number VARCHAR(50),
vat_registration VARCHAR(50),
drug_license VARCHAR(100), -- For pharma suppliers

-- Banking
bank_name VARCHAR(100),
bank_account VARCHAR(50),
bank_branch VARCHAR(100),

-- Terms
payment_terms INTEGER DEFAULT 30, -- Days
credit_limit DECIMAL(14, 2),

-- Rating
rating INTEGER CHECK (rating BETWEEN 1 AND 5),

-- Status
is_approved BOOLEAN DEFAULT FALSE,
approved_by UUID REFERENCES users(id),
approved_date DATE,

is_blacklisted BOOLEAN DEFAULT FALSE,
blacklist_reason TEXT,

is_active BOOLEAN DEFAULT TRUE,
created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

UNIQUE(organization_id, supplier_code)
);

-- Supplier Items (Items supplied by each supplier)
CREATE TABLE supplier_items (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  supplier_id UUID NOT NULL REFERENCES suppliers(id),
  item_id UUID NOT NULL REFERENCES item_master(id),

  supplier_item_code VARCHAR(50),
  supplier_item_name VARCHAR(255),

```

```

    unit_price DECIMAL(12, 2),
    currency VARCHAR(3) DEFAULT 'BDT',

    lead_time_days INTEGER,
    min_order_quantity INTEGER,

    is_preferred BOOLEAN DEFAULT FALSE,

    last_purchase_date DATE,
    last_purchase_price DECIMAL(12, 2),

    is_active BOOLEAN DEFAULT TRUE,
    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

    UNIQUE(supplier_id, item_id)
);

-- Stores/Warehouses
CREATE TABLE stores (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),
    branch_id UUID NOT NULL REFERENCES branches(id),

    store_code VARCHAR(20) NOT NULL,
    name VARCHAR(100) NOT NULL,

    store_type VARCHAR(50), -- central, sub_store, department_store, pharmacy

    location VARCHAR(255),
    floor VARCHAR(20),

    manager_id UUID REFERENCES users(id),

    is_active BOOLEAN DEFAULT TRUE,
    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

    UNIQUE(branch_id, store_code)
);

-- Store Inventory
CREATE TABLE store_inventory (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    store_id UUID NOT NULL REFERENCES stores(id),
    item_id UUID NOT NULL REFERENCES item_master(id),

    quantity_on_hand DECIMAL(12, 3) NOT NULL DEFAULT 0,
    quantity_reserved DECIMAL(12, 3) DEFAULT 0,
    quantity_available DECIMAL(12, 3) GENERATED ALWAYS AS (quantity_on_hand -
quantity_reserved) STORED,

    -- Location within store
    bin_location VARCHAR(50),
    shelf_number VARCHAR(20),
    rack_number VARCHAR(20),

```

```

-- Valuation
average_cost DECIMAL(12, 4),
last_purchase_price DECIMAL(12, 2),

-- Stock Levels
reorder_level DECIMAL(12, 3),
reorder_quantity DECIMAL(12, 3),

last_receipt_date TIMESTAMPTZ,
last_issue_date TIMESTAMPTZ,

updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

UNIQUE(store_id, item_id)
);

-- Item Batches
CREATE TABLE item_batches (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  store_id UUID NOT NULL REFERENCES stores(id),
  item_id UUID NOT NULL REFERENCES item_master(id),
  inventory_id UUID NOT NULL REFERENCES store_inventory(id),

  batch_number VARCHAR(50) NOT NULL,
  lot_number VARCHAR(50),
  serial_number VARCHAR(100),

  manufacturing_date DATE,
  expiry_date DATE,

  received_quantity DECIMAL(12, 3) NOT NULL,
  current_quantity DECIMAL(12, 3) NOT NULL,

  unit_cost DECIMAL(12, 4),

  supplier_id UUID REFERENCES suppliers(id),
  grn_id UUID,

  -- For sterilized items
  sterilization_date DATE,
  sterilization_expiry DATE,

  is_active BOOLEAN DEFAULT TRUE,
  created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

  UNIQUE(store_id, item_id, batch_number)
);

-- Purchase Requisitions
CREATE TABLE purchase_requisitions (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),
  branch_id UUID NOT NULL REFERENCES branches(id),

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pr_number VARCHAR(50) NOT NULL,

requesting_department_id UUID REFERENCES departments(id),
requesting_store_id UUID REFERENCES stores(id),

requested_by UUID NOT NULL REFERENCES users(id),
request_date DATE NOT NULL DEFAULT CURRENT_DATE,

required_date DATE,

priority VARCHAR(20) DEFAULT 'normal', -- low, normal, high, urgent

purpose TEXT,

-- Approval
status VARCHAR(20) DEFAULT 'draft', -- draft, submitted, approved, rejected,
converted, cancelled

approved_by UUID REFERENCES users(id),
approved_at TIMESTAMPTZ,
approval_remarks TEXT,

-- Conversion to PO
converted_to_po BOOLEAN DEFAULT FALSE,
po_id UUID,

total_estimated_amount DECIMAL(14, 2),

notes TEXT,

created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

UNIQUE(organization_id, pr_number)
);

-- Purchase Requisition Items
CREATE TABLE purchase_requisition_items (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  pr_id UUID NOT NULL REFERENCES purchase_requisitions(id) ON DELETE CASCADE,

  item_id UUID NOT NULL REFERENCES item_master(id),

  requested_quantity DECIMAL(12, 3) NOT NULL,
  unit VARCHAR(50),

  current_stock DECIMAL(12, 3),

  estimated_unit_price DECIMAL(12, 2),
  estimated_amount DECIMAL(12, 2),

  preferred_supplier_id UUID REFERENCES suppliers(id),

```

```

specifications TEXT,

-- Approval
approved_quantity DECIMAL(12, 3),

status VARCHAR(20) DEFAULT 'pending',

created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Request for Quotation
CREATE TABLE rfq (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),

  rfq_number VARCHAR(50) NOT NULL,

  pr_id UUID REFERENCES purchase_requisitions(id),

  rfq_date DATE NOT NULL DEFAULT CURRENT_DATE,
  submission_deadline TIMESTAMPTZ,

  status VARCHAR(20) DEFAULT 'draft', -- draft, sent, closed, evaluated

  terms_and_conditions TEXT,

  created_by UUID REFERENCES users(id),
  created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

  UNIQUE(organization_id, rfq_number)
);

-- RFQ Suppliers
CREATE TABLE rfq_suppliers (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  rfq_id UUID NOT NULL REFERENCES rfq(id) ON DELETE CASCADE,
  supplier_id UUID NOT NULL REFERENCES suppliers(id),

  sent_at TIMESTAMPTZ,
  sent_via VARCHAR(50), -- email, portal, manual

  response_received BOOLEAN DEFAULT FALSE,
  response_date TIMESTAMPTZ,

  total_quoted_amount DECIMAL(14, 2),

  is_selected BOOLEAN DEFAULT FALSE,

  UNIQUE(rfq_id, supplier_id)
);

-- Supplier Quotations
CREATE TABLE supplier_quotations (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),

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    rfq_supplier_id UUID NOT NULL REFERENCES rfq_suppliers(id),
    rfq_id UUID NOT NULL REFERENCES rfq(id),
    supplier_id UUID NOT NULL REFERENCES suppliers(id),

    quotation_number VARCHAR(50),
    quotation_date DATE,

    validity_date DATE,
    delivery_days INTEGER,
    payment_terms TEXT,

    total_amount DECIMAL(14, 2),

    terms_accepted BOOLEAN DEFAULT FALSE,

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Quotation Items
CREATE TABLE quotation_items (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    quotation_id UUID NOT NULL REFERENCES supplier_quotations(id) ON DELETE
    CASCADE,

    item_id UUID NOT NULL REFERENCES item_master(id),

    quantity DECIMAL(12, 3) NOT NULL,
    unit VARCHAR(50),

    unit_price DECIMAL(12, 2) NOT NULL,
    discount_percent DECIMAL(5, 2) DEFAULT 0,
    tax_percent DECIMAL(5, 2) DEFAULT 0,

    total_price DECIMAL(12, 2),

    brand_offered VARCHAR(100),
    delivery_days INTEGER,

    remarks TEXT
);

-- Purchase Orders
CREATE TABLE purchase_orders (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),
    branch_id UUID NOT NULL REFERENCES branches(id),

    po_number VARCHAR(50) NOT NULL,

    supplier_id UUID NOT NULL REFERENCES suppliers(id),

    pr_id UUID REFERENCES purchase_requisitions(id),
    quotation_id UUID REFERENCES supplier_quotations(id),

```

```

    po_date DATE NOT NULL DEFAULT CURRENT_DATE,
    expected_delivery_date DATE,

    delivery_address TEXT,
    delivery_store_id UUID REFERENCES stores(id),

    -- Amounts
    subtotal DECIMAL(14, 2),
    discount_amount DECIMAL(12, 2) DEFAULT 0,
    tax_amount DECIMAL(12, 2) DEFAULT 0,
    shipping_amount DECIMAL(10, 2) DEFAULT 0,
    total_amount DECIMAL(14, 2),

    currency VARCHAR(3) DEFAULT 'BDT',

    payment_terms TEXT,

    -- Status
    status VARCHAR(20) DEFAULT 'draft',
    -- draft, pending_approval, approved, sent, partial_received, received,
    cancelled

    -- Approval
    approved_by UUID REFERENCES users(id),
    approved_at TIMESTAMPTZ,

    -- Sending
    sent_at TIMESTAMPTZ,
    sent_by UUID REFERENCES users(id),

    terms_and_conditions TEXT,
    notes TEXT,

    created_by UUID REFERENCES users(id),
    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

    UNIQUE(organization_id, po_number)
);

-- Purchase Order Items
CREATE TABLE purchase_order_items (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    po_id UUID NOT NULL REFERENCES purchase_orders(id) ON DELETE CASCADE,

    item_id UUID NOT NULL REFERENCES item_master(id),

    ordered_quantity DECIMAL(12, 3) NOT NULL,
    unit VARCHAR(50),

    unit_price DECIMAL(12, 2) NOT NULL,
    discount_percent DECIMAL(5, 2) DEFAULT 0,
    tax_percent DECIMAL(5, 2) DEFAULT 0,

```



```

    line_total DECIMAL(12, 2),

    received_quantity DECIMAL(12, 3) DEFAULT 0,
    pending_quantity DECIMAL(12, 3),

    expected_delivery_date DATE,

    status VARCHAR(20) DEFAULT 'pending', -- pending, partial, received, cancelled

    remarks TEXT,

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Goods Receipt Note (GRN)
CREATE TABLE goods_receipt_notes (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),
    branch_id UUID NOT NULL REFERENCES branches(id),

    grn_number VARCHAR(50) NOT NULL,

    po_id UUID REFERENCES purchase_orders(id),
    supplier_id UUID NOT NULL REFERENCES suppliers(id),

    receiving_store_id UUID NOT NULL REFERENCES stores(id),

    receipt_date DATE NOT NULL DEFAULT CURRENT_DATE,

    -- Supplier Documents
    supplier_invoice_number VARCHAR(50),
    supplier_invoice_date DATE,
    supplier_challan_number VARCHAR(50),

    -- Inspection
    inspection_required BOOLEAN DEFAULT TRUE,
    inspection_status VARCHAR(20), -- pending, passed, failed, partial
    inspected_by UUID REFERENCES users(id),
    inspected_at TIMESTAMPTZ,

    -- Amounts
    invoice_amount DECIMAL(14, 2),
    received_amount DECIMAL(14, 2),
    variance_amount DECIMAL(12, 2),

    status VARCHAR(20) DEFAULT 'draft', -- draft, inspecting, approved, posted,
    rejected

    approved_by UUID REFERENCES users(id),
    approved_at TIMESTAMPTZ,

    posted_to_inventory BOOLEAN DEFAULT FALSE,
    posted_at TIMESTAMPTZ,

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    notes TEXT,

    created_by UUID REFERENCES users(id),
    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

    UNIQUE(organization_id, grn_number)
);

-- GRN Items
CREATE TABLE grn_items (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    grn_id UUID NOT NULL REFERENCES goods_receipt_notes(id) ON DELETE CASCADE,
    po_item_id UUID REFERENCES purchase_order_items(id),

    item_id UUID NOT NULL REFERENCES item_master(id),

    -- Quantities
    ordered_quantity DECIMAL(12, 3),
    received_quantity DECIMAL(12, 3) NOT NULL,
    accepted_quantity DECIMAL(12, 3),
    rejected_quantity DECIMAL(12, 3) DEFAULT 0,

    unit VARCHAR(50),

    -- Batch Info
    batch_number VARCHAR(50),
    manufacturing_date DATE,
    expiry_date DATE,

    -- Pricing
    unit_price DECIMAL(12, 2),
    line_total DECIMAL(12, 2),

    -- Inspection
    inspection_status VARCHAR(20),
    rejection_reason TEXT,

    -- Storage
    bin_location VARCHAR(50),

    remarks TEXT,

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Stock Issue (Department Indent)
CREATE TABLE stock_issues (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),
    branch_id UUID NOT NULL REFERENCES branches(id),

    issue_number VARCHAR(50) NOT NULL,

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    from_store_id UUID NOT NULL REFERENCES stores(id),
    to_department_id UUID REFERENCES departments(id),
    to_ward_id UUID REFERENCES wards(id),
    to_cost_center VARCHAR(100),

    indent_number VARCHAR(50),

    requested_by UUID REFERENCES users(id),
    request_date DATE,

    issue_date DATE NOT NULL DEFAULT CURRENT_DATE,
    issued_by UUID NOT NULL REFERENCES users(id),

    received_by UUID REFERENCES users(id),
    received_at TIMESTAMPTZ,

    total_value DECIMAL(14, 2),

    status VARCHAR(20) DEFAULT 'issued', -- draft, approved, issued, received,
cancelled

    notes TEXT,

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

    UNIQUE(organization_id, issue_number)
);

-- Stock Issue Items
CREATE TABLE stock_issue_items (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    issue_id UUID NOT NULL REFERENCES stock_issues(id) ON DELETE CASCADE,

    item_id UUID NOT NULL REFERENCES item_master(id),
    batch_id UUID REFERENCES item_batches(id),

    requested_quantity DECIMAL(12, 3),
    issued_quantity DECIMAL(12, 3) NOT NULL,

    unit VARCHAR(50),
    unit_cost DECIMAL(12, 4),
    line_value DECIMAL(12, 2),

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Stock Transfers
CREATE TABLE stock_transfers (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),

    transfer_number VARCHAR(50) NOT NULL,

    from_store_id UUID NOT NULL REFERENCES stores(id),

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to_store_id UUID NOT NULL REFERENCES stores(id),

transfer_date DATE NOT NULL DEFAULT CURRENT_DATE,

requested_by UUID REFERENCES users(id),
approved_by UUID REFERENCES users(id),

-- Dispatch
dispatched_by UUID REFERENCES users(id),
dispatched_at TIMESTAMPTZ,

-- Receipt
received_by UUID REFERENCES users(id),
received_at TIMESTAMPTZ,

total_value DECIMAL(14, 2),

status VARCHAR(20) DEFAULT 'pending', -- pending, approved, in_transit,
received, cancelled

notes TEXT,

created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

UNIQUE(organization_id, transfer_number)
);

-- Stock Transfer Items
CREATE TABLE stock_transfer_items (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  transfer_id UUID NOT NULL REFERENCES stock_transfers(id) ON DELETE CASCADE,

  item_id UUID NOT NULL REFERENCES item_master(id),
  batch_id UUID REFERENCES item_batches(id),

  requested_quantity DECIMAL(12, 3) NOT NULL,
  dispatched_quantity DECIMAL(12, 3),
  received_quantity DECIMAL(12, 3),

  unit VARCHAR(50),
  unit_cost DECIMAL(12, 4),

  variance_quantity DECIMAL(12, 3),
  variance_reason TEXT,

  created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Stock Adjustments
CREATE TABLE stock_adjustments (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),
  branch_id UUID NOT NULL REFERENCES branches(id),

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```

    adjustment_number VARCHAR(50) NOT NULL,

    store_id UUID NOT NULL REFERENCES stores(id),

    adjustment_type VARCHAR(50), -- physical_count, damage, expiry, theft,
    write_off, correction

    adjustment_date DATE NOT NULL DEFAULT CURRENT_DATE,

    reason TEXT NOT NULL,
    reference_document VARCHAR(100),

    total_adjustment_value DECIMAL(14, 2),

    status VARCHAR(20) DEFAULT 'draft', -- draft, pending_approval, approved,
    posted, rejected

    created_by UUID REFERENCES users(id),
    approved_by UUID REFERENCES users(id),
    approved_at TIMESTAMPTZ,

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

    UNIQUE(organization_id, adjustment_number)
);

-- Stock Adjustment Items
CREATE TABLE stock_adjustment_items (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    adjustment_id UUID NOT NULL REFERENCES stock_adjustments(id) ON DELETE
    CASCADE,

    item_id UUID NOT NULL REFERENCES item_master(id),
    batch_id UUID REFERENCES item_batches(id),

    system_quantity DECIMAL(12, 3) NOT NULL,
    physical_quantity DECIMAL(12, 3) NOT NULL,
    adjustment_quantity DECIMAL(12, 3) GENERATED ALWAYS AS (physical_quantity -
    system_quantity) STORED,

    unit VARCHAR(50),
    unit_cost DECIMAL(12, 4),
    adjustment_value DECIMAL(12, 2),

    reason TEXT,

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Stock Transactions
CREATE TABLE inventory_transactions (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),
    store_id UUID NOT NULL REFERENCES stores(id),

```

```

    item_id UUID NOT NULL REFERENCES item_master(id),
    batch_id UUID REFERENCES item_batches(id),

    transaction_type VARCHAR(50) NOT NULL,
    -- grn_receipt, po_return, issue, issue_return, transfer_out, transfer_in,
    -- adjustment_in, adjustment_out, opening_balance

    transaction_date TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,

    quantity DECIMAL(12, 3) NOT NULL,
    unit_cost DECIMAL(12, 4),
    total_value DECIMAL(14, 2),

    balance_quantity DECIMAL(12, 3),
    balance_value DECIMAL(14, 2),

    reference_type VARCHAR(50),
    reference_id UUID,
    reference_number VARCHAR(50),

    remarks TEXT,

    created_by UUID REFERENCES users(id),
    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Physical Inventory Count
CREATE TABLE physical_inventory_counts (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),
    branch_id UUID NOT NULL REFERENCES branches(id),

    count_number VARCHAR(50) NOT NULL,

    store_id UUID NOT NULL REFERENCES stores(id),

    count_date DATE NOT NULL,

    count_type VARCHAR(50), -- full, cycle, spot

    status VARCHAR(20) DEFAULT 'in_progress', -- in_progress, completed, approved,
    posted

    counted_by UUID[] ,
    supervised_by UUID REFERENCES users(id),

    approved_by UUID REFERENCES users(id),
    approved_at TIMESTAMPTZ,

    notes TEXT,

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

```

```

        UNIQUE(organization_id, count_number)
    );

-- Physical Count Items
CREATE TABLE physical_count_items (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    count_id UUID NOT NULL REFERENCES physical_inventory_counts(id) ON DELETE
    CASCADE,

    item_id UUID NOT NULL REFERENCES item_master(id),
    batch_id UUID REFERENCES item_batches(id),
    bin_location VARCHAR(50),

    system_quantity DECIMAL(12, 3),
    counted_quantity DECIMAL(12, 3),
    variance_quantity DECIMAL(12, 3),

    unit_cost DECIMAL(12, 4),
    variance_value DECIMAL(12, 2),

    count_1 DECIMAL(12, 3),
    count_2 DECIMAL(12, 3),
    count_3 DECIMAL(12, 3),

    remarks TEXT,

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Indexes for Inventory
CREATE INDEX idx_item_master_org ON item_master(organization_id);
CREATE INDEX idx_item_master_category ON item_master(category_id);
CREATE INDEX idx_item_master_code ON item_master(organization_id, item_code);

CREATE INDEX idx_suppliers_org ON suppliers(organization_id);
CREATE INDEX idx_suppliers_active ON suppliers(organization_id, is_active) WHERE
is_active = TRUE;

CREATE INDEX idx_store_inventory_store ON store_inventory(store_id);
CREATE INDEX idx_store_inventory_item ON store_inventory(item_id);
CREATE INDEX idx_store_inventory_low ON store_inventory(store_id) WHERE
quantity_available <= reorder_level;

CREATE INDEX idx_item_batches_store ON item_batches(store_id);
CREATE INDEX idx_item_batches_expiry ON item_batches(expiry_date);

CREATE INDEX idx_po_org ON purchase_orders(organization_id);
CREATE INDEX idx_po_supplier ON purchase_orders(supplier_id);
CREATE INDEX idx_po_status ON purchase_orders(status);

CREATE INDEX idx_grn_org ON goods_receipt_notes(organization_id);
CREATE INDEX idx_grn_po ON goods_receipt_notes(po_id);

CREATE INDEX idx_inv_trans_store ON inventory_transactions(store_id);

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CREATE INDEX idx_inv_trans_item ON inventory_transactions(item_id);
CREATE INDEX idx_inv_trans_date ON inventory_transactions(transaction_date);
```

PART 12: INSURANCE & CLAIM MANAGEMENT

SQL

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-- =====
-- SECTION 18: INSURANCE & CLAIM MANAGEMENT
-- =====

-- Insurance Companies
CREATE TABLE insurance_companies (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),

  company_code VARCHAR(20) NOT NULL,
  name VARCHAR(255) NOT NULL,
  short_name VARCHAR(50),

  company_type VARCHAR(50), -- insurance, tpa, corporate

  -- Contact
  contact_person VARCHAR(255),
  phone VARCHAR(20),
  email VARCHAR(255),
  website VARCHAR(255),

  -- Address
  address TEXT,
  city VARCHAR(100),

  -- Financial
  credit_limit DECIMAL(14, 2),
  current_outstanding DECIMAL(14, 2) DEFAULT 0,

  -- Agreement
  agreement_start_date DATE,
  agreement_end_date DATE,
  discount_percent DECIMAL(5, 2) DEFAULT 0,

  -- Processing
  claim_submission_email VARCHAR(255),
  claim_portal_url VARCHAR(255),

  -- Settlement
  typical_settlement_days INTEGER,

  is_active BOOLEAN DEFAULT TRUE,
  created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
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        updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

        UNIQUE(organization_id, company_code)
    );

-- Insurance Plans/Schemes
CREATE TABLE insurance_plans (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    insurance_company_id UUID NOT NULL REFERENCES insurance_companies(id),

    plan_code VARCHAR(50) NOT NULL,
    plan_name VARCHAR(255) NOT NULL,

    plan_type VARCHAR(50), -- individual, family, group, corporate

    -- Coverage
    coverage_type VARCHAR(50), -- cashless, reimbursement, both

    -- Limits
    sum_insured DECIMAL(14, 2),
    room_rent_limit DECIMAL(10, 2),
    room_rent_type VARCHAR(20), -- fixed, percent_of_sum
    icu_limit DECIMAL(10, 2),

    -- Co-pay
    copay_percent DECIMAL(5, 2) DEFAULT 0,
    copay_fixed DECIMAL(10, 2) DEFAULT 0,

    -- Deductible
    deductible_amount DECIMAL(10, 2) DEFAULT 0,

    -- Pre-existing
    pre_existing_waiting_years INTEGER DEFAULT 4,

    -- Coverage Details
    covered_services JSONB,
    exclusions JSONB,

    is_active BOOLEAN DEFAULT TRUE,
    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

    UNIQUE(insurance_company_id, plan_code)
);

-- Insurance Tariff (Agreed Rates)
CREATE TABLE insurance_tariffs (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    insurance_company_id UUID NOT NULL REFERENCES insurance_companies(id),

    service_type VARCHAR(50), -- room, procedure, investigation, package
    service_code VARCHAR(50),
    service_name VARCHAR(255),

    -- Reference

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    procedure_id UUID, -- Reference to procedures
    test_id UUID REFERENCES test_catalog(id),
    room_type VARCHAR(50),

    -- Rates
    hospital_rate DECIMAL(10, 2),
    agreed_rate DECIMAL(10, 2),
    discount_percent DECIMAL(5, 2),

    effective_from DATE,
    effective_to DATE,

    is_active BOOLEAN DEFAULT TRUE,
    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Patient Insurance Policies
CREATE TABLE patient_insurance_policies (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    patient_id UUID NOT NULL REFERENCES patients(id),
    insurance_company_id UUID NOT NULL REFERENCES insurance_companies(id),
    plan_id UUID REFERENCES insurance_plans(id),

    policy_number VARCHAR(100) NOT NULL,
    member_id VARCHAR(100),
    group_id VARCHAR(100),

    policy_holder_name VARCHAR(255),
    relationship_to_patient VARCHAR(50), -- self, spouse, child, parent

    -- Validity
    effective_date DATE NOT NULL,
    expiry_date DATE NOT NULL,

    -- Coverage
    sum_insured DECIMAL(14, 2),
    available_balance DECIMAL(14, 2),

    -- Corporate
    corporate_name VARCHAR(255),
    employee_id VARCHAR(50),

    -- Verification
    is_verified BOOLEAN DEFAULT FALSE,
    verified_at TIMESTAMPTZ,
    verified_by UUID REFERENCES users(id),
    verification_reference VARCHAR(100),

    -- Documents
    policy_document_url TEXT,
    card_front_url TEXT,
    card_back_url TEXT,

    is_primary BOOLEAN DEFAULT FALSE,

```

```

    is_active BOOLEAN DEFAULT TRUE,

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Pre-Authorization Requests
CREATE TABLE insurance_preauth (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),
    branch_id UUID NOT NULL REFERENCES branches(id),

    preauth_number VARCHAR(50) NOT NULL,

    patient_id UUID NOT NULL REFERENCES patients(id),
    policy_id UUID NOT NULL REFERENCES patient_insurance_policies(id),
    insurance_company_id UUID NOT NULL REFERENCES insurance_companies(id),

    admission_id UUID REFERENCES admissions(id),

    -- Request Details
    request_type VARCHAR(50), -- planned_admission, emergency, extension,
    enhancement

    admission_type VARCHAR(50), -- planned, emergency
    expected_admission_date DATE,
    expected_discharge_date DATE,
    expected_los_days INTEGER,

    -- Clinical
    diagnosis TEXT,
    icd_codes VARCHAR(100)[],

    proposed_treatment TEXT,
    proposed_procedures TEXT[],

    treating_doctor_id UUID REFERENCES doctors(id),

    -- Financial
    estimated_cost DECIMAL(14, 2),
    requested_amount DECIMAL(14, 2),

    -- Room
    room_type_requested VARCHAR(50),
    room_rate_per_day DECIMAL(10, 2),

    -- Documents
    documents JSONB, -- [{type, url, uploaded_at}]

    -- Status
    status claim_status_type DEFAULT 'draft',

    submitted_at TIMESTAMPTZ,
    submitted_by UUID REFERENCES users(id),

```

```

-- TPA Response
tpa_reference_number VARCHAR(100),

approved_amount DECIMAL(14, 2),
approved_los_days INTEGER,
approval_remarks TEXT,

rejection_reason TEXT,

responded_at TIMESTAMPTZ,

-- Extension/Enhancement
parent_preauth_id UUID REFERENCES insurance_preauth(id),

notes TEXT,

created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

UNIQUE(organization_id, preauth_number)
);

-- Pre-Auth Communication Log
CREATE TABLE preauth_communications (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  preauth_id UUID NOT NULL REFERENCES insurance_preauth(id),

  communication_type VARCHAR(50), -- query, response, update, escalation
  direction VARCHAR(20), -- incoming, outgoing

  communication_datetime TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

  subject TEXT,
  message TEXT,

  attachments JSONB,

  sent_by UUID REFERENCES users(id),
  received_from VARCHAR(255),

  created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Insurance Claims
CREATE TABLE insurance_claims (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),
  branch_id UUID NOT NULL REFERENCES branches(id),

  claim_number VARCHAR(50) NOT NULL,

  patient_id UUID NOT NULL REFERENCES patients(id),
  admission_id UUID REFERENCES admissions(id),

```

```
visit_id UUID,

policy_id UUID NOT NULL REFERENCES patient_insurance_policies(id),
insurance_company_id UUID NOT NULL REFERENCES insurance_companies(id),
preauth_id UUID REFERENCES insurance_preauth(id),

claim_type VARCHAR(50), -- cashless, reimbursement

-- Clinical
admission_date DATE,
discharge_date DATE,
los_days INTEGER,

diagnosis TEXT,
icd_codes VARCHAR(100)[],
procedures_performed TEXT,

-- Financial
total_bill_amount DECIMAL(14, 2),
non_payable_amount DECIMAL(12, 2) DEFAULT 0,
deduction_amount DECIMAL(12, 2) DEFAULT 0,
net_claim_amount DECIMAL(14, 2),

-- Pre-auth Amounts
preauth_amount DECIMAL(14, 2),

-- Status
status claim_status_type DEFAULT 'draft',

-- Submission
submitted_at TIMESTAMPTZ,
submitted_by UUID REFERENCES users(id),
submission_mode VARCHAR(50), -- online, physical, email

-- TPA
tpa_claim_number VARCHAR(100),
tpa_received_date DATE,

-- Settlement
approved_amount DECIMAL(14, 2),
disallowed_amount DECIMAL(12, 2),
disallowance_reasons JSONB,

settled_amount DECIMAL(14, 2),
settlement_date DATE,
settlement_reference VARCHAR(100),

-- Patient Responsibility
patient_payable DECIMAL(12, 2),
patient_paid DECIMAL(12, 2),

-- Deductions
tds_amount DECIMAL(10, 2),
```

```

-- Documents
claim_documents JSONB,

notes TEXT,

created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

UNIQUE(organization_id, claim_number)
);

-- Claim Items (Bill Breakdown)
CREATE TABLE claim_items (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  claim_id UUID NOT NULL REFERENCES insurance_claims(id) ON DELETE CASCADE,

  item_category VARCHAR(50), -- room, professional_fees, investigation,
  medicine, procedure, consumables
  item_description VARCHAR(500),

  service_date DATE,
  quantity INTEGER DEFAULT 1,

  billed_amount DECIMAL(12, 2),

  -- Adjudication
  approved_amount DECIMAL(12, 2),
  disallowed_amount DECIMAL(12, 2),
  disallowance_code VARCHAR(20),
  disallowance_reason TEXT,

  created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Claim Status History
CREATE TABLE claim_status_history (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  claim_id UUID NOT NULL REFERENCES insurance_claims(id),

  previous_status claim_status_type,
  new_status claim_status_type NOT NULL,

  changed_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
  changed_by UUID REFERENCES users(id),

  remarks TEXT
);

-- Corporate Accounts
CREATE TABLE corporate_accounts (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),

  company_code VARCHAR(20) NOT NULL,

```

```

company_name VARCHAR(255) NOT NULL,

-- Contact
contact_person VARCHAR(255),
phone VARCHAR(20),
email VARCHAR(255),

address TEXT,

-- Financial
credit_limit DECIMAL(14, 2),
current_outstanding DECIMAL(14, 2) DEFAULT 0,

-- Discount
discount_percent DECIMAL(5, 2) DEFAULT 0,

-- Agreement
agreement_start_date DATE,
agreement_end_date DATE,

-- Billing
billing_cycle VARCHAR(20), -- weekly, bi_weekly, monthly
payment_terms_days INTEGER DEFAULT 30,

is_active BOOLEAN DEFAULT TRUE,
created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

UNIQUE(organization_id, company_code)
);

-- Corporate Employees
CREATE TABLE corporate_employees (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  corporate_id UUID NOT NULL REFERENCES corporate_accounts(id),
  patient_id UUID NOT NULL REFERENCES patients(id),

  employee_id VARCHAR(50) NOT NULL,
  employee_name VARCHAR(255),
  department VARCHAR(100),
  designation VARCHAR(100),

  coverage_limit DECIMAL(12, 2),
  used_amount DECIMAL(12, 2) DEFAULT 0,

  valid_from DATE,
  valid_until DATE,

  is_active BOOLEAN DEFAULT TRUE,

  UNIQUE(corporate_id, employee_id)
);

-- Corporate Invoices

```

```

CREATE TABLE corporate_invoices (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),

  invoice_number VARCHAR(50) NOT NULL,
  corporate_id UUID NOT NULL REFERENCES corporate_accounts(id),

  invoice_date DATE NOT NULL,
  due_date DATE,

  period_from DATE,
  period_to DATE,

  -- Amounts
  subtotal DECIMAL(14, 2),
  discount_amount DECIMAL(12, 2) DEFAULT 0,
  tax_amount DECIMAL(12, 2) DEFAULT 0,
  total_amount DECIMAL(14, 2),

  paid_amount DECIMAL(14, 2) DEFAULT 0,
  balance_amount DECIMAL(14, 2),

  status VARCHAR(20) DEFAULT 'draft', -- draft, sent, partial_paid, paid,
  overdue, disputed

  sent_at TIMESTAMPTZ,

  notes TEXT,

  created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
  updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

  UNIQUE(organization_id, invoice_number)
);

-- Indexes
CREATE INDEX idx_insurance_companies_org ON insurance_companies(organization_id);
CREATE INDEX idx_patient_policies_patient ON
patient_insurance_policies(patient_id);
CREATE INDEX idx_patient_policies_company ON
patient_insurance_policies(insurance_company_id);

CREATE INDEX idx_preauth_org ON insurance_preauth(organization_id);
CREATE INDEX idx_preauth_patient ON insurance_preauth(patient_id);
CREATE INDEX idx_preauth_status ON insurance_preauth(status);

CREATE INDEX idx_claims_org ON insurance_claims(organization_id);
CREATE INDEX idx_claims_patient ON insurance_claims(patient_id);
CREATE INDEX idx_claims_admission ON insurance_claims(admission_id);
CREATE INDEX idx_claims_status ON insurance_claims(status);
CREATE INDEX idx_claims_company ON insurance_claims(insurance_company_id);

```


📁 PART 13: HOUSEKEEPING MANAGEMENT

SQL

```
-- =====
-- SECTION 19: HOUSEKEEPING MANAGEMENT
-- =====

-- Housekeeping Staff
CREATE TABLE housekeeping_staff (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    user_id UUID NOT NULL REFERENCES users(id),
    organization_id UUID NOT NULL REFERENCES organizations(id),
    branch_id UUID NOT NULL REFERENCES branches(id),

    staff_code VARCHAR(20),

    designation VARCHAR(50), -- cleaner, supervisor, manager

    assigned_zone_id UUID,
    current_shift shift_type,

    is_active BOOLEAN DEFAULT TRUE,
    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

    UNIQUE(organization_id, staff_code)
);

-- Cleaning Zones
CREATE TABLE cleaning_zones (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),
    branch_id UUID NOT NULL REFERENCES branches(id),

    zone_code VARCHAR(20) NOT NULL,
    zone_name VARCHAR(100) NOT NULL,

    zone_type VARCHAR(50), -- ward, ot, icu, public_area, office, corridor

    -- Coverage
    buildings TEXT[],
    floors TEXT[],

    -- Risk Level
    risk_level VARCHAR(20), -- high, medium, low
    color_code VARCHAR(20), -- red, yellow, green

    -- Cleaning Frequency
    cleaning_frequency VARCHAR(50), -- hourly, twice_daily, daily, weekly

    supervisor_id UUID REFERENCES housekeeping_staff(id),
```

```

    special_instructions TEXT,

    is_active BOOLEAN DEFAULT TRUE,
    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

    UNIQUE(branch_id, zone_code)
);

-- Zone Area Mapping
CREATE TABLE zone_areas (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    zone_id UUID NOT NULL REFERENCES cleaning_zones(id),

    area_type VARCHAR(50), -- ward, room, corridor, toilet, nurses_station,
ot_room
    area_id UUID, -- Reference to ward/room/bed
    area_name VARCHAR(100),

    is_active BOOLEAN DEFAULT TRUE
);

-- Cleaning Schedules
CREATE TABLE cleaning_schedules (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),
    branch_id UUID NOT NULL REFERENCES branches(id),

    schedule_type VARCHAR(50), -- routine, discharge, terminal, deep_cleaning

    zone_id UUID REFERENCES cleaning_zones(id),
    area_type VARCHAR(50),
    area_id UUID,
    bed_id UUID REFERENCES beds(id),

    -- Schedule
    scheduled_date DATE NOT NULL,
    scheduled_time TIME,

    frequency VARCHAR(50), -- once, daily, weekly, monthly

    -- Assignment
    assigned_to UUID REFERENCES housekeeping_staff(id),
    assigned_at TIMESTAMPTZ,

    -- Trigger
    triggered_by VARCHAR(50), -- schedule, discharge, transfer, manual, request
    trigger_reference_id UUID,

    priority VARCHAR(20) DEFAULT 'normal',

    -- Checklist
    checklist_id UUID,

    -- Progress

```

```

    status VARCHAR(20) DEFAULT 'pending', -- pending, assigned, in_progress,
    completed, verified

    started_at TIMESTAMPTZ,
    completed_at TIMESTAMPTZ,
    completed_by UUID REFERENCES housekeeping_staff(id),

    -- QA
    verified_by UUID REFERENCES users(id),
    verified_at TIMESTAMPTZ,
    verification_status VARCHAR(20), -- passed, failed
    verification_remarks TEXT,

    -- Photos
    before_photos TEXT[],
    after_photos TEXT[],

    notes TEXT,

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Cleaning Checklists
CREATE TABLE cleaning_checklists (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),

    checklist_code VARCHAR(20) NOT NULL,
    name VARCHAR(100) NOT NULL,

    cleaning_type VARCHAR(50), -- routine, discharge, terminal, ot, isolation
    area_type VARCHAR(50),

    items JSONB NOT NULL, -- [{item, is_mandatory, instructions}]

    estimated_duration_minutes INTEGER,

    is_active BOOLEAN DEFAULT TRUE,
    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

    UNIQUE(organization_id, checklist_code)
);

-- Cleaning Task Execution
CREATE TABLE cleaning_task_execution (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    schedule_id UUID NOT NULL REFERENCES cleaning_schedules(id),
    checklist_id UUID REFERENCES cleaning_checklists(id),

    executed_by UUID NOT NULL REFERENCES housekeeping_staff(id),

    checklist_responses JSONB, -- [{item, completed, time, remarks}]

```

```

chemicals_used JSONB, -- [{chemical, quantity}]

execution_time_minutes INTEGER,

issues_found TEXT,
action_taken TEXT,

created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Linen Management
CREATE TABLE linen_types (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),

  code VARCHAR(20) NOT NULL,
  name VARCHAR(100) NOT NULL,

  category VARCHAR(50), -- bed_sheet, pillow_cover, blanket, towel, curtain,
  patient_gown

  par_level_per_bed INTEGER,

  is_active BOOLEAN DEFAULT TRUE,

  UNIQUE(organization_id, code)
);

-- Linen Inventory
CREATE TABLE linen_inventory (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),
  branch_id UUID NOT NULL REFERENCES branches(id),

  linen_type_id UUID NOT NULL REFERENCES linen_types(id),

  total_quantity INTEGER DEFAULT 0,
  available_quantity INTEGER DEFAULT 0,
  in_use_quantity INTEGER DEFAULT 0,
  in_laundry_quantity INTEGER DEFAULT 0,
  condemned_quantity INTEGER DEFAULT 0,

  last_count_date DATE,

  updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

  UNIQUE(branch_id, linen_type_id)
);

-- Linen Transactions
CREATE TABLE linen_transactions (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),
  branch_id UUID NOT NULL REFERENCES branches(id),

```

```

        linen_type_id UUID NOT NULL REFERENCES linen_types(id),

        transaction_type VARCHAR(50), -- issue, return, send_laundry, receive_laundry,
        condemn, purchase

        quantity INTEGER NOT NULL,

        -- For issue/return
        ward_id UUID REFERENCES wards(id),
        bed_id UUID REFERENCES beds(id),

        transaction_date TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
        transacted_by UUID REFERENCES users(id),

        remarks TEXT,

        created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
    );

-- Disinfection Log
CREATE TABLE disinfection_log (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),
    branch_id UUID NOT NULL REFERENCES branches(id),

    disinfection_type VARCHAR(50), -- surface, fumigation, uv, fogging

    area_type VARCHAR(50),
    area_id UUID,
    area_name VARCHAR(255),

    -- Schedule
    scheduled_datetime TIMESTAMPTZ,

    -- Execution
    started_at TIMESTAMPTZ,
    completed_at TIMESTAMPTZ,

    performed_by UUID REFERENCES users(id),

    -- Details
    chemicals_used JSONB,
    concentration VARCHAR(50),
    contact_time_minutes INTEGER,

    -- For fumigation
    room_sealed BOOLEAN,
    ventilation_time_minutes INTEGER,

    -- Verification
    verification_done BOOLEAN DEFAULT FALSE,
    verified_by UUID REFERENCES users(id),
    verification_method VARCHAR(50),

```

```

        verification_result VARCHAR(50),

        notes TEXT,

        created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
    );

-- Cleanliness Complaints
CREATE TABLE cleanliness_complaints (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),
    branch_id UUID NOT NULL REFERENCES branches(id),

    complaint_number VARCHAR(50) NOT NULL,

    location VARCHAR(255),
    area_type VARCHAR(50),
    area_id UUID,

    complaint_type VARCHAR(50), -- dirty_floor, dirty_toilet, bad_smell,
    waste_not_cleared, linen_dirty

    description TEXT,

    reported_by VARCHAR(255),
    reported_by_user_id UUID REFERENCES users(id),
    reported_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

    priority VARCHAR(20) DEFAULT 'normal',

    -- Assignment
    assigned_to UUID REFERENCES housekeeping_staff(id),
    assigned_at TIMESTAMPTZ,

    -- Resolution
    status VARCHAR(20) DEFAULT 'open', -- open, assigned, in_progress, resolved,
    closed

    resolved_at TIMESTAMPTZ,
    resolved_by UUID REFERENCES users(id),
    resolution_notes TEXT,

    response_time_minutes INTEGER,
    resolution_time_minutes INTEGER,

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

    UNIQUE(organization_id, complaint_number)
);

-- Indexes
CREATE INDEX idx_cleaning_schedules_branch ON cleaning_schedules(branch_id);
CREATE INDEX idx_cleaning_schedules_date ON cleaning_schedules(scheduled_date);
CREATE INDEX idx_cleaning_schedules_status ON cleaning_schedules(status);

```

```
CREATE INDEX idx_cleaning_schedules_assigned ON cleaning_schedules(assigned_to);

CREATE INDEX idx_linen_transactions_branch ON linen_transactions(branch_id);
CREATE INDEX idx_linen_transactions_type ON linen_transactions(linen_type_id);
```

PART 14: HR & PAYROLL

SQL

```
-- =====
-- SECTION 20: HR & PAYROLL
-- =====

-- Employee Master
CREATE TABLE employees (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    user_id UUID REFERENCES users(id),
    organization_id UUID NOT NULL REFERENCES organizations(id),
    branch_id UUID NOT NULL REFERENCES branches(id),

    employee_code VARCHAR(20) NOT NULL,

    -- Personal Info
    first_name VARCHAR(100) NOT NULL,
    last_name VARCHAR(100),
    full_name VARCHAR(255),
    name_bangla VARCHAR(255),

    father_name VARCHAR(255),
    mother_name VARCHAR(255),
    spouse_name VARCHAR(255),

    gender gender_type,
    date_of_birth DATE,
    blood_group blood_group_type,
    marital_status marital_status_type,
    nationality VARCHAR(100) DEFAULT 'Bangladeshi',
    religion VARCHAR(50),

    -- Contact
    personal_phone VARCHAR(20),
    personal_email VARCHAR(255),
    emergency_contact_name VARCHAR(255),
    emergency_contact_phone VARCHAR(20),
    emergency_contact_relation VARCHAR(50),

    -- Identity
    nid VARCHAR(20),
    passport_no VARCHAR(50),
    passport_expiry DATE,
```

```
driving_license VARCHAR(50),

-- Address
present_address TEXT,
permanent_address TEXT,

-- Photo
photo_url TEXT,

-- Employment
department_id UUID REFERENCES departments(id),
designation VARCHAR(100),
job_title VARCHAR(100),
grade VARCHAR(20),

reporting_manager_id UUID REFERENCES employees(id),

employment_type employment_type DEFAULT 'permanent',

joining_date DATE NOT NULL,
confirmation_date DATE,
probation_end_date DATE,
resignation_date DATE,
last_working_date DATE,

-- Work Details
work_location VARCHAR(100),
shift_id UUID,

-- Bank
bank_name VARCHAR(100),
bank_branch VARCHAR(100),
bank_account_number VARCHAR(50),
bank_routing_number VARCHAR(50),

-- Mobile Banking
mobile_banking_provider VARCHAR(50), -- bkaash, nagad, rocket
mobile_banking_number VARCHAR(20),

-- Tax
tin_number VARCHAR(50),

-- Salary
basic_salary DECIMAL(12, 2),
gross_salary DECIMAL(12, 2),

-- PF
pf_enrolled BOOLEAN DEFAULT FALSE,
pf_number VARCHAR(50),
pf_contribution_percent DECIMAL(5, 2),

-- Status
employment_status VARCHAR(20) DEFAULT 'active', -- active, on_notice,
resigned, terminated, retired
```



```

    is_active BOOLEAN DEFAULT TRUE,
    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

    UNIQUE(organization_id, employee_code)
);

-- Employee Qualifications
CREATE TABLE employee_qualifications (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    employee_id UUID NOT NULL REFERENCES employees(id),

    qualification_type VARCHAR(50), -- education, certification, license

    -- Education
    degree VARCHAR(100),
    field_of_study VARCHAR(100),
    institution VARCHAR(255),
    board_university VARCHAR(255),
    year_of_passing INTEGER,
    result VARCHAR(50),

    -- Certification/License
    certificate_name VARCHAR(255),
    issuing_authority VARCHAR(255),
    certificate_number VARCHAR(100),
    issue_date DATE,
    expiry_date DATE,

    -- Document
    document_url TEXT,

    is_verified BOOLEAN DEFAULT FALSE,
    verified_by UUID REFERENCES users(id),

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Employee Documents
CREATE TABLE employee_documents (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    employee_id UUID NOT NULL REFERENCES employees(id),

    document_type VARCHAR(50), -- nid, passport, photo, certificate,
    appointment_letter, contract, other
    document_name VARCHAR(255),

    file_url TEXT NOT NULL,
    file_size BIGINT,

    expiry_date DATE,

    uploaded_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

```

```

        uploaded_by UUID REFERENCES users(id)
    );

-- Shifts
CREATE TABLE shifts (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),

    shift_code VARCHAR(20) NOT NULL,
    name VARCHAR(100) NOT NULL,

    shift_type shift_type,

    start_time TIME NOT NULL,
    end_time TIME NOT NULL,

    break_duration_minutes INTEGER DEFAULT 60,

    -- Grace Period
    late_grace_minutes INTEGER DEFAULT 15,
    early_leave_grace_minutes INTEGER DEFAULT 15,

    -- Overtime
    ot_threshold_minutes INTEGER DEFAULT 30,

    is_night_shift BOOLEAN DEFAULT FALSE,

    applicable_days INTEGER[] DEFAULT ARRAY[0,1,2,3,4,5,6], -- 0=Sunday

    is_active BOOLEAN DEFAULT TRUE,
    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

    UNIQUE(organization_id, shift_code)
);

-- Attendance
CREATE TABLE attendance (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),
    branch_id UUID NOT NULL REFERENCES branches(id),

    employee_id UUID NOT NULL REFERENCES employees(id),
    attendance_date DATE NOT NULL,

    shift_id UUID REFERENCES shifts(id),

    -- Punch Times
    punch_in_time TIMESTAMPTZ,
    punch_out_time TIMESTAMPTZ,

    -- Source
    punch_in_source VARCHAR(50), -- biometric, rfid, manual, geo_fence,
    face_recognition
    punch_out_source VARCHAR(50),

```

```

    punch_in_device VARCHAR(100),
    punch_out_device VARCHAR(100),

    punch_in_location POINT, -- For geo-fence
    punch_out_location POINT,

    -- Calculated
    working_hours DECIMAL(4, 2),
    overtime_hours DECIMAL(4, 2),

    -- Status
    status VARCHAR(20), -- present, absent, half_day, leave, holiday, week_off,
late, early_leave

    late_minutes INTEGER DEFAULT 0,
    early_leave_minutes INTEGER DEFAULT 0,

    -- Regularization
    is_regularized BOOLEAN DEFAULT FALSE,
    regularization_reason TEXT,
    regularized_by UUID REFERENCES users(id),

    remarks TEXT,

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

    UNIQUE(employee_id, attendance_date)
);

-- Leave Types
CREATE TABLE leave_types (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),

    leave_code VARCHAR(20) NOT NULL,
    name VARCHAR(100) NOT NULL,

    leave_category VARCHAR(50), -- earned, casual, sick, maternity, paternity,
unpaid, compensatory

    -- Annual Quota
    annual_quota INTEGER,

    -- Rules
    is_paid BOOLEAN DEFAULT TRUE,
    is_carry_forward BOOLEAN DEFAULT FALSE,
    max_carry_forward INTEGER,

    is_encashable BOOLEAN DEFAULT FALSE,
    max_encashment INTEGER,

    min_days_per_application DECIMAL(3,1) DEFAULT 0.5,

```

```

max_days_per_application INTEGER,

requires_document BOOLEAN DEFAULT FALSE, -- Medical certificate etc.

advance_days_required INTEGER DEFAULT 0, -- Days before leave

-- Applicability
applicable_gender gender_type, -- NULL for all
min_service_months INTEGER DEFAULT 0,

is_active BOOLEAN DEFAULT TRUE,
created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

UNIQUE(organization_id, leave_code)
);

-- Leave Balance
CREATE TABLE leave_balances (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  employee_id UUID NOT NULL REFERENCES employees(id),
  leave_type_id UUID NOT NULL REFERENCES leave_types(id),

  year INTEGER NOT NULL,

  opening_balance DECIMAL(5, 1) DEFAULT 0,
  accrued DECIMAL(5, 1) DEFAULT 0,
  taken DECIMAL(5, 1) DEFAULT 0,
  adjustment DECIMAL(5, 1) DEFAULT 0,
  carry_forward DECIMAL(5, 1) DEFAULT 0,
  encashed DECIMAL(5, 1) DEFAULT 0,

  available_balance DECIMAL(5, 1) GENERATED ALWAYS AS (
    opening_balance + accrued + carry_forward + adjustment - taken - encashed
  ) STORED,

  updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

  UNIQUE(employee_id, leave_type_id, year)
);

-- Leave Applications
CREATE TABLE leave_applications (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),

  application_number VARCHAR(50) NOT NULL,

  employee_id UUID NOT NULL REFERENCES employees(id),
  leave_type_id UUID NOT NULL REFERENCES leave_types(id),

  from_date DATE NOT NULL,
  to_date DATE NOT NULL,

  number_of_days DECIMAL(4, 1) NOT NULL,

```

```

is_half_day BOOLEAN DEFAULT FALSE,
half_day_type VARCHAR(20), -- first_half, second_half

reason TEXT,

-- Contact during leave
contact_number VARCHAR(20),
address_during_leave TEXT,

-- Documents
supporting_documents TEXT[],

-- Substitute
substitute_employee_id UUID REFERENCES employees(id),

-- Status
status leave_status_type DEFAULT 'pending',

-- Approval
applied_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

-- Level 1
approver_1_id UUID REFERENCES users(id),
approver_1_status leave_status_type,
approver_1_at TIMESTAMPTZ,
approver_1_remarks TEXT,

-- Level 2
approver_2_id UUID REFERENCES users(id),
approver_2_status leave_status_type,
approver_2_at TIMESTAMPTZ,
approver_2_remarks TEXT,

cancelled_at TIMESTAMPTZ,
cancellation_reason TEXT,

created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

UNIQUE(organization_id, application_number)
);

-- Salary Components
CREATE TABLE salary_components (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),

  component_code VARCHAR(20) NOT NULL,
  name VARCHAR(100) NOT NULL,

  component_type VARCHAR(20) NOT NULL, -- earning, deduction

  calculation_type VARCHAR(20), -- fixed, percent_of_basic, percent_of_gross,

```

```

formula
  calculation_value DECIMAL(12, 2),
  calculation_formula TEXT,

  is_taxable BOOLEAN DEFAULT FALSE,
  is_statutory BOOLEAN DEFAULT FALSE,

  applicable_to VARCHAR(20), -- all, specific_grades, specific_employees

  display_order INTEGER,

  is_active BOOLEAN DEFAULT TRUE,
  created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

  UNIQUE(organization_id, component_code)
);

-- Employee Salary Structure
CREATE TABLE employee_salary_structures (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  employee_id UUID NOT NULL REFERENCES employees(id),

  effective_from DATE NOT NULL,
  effective_to DATE,

  basic_salary DECIMAL(12, 2) NOT NULL,

  -- Components
  components JSONB NOT NULL, -- [{component_id, amount}]

  gross_salary DECIMAL(12, 2),

  ctc DECIMAL(12, 2), -- Cost to Company

  is_current BOOLEAN DEFAULT TRUE,

  approved_by UUID REFERENCES users(id),
  approved_at TIMESTAMPTZ,

  created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Payroll Runs
CREATE TABLE payroll_runs (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),
  branch_id UUID REFERENCES branches(id), -- NULL for all branches

  payroll_month INTEGER NOT NULL,
  payroll_year INTEGER NOT NULL,

  pay_period_from DATE NOT NULL,
  pay_period_to DATE NOT NULL,

```

```

    status VARCHAR(20) DEFAULT 'draft', -- draft, processing, computed, approved,
    paid, cancelled

    -- Counts
    total_employees INTEGER,
    processed_employees INTEGER,

    -- Totals
    total_gross DECIMAL(14, 2),
    total_deductions DECIMAL(14, 2),
    total_net_pay DECIMAL(14, 2),

    -- Approval
    approved_by UUID REFERENCES users(id),
    approved_at TIMESTAMPTZ,

    -- Payment
    payment_date DATE,
    bank_file_generated BOOLEAN DEFAULT FALSE,
    bank_file_url TEXT,

    created_by UUID REFERENCES users(id),
    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

    UNIQUE(organization_id, branch_id, payroll_year, payroll_month)
);

-- Payroll Details (Payslips)
CREATE TABLE payroll_details (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    payroll_run_id UUID NOT NULL REFERENCES payroll_runs(id),
    employee_id UUID NOT NULL REFERENCES employees(id),

    -- Attendance Summary
    working_days INTEGER,
    present_days DECIMAL(4, 1),
    leave_days DECIMAL(4, 1),
    absent_days DECIMAL(4, 1),
    holidays INTEGER,
    week_offs INTEGER,

    overtime_hours DECIMAL(6, 2),
    late_count INTEGER,

    -- Earnings
    basic_salary DECIMAL(12, 2),
    earnings JSONB, -- [{component_code, component_name, amount}]
    total_earnings DECIMAL(12, 2),

    -- Deductions
    deductions JSONB, -- [{component_code, component_name, amount}]
    total_deductions DECIMAL(12, 2),

```

```

-- Net Pay
gross_salary DECIMAL(12, 2),
net_pay DECIMAL(12, 2),

-- Tax
tds_amount DECIMAL(10, 2),

-- PF
employee_pf DECIMAL(10, 2),
employer_pf DECIMAL(10, 2),

-- Loans/Advances
loan_deduction DECIMAL(10, 2),
advance_deduction DECIMAL(10, 2),

-- Payment
payment_mode VARCHAR(50), -- bank_transfer, cash, cheque, mobile_banking
bank_account VARCHAR(50),
payment_reference VARCHAR(100),
payment_date DATE,

-- Status
status VARCHAR(20) DEFAULT 'computed', -- computed, approved, paid, hold

hold_reason TEXT,

payslip_pdf_url TEXT,

created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

UNIQUE(payroll_run_id, employee_id)
);

-- Salary Advances
CREATE TABLE salary_advances (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),

  advance_number VARCHAR(50) NOT NULL,

  employee_id UUID NOT NULL REFERENCES employees(id),

  advance_type VARCHAR(50), -- salary_advance, travel_advance, other

  requested_amount DECIMAL(12, 2) NOT NULL,
  approved_amount DECIMAL(12, 2),

  reason TEXT,

-- Repayment
repayment_months INTEGER DEFAULT 1,
monthly_deduction DECIMAL(10, 2),

-- Status

```



```

    status VARCHAR(20) DEFAULT 'pending', -- pending, approved, rejected,
    disbursed, recovered

    applied_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

    approved_by UUID REFERENCES users(id),
    approved_at TIMESTAMPTZ,

    disbursed_at TIMESTAMPTZ,
    disbursement_mode VARCHAR(50),

    recovered_amount DECIMAL(12, 2) DEFAULT 0,
    balance_amount DECIMAL(12, 2),

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

    UNIQUE(organization_id, advance_number)
);

-- Employee Loans
CREATE TABLE employee_loans (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),

    loan_number VARCHAR(50) NOT NULL,

    employee_id UUID NOT NULL REFERENCES employees(id),

    loan_type VARCHAR(50), -- personal, housing, vehicle, education, medical

    loan_amount DECIMAL(14, 2) NOT NULL,
    interest_rate DECIMAL(5, 2) DEFAULT 0,

    tenure_months INTEGER NOT NULL,
    emi_amount DECIMAL(10, 2),

    -- Disbursement
    disbursement_date DATE,

    -- Status
    status VARCHAR(20) DEFAULT 'pending', -- pending, approved, disbursed, active,
    closed

    approved_by UUID REFERENCES users(id),
    approved_at TIMESTAMPTZ,

    -- Recovery
    total_recovered DECIMAL(14, 2) DEFAULT 0,
    outstanding_amount DECIMAL(14, 2),
    emis_paid INTEGER DEFAULT 0,

    start_deduction_month DATE,

    guarantor_name VARCHAR(255),

```

```

    guarantor_employee_id UUID REFERENCES employees(id),

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

    UNIQUE(organization_id, loan_number)
);

-- Loan EMI Schedule
CREATE TABLE loan_emi_schedule (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    loan_id UUID NOT NULL REFERENCES employee_loans(id),

    emi_number INTEGER NOT NULL,
    due_date DATE NOT NULL,

    principal_amount DECIMAL(10, 2),
    interest_amount DECIMAL(10, 2),
    emi_amount DECIMAL(10, 2),

    status VARCHAR(20) DEFAULT 'pending', -- pending, paid, overdue, waived

    paid_date DATE,
    paid_amount DECIMAL(10, 2),
    payroll_detail_id UUID REFERENCES payroll_details(id),

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

    UNIQUE(loan_id, emi_number)
);

-- Indexes
CREATE INDEX idx_employees_org ON employees(organization_id);
CREATE INDEX idx_employees_branch ON employees(branch_id);
CREATE INDEX idx_employees_department ON employees(department_id);
CREATE INDEX idx_employees_code ON employees(organization_id, employee_code);

CREATE INDEX idx_attendance_employee ON attendance(employee_id);
CREATE INDEX idx_attendance_date ON attendance(attendance_date);
CREATE INDEX idx_attendance_employee_date ON attendance(employee_id, attendance_date);

CREATE INDEX idx_leave_apps_employee ON leave_applications(employee_id);
CREATE INDEX idx_leave_apps_dates ON leave_applications(from_date, to_date);
CREATE INDEX idx_leave_apps_status ON leave_applications(status);

CREATE INDEX idx_payroll_details_employee ON payroll_details(employee_id);
CREATE INDEX idx_payroll_details_run ON payroll_details(payroll_run_id);

```

PART 15: DOCTOR DUTY ROSTER & BUILDING/MAINTENANCE

SQL

```

-- =====
-- SECTION 21: DOCTOR DUTY ROSTER
-- =====

-- Doctor Duty Roster
CREATE TABLE doctor_duty_roster (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),
  branch_id UUID NOT NULL REFERENCES branches(id),

  doctor_id UUID NOT NULL REFERENCES doctors(id),

  duty_date DATE NOT NULL,

  duty_type VARCHAR(50), -- opd, ipd_round, emergency, icu, on_call, ot

  shift shift_type,

  start_time TIME NOT NULL,
  end_time TIME NOT NULL,

  -- Ward/Location Assignment
  ward_id UUID REFERENCES wards(id),
  department_id UUID REFERENCES departments(id),

  -- OPD Room
  opd_room VARCHAR(50),

  -- On-Call
  is_on_call BOOLEAN DEFAULT FALSE,
  on_call_priority INTEGER,

  -- Status
  status VARCHAR(20) DEFAULT 'scheduled', -- scheduled, confirmed, completed,
  cancelled, swapped

  -- Swap
  swapped_with_doctor_id UUID REFERENCES doctors(id),
  swap_reason TEXT,
  swap_approved_by UUID REFERENCES users(id),

  notes TEXT,

  created_by UUID REFERENCES users(id),
  created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
  updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

  UNIQUE(doctor_id, duty_date, duty_type, shift)
);

-- Doctor On-Call Schedule
CREATE TABLE doctor_oncall_schedule (

```

```

    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),
    branch_id UUID NOT NULL REFERENCES branches(id),

    department_id UUID REFERENCES departments(id),
    specialization VARCHAR(100),

    oncall_date DATE NOT NULL,

    -- Time period
    start_time TIME NOT NULL,
    end_time TIME NOT NULL,

    primary_doctor_id UUID NOT NULL REFERENCES doctors(id),
    backup_doctor_id UUID REFERENCES doctors(id),

    is_active BOOLEAN DEFAULT TRUE,

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

    UNIQUE(branch_id, department_id, oncall_date, start_time)
);

-- =====
-- SECTION 22: BUILDING MANAGEMENT
-- =====

-- Buildings
CREATE TABLE buildings (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),
    branch_id UUID NOT NULL REFERENCES branches(id),

    building_code VARCHAR(20) NOT NULL,
    name VARCHAR(100) NOT NULL,

    address TEXT,

    -- Specifications
    total_floors INTEGER,
    total_area_sqft DECIMAL(12, 2),
    construction_year INTEGER,

    -- Utilities
    has_generator BOOLEAN DEFAULT FALSE,
    has_central_ac BOOLEAN DEFAULT FALSE,
    has_lift BOOLEAN DEFAULT FALSE,
    number_of_lifts INTEGER,

    floor_plan_url TEXT,

    is_active BOOLEAN DEFAULT TRUE,
    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

```

```

        UNIQUE(branch_id, building_code)
    );

-- Floors
CREATE TABLE floors (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    building_id UUID NOT NULL REFERENCES buildings(id),

    floor_code VARCHAR(20) NOT NULL,
    floor_number INTEGER NOT NULL,
    name VARCHAR(100),

    area_sqft DECIMAL(10, 2),

    floor_plan_url TEXT,

    is_active BOOLEAN DEFAULT TRUE,

    UNIQUE(building_id, floor_code)
);

-- Room Registry (Non-clinical rooms)
CREATE TABLE facility_rooms (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),
    branch_id UUID NOT NULL REFERENCES branches(id),
    building_id UUID REFERENCES buildings(id),
    floor_id UUID REFERENCES floors(id),

    room_code VARCHAR(20) NOT NULL,
    name VARCHAR(100),

    room_type VARCHAR(50), -- office, conference, store, server_room, canteen,
    rest_room, utility

    area_sqft DECIMAL(8, 2),
    capacity INTEGER,

    assigned_department_id UUID REFERENCES departments(id),

    -- Amenities
    has_ac BOOLEAN DEFAULT FALSE,
    has_projector BOOLEAN DEFAULT FALSE,
    has_phone BOOLEAN DEFAULT FALSE,

    is_bookable BOOLEAN DEFAULT FALSE, -- For conference rooms

    is_active BOOLEAN DEFAULT TRUE,
    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

    UNIQUE(branch_id, room_code)
);

-- Utility Meters

```

```

CREATE TABLE utility_meters (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),
  branch_id UUID NOT NULL REFERENCES branches(id),
  building_id UUID REFERENCES buildings(id),

  meter_code VARCHAR(50) NOT NULL,
  meter_type VARCHAR(50), -- electricity, water, gas

  meter_number VARCHAR(100),
  account_number VARCHAR(100),

  location VARCHAR(255),

  is_main_meter BOOLEAN DEFAULT FALSE,
  parent_meter_id UUID REFERENCES utility_meters(id),

  is_active BOOLEAN DEFAULT TRUE,
  created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

  UNIQUE(branch_id, meter_code)
);

-- Utility Readings
CREATE TABLE utility_readings (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  meter_id UUID NOT NULL REFERENCES utility_meters(id),

  reading_date DATE NOT NULL,
  reading_value DECIMAL(12, 2) NOT NULL,

  previous_reading DECIMAL(12, 2),
  consumption DECIMAL(12, 2),

  billing_period_from DATE,
  billing_period_to DATE,

  recorded_by UUID REFERENCES users(id),

  photo_url TEXT,

  created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

  UNIQUE(meter_id, reading_date)
);

-- Utility Bills
CREATE TABLE utility_bills (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),
  branch_id UUID NOT NULL REFERENCES branches(id),
  meter_id UUID REFERENCES utility_meters(id),

  bill_number VARCHAR(50) NOT NULL,

```

```

    bill_type VARCHAR(50), -- electricity, water, gas, internet, phone

    billing_period_from DATE,
    billing_period_to DATE,

    units_consumed DECIMAL(12, 2),
    rate_per_unit DECIMAL(8, 4),

    bill_amount DECIMAL(12, 2) NOT NULL,
    tax_amount DECIMAL(10, 2),
    total_amount DECIMAL(12, 2),

    due_date DATE,

    -- Payment
    paid_amount DECIMAL(12, 2),
    payment_date DATE,
    payment_reference VARCHAR(100),

    status VARCHAR(20) DEFAULT 'pending', -- pending, paid, overdue

    bill_document_url TEXT,

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

    UNIQUE(organization_id, bill_number)
);

-- =====
-- SECTION 23: MAINTENANCE MANAGEMENT
-- =====

-- Equipment Registry
CREATE TABLE equipment_registry (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),
    branch_id UUID NOT NULL REFERENCES branches(id),

    equipment_code VARCHAR(50) NOT NULL,
    name VARCHAR(255) NOT NULL,

    category VARCHAR(50), -- medical_equipment, non_medical, it_equipment, hvac,
electrical
    sub_category VARCHAR(100),

    -- Details
    manufacturer VARCHAR(255),
    model VARCHAR(100),
    serial_number VARCHAR(100),

    -- Purchase
    purchase_date DATE,
    purchase_price DECIMAL(14, 2),
    supplier_id UUID REFERENCES suppliers(id),

```

```

    po_number VARCHAR(50),

    -- Warranty
    warranty_start_date DATE,
    warranty_end_date DATE,
    warranty_terms TEXT,

    -- AMC
    amc_vendor_id UUID REFERENCES suppliers(id),
    amc_start_date DATE,
    amc_end_date DATE,
    amc_cost DECIMAL(12, 2),

    -- Location
    department_id UUID REFERENCES departments(id),
    building_id UUID REFERENCES buildings(id),
    floor_id UUID REFERENCES floors(id),
    room_id UUID,
    location_description VARCHAR(255),

    -- Technical
    power_requirement VARCHAR(100),
    installation_requirements TEXT,

    -- Calibration (for medical equipment)
    requires_calibration BOOLEAN DEFAULT FALSE,
    calibration_frequency_months INTEGER,
    last_calibration_date DATE,
    next_calibration_due DATE,

    -- Maintenance
    pm_frequency_months INTEGER, -- Preventive maintenance frequency
    last_pm_date DATE,
    next_pm_due DATE,

    -- Status
    status VARCHAR(20) DEFAULT 'active', -- active, under_maintenance,
    under_repair, condemned, disposed

    -- Asset
    asset_tag VARCHAR(50),
    depreciation_rate DECIMAL(5, 2),
    current_value DECIMAL(14, 2),

    -- Documents
    manual_url TEXT,

    is_critical BOOLEAN DEFAULT FALSE,

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

    UNIQUE(organization_id, equipment_code)
);

```



```

-- Preventive Maintenance Schedule
CREATE TABLE pm_schedules (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),

  equipment_id UUID NOT NULL REFERENCES equipment_registry(id),

  schedule_type VARCHAR(50), -- daily, weekly, monthly, quarterly, semi_annual,
annual

  checklist JSONB, -- [{item, instructions}]

  last_performed_date DATE,
  next_due_date DATE NOT NULL,

  assigned_technician_id UUID REFERENCES users(id),

  is_active BOOLEAN DEFAULT TRUE,
  created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- PM Completion Records
CREATE TABLE pm_records (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  schedule_id UUID NOT NULL REFERENCES pm_schedules(id),
  equipment_id UUID NOT NULL REFERENCES equipment_registry(id),

  pm_date DATE NOT NULL,

  performed_by UUID NOT NULL REFERENCES users(id),

  checklist_responses JSONB,

  findings TEXT,
  actions_taken TEXT,

  -- Parts Used
  parts_used JSONB,
  parts_cost DECIMAL(10, 2),

  labor_hours DECIMAL(4, 2),
  labor_cost DECIMAL(10, 2),

  total_cost DECIMAL(10, 2),

  equipment_status_after VARCHAR(50),

  photos TEXT[],

  next_pm_due DATE,

  reviewed_by UUID REFERENCES users(id),
  reviewed_at TIMESTAMPTZ,

```

```

        created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
    );

-- Calibration Records
CREATE TABLE calibration_records (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    equipment_id UUID NOT NULL REFERENCES equipment_registry(id),

    calibration_date DATE NOT NULL,
    calibration_due_date DATE,

    calibration_type VARCHAR(50), -- internal, external

    calibration_agency VARCHAR(255),
    technician_name VARCHAR(255),

-- Results
    parameters_checked JSONB,

    result VARCHAR(20), -- pass, fail, conditional_pass

    deviations_found TEXT,
    corrective_actions TEXT,

    certificate_number VARCHAR(100),
    certificate_url TEXT,

    next_calibration_date DATE,

    cost DECIMAL(10, 2),

    created_by UUID REFERENCES users(id),
    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Maintenance Tickets (Breakdown)
CREATE TABLE maintenance_tickets (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),
    branch_id UUID NOT NULL REFERENCES branches(id),

    ticket_number VARCHAR(50) NOT NULL,

-- Type
    ticket_type VARCHAR(50), -- breakdown, corrective, request
    category VARCHAR(50), -- equipment, civil, electrical, plumbing, hvac, it,
    other

-- Equipment (if applicable)
    equipment_id UUID REFERENCES equipment_registry(id),

-- Location
    building_id UUID REFERENCES buildings(id),

```

```

    floor_id UUID REFERENCES floors(id),
    location_description VARCHAR(255),
    department_id UUID REFERENCES departments(id),

    -- Problem
    problem_title VARCHAR(255) NOT NULL,
    problem_description TEXT,

    priority maintenance_priority_type DEFAULT 'normal',

    -- Reported
    reported_by UUID REFERENCES users(id),
    reported_datetime TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

    -- Impact
    patient_impact BOOLEAN DEFAULT FALSE,
    safety_impact BOOLEAN DEFAULT FALSE,

    -- Assignment
    assigned_to UUID REFERENCES users(id),
    assigned_at TIMESTAMPTZ,

    -- Status
    status ticket_status_type DEFAULT 'open',

    -- Work
    acknowledged_at TIMESTAMPTZ,
    work_started_at TIMESTAMPTZ,
    work_completed_at TIMESTAMPTZ,

    -- Resolution
    root_cause TEXT,
    resolution TEXT,
    resolution_type VARCHAR(50), -- repaired, replaced, temporary_fix,
    external_service

    -- Parts
    spare_parts_used JSONB,
    parts_cost DECIMAL(10, 2),

    -- Labor
    labor_hours DECIMAL(5, 2),
    labor_cost DECIMAL(10, 2),

    external_vendor_id UUID REFERENCES suppliers(id),
    external_cost DECIMAL(10, 2),

    total_cost DECIMAL(12, 2),

    -- Equipment Status
    equipment_status_after VARCHAR(50),
    equipment_downtime_hours DECIMAL(6, 2),

    -- Closure

```

```

        closed_at TIMESTAMPTZ,
        closed_by UUID REFERENCES users(id),

        -- Feedback
        user_feedback TEXT,
        user_rating INTEGER CHECK (user_rating BETWEEN 1 AND 5),

        photos TEXT[],

        created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
        updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

        UNIQUE(organization_id, ticket_number)
    );

-- Ticket Work Log
CREATE TABLE ticket_work_log (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    ticket_id UUID NOT NULL REFERENCES maintenance_tickets(id),

    log_datetime TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
    logged_by UUID NOT NULL REFERENCES users(id),

    work_type VARCHAR(50), -- diagnosis, repair, parts_request, waiting, completed

    description TEXT,

    hours_spent DECIMAL(4, 2),

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Spare Parts Inventory
CREATE TABLE spare_parts (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),
    branch_id UUID NOT NULL REFERENCES branches(id),

    part_code VARCHAR(50) NOT NULL,
    name VARCHAR(255) NOT NULL,

    category VARCHAR(100),

    -- Compatibility
    compatible_equipment_ids UUID[],

    -- Stock
    quantity_on_hand INTEGER DEFAULT 0,
    reorder_level INTEGER,

    unit_cost DECIMAL(10, 2),

    supplier_id UUID REFERENCES suppliers(id),

```

```

    storage_location VARCHAR(100),

    is_critical BOOLEAN DEFAULT FALSE,

    is_active BOOLEAN DEFAULT TRUE,
    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

    UNIQUE(organization_id, part_code)
);

-- Indexes
CREATE INDEX idx_equipment_org ON equipment_registry(organization_id);
CREATE INDEX idx_equipment_branch ON equipment_registry(branch_id);
CREATE INDEX idx_equipment_category ON equipment_registry(category);
CREATE INDEX idx_equipment_status ON equipment_registry(status);
CREATE INDEX idx_equipment_pm_due ON equipment_registry(next_pm_due);
CREATE INDEX idx_equipment_calibration_due ON
equipment_registry(next_calibration_due);

CREATE INDEX idx_maint_tickets_org ON maintenance_tickets(organization_id);
CREATE INDEX idx_maint_tickets_status ON maintenance_tickets(status);
CREATE INDEX idx_maint_tickets_priority ON maintenance_tickets(priority);
CREATE INDEX idx_maint_tickets_assigned ON maintenance_tickets(assigned_to);
CREATE INDEX idx_maint_tickets_equipment ON maintenance_tickets(equipment_id);

```

PART 16: DIET, TELEMEDICINE, COMPLAINTS & CONSENT

SQL

```

-- =====
-- SECTION 24: DIET MANAGEMENT
-- =====

-- Diet Types
CREATE TABLE diet_types (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),

    diet_code VARCHAR(20) NOT NULL,
    name VARCHAR(100) NOT NULL,
    name_bangla VARCHAR(100),

    description TEXT,

    diet_category VARCHAR(50), -- regular, therapeutic, special

    -- Nutritional Targets
    calories_min INTEGER,
    calories_max INTEGER,
    protein_grams INTEGER,

```

```

    carbs_grams INTEGER,
    fat_grams INTEGER,
    sodium_mg INTEGER,
    potassium_mg INTEGER,

    -- Restrictions
    allowed_foods TEXT,
    restricted_foods TEXT,

    special_instructions TEXT,

    is_active BOOLEAN DEFAULT TRUE,
    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

    UNIQUE(organization_id, diet_code)
);

-- Patient Diet Orders
CREATE TABLE patient_diet_orders (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),

    patient_id UUID NOT NULL REFERENCES patients(id),
    admission_id UUID NOT NULL REFERENCES admissions(id),

    diet_type_id UUID NOT NULL REFERENCES diet_types(id),

    order_datetime TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
    ordered_by UUID NOT NULL REFERENCES users(id),

    -- Custom modifications
    modifications TEXT,
    allergies_noted TEXT,

    -- Texture
    texture VARCHAR(50), -- regular, soft, pureed, liquid

    -- Fluid restriction
    fluid_restriction_ml INTEGER,

    -- Timing
    effective_from TIMESTAMPTZ,
    effective_until TIMESTAMPTZ,

    -- NPO (Nothing by mouth)
    is_npo BOOLEAN DEFAULT FALSE,
    npo_reason TEXT,

    -- Tube feeding
    is_tube_feeding BOOLEAN DEFAULT FALSE,
    tube_type VARCHAR(50), -- ng, peg, nj
    formula VARCHAR(100),
    rate_ml_per_hour INTEGER,

```

```

    status VARCHAR(20) DEFAULT 'active', -- active, discontinued, modified

    discontinued_at TIMESTAMPTZ,
    discontinued_by UUID REFERENCES users(id),
    discontinuation_reason TEXT,

    notes TEXT,

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Meal Plans
CREATE TABLE meal_plans (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),

    diet_type_id UUID NOT NULL REFERENCES diet_types(id),

    day_of_week INTEGER, -- 0-6, NULL for everyday

    meal_type VARCHAR(20) NOT NULL, -- breakfast, lunch, dinner, snack
    meal_time TIME,

    menu_items JSONB, -- [{item_name, quantity, calories, notes}]

    total_calories INTEGER,

    is_active BOOLEAN DEFAULT TRUE,
    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Patient Meal Delivery
CREATE TABLE meal_deliveries (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),

    patient_id UUID NOT NULL REFERENCES patients(id),
    admission_id UUID NOT NULL REFERENCES admissions(id),
    diet_order_id UUID REFERENCES patient_diet_orders(id),

    meal_date DATE NOT NULL,
    meal_type VARCHAR(20) NOT NULL,

    ward_id UUID REFERENCES wards(id),
    bed_id UUID REFERENCES beds(id),

    -- Menu
    menu_items JSONB,
    special_requests TEXT,

    -- Delivery
    prepared_at TIMESTAMPTZ,
    prepared_by UUID REFERENCES users(id),

```

```

    delivered_at TIMESTAMPTZ,
    delivered_by UUID REFERENCES users(id),

    -- Consumption
    consumption_status VARCHAR(50), -- full, partial, refused, npo
    percentage_consumed INTEGER,

    -- Feedback
    patient_feedback TEXT,

    notes TEXT,

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Nutritional Assessment
CREATE TABLE nutritional_assessments (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    patient_id UUID NOT NULL REFERENCES patients(id),
    admission_id UUID REFERENCES admissions(id),

    assessment_date DATE NOT NULL,
    assessed_by UUID NOT NULL REFERENCES users(id), -- Dietitian

    -- Anthropometry
    height_cm DECIMAL(5, 2),
    weight_kg DECIMAL(5, 2),
    bmi DECIMAL(4, 1),
    ideal_body_weight DECIMAL(5, 2),

    waist_circumference_cm DECIMAL(5, 1),
    mid_arm_circumference_cm DECIMAL(4, 1),

    -- Screening Score
    screening_tool VARCHAR(50), -- MST, NRS-2002, MUST
    screening_score INTEGER,
    nutrition_risk_level VARCHAR(20),

    -- Dietary History
    dietary_history TEXT,
    recent_weight_change TEXT,
    appetite VARCHAR(50),

    -- Lab Values
    albumin DECIMAL(4, 2),
    prealbumin DECIMAL(4, 2),
    hemoglobin DECIMAL(4, 1),

    -- Assessment
    nutrition_diagnosis TEXT,

    -- Recommendations
    calorie_requirement INTEGER,
    protein_requirement INTEGER,

```



```

    fluid_requirement INTEGER,

    recommendations TEXT,

    -- Follow-up
    followup_date DATE,

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- =====
-- SECTION 25: TELEMEDICINE
-- =====

-- Telemedicine Sessions
CREATE TABLE telemedicine_sessions (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),
    branch_id UUID NOT NULL REFERENCES branches(id),

    session_number VARCHAR(50) NOT NULL,

    appointment_id UUID REFERENCES appointments(id),

    patient_id UUID NOT NULL REFERENCES patients(id),
    doctor_id UUID NOT NULL REFERENCES doctors(id),

    -- Schedule
    scheduled_datetime TIMESTAMPTZ NOT NULL,
    scheduled_duration_minutes INTEGER DEFAULT 15,

    -- Video Platform
    platform VARCHAR(50), -- jitsi, zoom, custom
    meeting_url TEXT,
    meeting_id VARCHAR(100),
    meeting_password VARCHAR(50),

    -- Consent
    patient_consent_obtained BOOLEAN DEFAULT FALSE,
    consent_datetime TIMESTAMPTZ,

    -- Status
    status VARCHAR(50) DEFAULT 'scheduled',
    -- scheduled, patient_waiting, in_progress, completed, cancelled, no_show,
    technical_issue

    -- Actual Times
    patient_joined_at TIMESTAMPTZ,
    doctor_joined_at TIMESTAMPTZ,
    session_started_at TIMESTAMPTZ,
    session_ended_at TIMESTAMPTZ,

    actual_duration_minutes INTEGER,

```

```

-- Technical
patient_device_type VARCHAR(50),
connection_quality VARCHAR(20), -- excellent, good, fair, poor
technical_issues TEXT,

-- Clinical
visit_id UUID, -- Reference to OPD visit created
prescription_id UUID REFERENCES prescriptions(id),

-- Recording
is_recorded BOOLEAN DEFAULT FALSE,
recording_url TEXT,
recording_consent BOOLEAN DEFAULT FALSE,

-- Billing
consultation_fee DECIMAL(10, 2),
payment_status payment_status_type DEFAULT 'pending',

-- Feedback
patient_rating INTEGER CHECK (patient_rating BETWEEN 1 AND 5),
patient_feedback TEXT,

notes TEXT,

created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

UNIQUE(organization_id, session_number)
);

-- =====
-- SECTION 26: COMPLAINT & FEEDBACK
-- =====

-- Complaint Categories
CREATE TABLE complaint_categories (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),

  code VARCHAR(20) NOT NULL,
  name VARCHAR(100) NOT NULL,

  parent_category_id UUID REFERENCES complaint_categories(id),

  sla_hours INTEGER, -- Response SLA
  resolution_sla_hours INTEGER,

  escalation_level_1_user_id UUID REFERENCES users(id),
  escalation_level_2_user_id UUID REFERENCES users(id),

  is_active BOOLEAN DEFAULT TRUE,

  UNIQUE(organization_id, code)
);

```

```

-- Complaints
CREATE TABLE complaints (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),
  branch_id UUID NOT NULL REFERENCES branches(id),

  complaint_number VARCHAR(50) NOT NULL,

  -- Complainant
  complainant_type VARCHAR(50), -- patient, attendant, staff, visitor, other
  patient_id UUID REFERENCES patients(id),
  complainant_name VARCHAR(255),
  complainant_phone VARCHAR(20),
  complainant_email VARCHAR(255),

  -- Context
  visit_id UUID,
  admission_id UUID REFERENCES admissions(id),

  -- Complaint Details
  category_id UUID REFERENCES complaint_categories(id),
  sub_category VARCHAR(100),

  subject VARCHAR(255) NOT NULL,
  description TEXT NOT NULL,

  -- Against
  against_department_id UUID REFERENCES departments(id),
  against_staff_id UUID REFERENCES users(id),

  -- Source
  received_via VARCHAR(50), -- walk_in, phone, email, app, website, social_media
  received_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
  received_by UUID REFERENCES users(id),

  priority VARCHAR(20) DEFAULT 'normal', -- low, normal, high, critical

  -- Status
  status VARCHAR(50) DEFAULT 'open',
  -- open, acknowledged, investigating, pending_response, resolved, closed,
  reopened

  -- Acknowledgment
  acknowledged_at TIMESTAMPTZ,
  acknowledgment_sent BOOLEAN DEFAULT FALSE,

  -- Assignment
  assigned_to UUID REFERENCES users(id),
  assigned_department_id UUID REFERENCES departments(id),
  assigned_at TIMESTAMPTZ,

  -- Investigation
  investigation_notes TEXT,

```

```

    root_cause TEXT,

    -- Resolution
    resolution TEXT,
    corrective_action TEXT,
    preventive_action TEXT,

    resolved_at TIMESTAMPTZ,
    resolved_by UUID REFERENCES users(id),

    -- Closure
    closed_at TIMESTAMPTZ,
    closed_by UUID REFERENCES users(id),
    closure_remarks TEXT,

    -- Verification
    complainant_satisfied BOOLEAN,
    satisfaction_rating INTEGER CHECK (satisfaction_rating BETWEEN 1 AND 5),

    -- SLA
    response_sla_breached BOOLEAN DEFAULT FALSE,
    resolution_sla_breached BOOLEAN DEFAULT FALSE,

    -- Escalation
    escalation_level INTEGER DEFAULT 0,
    escalated_at TIMESTAMPTZ,
    escalated_to UUID REFERENCES users(id),

    attachments TEXT[],

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

    UNIQUE(organization_id, complaint_number)
);

-- Complaint Communications
CREATE TABLE complaint_communications (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    complaint_id UUID NOT NULL REFERENCES complaints(id),

    communication_type VARCHAR(50), -- note, email, call, sms, meeting
    direction VARCHAR(20), -- internal, to_complainant, from_complainant

    communication_datetime TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

    subject TEXT,
    content TEXT,

    communicated_by UUID REFERENCES users(id),

    attachments TEXT[],

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP

```

```

);

-- Feedback Forms
CREATE TABLE feedback_forms (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),

  form_code VARCHAR(20) NOT NULL,
  name VARCHAR(100) NOT NULL,

  form_type VARCHAR(50), -- discharge, opd, service_specific

  questions JSONB NOT NULL, -- [{question, type, options, is_required}]

  is_active BOOLEAN DEFAULT TRUE,
  created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

  UNIQUE(organization_id, form_code)
);

-- Feedback Responses
CREATE TABLE feedback_responses (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),

  form_id UUID NOT NULL REFERENCES feedback_forms(id),

  patient_id UUID REFERENCES patients(id),
  visit_id UUID,
  admission_id UUID REFERENCES admissions(id),

  respondent_type VARCHAR(50), -- patient, attendant
  respondent_name VARCHAR(255),
  respondent_phone VARCHAR(20),

  responses JSONB NOT NULL,

  -- Calculated Scores
  overall_score DECIMAL(4, 2),
  nps_score INTEGER,

  -- Collection
  collected_via VARCHAR(50), -- kiosk, app, sms, email, call, paper
  collected_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
  collected_by UUID REFERENCES users(id),

  comments TEXT,

  -- Follow-up
  requires_followup BOOLEAN DEFAULT FALSE,
  followup_notes TEXT,
  followed_up_by UUID REFERENCES users(id),
  followed_up_at TIMESTAMPTZ,

```

```

        created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
    );

-- =====
-- SECTION 27: CONSENT FORM MANAGEMENT
-- =====

-- Consent Templates
CREATE TABLE consent_templates (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),

    template_code VARCHAR(20) NOT NULL,
    name VARCHAR(255) NOT NULL,
    name_bangla VARCHAR(255),

    consent_type VARCHAR(50),
    -- general_treatment, surgery, anesthesia, blood_transfusion, procedure,
    -- telemedicine, research, photography, data_sharing, dnar

    version VARCHAR(20),
    version_date DATE,

    -- Content
    content_english TEXT NOT NULL,
    content_bangla TEXT,

    -- Structured content
    sections JSONB, -- [{title, content, requires_initial}]

    -- Requirements
    requires_witness BOOLEAN DEFAULT FALSE,
    requires_guardian BOOLEAN DEFAULT FALSE, -- For minors
    requires_interpreter BOOLEAN DEFAULT FALSE,

    -- Legal Review
    legal_reviewed BOOLEAN DEFAULT FALSE,
    legal_reviewed_by VARCHAR(255),
    legal_reviewed_date DATE,

    is_active BOOLEAN DEFAULT TRUE,
    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

    UNIQUE(organization_id, template_code, version)
);

-- Patient Consents
CREATE TABLE patient_consents (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),
    branch_id UUID NOT NULL REFERENCES branches(id),

    consent_number VARCHAR(50) NOT NULL,

```

```
template_id UUID NOT NULL REFERENCES consent_templates(id),

patient_id UUID NOT NULL REFERENCES patients(id),

-- Context
visit_id UUID,
admission_id UUID REFERENCES admissions(id),
surgery_id UUID REFERENCES surgeries(id),
procedure_name VARCHAR(255),

-- Consent Process
explained_by UUID NOT NULL REFERENCES users(id), -- Doctor who explained
explanation_datetime TIMESTAMPTZ,

language_used VARCHAR(50),
interpreter_used BOOLEAN DEFAULT FALSE,
interpreter_name VARCHAR(255),

-- Patient Understanding
questions_asked TEXT,
questions_answered BOOLEAN DEFAULT TRUE,

-- Decision
consent_status consent_status_type DEFAULT 'pending',

consent_datetime TIMESTAMPTZ,

-- Signatures
patient_signature_url TEXT,
patient_signed_at TIMESTAMPTZ,

-- For minors/incapacitated
guardian_name VARCHAR(255),
guardian_relationship VARCHAR(50),
guardian_signature_url TEXT,
guardian_signed_at TIMESTAMPTZ,
guardian_id_number VARCHAR(50),

-- Witness
witness_name VARCHAR(255),
witness_signature_url TEXT,
witness_signed_at TIMESTAMPTZ,

-- If refused
refusal_reason TEXT,
refusal_acknowledged BOOLEAN DEFAULT FALSE,

-- PDF
consent_pdf_url TEXT,

-- Withdrawal
is_withdrawn BOOLEAN DEFAULT FALSE,
withdrawn_at TIMESTAMPTZ,
```

```

        withdrawn_by VARCHAR(255),
        withdrawal_reason TEXT,

        notes TEXT,

        created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
        updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

        UNIQUE(organization_id, consent_number)
    );

-- Consent Access Log
CREATE TABLE consent_access_log (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    consent_id UUID NOT NULL REFERENCES patient_consents(id),

    accessed_by UUID NOT NULL REFERENCES users(id),
    accessed_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

    access_type VARCHAR(50), -- view, print, download

    ip_address INET,
    user_agent TEXT
);

-- Indexes
CREATE INDEX idx_diet_orders_patient ON patient_diet_orders(patient_id);
CREATE INDEX idx_diet_orders_admission ON patient_diet_orders(admission_id);
CREATE INDEX idx_diet_orders_active ON patient_diet_orders(admission_id, status)
WHERE status = 'active';

CREATE INDEX idx_meal_deliveries_date ON meal_deliveries(meal_date);
CREATE INDEX idx_meal_deliveries_ward ON meal_deliveries(ward_id, meal_date);

CREATE INDEX idx_telemedicine_patient ON telemedicine_sessions(patient_id);
CREATE INDEX idx_telemedicine_doctor ON telemedicine_sessions(doctor_id);
CREATE INDEX idx_telemedicine_datetime ON
telemedicine_sessions(scheduled_datetime);

CREATE INDEX idx_complaints_org ON complaints(organization_id);
CREATE INDEX idx_complaints_status ON complaints(status);
CREATE INDEX idx_complaints_patient ON complaints(patient_id);
CREATE INDEX idx_complaints_assigned ON complaints(assigned_to);

CREATE INDEX idx_consents_patient ON patient_consents(patient_id);
CREATE INDEX idx_consents_admission ON patient_consents(admission_id);
CREATE INDEX idx_consents_surgery ON patient_consents(surgery_id);

```

PART 17: BILLING, REPORTS & NOTIFICATIONS

SQL


```

-- =====
-- SECTION 28: BILLING SYSTEM
-- =====

-- Service Master
CREATE TABLE service_master (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),

  service_code VARCHAR(50) NOT NULL,
  name VARCHAR(255) NOT NULL,
  name_bangla VARCHAR(255),

  category VARCHAR(50), -- consultation, procedure, room, investigation,
  pharmacy, other
  sub_category VARCHAR(100),
  department_id UUID REFERENCES departments(id),

  -- Pricing
  base_price DECIMAL(12, 2) NOT NULL,

  -- Tax
  is_taxable BOOLEAN DEFAULT FALSE,
  tax_percent DECIMAL(5, 2) DEFAULT 0,

  -- Revenue Sharing
  doctor_share_percent DECIMAL(5, 2),
  hospital_share_percent DECIMAL(5, 2),

  -- CPT/HCPCS
  cpt_code VARCHAR(20),

  is_active BOOLEAN DEFAULT TRUE,
  created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

  UNIQUE(organization_id, service_code)
);

-- Price Lists
CREATE TABLE price_lists (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),

  price_list_code VARCHAR(20) NOT NULL,
  name VARCHAR(100) NOT NULL,

  price_list_type VARCHAR(50), -- standard, insurance, corporate, package

  applicable_insurance_id UUID REFERENCES insurance_companies(id),
  applicable_corporate_id UUID REFERENCES corporate_accounts(id),

  effective_from DATE,

```

```

    effective_to DATE,

    is_default BOOLEAN DEFAULT FALSE,
    is_active BOOLEAN DEFAULT TRUE,

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

    UNIQUE(organization_id, price_list_code)
);

-- Price List Items
CREATE TABLE price_list_items (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    price_list_id UUID NOT NULL REFERENCES price_lists(id),
    service_id UUID NOT NULL REFERENCES service_master(id),

    price DECIMAL(12, 2) NOT NULL,
    discount_percent DECIMAL(5, 2) DEFAULT 0,

    is_active BOOLEAN DEFAULT TRUE,

    UNIQUE(price_list_id, service_id)
);

-- Bills
CREATE TABLE bills (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),
    branch_id UUID NOT NULL REFERENCES branches(id),

    bill_number VARCHAR(50) NOT NULL,

    patient_id UUID NOT NULL REFERENCES patients(id),

    -- Context
    bill_type VARCHAR(50), -- opd, ipd, pharmacy, emergency
    visit_id UUID,
    admission_id UUID REFERENCES admissions(id),

    bill_date DATE NOT NULL DEFAULT CURRENT_DATE,
    bill_datetime TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

    -- Price List
    price_list_id UUID REFERENCES price_lists(id),

    -- Amounts
    gross_amount DECIMAL(14, 2) NOT NULL,
    discount_amount DECIMAL(12, 2) DEFAULT 0,
    discount_reason TEXT,
    discount_approved_by UUID REFERENCES users(id),

    tax_amount DECIMAL(12, 2) DEFAULT 0,

    net_amount DECIMAL(14, 2) NOT NULL,

```

```

-- Adjustments
advance_adjusted DECIMAL(12, 2) DEFAULT 0,
insurance_amount DECIMAL(12, 2) DEFAULT 0,
corporate_amount DECIMAL(12, 2) DEFAULT 0,

patient_payable DECIMAL(14, 2),

-- Payment
paid_amount DECIMAL(14, 2) DEFAULT 0,
balance_amount DECIMAL(14, 2),

payment_status payment_status_type DEFAULT 'pending',

-- Insurance
is_insurance BOOLEAN DEFAULT FALSE,
insurance_policy_id UUID REFERENCES patient_insurance_policies(id),
preauth_id UUID REFERENCES insurance_preauth(id),

-- Corporate
is_corporate BOOLEAN DEFAULT FALSE,
corporate_id UUID REFERENCES corporate_accounts(id),
corporate_employee_id UUID REFERENCES corporate_employees(id),

-- For IPD - Final Bill
is_final_bill BOOLEAN DEFAULT FALSE,

-- Cancellation
is_cancelled BOOLEAN DEFAULT FALSE,
cancelled_at TIMESTAMPTZ,
cancelled_by UUID REFERENCES users(id),
cancellation_reason TEXT,

-- PDF
bill_pdf_url TEXT,

notes TEXT,

created_by UUID REFERENCES users(id),
created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

UNIQUE(organization_id, bill_number)
);

-- Bill Items
CREATE TABLE bill_items (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  bill_id UUID NOT NULL REFERENCES bills(id) ON DELETE CASCADE,

  service_id UUID REFERENCES service_master(id),
  service_code VARCHAR(50),
  service_name VARCHAR(255) NOT NULL,

```

```

    category VARCHAR(50),

    -- Reference
    reference_type VARCHAR(50), -- lab_order, radiology_order, prescription,
    procedure, bed_charge
    reference_id UUID,

    service_date DATE,

    -- Doctor (for consultation/procedure)
    doctor_id UUID REFERENCES doctors(id),

    -- Quantity & Pricing
    quantity DECIMAL(10, 3) DEFAULT 1,
    unit_price DECIMAL(12, 2) NOT NULL,

    discount_percent DECIMAL(5, 2) DEFAULT 0,
    discount_amount DECIMAL(10, 2) DEFAULT 0,

    tax_percent DECIMAL(5, 2) DEFAULT 0,
    tax_amount DECIMAL(10, 2) DEFAULT 0,

    net_amount DECIMAL(12, 2) NOT NULL,

    -- Insurance/Corporate
    insurance_applicable BOOLEAN DEFAULT TRUE,
    copay_amount DECIMAL(10, 2) DEFAULT 0,

    -- Status
    is_cancelled BOOLEAN DEFAULT FALSE,

    remarks TEXT,

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Payments
CREATE TABLE payments (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),
    branch_id UUID NOT NULL REFERENCES branches(id),

    receipt_number VARCHAR(50) NOT NULL,

    patient_id UUID NOT NULL REFERENCES patients(id),
    bill_id UUID REFERENCES bills(id),

    payment_type VARCHAR(50), -- advance, bill_payment, refund

    payment_date DATE NOT NULL DEFAULT CURRENT_DATE,
    payment_datetime TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

    amount DECIMAL(14, 2) NOT NULL,

```

```

    payment_method payment_method_type NOT NULL,

    -- For Card
    card_type VARCHAR(20),
    card_last_four VARCHAR(4),

    -- For Cheque
    cheque_number VARCHAR(50),
    cheque_date DATE,
    bank_name VARCHAR(100),

    -- For Bank Transfer
    transaction_reference VARCHAR(100),

    -- For Mobile Banking
    mobile_banking_provider VARCHAR(50),
    mobile_number VARCHAR(20),

    -- Status
    status VARCHAR(20) DEFAULT 'completed', -- completed, pending, failed,
refunded

    -- For Refunds
    is_refund BOOLEAN DEFAULT FALSE,
    original_payment_id UUID REFERENCES payments(id),
    refund_reason TEXT,

    received_by UUID REFERENCES users(id),

    receipt_pdf_url TEXT,

    notes TEXT,

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

    UNIQUE(organization_id, receipt_number)
);

-- Payment Allocations
CREATE TABLE payment_allocations (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    payment_id UUID NOT NULL REFERENCES payments(id),
    bill_id UUID NOT NULL REFERENCES bills(id),

    allocated_amount DECIMAL(14, 2) NOT NULL,

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- =====
-- SECTION 29: REPORTS & DASHBOARDS
-- =====

-- Report Definitions

```

```

CREATE TABLE report_definitions (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID REFERENCES organizations(id), -- NULL for system reports

  report_code VARCHAR(50) NOT NULL,
  name VARCHAR(255) NOT NULL,

  category VARCHAR(50), -- clinical, financial, operational, quality, hr
  sub_category VARCHAR(100),

  description TEXT,

  -- Query
  base_query TEXT,

  -- Parameters
  parameters JSONB, -- [{name, type, required, default_value}]

  -- Columns
  columns JSONB, -- [{field, header, type, format}]

  -- Grouping/Sorting
  default_group_by TEXT[],
  default_sort TEXT[],

  -- Access
  allowed_roles UUID[],

  -- Schedule
  can_schedule BOOLEAN DEFAULT FALSE,

  is_active BOOLEAN DEFAULT TRUE,
  created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Scheduled Reports
CREATE TABLE scheduled_reports (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),

  report_id UUID NOT NULL REFERENCES report_definitions(id),

  schedule_name VARCHAR(255) NOT NULL,

  -- Parameters
  parameters JSONB,

  -- Schedule
  frequency VARCHAR(20), -- daily, weekly, monthly
  day_of_week INTEGER, -- 0-6 for weekly
  day_of_month INTEGER, -- 1-31 for monthly
  time_of_day TIME,

  -- Output

```

```

output_format VARCHAR(20), -- pdf, excel, csv

-- Recipients
email_recipients TEXT[],

-- Status
is_active BOOLEAN DEFAULT TRUE,
last_run_at TIMESTAMPTZ,
next_run_at TIMESTAMPTZ,

created_by UUID REFERENCES users(id),
created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Dashboard Widgets
CREATE TABLE dashboard_widgets (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID REFERENCES organizations(id),

  widget_code VARCHAR(50) NOT NULL,
  name VARCHAR(100) NOT NULL,

  widget_type VARCHAR(50), -- kpi_card, chart, table, gauge

  -- Data Source
  data_query TEXT,

  -- Configuration
  config JSONB, -- Chart type, colors, etc.

  -- Refresh
  refresh_interval_minutes INTEGER DEFAULT 15,

  allowed_roles UUID[],

  is_active BOOLEAN DEFAULT TRUE,
  created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- User Dashboards
CREATE TABLE user_dashboards (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  user_id UUID NOT NULL REFERENCES users(id),

  dashboard_name VARCHAR(100) DEFAULT 'My Dashboard',

  -- Widget Layout
  layout JSONB, -- [{widget_id, position, size}]

  is_default BOOLEAN DEFAULT FALSE,

  created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
  updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

```

```

-- =====
-- SECTION 30: NOTIFICATION ENGINE
-- =====

-- Notification Templates
CREATE TABLE notification_templates (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID REFERENCES organizations(id),

    template_code VARCHAR(50) NOT NULL,
    name VARCHAR(100) NOT NULL,

    event_type VARCHAR(100), -- appointment_reminder, lab_result, bill_generated,
etc.

    channel notification_channel_type NOT NULL,

    -- Content
    subject VARCHAR(255),
    body_template TEXT NOT NULL, -- With placeholders like {{patient_name}}
    body_template_bangla TEXT,

    -- For WhatsApp
    whatsapp_template_id VARCHAR(100),

    is_active BOOLEAN DEFAULT TRUE,
    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

    UNIQUE(organization_id, template_code, channel)
);

-- Notification Queue
CREATE TABLE notification_queue (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),

    template_id UUID REFERENCES notification_templates(id),

    channel notification_channel_type NOT NULL,

    -- Recipient
    recipient_type VARCHAR(50), -- patient, user, doctor
    recipient_id UUID,
    recipient_name VARCHAR(255),
    recipient_contact VARCHAR(255) NOT NULL, -- Phone/Email

    -- Content
    subject VARCHAR(255),
    body TEXT NOT NULL,

    -- Reference
    reference_type VARCHAR(50),
    reference_id UUID,

```



```

-- Schedule
scheduled_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

-- Status
status VARCHAR(20) DEFAULT 'pending', -- pending, sent, delivered, failed,
cancelled

-- Attempts
attempts INTEGER DEFAULT 0,
max_attempts INTEGER DEFAULT 3,
last_attempt_at TIMESTAMPTZ,

-- Result
sent_at TIMESTAMPTZ,
delivered_at TIMESTAMPTZ,

error_message TEXT,

-- Provider Response
provider_response JSONB,
provider_message_id VARCHAR(255),

created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Notification Log
CREATE TABLE notification_log (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  queue_id UUID REFERENCES notification_queue(id),

  channel notification_channel_type NOT NULL,
  recipient VARCHAR(255) NOT NULL,

  status VARCHAR(20) NOT NULL,

  provider VARCHAR(50), -- twilio, msg91, sendgrid, firebase
  provider_response JSONB,

  sent_at TIMESTAMPTZ,
  delivered_at TIMESTAMPTZ,
  read_at TIMESTAMPTZ,

  error_message TEXT,

  created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Indexes for Billing & Notifications
CREATE INDEX idx_bills_org ON bills(organization_id);
CREATE INDEX idx_bills_patient ON bills(patient_id);
CREATE INDEX idx_bills_admission ON bills(admission_id);
CREATE INDEX idx_bills_date ON bills(bill_date);

```

```

CREATE INDEX idx_bills_status ON bills(payment_status);

CREATE INDEX idx_payments_org ON payments(organization_id);
CREATE INDEX idx_payments_patient ON payments(patient_id);
CREATE INDEX idx_payments_bill ON payments(bill_id);
CREATE INDEX idx_payments_date ON payments(payment_date);

CREATE INDEX idx_notification_queue_status ON notification_queue(status);
CREATE INDEX idx_notification_queue_scheduled ON notification_queue(scheduled_at)
WHERE status = 'pending';

```

📁 PART 18: FINAL - FUNCTIONS, TRIGGERS & ROW LEVEL SECURITY

SQL

```

-- =====
-- SECTION 31: HELPER FUNCTIONS
-- =====

-- Function to generate sequential numbers
CREATE OR REPLACE FUNCTION generate_sequential_number(
    p_organization_id UUID,
    p_prefix VARCHAR,
    p_sequence_name VARCHAR
) RETURNS VARCHAR AS $$
DECLARE
    v_sequence_table VARCHAR := 'sequence_numbers';
    v_current_number INTEGER;
    v_result VARCHAR;
BEGIN
    -- Create sequence table if not exists
    CREATE TABLE IF NOT EXISTS sequence_numbers (
        organization_id UUID,
        sequence_name VARCHAR(100),
        prefix VARCHAR(20),
        current_number INTEGER DEFAULT 0,
        last_reset_date DATE,
        PRIMARY KEY (organization_id, sequence_name)
    );

    -- Get and increment
    INSERT INTO sequence_numbers (organization_id, sequence_name, prefix,
current_number, last_reset_date)
VALUES (p_organization_id, p_sequence_name, p_prefix, 1, CURRENT_DATE)
ON CONFLICT (organization_id, sequence_name)
DO UPDATE SET current_number = sequence_numbers.current_number + 1
RETURNING current_number INTO v_current_number;

    -- Format: PREFIX-YYYYMMDD-NNNNN
    v_result := p_prefix || '-' || TO_CHAR(CURRENT_DATE, 'YYYYMMDD') || '-' ||
LPAD(v_current_number::TEXT, 5, '0');

```

```

    RETURN v_result;
END;
$$ LANGUAGE plpgsql;

-- Function to calculate age
CREATE OR REPLACE FUNCTION calculate_age(birth_date DATE)
RETURNS TEXT AS $$
DECLARE
    years INTEGER;
    months INTEGER;
    days INTEGER;
    age_interval INTERVAL;
BEGIN
    IF birth_date IS NULL THEN
        RETURN NULL;
    END IF;

    age_interval := age(CURRENT_DATE, birth_date);
    years := EXTRACT(YEAR FROM age_interval);
    months := EXTRACT(MONTH FROM age_interval);
    days := EXTRACT(DAY FROM age_interval);

    IF years > 0 THEN
        RETURN years || ' years';
    ELSIF months > 0 THEN
        RETURN months || ' months';
    ELSE
        RETURN days || ' days';
    END IF;
END;
$$ LANGUAGE plpgsql IMMUTABLE;

-- Function to check drug interactions
CREATE OR REPLACE FUNCTION check_drug_interactions(
    p_patient_id UUID,
    p_new_drug_id UUID
) RETURNS TABLE(
    interacting_drug_name VARCHAR,
    severity VARCHAR,
    description TEXT
) AS $$
BEGIN
    RETURN QUERY
    SELECT
        dm.generic_name,
        di.severity,
        di.description
    FROM patient_medications pm
    JOIN drug_interactions di ON (
        (di.drug_id_1 = pm.drug_id AND di.drug_id_2 = p_new_drug_id)
        OR (di.drug_id_2 = pm.drug_id AND di.drug_id_1 = p_new_drug_id)
    )
    JOIN drug_master dm ON dm.id = pm.drug_id

```

```

        WHERE pm.patient_id = p_patient_id
        AND pm.is_current = TRUE
        AND di.is_active = TRUE;
END;
$$ LANGUAGE plpgsql;

-- Function to check patient allergies against drug
CREATE OR REPLACE FUNCTION check_drug_allergy(
    p_patient_id UUID,
    p_drug_id UUID
) RETURNS TABLE(
    allergen_name VARCHAR,
    severity VARCHAR,
    reaction TEXT
) AS $$
BEGIN
    RETURN QUERY
    SELECT
        pa.allergen_name,
        pa.severity,
        pa.reaction
    FROM patient_allergies pa
    WHERE pa.patient_id = p_patient_id
    AND pa.is_active = TRUE
    AND (
        pa.allergen_name ILIKE '%' || (SELECT generic_name FROM drug_master WHERE
id = p_drug_id) || '%'
        OR pa.allergen_id IN (
            SELECT allergen_id FROM drug_allergen_mapping WHERE drug_id =
p_drug_id
        )
    );
END;
$$ LANGUAGE plpgsql;

-- =====
-- SECTION 32: TRIGGERS
-- =====

-- Trigger function for updated_at timestamp
CREATE OR REPLACE FUNCTION update_updated_at_column()
RETURNS TRIGGER AS $$
BEGIN
    NEW.updated_at = CURRENT_TIMESTAMP;
    RETURN NEW;
END;
$$ LANGUAGE plpgsql;

-- Apply updated_at trigger to relevant tables
DO $$
DECLARE
    t TEXT;
BEGIN
    FOR t IN

```

```

        SELECT table_name
        FROM information_schema.columns
        WHERE column_name = 'updated_at'
        AND table_schema = 'public'
    LOOP
        EXECUTE format('
            DROP TRIGGER IF EXISTS update_%s_updated_at ON %s;
            CREATE TRIGGER update_%s_updated_at
            BEFORE UPDATE ON %s
            FOR EACH ROW
            EXECUTE FUNCTION update_updated_at_column();
        ', t, t, t, t, t);
    END LOOP;
END;
$$;

-- Trigger for audit logging
CREATE OR REPLACE FUNCTION audit_trigger_function()
RETURNS TRIGGER AS $$
DECLARE
    v_old_data JSONB;
    v_new_data JSONB;
    v_changed_fields TEXT[];
BEGIN
    IF TG_OP = 'UPDATE' THEN
        v_old_data := to_jsonb(OLD);
        v_new_data := to_jsonb(NEW);

        -- Get changed fields
        SELECT array_agg(key) INTO v_changed_fields
        FROM jsonb_each(v_new_data)
        WHERE v_old_data -> key IS DISTINCT FROM v_new_data -> key
        AND key NOT IN ('updated_at', 'created_at');

        IF v_changed_fields IS NOT NULL THEN
            INSERT INTO audit_logs (
                organization_id,
                user_id,
                action,
                table_name,
                record_id,
                old_values,
                new_values,
                changed_fields
            ) VALUES (
                COALESCE(NEW.organization_id,
current_setting('app.organization_id', TRUE)::UUID),
                current_setting('app.user_id', TRUE)::UUID,
                'UPDATE',
                TG_TABLE_NAME,
                NEW.id,
                v_old_data,
                v_new_data,
                v_changed_fields

```

```

    );
    END IF;
    RETURN NEW;

ELSIF TG_OP = 'DELETE' THEN
    v_old_data := to_jsonb(OLD);

    INSERT INTO audit_logs (
        organization_id,
        user_id,
        action,
        table_name,
        record_id,
        old_values
    ) VALUES (
        COALESCE(OLD.organization_id, current_setting('app.organization_id',
TRUE)::UUID),
        current_setting('app.user_id', TRUE)::UUID,
        'DELETE',
        TG_TABLE_NAME,
        OLD.id,
        v_old_data
    );
    RETURN OLD;

ELSIF TG_OP = 'INSERT' THEN
    v_new_data := to_jsonb(NEW);

    INSERT INTO audit_logs (
        organization_id,
        user_id,
        action,
        table_name,
        record_id,
        new_values
    ) VALUES (
        COALESCE(NEW.organization_id, current_setting('app.organization_id',
TRUE)::UUID),
        current_setting('app.user_id', TRUE)::UUID,
        'INSERT',
        TG_TABLE_NAME,
        NEW.id,
        v_new_data
    );
    RETURN NEW;
END IF;
END;
$$ LANGUAGE plpgsql;

-- Trigger to update bed status on admission
CREATE OR REPLACE FUNCTION update_bed_on_allocation()
RETURNS TRIGGER AS $$
BEGIN
    IF TG_OP = 'INSERT' THEN

```

```

        UPDATE beds
        SET status = 'occupied',
            current_patient_id = (SELECT patient_id FROM admissions WHERE id =
NEW.admission_id),
            current_admission_id = NEW.admission_id,
            updated_at = CURRENT_TIMESTAMP
        WHERE id = NEW.bed_id;

    ELSIF TG_OP = 'UPDATE' AND NEW.release_datetime IS NOT NULL AND
OLD.release_datetime IS NULL THEN
        UPDATE beds
        SET status = 'cleaning',
            current_patient_id = NULL,
            current_admission_id = NULL,
            updated_at = CURRENT_TIMESTAMP
        WHERE id = OLD.bed_id;

        -- Create cleaning task
        INSERT INTO bed_cleaning_tasks (
            organization_id, branch_id, bed_id, ward_id,
            cleaning_type, triggered_by, previous_patient_id
        )
        SELECT
            b.organization_id, b.branch_id, b.id, b.ward_id,
            'discharge', 'discharge', OLD.admission_id
        FROM beds b WHERE b.id = OLD.bed_id;
    END IF;

    RETURN NEW;
END;
$$ LANGUAGE plpgsql;

CREATE TRIGGER trg_bed_allocation_status
AFTER INSERT OR UPDATE ON bed_allocations
FOR EACH ROW
EXECUTE FUNCTION update_bed_on_allocation();

-- Trigger to update inventory on dispensing
CREATE OR REPLACE FUNCTION update_inventory_on_dispensing()
RETURNS TRIGGER AS $$
BEGIN
    -- Update batch quantity
    UPDATE medicine_batches
    SET current_quantity = current_quantity - NEW.quantity_dispensed,
        updated_at = CURRENT_TIMESTAMP
    WHERE id = NEW.batch_id;

    -- Update inventory
    UPDATE medicine_inventory mi
    SET quantity_on_hand = quantity_on_hand - NEW.quantity_dispensed,
        updated_at = CURRENT_TIMESTAMP
    WHERE mi.drug_id = NEW.drug_id
    AND mi.pharmacy_id = (SELECT pharmacy_id FROM medicine_dispensing WHERE id =
NEW.dispensing_id);

```

```

-- Create transaction record
INSERT INTO medicine_stock_transactions (
    organization_id, pharmacy_id, drug_id, batch_id,
    transaction_type, transaction_date, quantity,
    unit_price, reference_type, reference_id
)
SELECT
    md.organization_id, md.pharmacy_id, NEW.drug_id, NEW.batch_id,
    'sale', CURRENT_TIMESTAMP, -NEW.quantity_dispensed,
    NEW.unit_price, 'dispensing', NEW.dispensing_id
FROM medicine_dispensing md WHERE md.id = NEW.dispensing_id;

RETURN NEW;
END;
$$ LANGUAGE plpgsql;

CREATE TRIGGER trg_dispensing_inventory
AFTER INSERT ON medicine_dispensing_items
FOR EACH ROW
EXECUTE FUNCTION update_inventory_on_dispensing();

-- =====
-- SECTION 33: ROW LEVEL SECURITY
-- =====

-- Enable RLS on multi-tenant tables
ALTER TABLE patients ENABLE ROW LEVEL SECURITY;
ALTER TABLE admissions ENABLE ROW LEVEL SECURITY;
ALTER TABLE appointments ENABLE ROW LEVEL SECURITY;
ALTER TABLE bills ENABLE ROW LEVEL SECURITY;
ALTER TABLE prescriptions ENABLE ROW LEVEL SECURITY;

-- Policy for organization isolation
CREATE POLICY organization_isolation_policy ON patients
    FOR ALL
    USING (organization_id = current_setting('app.organization_id', TRUE)::UUID);

CREATE POLICY organization_isolation_policy ON admissions
    FOR ALL
    USING (organization_id = current_setting('app.organization_id', TRUE)::UUID);

CREATE POLICY organization_isolation_policy ON appointments
    FOR ALL
    USING (organization_id = current_setting('app.organization_id', TRUE)::UUID);

CREATE POLICY organization_isolation_policy ON bills
    FOR ALL
    USING (organization_id = current_setting('app.organization_id', TRUE)::UUID);

CREATE POLICY organization_isolation_policy ON prescriptions
    FOR ALL
    USING (organization_id = current_setting('app.organization_id', TRUE)::UUID);

```



```

-- =====
-- SECTION 34: MATERIALIZED VIEWS FOR REPORTING
-- =====

-- Daily Census View
CREATE MATERIALIZED VIEW mv_daily_census AS
SELECT
    a.organization_id,
    a.branch_id,
    b.ward_id,
    DATE(a.admission_date) as census_date,
    COUNT(DISTINCT a.id) as total_admissions,
    COUNT(DISTINCT CASE WHEN a.status = 'admitted' THEN a.id END) as
current_admitted,
    COUNT(DISTINCT CASE WHEN a.actual_discharge_date = DATE(a.admission_date) THEN
a.id END) as same_day_discharges
FROM admissions a
JOIN bed_allocations b ON a.id = b.admission_id
GROUP BY a.organization_id, a.branch_id, b.ward_id, DATE(a.admission_date)
WITH DATA;

CREATE UNIQUE INDEX idx_mv_daily_census ON mv_daily_census(organization_id,
branch_id, ward_id, census_date);

-- Revenue Summary View
CREATE MATERIALIZED VIEW mv_revenue_summary AS
SELECT

```

Complete Hospital ERP PostgreSQL Schema (Continued - Final Part)

PART 18: FINAL - FUNCTIONS, TRIGGERS & ROW LEVEL SECURITY (Continued)

SQL

```

-- =====
-- SECTION 34: MATERIALIZED VIEWS FOR REPORTING (Continued)
-- =====

-- Revenue Summary View
CREATE MATERIALIZED VIEW mv_revenue_summary AS
SELECT
    b.organization_id,
    b.branch_id,
    DATE(b.bill_date) as revenue_date,
    b.bill_type,
    bi.category as service_category,
    COUNT(DISTINCT b.id) as bill_count,

```

```

        SUM(bi.net_amount) as gross_revenue,
        SUM(bi.discount_amount) as total_discount,
        SUM(CASE WHEN b.is_insurance THEN bi.net_amount ELSE 0 END) as
insurance_revenue,
        SUM(CASE WHEN b.is_corporate THEN bi.net_amount ELSE 0 END) as
corporate_revenue,
        SUM(CASE WHEN NOT b.is_insurance AND NOT b.is_corporate THEN bi.net_amount
ELSE 0 END) as self_pay_revenue
FROM bills b
JOIN bill_items bi ON b.id = bi.bill_id
WHERE b.is_cancelled = FALSE
GROUP BY b.organization_id, b.branch_id, DATE(b.bill_date), b.bill_type,
bi.category
WITH DATA;

CREATE UNIQUE INDEX idx_mv_revenue_summary ON mv_revenue_summary(organization_id,
branch_id, revenue_date, bill_type, service_category);

-- Doctor Performance View
CREATE MATERIALIZED VIEW mv_doctor_performance AS
SELECT
    d.organization_id,
    d.id as doctor_id,
    d.user_id,
    u.full_name as doctor_name,
    d.specialization,
    DATE_TRUNC('month', a.appointment_date) as month,
    COUNT(DISTINCT a.id) as total_appointments,
    COUNT(DISTINCT CASE WHEN a.status = 'completed' THEN a.id END) as
completed_appointments,
    COUNT(DISTINCT CASE WHEN a.status = 'no_show' THEN a.id END) as no_shows,
    COUNT(DISTINCT adm.id) as admissions,
    COUNT(DISTINCT s.id) as surgeries,
    SUM(bi.net_amount) as revenue_generated,
    AVG(EXTRACT(EPOCH FROM (ov.consultation_end_at -
ov.consultation_start_at))/60) as avg_consultation_minutes
FROM doctors d
JOIN users u ON d.user_id = u.id
LEFT JOIN appointments a ON d.id = a.doctor_id
LEFT JOIN opd_visits ov ON a.id = ov.appointment_id
LEFT JOIN admissions adm ON d.id = adm.primary_doctor_id
LEFT JOIN surgeries s ON d.id = s.primary_surgeon_id
LEFT JOIN bills b ON b.visit_id::text = a.id::text OR b.admission_id = adm.id
LEFT JOIN bill_items bi ON b.id = bi.bill_id AND bi.doctor_id = d.id
GROUP BY d.organization_id, d.id, d.user_id, u.full_name, d.specialization,
DATE_TRUNC('month', a.appointment_date)
WITH DATA;

CREATE INDEX idx_mv_doctor_performance ON mv_doctor_performance(organization_id,
doctor_id, month);

-- Bed Occupancy View
CREATE MATERIALIZED VIEW mv_bed_occupancy AS
SELECT

```

```

    w.organization_id,
    w.branch_id,
    w.id as ward_id,
    w.name as ward_name,
    w.ward_type,
    DATE(ba.allocation_datetime) as occupancy_date,
    COUNT(DISTINCT b.id) as total_beds,
    COUNT(DISTINCT CASE WHEN b.status = 'occupied' THEN b.id END) as
occupied_beds,
    COUNT(DISTINCT CASE WHEN b.status = 'available' THEN b.id END) as
available_beds,
    ROUND(COUNT(DISTINCT CASE WHEN b.status = 'occupied' THEN b.id END)::NUMERIC /
    NULLIF(COUNT(DISTINCT b.id), 0) * 100, 2) as occupancy_rate,
    AVG(EXTRACT(EPOCH FROM (COALESCE(ba.release_datetime, CURRENT_TIMESTAMP) -
ba.allocation_datetime))/86400) as avg_los_days
FROM wards w
JOIN beds b ON w.id = b.ward_id
LEFT JOIN bed_allocations ba ON b.id = ba.bed_id
WHERE b.is_active = TRUE
GROUP BY w.organization_id, w.branch_id, w.id, w.name, w.ward_type,
DATE(ba.allocation_datetime)
WITH DATA;

CREATE INDEX idx_mv_bed_occupancy ON mv_bed_occupancy(organization_id, branch_id,
ward_id, occupancy_date);

-- Lab TAT View
CREATE MATERIALIZED VIEW mv_lab_tat AS
SELECT
    lo.organization_id,
    lo.branch_id,
    DATE(lo.order_datetime) as order_date,
    tc.department,
    tc.category,
    lo.priority,
    COUNT(DISTINCT loi.id) as total_tests,
    AVG(EXTRACT(EPOCH FROM (lr.result_datetime - lo.order_datetime))/3600) as
avg_tat_hours,
    PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY EXTRACT(EPOCH FROM
(lr.result_datetime - lo.order_datetime))/3600) as median_tat_hours,
    COUNT(DISTINCT CASE WHEN lr.is_critical THEN lr.id END) as critical_results,
    COUNT(DISTINCT CASE WHEN
        EXTRACT(EPOCH FROM (lr.result_datetime - lo.order_datetime))/3600 >
tc.tat_routine_hours
        THEN loi.id END) as tat_breached
FROM lab_orders lo
JOIN lab_order_items loi ON lo.id = loi.order_id
JOIN test_catalog tc ON loi.test_id = tc.id
LEFT JOIN lab_results lr ON loi.id = lr.order_item_id
WHERE lo.status = 'completed'
GROUP BY lo.organization_id, lo.branch_id, DATE(lo.order_datetime), tc.department,
tc.category, lo.priority
WITH DATA;

```

```

CREATE INDEX idx_mv_lab_tat ON mv_lab_tat(organization_id, branch_id, order_date);

-- Insurance Claims Summary View
CREATE MATERIALIZED VIEW mv_insurance_claims_summary AS
SELECT
    ic.organization_id,
    ic.insurance_company_id,
    ins.name as insurance_company_name,
    DATE_TRUNC('month', ic.created_at) as claim_month,
    ic.status,
    COUNT(DISTINCT ic.id) as claim_count,
    SUM(ic.net_claim_amount) as total_claimed,
    SUM(ic.approved_amount) as total_approved,
    SUM(ic.settled_amount) as total_settled,
    SUM(ic.disallowed_amount) as total_disallowed,
    AVG(EXTRACT(DAY FROM (ic.settlement_date - ic.submitted_at::date))) as
avg_settlement_days
FROM insurance_claims ic
JOIN insurance_companies ins ON ic.insurance_company_id = ins.id
GROUP BY ic.organization_id, ic.insurance_company_id, ins.name,
DATE_TRUNC('month', ic.created_at), ic.status
WITH DATA;

CREATE INDEX idx_mv_insurance_claims ON
mv_insurance_claims_summary(organization_id, insurance_company_id, claim_month);

-- Refresh function for materialized views
CREATE OR REPLACE FUNCTION refresh_all_materialized_views()
RETURNS VOID AS $$
BEGIN
    REFRESH MATERIALIZED VIEW CONCURRENTLY mv_daily_census;
    REFRESH MATERIALIZED VIEW CONCURRENTLY mv_revenue_summary;
    REFRESH MATERIALIZED VIEW CONCURRENTLY mv_doctor_performance;
    REFRESH MATERIALIZED VIEW CONCURRENTLY mv_bed_occupancy;
    REFRESH MATERIALIZED VIEW CONCURRENTLY mv_lab_tat;
    REFRESH MATERIALIZED VIEW CONCURRENTLY mv_insurance_claims_summary;
END;
$$ LANGUAGE plpgsql;

-- =====
-- SECTION 35: ADDITIONAL TABLES FOR REMAINING MODULES
-- =====

-- Ambulance Management
CREATE TABLE ambulances (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),
    branch_id UUID NOT NULL REFERENCES branches(id),

    ambulance_code VARCHAR(20) NOT NULL,
    registration_number VARCHAR(50) NOT NULL,

    ambulance_type VARCHAR(50), -- basic, advanced, icu, neonatal

```

```

-- Vehicle Details
make VARCHAR(100),
model VARCHAR(100),
year INTEGER,

-- Equipment
equipment_list JSONB,
has_ventilator BOOLEAN DEFAULT FALSE,
has_defibrillator BOOLEAN DEFAULT FALSE,
has_oxygen BOOLEAN DEFAULT TRUE,

-- Capacity
patient_capacity INTEGER DEFAULT 1,

-- Status
status VARCHAR(20) DEFAULT 'available', -- available, on_call, maintenance,
out_of_service
current_location POINT,

-- Insurance
insurance_valid_until DATE,
fitness_valid_until DATE,

-- Assigned Staff
current_driver_id UUID REFERENCES employees(id),
current_paramedic_id UUID REFERENCES employees(id),

is_active BOOLEAN DEFAULT TRUE,
created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

UNIQUE(organization_id, ambulance_code)
);

-- Ambulance Trips
CREATE TABLE ambulance_trips (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),
  branch_id UUID NOT NULL REFERENCES branches(id),

  trip_number VARCHAR(50) NOT NULL,

  ambulance_id UUID NOT NULL REFERENCES ambulances(id),

  trip_type VARCHAR(50), -- emergency_pickup, inter_hospital_transfer,
  patient_drop, dead_body

-- Patient
patient_id UUID REFERENCES patients(id),
patient_name VARCHAR(255),
patient_phone VARCHAR(20),

-- Pickup
pickup_address TEXT,
pickup_latitude DECIMAL(10, 8),

```

```

pickup_longitude DECIMAL(11, 8),

-- Destination
destination_address TEXT,
destination_latitude DECIMAL(10, 8),
destination_longitude DECIMAL(11, 8),

-- Timing
request_datetime TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
dispatched_at TIMESTAMPTZ,
reached_pickup_at TIMESTAMPTZ,
left_pickup_at TIMESTAMPTZ,
reached_destination_at TIMESTAMPTZ,
trip_completed_at TIMESTAMPTZ,

-- Distance & Duration
distance_km DECIMAL(8, 2),
duration_minutes INTEGER,

-- Staff
driver_id UUID REFERENCES employees(id),
paramedic_id UUID REFERENCES employees(id),

-- Clinical
patient_condition_at_pickup TEXT,
vitals_during_transport JSONB,
treatment_given TEXT,

-- Billing
trip_charge DECIMAL(10, 2),
additional_charges DECIMAL(10, 2),
total_charge DECIMAL(10, 2),

bill_id UUID REFERENCES bills(id),

status VARCHAR(20) DEFAULT 'requested', -- requested, dispatched, in_progress,
completed, cancelled

cancellation_reason TEXT,

notes TEXT,

created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

UNIQUE(organization_id, trip_number)
);

-- Blood Bank Management
CREATE TABLE blood_bank_inventory (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),
  branch_id UUID NOT NULL REFERENCES branches(id),

  blood_group blood_group_type NOT NULL,

```

```

component_type VARCHAR(50) NOT NULL, -- whole_blood, prbc, ffp, platelets,
cryo

unit_number VARCHAR(50) NOT NULL,
bag_number VARCHAR(50),

-- Donor Info
donor_id UUID,
donor_name VARCHAR(255),
donation_date DATE,

-- Collection
collection_date DATE NOT NULL,
collection_center VARCHAR(255),

-- Testing
testing_status VARCHAR(20) DEFAULT 'pending', -- pending, passed, failed
tested_by UUID REFERENCES users(id),
tested_at TIMESTAMPTZ,

hiv_result VARCHAR(20),
hbsag_result VARCHAR(20),
hcv_result VARCHAR(20),
vdrl_result VARCHAR(20),
malaria_result VARCHAR(20),

-- Storage
storage_location VARCHAR(100),
storage_temperature DECIMAL(4, 1),

-- Expiry
expiry_date DATE NOT NULL,

-- Status
status VARCHAR(20) DEFAULT 'available', -- available, reserved, issued,
expired, discarded

-- If issued
issued_to_patient_id UUID REFERENCES patients(id),
issued_for_admission_id UUID REFERENCES admissions(id),
issued_at TIMESTAMPTZ,
issued_by UUID REFERENCES users(id),

-- Cross Match
cross_match_done BOOLEAN DEFAULT FALSE,
cross_match_result VARCHAR(20),
cross_matched_by UUID REFERENCES users(id),

notes TEXT,

created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

UNIQUE(organization_id, unit_number)

```

```

);

-- Blood Requisitions
CREATE TABLE blood_requisitions (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),
  branch_id UUID NOT NULL REFERENCES branches(id),

  requisition_number VARCHAR(50) NOT NULL,

  patient_id UUID NOT NULL REFERENCES patients(id),
  admission_id UUID REFERENCES admissions(id),
  surgery_id UUID REFERENCES surgeries(id),

  blood_group blood_group_type NOT NULL,

  -- Request
  component_type VARCHAR(50) NOT NULL,
  units_requested INTEGER NOT NULL,

  indication TEXT,
  urgency VARCHAR(20) DEFAULT 'routine', -- routine, urgent, emergency

  requested_by UUID NOT NULL REFERENCES users(id),
  requested_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
  required_by TIMESTAMPTZ,

  -- Approval
  status VARCHAR(20) DEFAULT 'pending', -- pending, approved, partial_issued,
  issued, cancelled

  approved_by UUID REFERENCES users(id),
  approved_at TIMESTAMPTZ,

  units_approved INTEGER,
  units_issued INTEGER DEFAULT 0,

  -- Cross Match
  sample_collected BOOLEAN DEFAULT FALSE,
  sample_collected_at TIMESTAMPTZ,

  notes TEXT,

  created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

  UNIQUE(organization_id, requisition_number)
);

-- Transfusion Records
CREATE TABLE transfusion_records (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  requisition_id UUID NOT NULL REFERENCES blood_requisitions(id),
  blood_unit_id UUID NOT NULL REFERENCES blood_bank_inventory(id),

```



```

    patient_id UUID NOT NULL REFERENCES patients(id),
    admission_id UUID REFERENCES admissions(id),

    -- Pre-Transfusion
    pre_transfusion_vitals JSONB,
    pre_transfusion_check_by UUID REFERENCES users(id),

    -- Transfusion
    transfusion_started_at TIMESTAMPTZ,
    transfusion_completed_at TIMESTAMPTZ,

    administered_by UUID REFERENCES users(id),
    supervised_by UUID REFERENCES users(id),

    -- Monitoring
    monitoring_chart JSONB, -- Vitals every 15 mins

    -- Reaction
    reaction_occurred BOOLEAN DEFAULT FALSE,
    reaction_type VARCHAR(100),
    reaction_severity VARCHAR(20),
    reaction_time TIMESTAMPTZ,
    action_taken TEXT,

    -- Post-Transfusion
    post_transfusion_vitals JSONB,

    outcome VARCHAR(50), -- completed, stopped_reaction, stopped_other

    notes TEXT,

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Waste Management
CREATE TABLE waste_categories (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),

    code VARCHAR(20) NOT NULL,
    name VARCHAR(100) NOT NULL,

    color_code VARCHAR(20), -- yellow, red, blue, black, white
    container_type VARCHAR(50),

    handling_instructions TEXT,
    disposal_method TEXT,

    is_hazardous BOOLEAN DEFAULT FALSE,

    is_active BOOLEAN DEFAULT TRUE,

    UNIQUE(organization_id, code)
);

```

```

-- Waste Collection Log
CREATE TABLE waste_collection_log (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),
  branch_id UUID NOT NULL REFERENCES branches(id),

  collection_date DATE NOT NULL,
  collection_time TIME,

  category_id UUID NOT NULL REFERENCES waste_categories(id),

  -- Source
  source_department_id UUID REFERENCES departments(id),
  source_ward_id UUID REFERENCES wards(id),
  source_location VARCHAR(255),

  -- Quantity
  quantity_kg DECIMAL(8, 2) NOT NULL,
  number_of_bags INTEGER,

  -- Collection
  collected_by UUID REFERENCES users(id),

  -- Handover
  handed_over_to VARCHAR(255), -- Disposal company/person
  handover_time TIMESTAMPTZ,
  handover_receipt VARCHAR(100),

  notes TEXT,

  created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Visitor Management
CREATE TABLE visitor_passes (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),
  branch_id UUID NOT NULL REFERENCES branches(id),

  pass_number VARCHAR(50) NOT NULL,

  -- Visitor Info
  visitor_name VARCHAR(255) NOT NULL,
  visitor_phone VARCHAR(20),
  visitor_id_type VARCHAR(50),
  visitor_id_number VARCHAR(100),

  visitor_photo_url TEXT,

  -- Purpose
  purpose VARCHAR(50), -- patient_visit, meeting, vendor, contractor, other

  -- If visiting patient

```

```

    patient_id UUID REFERENCES patients(id),
    admission_id UUID REFERENCES admissions(id),
    ward_id UUID REFERENCES wards(id),
    relationship_to_patient VARCHAR(50),

    -- If meeting
    meeting_with_employee_id UUID REFERENCES employees(id),
    meeting_with_department_id UUID REFERENCES departments(id),

    -- Validity
    valid_date DATE NOT NULL,
    valid_from_time TIME,
    valid_until_time TIME,

    -- Check In/Out
    checked_in_at TIMESTAMPTZ,
    checked_out_at TIMESTAMPTZ,

    -- Pass Type
    pass_type VARCHAR(20) DEFAULT 'single', -- single, daily, weekly, permanent

    -- Status
    status VARCHAR(20) DEFAULT 'active', -- active, expired, cancelled,
    blacklisted

    -- Belongings
    belongings_deposited TEXT,
    belongings_token VARCHAR(50),

    issued_by UUID REFERENCES users(id),

    notes TEXT,

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

    UNIQUE(organization_id, pass_number)
);

-- Canteen Management
CREATE TABLE canteen_menu_items (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),
    branch_id UUID NOT NULL REFERENCES branches(id),

    item_code VARCHAR(20) NOT NULL,
    name VARCHAR(255) NOT NULL,
    name_bangla VARCHAR(255),

    category VARCHAR(50), -- breakfast, lunch, dinner, snacks, beverages

    description TEXT,

    price DECIMAL(8, 2) NOT NULL,

```

```

-- Availability
available_days INTEGER[] DEFAULT ARRAY[0,1,2,3,4,5,6],
available_from TIME,
available_until TIME,

-- Dietary Info
is_vegetarian BOOLEAN DEFAULT FALSE,
is_vegan BOOLEAN DEFAULT FALSE,
calories INTEGER,

image_url TEXT,

is_available BOOLEAN DEFAULT TRUE,
is_active BOOLEAN DEFAULT TRUE,

created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

UNIQUE(branch_id, item_code)
);

-- Canteen Orders
CREATE TABLE canteen_orders (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),
  branch_id UUID NOT NULL REFERENCES branches(id),

  order_number VARCHAR(50) NOT NULL,

-- Customer
customer_type VARCHAR(50), -- staff, visitor, patient_attendant
employee_id UUID REFERENCES employees(id),
customer_name VARCHAR(255),

order_datetime TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

-- Items
items JSONB NOT NULL, -- [{item_id, quantity, price}]

subtotal DECIMAL(10, 2),
discount DECIMAL(8, 2) DEFAULT 0,
total_amount DECIMAL(10, 2),

-- Payment
payment_method payment_method_type,
payment_status payment_status_type DEFAULT 'pending',

-- Fulfillment
status VARCHAR(20) DEFAULT 'pending', -- pending, preparing, ready, delivered,
cancelled

prepared_at TIMESTAMPTZ,
delivered_at TIMESTAMPTZ,

notes TEXT,

```

```

        created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

        UNIQUE(organization_id, order_number)
    );

-- Kitchen Management
CREATE TABLE kitchen_orders (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),
    branch_id UUID NOT NULL REFERENCES branches(id),

    order_number VARCHAR(50) NOT NULL,

    order_type VARCHAR(50), -- patient_meal, canteen, event

    order_date DATE NOT NULL,
    meal_type VARCHAR(20), -- breakfast, lunch, dinner, snack

    -- Source
    source_type VARCHAR(50), -- diet_order, canteen_order, manual
    source_id UUID,

    -- Items
    items JSONB NOT NULL,

    -- Quantities
    total_portions INTEGER,

    -- Timing
    required_by TIMESTAMPTZ,

    -- Status
    status VARCHAR(20) DEFAULT 'pending', -- pending, preparing, ready, dispatched

    started_at TIMESTAMPTZ,
    completed_at TIMESTAMPTZ,

    prepared_by UUID REFERENCES users(id),

    notes TEXT,

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

    UNIQUE(organization_id, order_number)
);

-- =====
-- SECTION 36: INDEXES FOR ADDITIONAL TABLES
-- =====

CREATE INDEX idx_ambulances_status ON ambulances(status);
CREATE INDEX idx_ambulance_trips_ambulance ON ambulance_trips(ambulance_id);
CREATE INDEX idx_ambulance_trips_date ON ambulance_trips(request_datetime);

```

```

CREATE INDEX idx_blood_inventory_blood_group ON blood_bank_inventory(blood_group,
component_type);
CREATE INDEX idx_blood_inventory_status ON blood_bank_inventory(status);
CREATE INDEX idx_blood_inventory_expiry ON blood_bank_inventory(expiry_date);

CREATE INDEX idx_blood_requisitions_patient ON blood_requisitions(patient_id);
CREATE INDEX idx_blood_requisitions_status ON blood_requisitions(status);

CREATE INDEX idx_visitor_passes_date ON visitor_passes(valid_date);
CREATE INDEX idx_visitor_passes_patient ON visitor_passes(patient_id);

-- =====
-- SECTION 37: DATABASE MAINTENANCE PROCEDURES
-- =====

-- Procedure to archive old data
CREATE OR REPLACE PROCEDURE archive_old_data(
    p_organization_id UUID,
    p_archive_before_date DATE
)
LANGUAGE plpgsql
AS $$
BEGIN
    -- Archive old audit logs
    INSERT INTO audit_logs_archive
    SELECT * FROM audit_logs
    WHERE organization_id = p_organization_id
    AND created_at < p_archive_before_date;

    DELETE FROM audit_logs
    WHERE organization_id = p_organization_id
    AND created_at < p_archive_before_date;

    -- Archive old notification logs
    INSERT INTO notification_log_archive
    SELECT * FROM notification_log
    WHERE created_at < p_archive_before_date;

    DELETE FROM notification_log
    WHERE created_at < p_archive_before_date;

    RAISE NOTICE 'Data archived successfully for organization %',
p_organization_id;
END;
$$;

-- Procedure to clean up expired sessions
CREATE OR REPLACE PROCEDURE cleanup_expired_sessions()
LANGUAGE plpgsql
AS $$
DECLARE
    v_deleted_count INTEGER;
BEGIN

```

```

DELETE FROM user_sessions
WHERE expires_at < CURRENT_TIMESTAMP
OR (last_activity_at < CURRENT_TIMESTAMP - INTERVAL '24 hours' AND is_active =
TRUE);

GET DIAGNOSTICS v_deleted_count = ROW_COUNT;

RAISE NOTICE 'Cleaned up % expired sessions', v_deleted_count;
END;
$$;

-- Procedure to update bed census
CREATE OR REPLACE PROCEDURE update_daily_census(
    p_organization_id UUID,
    p_census_date DATE DEFAULT CURRENT_DATE
)
LANGUAGE plpgsql
AS $$
BEGIN
    INSERT INTO ward_census (
        organization_id, branch_id, ward_id, census_date,
        total_beds, occupied_beds, available_beds, reserved_beds,
        cleaning_beds, maintenance_beds, occupancy_percentage,
        admissions_today, discharges_today, transfers_in, transfers_out,
        deaths_today
    )
    SELECT
        w.organization_id,
        w.branch_id,
        w.id as ward_id,
        p_census_date,
        COUNT(b.id) as total_beds,
        COUNT(CASE WHEN b.status = 'occupied' THEN 1 END) as occupied_beds,
        COUNT(CASE WHEN b.status = 'available' THEN 1 END) as available_beds,
        COUNT(CASE WHEN b.status = 'reserved' THEN 1 END) as reserved_beds,
        COUNT(CASE WHEN b.status = 'cleaning' THEN 1 END) as cleaning_beds,
        COUNT(CASE WHEN b.status = 'maintenance' THEN 1 END) as maintenance_beds,
        ROUND(COUNT(CASE WHEN b.status = 'occupied' THEN 1 END)::NUMERIC /
        NULLIF(COUNT(b.id), 0) * 100, 2),
        (SELECT COUNT(*) FROM admissions a WHERE a.admission_date = p_census_date
        AND a.branch_id = w.branch_id),
        (SELECT COUNT(*) FROM admissions a WHERE a.actual_discharge_date =
        p_census_date AND a.branch_id = w.branch_id),
        (SELECT COUNT(*) FROM bed_transfer_requests btr WHERE
        DATE(btr.transferred_at) = p_census_date AND btr.to_ward_id = w.id AND btr.status
        = 'completed'),
        (SELECT COUNT(*) FROM bed_transfer_requests btr WHERE
        DATE(btr.transferred_at) = p_census_date AND btr.from_ward_id = w.id AND
        btr.status = 'completed'),
        (SELECT COUNT(*) FROM admissions a WHERE a.actual_discharge_date =
        p_census_date AND a.discharge_type = 'expired' AND a.branch_id = w.branch_id)
    FROM wards w
    LEFT JOIN beds b ON w.id = b.ward_id AND b.is_active = TRUE
    WHERE w.organization_id = p_organization_id

```

```

AND w.is_active = TRUE
GROUP BY w.organization_id, w.branch_id, w.id
ON CONFLICT (ward_id, census_date)
DO UPDATE SET
    total_beds = EXCLUDED.total_beds,
    occupied_beds = EXCLUDED.occupied_beds,
    available_beds = EXCLUDED.available_beds,
    reserved_beds = EXCLUDED.reserved_beds,
    cleaning_beds = EXCLUDED.cleaning_beds,
    maintenance_beds = EXCLUDED.maintenance_beds,
    occupancy_percentage = EXCLUDED.occupancy_percentage,
    admissions_today = EXCLUDED.admissions_today,
    discharges_today = EXCLUDED.discharges_today,
    transfers_in = EXCLUDED.transfers_in,
    transfers_out = EXCLUDED.transfers_out,
    deaths_today = EXCLUDED.deaths_today;

END;
$$;

-- =====
-- SECTION 38: CRON JOBS (Using pg_cron extension)
-- =====

-- Note: These require pg_cron extension to be installed

-- Schedule daily census update
-- SELECT cron.schedule('update-daily-census', '0 0 * * *', 'CALL
update_daily_census(NULL, CURRENT_DATE - 1)');

-- Schedule session cleanup
-- SELECT cron.schedule('cleanup-sessions', '0 */6 * * *', 'CALL
cleanup_expired_sessions()');

-- Schedule materialized view refresh
-- SELECT cron.schedule('refresh-mv', '0 */4 * * *', 'SELECT
refresh_all_materialized_views()');

-- =====
-- SECTION 39: INITIAL DATA SEEDING
-- =====

-- Insert system roles
INSERT INTO roles (id, code, name, role_type, is_system_role, is_active) VALUES
    (uuid_generate_v4(), 'SUPER_ADMIN', 'Super Administrator', 'system', TRUE,
TRUE),
    (uuid_generate_v4(), 'ORG_ADMIN', 'Organization Administrator', 'system',
TRUE, TRUE),
    (uuid_generate_v4(), 'BRANCH_ADMIN', 'Branch Administrator', 'system', TRUE,
TRUE),
    (uuid_generate_v4(), 'DOCTOR', 'Doctor', 'system', TRUE, TRUE),
    (uuid_generate_v4(), 'NURSE', 'Nurse', 'system', TRUE, TRUE),
    (uuid_generate_v4(), 'PHARMACIST', 'Pharmacist', 'system', TRUE, TRUE),
    (uuid_generate_v4(), 'LAB_TECHNICIAN', 'Lab Technician', 'system', TRUE,
TRUE),

```



```

        (uuid_generate_v4(), 'RADIOLOGIST', 'Radiologist', 'system', TRUE, TRUE),
        (uuid_generate_v4(), 'RECEPTIONIST', 'Receptionist', 'system', TRUE, TRUE),
        (uuid_generate_v4(), 'CASHIER', 'Cashier', 'system', TRUE, TRUE),
        (uuid_generate_v4(), 'ACCOUNTANT', 'Accountant', 'system', TRUE, TRUE),
        (uuid_generate_v4(), 'HR_MANAGER', 'HR Manager', 'system', TRUE, TRUE),
        (uuid_generate_v4(), 'STORE_MANAGER', 'Store Manager', 'system', TRUE, TRUE)
ON CONFLICT DO NOTHING;

-- Insert Bangladesh divisions
INSERT INTO countries (id, code, name, phone_code, currency_code) VALUES
    (uuid_generate_v4(), 'BD', 'Bangladesh', '+880', 'BDT')
ON CONFLICT DO NOTHING;

INSERT INTO divisions (country_id, code, name, name_bangla)
SELECT c.id, d.code, d.name, d.name_bangla
FROM countries c,
(VALUE
    ('DHK', 'Dhaka', 'ঢাকা'),
    ('CTG', 'Chattogram', 'চট্টগ্রাম'),
    ('RAJ', 'Rajshahi', 'রাজশাহী'),
    ('KHU', 'Khulna', 'খুলনা'),
    ('BAR', 'Barishal', 'বরিশাল'),
    ('SYL', 'Sylhet', 'সিলেট'),
    ('RAN', 'Rangpur', 'রংপুর'),
    ('MYM', 'Mymensingh', 'ময়মনসিংহ')
) AS d(code, name, name_bangla)
WHERE c.code = 'BD'
ON CONFLICT DO NOTHING;

-- Insert common allergens
INSERT INTO allergens (code, name, category) VALUES
    ('PEN', 'Penicillin', 'drug'),
    ('SUL', 'Sulfa Drugs', 'drug'),
    ('ASP', 'Aspirin', 'drug'),
    ('IBU', 'Ibuprofen', 'drug'),
    ('NSAID', 'NSAIDs', 'drug'),
    ('LATEX', 'Latex', 'environment'),
    ('PEANUT', 'Peanuts', 'food'),
    ('SHELLFISH', 'Shellfish', 'food'),
    ('EGG', 'Eggs', 'food'),
    ('MILK', 'Milk/Dairy', 'food'),
    ('SOY', 'Soy', 'food'),
    ('WHEAT', 'Wheat/Gluten', 'food'),
    ('CONTRAST', 'IV Contrast Dye', 'drug'),
    ('CODEINE', 'Codeine', 'drug'),
    ('MORPHINE', 'Morphine', 'drug')
ON CONFLICT DO NOTHING;

-- =====
-- SECTION 40: GRANT PERMISSIONS
-- =====

-- Create application roles
DO $$

```

```

BEGIN
    IF NOT EXISTS (SELECT FROM pg_roles WHERE rolname = 'hospital_app') THEN
        CREATE ROLE hospital_app LOGIN PASSWORD 'secure_password_here';
    END IF;

    IF NOT EXISTS (SELECT FROM pg_roles WHERE rolname = 'hospital_readonly') THEN
        CREATE ROLE hospital_readonly;
    END IF;
END
$$;

-- Grant permissions to application role
GRANT USAGE ON SCHEMA public TO hospital_app;
GRANT SELECT, INSERT, UPDATE, DELETE ON ALL TABLES IN SCHEMA public TO
hospital_app;
GRANT USAGE, SELECT ON ALL SEQUENCES IN SCHEMA public TO hospital_app;
GRANT EXECUTE ON ALL FUNCTIONS IN SCHEMA public TO hospital_app;

-- Grant readonly permissions
GRANT USAGE ON SCHEMA public TO hospital_readonly;
GRANT SELECT ON ALL TABLES IN SCHEMA public TO hospital_readonly;

-- Set default privileges for future objects
ALTER DEFAULT PRIVILEGES IN SCHEMA public GRANT SELECT, INSERT, UPDATE, DELETE ON
TABLES TO hospital_app;
ALTER DEFAULT PRIVILEGES IN SCHEMA public GRANT USAGE, SELECT ON SEQUENCES TO
hospital_app;
ALTER DEFAULT PRIVILEGES IN SCHEMA public GRANT SELECT ON TABLES TO
hospital_readonly;

-- =====
-- SCHEMA COMPLETE
-- =====

-- Final statistics
DO $$
DECLARE
    table_count INTEGER;
    index_count INTEGER;
    function_count INTEGER;
    trigger_count INTEGER;
BEGIN
    SELECT COUNT(*) INTO table_count FROM information_schema.tables WHERE
table_schema = 'public' AND table_type = 'BASE TABLE';
    SELECT COUNT(*) INTO index_count FROM pg_indexes WHERE schemaname = 'public';
    SELECT COUNT(*) INTO function_count FROM information_schema.routines WHERE
routine_schema = 'public';
    SELECT COUNT(*) INTO trigger_count FROM information_schema.triggers WHERE
trigger_schema = 'public';

    RAISE NOTICE '=====';
    RAISE NOTICE 'Hospital ERP Schema Creation Complete';
    RAISE NOTICE '=====';
    RAISE NOTICE 'Tables created: %', table_count;

```

```
RAISE NOTICE 'Indexes created: %', index_count;
RAISE NOTICE 'Functions created: %', function_count;
RAISE NOTICE 'Triggers created: %', trigger_count;
RAISE NOTICE '=====';
END
$$;
```

 SCHEMA SUMMARY

Total Tables Created: ~180+

Category	Table Count	Key Tables
Foundation	15	organizations, branches, departments, users, roles, permissions
Patient Management	12	patients, patient_allergies, patient_medical_history, patient_documents
Appointments & Visits	8	appointments, opd_visits, emergency_visits, doctors
Admissions	6	admissions, bed_allocations, discharge_summaries
EMR/Clinical	10	patient_vitals, clinical_notes, clinical_diagnoses, clinical_procedures
Prescriptions	5	prescriptions, prescription_items, narcotic_prescriptions
Nurse Station	12	nurse_duty_roster, doctor_orders, medication_administration_records, nursing_assessments
Bed Management	8	wards, rooms, beds, bed_reservations, bed_cleaning_tasks
OT Management	10	ot_rooms, surgeries, surgery_preop_checklist, anesthesia_records
Diagnostics	15	test_catalog, lab_orders, lab_results, radiology_orders, radiology_reports
Pharmacy	12	medicine_inventory, medicine_batches, medicine_dispensing, narcotic_register
Inventory	18	item_master, suppliers, stores, purchase_orders, goods_receipt_notes
Insurance	10	insurance_companies, patient_insurance_policies, insurance_preauth, insurance_claims
Housekeeping	8	cleaning_zones, cleaning_schedules, linen_inventory, disinfection_log
HR & Payroll	15	employees, attendance, leave_applications, payroll_runs, payroll_details
Maintenance	8	equipment_registry, maintenance_tickets, pm_schedules, calibration_records
Diet & Kitchen	6	diet_types, patient_diet_orders, meal_deliveries, kitchen_orders

Category	Table Count	Key Tables
Telemedicine	2	telemedicine_sessions
Complaints	5	complaints, complaint_categories, feedback_forms, feedback_responses
Consent	3	consent_templates, patient_consents
Billing	6	service_master, bills, bill_items, payments
Additional	10	ambulances, blood_bank_inventory, visitor_passes, waste_collection_log
System	8	audit_logs, notification_queue, report_definitions, dashboard_widgets

Key Features Implemented:

- ✔ **Multi-Tenancy** - Organization → Branch → Department hierarchy
- ✔ **Role-Based Access Control (RBAC)** - Granular permissions
- ✔ **Row-Level Security (RLS)** - Data isolation per organization
- ✔ **Audit Logging** - Complete change tracking
- ✔ **Materialized Views** - For reporting performance
- ✔ **Triggers** - Automated workflows
- ✔ **Stored Procedures** - Business logic
- ✔ **Sequential Number Generation** - For all document numbers
- ✔ **Full-Text Search Support** - Using pg_trgm
- ✔ **JSONB for Flexibility** - For dynamic fields

Database Extensions Required:

atabase Extensions Required:

SQL

```
CREATE EXTENSION IF NOT EXISTS "uuid-oss";
CREATE EXTENSION IF NOT EXISTS "pgcrypto";
CREATE EXTENSION IF NOT EXISTS "pg_trgm";
CREATE EXTENSION IF NOT EXISTS "btree_gist";
-- Optional: CREATE EXTENSION IF NOT EXISTS "pg_cron";
```

Complete Hospital ERP PostgreSQL Schema (Additional Components)

📁 PART 19: ADVANCED FEATURES & OPTIMIZATIONS

SQL

```

-- =====
-- SECTION 41: TABLE PARTITIONING FOR LARGE TABLES
-- =====

-- Partition audit_logs by month for better performance
CREATE TABLE audit_logs_partitioned (
    id UUID DEFAULT uuid_generate_v4(),
    organization_id UUID REFERENCES organizations(id),
    branch_id UUID REFERENCES branches(id),
    user_id UUID REFERENCES users(id),
    username VARCHAR(100),
    ip_address INET,
    user_agent TEXT,
    action VARCHAR(50) NOT NULL,
    module VARCHAR(100),
    sub_module VARCHAR(100),
    table_name VARCHAR(100),
    record_id UUID,
    old_values JSONB,
    new_values JSONB,
    changed_fields TEXT[],
    description TEXT,
    metadata JSONB,
    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
    PRIMARY KEY (id, created_at)
) PARTITION BY RANGE (created_at);

-- Create partitions for audit logs
CREATE TABLE audit_logs_y2024m01 PARTITION OF audit_logs_partitioned
    FOR VALUES FROM ('2024-01-01') TO ('2024-02-01');
CREATE TABLE audit_logs_y2024m02 PARTITION OF audit_logs_partitioned
    FOR VALUES FROM ('2024-02-01') TO ('2024-03-01');
CREATE TABLE audit_logs_y2024m03 PARTITION OF audit_logs_partitioned
    FOR VALUES FROM ('2024-03-01') TO ('2024-04-01');
CREATE TABLE audit_logs_y2024m04 PARTITION OF audit_logs_partitioned
    FOR VALUES FROM ('2024-04-01') TO ('2024-05-01');
CREATE TABLE audit_logs_y2024m05 PARTITION OF audit_logs_partitioned
    FOR VALUES FROM ('2024-05-01') TO ('2024-06-01');
CREATE TABLE audit_logs_y2024m06 PARTITION OF audit_logs_partitioned
    FOR VALUES FROM ('2024-06-01') TO ('2024-07-01');
CREATE TABLE audit_logs_y2024m07 PARTITION OF audit_logs_partitioned
    FOR VALUES FROM ('2024-07-01') TO ('2024-08-01');
CREATE TABLE audit_logs_y2024m08 PARTITION OF audit_logs_partitioned
    FOR VALUES FROM ('2024-08-01') TO ('2024-09-01');
CREATE TABLE audit_logs_y2024m09 PARTITION OF audit_logs_partitioned
    FOR VALUES FROM ('2024-09-01') TO ('2024-10-01');
CREATE TABLE audit_logs_y2024m10 PARTITION OF audit_logs_partitioned
    FOR VALUES FROM ('2024-10-01') TO ('2024-11-01');
CREATE TABLE audit_logs_y2024m11 PARTITION OF audit_logs_partitioned
    FOR VALUES FROM ('2024-11-01') TO ('2024-12-01');
CREATE TABLE audit_logs_y2024m12 PARTITION OF audit_logs_partitioned
    FOR VALUES FROM ('2024-12-01') TO ('2025-01-01');

```

```

-- Function to create future partitions automatically
CREATE OR REPLACE FUNCTION create_audit_partition()
RETURNS VOID AS $$
DECLARE
    partition_date DATE;
    partition_name TEXT;
    start_date DATE;
    end_date DATE;
BEGIN
    -- Create partitions for next 3 months
    FOR i IN 0..2 LOOP
        partition_date := DATE_TRUNC('month', CURRENT_DATE + (i || '
months')::INTERVAL);
        partition_name := 'audit_logs_y' || TO_CHAR(partition_date, 'YYYY') || 'm'
|| TO_CHAR(partition_date, 'MM');
        start_date := partition_date;
        end_date := partition_date + INTERVAL '1 month';

        -- Check if partition exists
        IF NOT EXISTS (
            SELECT 1 FROM pg_class WHERE relname = partition_name
        ) THEN
            EXECUTE format(
                'CREATE TABLE %I PARTITION OF audit_logs_partitioned FOR VALUES
FROM (%L) TO (%L)',
                partition_name, start_date, end_date
            );
            RAISE NOTICE 'Created partition: %', partition_name;
        END IF;
    END LOOP;
END;
$$ LANGUAGE plpgsql;

-- Partition lab_results by month
CREATE TABLE lab_results_partitioned (
    id UUID DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL,
    order_item_id UUID NOT NULL,
    order_id UUID NOT NULL,
    patient_id UUID NOT NULL,
    test_id UUID NOT NULL,
    result_datetime TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
    parameter_name VARCHAR(255),
    parameter_code VARCHAR(50),
    result_value VARCHAR(500),
    result_numeric DECIMAL(15, 5),
    result_unit VARCHAR(50),
    reference_min DECIMAL(15, 5),
    reference_max DECIMAL(15, 5),
    reference_text VARCHAR(255),
    abnormal_flag VARCHAR(10),
    is_critical BOOLEAN DEFAULT FALSE,
    critical_notified BOOLEAN DEFAULT FALSE,
    critical_notified_to UUID,

```

```

critical_notified_at TIMESTAMPTZ,
previous_value DECIMAL(15, 5),
previous_date DATE,
delta_flag BOOLEAN DEFAULT FALSE,
result_comment TEXT,
technical_comment TEXT,
entered_by UUID,
entered_at TIMESTAMPTZ,
verified_by UUID,
verified_at TIMESTAMPTZ,
status_result_status_type DEFAULT 'preliminary',
analyzer_name VARCHAR(100),
analyzer_result_id VARCHAR(100),
created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
PRIMARY KEY (id, result_datetime)
) PARTITION BY RANGE (result_datetime);

-- =====
-- SECTION 42: FULL-TEXT SEARCH CONFIGURATION
-- =====

-- Create custom text search configuration for Bangla/English
CREATE TEXT SEARCH CONFIGURATION hospital_search (COPY = pg_catalog.english);

-- Add full-text search columns to patients
ALTER TABLE patients ADD COLUMN IF NOT EXISTS search_vector tsvector;

-- Function to update patient search vector
CREATE OR REPLACE FUNCTION update_patient_search_vector()
RETURNS TRIGGER AS $$
BEGIN
    NEW.search_vector :=
        setweight(to_tsvector('english', COALESCE(NEW.mrn, '')), 'A') ||
        setweight(to_tsvector('english', COALESCE(NEW.first_name, '')), 'A') ||
        setweight(to_tsvector('english', COALESCE(NEW.last_name, '')), 'A') ||
        setweight(to_tsvector('english', COALESCE(NEW.primary_phone, '')), 'B') ||
        setweight(to_tsvector('english', COALESCE(NEW.nid, '')), 'B') ||
        setweight(to_tsvector('english', COALESCE(NEW.email, '')), 'C') ||
        setweight(to_tsvector('english', COALESCE(NEW.present_address_line1, '')),
'D');
    RETURN NEW;
END;
$$ LANGUAGE plpgsql;

CREATE TRIGGER trg_patient_search_vector
BEFORE INSERT OR UPDATE ON patients
FOR EACH ROW
EXECUTE FUNCTION update_patient_search_vector();

-- Create GIN index for full-text search
CREATE INDEX idx_patients_search ON patients USING GIN(search_vector);

-- Full-text search function for patients

```

```

CREATE OR REPLACE FUNCTION search_patients(
    p_organization_id UUID,
    p_search_term TEXT,
    p_limit INTEGER DEFAULT 50
)
RETURNS TABLE(
    id UUID,
    mrn VARCHAR,
    full_name VARCHAR,
    primary_phone VARCHAR,
    date_of_birth DATE,
    gender gender_type,
    rank REAL
) AS $$
BEGIN
    RETURN QUERY
    SELECT
        p.id,
        p.mrn,
        p.full_name,
        p.primary_phone,
        p.date_of_birth,
        p.gender,
        ts_rank(p.search_vector, plainto_tsquery('english', p_search_term)) as
rank
    FROM patients p
    WHERE p.organization_id = p_organization_id
    AND p.is_active = TRUE
    AND (
        p.search_vector @@ plainto_tsquery('english', p_search_term)
        OR p.mrn ILIKE '%' || p_search_term || '%'
        OR p.primary_phone ILIKE '%' || p_search_term || '%'
        OR p.nid ILIKE '%' || p_search_term || '%'
        OR p.full_name ILIKE '%' || p_search_term || '%'
    )
    ORDER BY rank DESC, p.full_name
    LIMIT p_limit;
END;
$$ LANGUAGE plpgsql;

-- Add search vector to drug_master
ALTER TABLE drug_master ADD COLUMN IF NOT EXISTS search_vector tsvector;

CREATE OR REPLACE FUNCTION update_drug_search_vector()
RETURNS TRIGGER AS $$
BEGIN
    NEW.search_vector :=
        setweight(to_tsvector('english', COALESCE(NEW.generic_name, '')), 'A') ||
        setweight(to_tsvector('english', COALESCE(NEW.brand_name, '')), 'A') ||
        setweight(to_tsvector('english', COALESCE(NEW.manufacturer, '')), 'B') ||
        setweight(to_tsvector('english', COALESCE(NEW.therapeutic_class, '')),
'C');
    RETURN NEW;
END;

```



```

$$ LANGUAGE plpgsql;

CREATE TRIGGER trg_drug_search_vector
BEFORE INSERT OR UPDATE ON drug_master
FOR EACH ROW
EXECUTE FUNCTION update_drug_search_vector();

CREATE INDEX idx_drug_search ON drug_master USING GIN(search_vector);

-- =====
-- SECTION 43: API HELPER FUNCTIONS
-- =====

-- Function to get patient summary
CREATE OR REPLACE FUNCTION get_patient_summary(p_patient_id UUID)
RETURNS JSONB AS $$
DECLARE
    v_result JSONB;
BEGIN
    SELECT jsonb_build_object(
        'patient', jsonb_build_object(
            'id', p.id,
            'mrn', p.mrn,
            'full_name', p.full_name,
            'date_of_birth', p.date_of_birth,
            'age', calculate_age(p.date_of_birth),
            'gender', p.gender,
            'blood_group', p.blood_group,
            'phone', p.primary_phone,
            'email', p.email,
            'photo_url', p.photo_url,
            'is_vip', p.is_vip
        ),
        'allergies', (
            SELECT COALESCE(jsonb_agg(jsonb_build_object(
                'allergen', pa.allergen_name,
                'severity', pa.severity,
                'reaction', pa.reaction
            )), '[]'::jsonb)
            FROM patient_allergies pa
            WHERE pa.patient_id = p.id AND pa.is_active = TRUE
        ),
        'chronic_conditions', (
            SELECT COALESCE(jsonb_agg(jsonb_build_object(
                'condition', pcc.condition_name,
                'diagnosed_date', pcc.diagnosis_date
            )), '[]'::jsonb)
            FROM patient_chronic_conditions pcc
            WHERE pcc.patient_id = p.id AND pcc.is_active = TRUE
        ),
        'current_medications', (
            SELECT COALESCE(jsonb_agg(jsonb_build_object(
                'medication', pm.medication_name,
                'dose', pm.dose,

```

```

        'frequency', pm.frequency
    )), '[]'::jsonb)
    FROM patient_medications pm
    WHERE pm.patient_id = p.id AND pm.is_current = TRUE
),
'recent_visits', (
    SELECT COALESCE(jsonb_agg(jsonb_build_object(
        'date', ov.visit_date,
        'doctor', u.full_name,
        'chief_complaint', ov.chief_complaint
    ) ORDER BY ov.visit_date DESC), '[]'::jsonb)
    FROM opd_visits ov
    JOIN doctors d ON ov.doctor_id = d.id
    JOIN users u ON d.user_id = u.id
    WHERE ov.patient_id = p.id
    LIMIT 5
),
'active_admission', (
    SELECT jsonb_build_object(
        'admission_number', a.admission_number,
        'admission_date', a.admission_date,
        'ward', w.name,
        'bed', b.bed_number,
        'doctor', u.full_name,
        'diagnosis', a.provisional_diagnosis
    )
    FROM admissions a
    LEFT JOIN bed_allocations ba ON a.id = ba.admission_id AND
ba.release_datetime IS NULL
    LEFT JOIN beds b ON ba.bed_id = b.id
    LEFT JOIN wards w ON b.ward_id = w.id
    LEFT JOIN doctors d ON a.primary_doctor_id = d.id
    LEFT JOIN users u ON d.user_id = u.id
    WHERE a.patient_id = p.id AND a.status = 'admitted'
    LIMIT 1
)
) INTO v_result
FROM patients p
WHERE p.id = p_patient_id;

RETURN v_result;
END;
$$ LANGUAGE plpgsql;

-- Function to get admission details
CREATE OR REPLACE FUNCTION get_admission_details(p_admission_id UUID)
RETURNS JSONB AS $$
DECLARE
    v_result JSONB;
BEGIN
    SELECT jsonb_build_object(
        'admission', jsonb_build_object(
            'id', a.id,
            'admission_number', a.admission_number,

```

```

        'admission_date', a.admission_date,
        'admission_time', a.admission_time,
        'status', a.status,
        'admission_type', a.admission_type,
        'provisional_diagnosis', a.provisional_diagnosis
    ),
    'patient', jsonb_build_object(
        'id', p.id,
        'mrn', p.mrn,
        'full_name', p.full_name,
        'age', calculate_age(p.date_of_birth),
        'gender', p.gender,
        'blood_group', p.blood_group
    ),
    'bed', (
        SELECT jsonb_build_object(
            'ward', w.name,
            'room', r.room_number,
            'bed', b.bed_number,
            'bed_type', b.bed_type,
            'daily_rate', b.daily_rate
        )
        FROM bed_allocations ba
        JOIN beds b ON ba.bed_id = b.id
        JOIN wards w ON b.ward_id = w.id
        LEFT JOIN rooms r ON b.room_id = r.id
        WHERE ba.admission_id = a.id AND ba.release_datetime IS NULL
        LIMIT 1
    ),
    'primary_doctor', (
        SELECT jsonb_build_object(
            'id', d.id,
            'name', u.full_name,
            'specialization', d.specialization
        )
        FROM doctors d
        JOIN users u ON d.user_id = u.id
        WHERE d.id = a.primary_doctor_id
    ),
    'vitals', (
        SELECT jsonb_build_object(
            'latest', (
                SELECT jsonb_build_object(
                    'recorded_at', pv.recorded_datetime,
                    'temperature', pv.temperature,
                    'pulse_rate', pv.pulse_rate,
                    'bp_systolic', pv.blood_pressure_systolic,
                    'bp_diastolic', pv.blood_pressure_diastolic,
                    'spo2', pv.spo2,
                    'respiratory_rate', pv.respiratory_rate
                )
                FROM patient_vitals pv
                WHERE pv.admission_id = a.id
                ORDER BY pv.recorded_datetime DESC
            )
        )
    )

```

```

        LIMIT 1
    )
)
),
'pending_orders', (
    SELECT COALESCE(jsonb_agg(jsonb_build_object(
        'order_number', do.order_number,
        'order_type', do.order_type,
        'order_text', do.order_text,
        'priority', do.priority,
        'ordered_at', do.order_datetime
    )), '[]'::jsonb)
    FROM doctor_orders do
    WHERE do.admission_id = a.id AND do.status = 'pending'
),
'pending_labs', (
    SELECT COUNT(*)
    FROM lab_order_items loi
    JOIN lab_orders lo ON loi.order_id = lo.id
    WHERE lo.admission_id = a.id AND loi.status = 'pending'
),
'bill_summary', (
    SELECT jsonb_build_object(
        'total_charges', COALESCE(SUM(bi.net_amount), 0),
        'total_paid', COALESCE(SUM(p.amount), 0),
        'balance', COALESCE(SUM(bi.net_amount), 0) -
COALESCE(SUM(p.amount), 0)
    )
    FROM bills b
    LEFT JOIN bill_items bi ON b.id = bi.bill_id
    LEFT JOIN payments p ON b.id = p.bill_id AND p.status = 'completed'
    WHERE b.admission_id = a.id AND b.is_cancelled = FALSE
)
) INTO v_result
FROM admissions a
JOIN patients p ON a.patient_id = p.id
WHERE a.id = p_admission_id;

RETURN v_result;
END;
$$ LANGUAGE plpgsql;

-- Function to get ward dashboard
CREATE OR REPLACE FUNCTION get_ward_dashboard(p_ward_id UUID)
RETURNS JSONB AS $$
DECLARE
    v_result JSONB;
BEGIN
    SELECT jsonb_build_object(
        'ward_info', jsonb_build_object(
            'id', w.id,
            'name', w.name,
            'ward_type', w.ward_type,
            'total_beds', w.total_beds

```

```

    ),
    'bed_summary', jsonb_build_object(
        'total', COUNT(b.id),
        'occupied', COUNT(CASE WHEN b.status = 'occupied' THEN 1 END),
        'available', COUNT(CASE WHEN b.status = 'available' THEN 1 END),
        'reserved', COUNT(CASE WHEN b.status = 'reserved' THEN 1 END),
        'cleaning', COUNT(CASE WHEN b.status = 'cleaning' THEN 1 END),
        'maintenance', COUNT(CASE WHEN b.status = 'maintenance' THEN 1 END),
        'occupancy_rate', ROUND(
            COUNT(CASE WHEN b.status = 'occupied' THEN 1 END)::NUMERIC /
            NULLIF(COUNT(b.id), 0) * 100, 1
        )
    ),
    'patients', (
        SELECT COALESCE(jsonb_agg(jsonb_build_object(
            'bed_number', bed.bed_number,
            'patient_name', p.full_name,
            'mrn', p.mrn,
            'age', calculate_age(p.date_of_birth),
            'gender', p.gender,
            'admission_date', a.admission_date,
            'los_days', CURRENT_DATE - a.admission_date,
            'doctor', u.full_name,
            'diagnosis', a.provisional_diagnosis,
            'latest_vitals', (
                SELECT jsonb_build_object(
                    'temp', pv.temperature,
                    'pulse', pv.pulse_rate,
                    'bp', pv.blood_pressure_systolic || '/' ||
pv.blood_pressure_diastolic,
                    'spo2', pv.spo2
                )
                FROM patient_vitals pv
                WHERE pv.admission_id = a.id
                ORDER BY pv.recorded_datetime DESC
                LIMIT 1
            ),
            'pending_orders', (
                SELECT COUNT(*)
                FROM doctor_orders do
                WHERE do.admission_id = a.id AND do.status = 'pending'
            )
        ) ORDER BY bed.bed_number), '[]'::jsonb)
    FROM beds bed
    JOIN admissions a ON bed.current_admission_id = a.id
    JOIN patients p ON a.patient_id = p.id
    LEFT JOIN doctors d ON a.primary_doctor_id = d.id
    LEFT JOIN users u ON d.user_id = u.id
    WHERE bed.ward_id = w.id AND bed.status = 'occupied'
    ),
    'today_stats', jsonb_build_object(
        'admissions', (
            SELECT COUNT(*) FROM admissions
            WHERE admission_date = CURRENT_DATE

```

```

        AND branch_id = w.branch_id
    ),
    'discharges', (
        SELECT COUNT(*) FROM admissions
        WHERE actual_discharge_date = CURRENT_DATE
        AND branch_id = w.branch_id
    ),
    'critical_patients', (
        SELECT COUNT(*)
        FROM admissions a
        JOIN bed_allocations ba ON a.id = ba.admission_id
        JOIN beds b ON ba.bed_id = b.id
        WHERE b.ward_id = w.id
        AND a.status = 'admitted'
        AND EXISTS (
            SELECT 1 FROM patient_vitals pv
            WHERE pv.admission_id = a.id
            AND pv.recorded_datetime > CURRENT_TIMESTAMP - INTERVAL '24
hours'

            AND (pv.blood_pressure_systolic < 90 OR pv.spo2 < 90)
        )
    )
),
'nurses_on_duty', (
    SELECT COALESCE(jsonb_agg(jsonb_build_object(
        'name', u.full_name,
        'shift', ndr.shift,
        'is_head_nurse', n.is_head_nurse
    )), '[]'::jsonb)
    FROM nurse_duty_roster ndr
    JOIN nurses n ON ndr.nurse_id = n.id
    JOIN users u ON n.user_id = u.id
    WHERE ndr.ward_id = w.id
    AND ndr.duty_date = CURRENT_DATE
    AND ndr.status = 'scheduled'
)
) INTO v_result
FROM wards w
LEFT JOIN beds b ON w.id = b.ward_id AND b.is_active = TRUE
WHERE w.id = p_ward_id
GROUP BY w.id, w.name, w.ward_type, w.total_beds, w.branch_id;

RETURN v_result;
END;
$$ LANGUAGE plpgsql;

-- Function to get doctor schedule
CREATE OR REPLACE FUNCTION get_doctor_schedule(
    p_doctor_id UUID,
    p_date DATE DEFAULT CURRENT_DATE
)
RETURNS JSONB AS $$
DECLARE
    v_result JSONB;

```

```

BEGIN
    SELECT jsonb_build_object(
        'doctor', jsonb_build_object(
            'id', d.id,
            'name', u.full_name,
            'specialization', d.specialization,
            'opd_fee', d.opd_fee
        ),
        'date', p_date,
        'schedule', (
            SELECT COALESCE(jsonb_agg(jsonb_build_object(
                'branch', br.name,
                'session1_start', dst.session1_start,
                'session1_end', dst.session1_end,
                'session1_room', dst.session1_room,
                'session2_start', dst.session2_start,
                'session2_end', dst.session2_end,
                'session2_room', dst.session2_room
            )), '[]'::jsonb)
            FROM doctor_schedule_templates dst
            JOIN branches br ON dst.branch_id = br.id
            WHERE dst.doctor_id = d.id
            AND dst.day_of_week = EXTRACT(DOW FROM p_date)
            AND dst.is_active = TRUE
            AND NOT EXISTS (
                SELECT 1 FROM doctor_schedule_overrides dso
                WHERE dso.doctor_id = d.id
                AND dso.override_date = p_date
                AND dso.override_type = 'leave'
            )
        ),
        'appointments', (
            SELECT COALESCE(jsonb_agg(jsonb_build_object(
                'time', a.appointment_time,
                'patient_name', p.full_name,
                'patient_phone', p.primary_phone,
                'type', a.appointment_type,
                'status', a.status,
                'token', a.token_number
            ) ORDER BY a.appointment_time), '[]'::jsonb)
            FROM appointments a
            JOIN patients p ON a.patient_id = p.id
            WHERE a.doctor_id = d.id
            AND a.appointment_date = p_date
            AND a.status NOT IN ('cancelled')
        ),
        'appointment_summary', jsonb_build_object(
            'total', (
                SELECT COUNT(*) FROM appointments
                WHERE doctor_id = d.id AND appointment_date = p_date
                AND status NOT IN ('cancelled')
            ),
            'completed', (
                SELECT COUNT(*) FROM appointments

```

```

        WHERE doctor_id = d.id AND appointment_date = p_date
        AND status = 'completed'
    ),
    'waiting', (
        SELECT COUNT(*) FROM appointments
        WHERE doctor_id = d.id AND appointment_date = p_date
        AND status IN ('scheduled', 'checked_in')
    ),
    'no_show', (
        SELECT COUNT(*) FROM appointments
        WHERE doctor_id = d.id AND appointment_date = p_date
        AND status = 'no_show'
    )
),
'ipd_patients', (
    SELECT COALESCE(jsonb_agg(jsonb_build_object(
        'patient_name', p.full_name,
        'ward', w.name,
        'bed', b.bed_number,
        'diagnosis', adm.provisional_diagnosis,
        'los_days', CURRENT_DATE - adm.admission_date
    )), '[]'::jsonb)
    FROM admissions adm
    JOIN patients p ON adm.patient_id = p.id
    JOIN bed_allocations ba ON adm.id = ba.admission_id AND
ba.release_datetime IS NULL
    JOIN beds b ON ba.bed_id = b.id
    JOIN wards w ON b.ward_id = w.id
    WHERE adm.primary_doctor_id = d.id
    AND adm.status = 'admitted'
),
'surgeries', (
    SELECT COALESCE(jsonb_agg(jsonb_build_object(
        'time', s.scheduled_start_time,
        'patient_name', p.full_name,
        'surgery', s.surgery_name,
        'ot_room', otr.room_number,
        'status', s.status
    ) ORDER BY s.scheduled_start_time), '[]'::jsonb)
    FROM surgeries s
    JOIN patients p ON s.patient_id = p.id
    JOIN ot_rooms otr ON s.ot_room_id = otr.id
    WHERE s.primary_surgeon_id = d.id
    AND s.scheduled_date = p_date
    AND s.status NOT IN ('cancelled')
)
) INTO v_result
FROM doctors d
JOIN users u ON d.user_id = u.id
WHERE d.id = p_doctor_id;

RETURN v_result;
END;
$$ LANGUAGE plpgsql;

```



```

-- =====
-- SECTION 44: VALIDATION FUNCTIONS
-- =====

-- Validate NID format (Bangladesh)
CREATE OR REPLACE FUNCTION validate_nid(p_nid VARCHAR)
RETURNS BOOLEAN AS $$
BEGIN
    -- NID can be 10, 13, or 17 digits
    IF p_nid IS NULL THEN
        RETURN TRUE;
    END IF;

    IF p_nid ~ '^d{10}$' OR p_nid ~ '^d{13}$' OR p_nid ~ '^d{17}$' THEN
        RETURN TRUE;
    END IF;

    RETURN FALSE;
END;
$$ LANGUAGE plpgsql IMMUTABLE;

-- Validate phone number (Bangladesh)
CREATE OR REPLACE FUNCTION validate_phone_bd(p_phone VARCHAR)
RETURNS BOOLEAN AS $$
BEGIN
    IF p_phone IS NULL THEN
        RETURN TRUE;
    END IF;

    -- Accepts: +8801XXXXXXXX, 01XXXXXXXX, 8801XXXXXXXX
    IF p_phone ~ '^[\+?880?1[3-9]\d{8}]$' THEN
        RETURN TRUE;
    END IF;

    RETURN FALSE;
END;
$$ LANGUAGE plpgsql IMMUTABLE;

-- Validate email
CREATE OR REPLACE FUNCTION validate_email(p_email VARCHAR)
RETURNS BOOLEAN AS $$
BEGIN
    IF p_email IS NULL THEN
        RETURN TRUE;
    END IF;

    IF p_email ~ '^([A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,})$' THEN
        RETURN TRUE;
    END IF;

    RETURN FALSE;
END;
$$ LANGUAGE plpgsql IMMUTABLE;

```

```

-- Add constraints using validation functions
ALTER TABLE patients ADD CONSTRAINT chk_patient_nid CHECK (validate_nid(nid));
ALTER TABLE patients ADD CONSTRAINT chk_patient_phone CHECK
(validate_phone_bd(primary_phone));
ALTER TABLE patients ADD CONSTRAINT chk_patient_email CHECK
(validate_email(email));

-- Validate age for procedures
CREATE OR REPLACE FUNCTION check_age_restriction(
    p_patient_id UUID,
    p_procedure_code VARCHAR,
    p_min_age_years INTEGER DEFAULT NULL,
    p_max_age_years INTEGER DEFAULT NULL
)
RETURNS BOOLEAN AS $$
DECLARE
    v_age_years INTEGER;
BEGIN
    SELECT EXTRACT(YEAR FROM age(CURRENT_DATE, date_of_birth))
    INTO v_age_years
    FROM patients
    WHERE id = p_patient_id;

    IF p_min_age_years IS NOT NULL AND v_age_years < p_min_age_years THEN
        RETURN FALSE;
    END IF;

    IF p_max_age_years IS NOT NULL AND v_age_years > p_max_age_years THEN
        RETURN FALSE;
    END IF;

    RETURN TRUE;
END;
$$ LANGUAGE plpgsql;

-- =====
-- SECTION 45: BUSINESS LOGIC FUNCTIONS
-- =====

-- Function to calculate bill total
CREATE OR REPLACE FUNCTION calculate_bill_total(p_bill_id UUID)
RETURNS TABLE(
    gross_amount DECIMAL,
    discount_amount DECIMAL,
    tax_amount DECIMAL,
    net_amount DECIMAL,
    insurance_amount DECIMAL,
    patient_payable DECIMAL
) AS $$
BEGIN
    RETURN QUERY
    WITH bill_calc AS (
        SELECT

```

```

        COALESCE(SUM(bi.net_amount), 0) as gross,
        COALESCE(SUM(bi.discount_amount), 0) as discount,
        COALESCE(SUM(bi.tax_amount), 0) as tax
    FROM bill_items bi
    WHERE bi.bill_id = p_bill_id
    AND bi.is_cancelled = FALSE
),
insurance_calc AS (
    SELECT COALESCE(ip.approved_amount, 0) as ins_amount
    FROM bills b
    LEFT JOIN insurance_preauth ip ON b.preauth_id = ip.id
    WHERE b.id = p_bill_id
)
SELECT
    bc.gross,
    bc.discount,
    bc.tax,
    bc.gross - bc.discount + bc.tax as net,
    ic.ins_amount,
    GREATEST(bc.gross - bc.discount + bc.tax - ic.ins_amount, 0) as
patient_pay
    FROM bill_calc bc, insurance_calc ic;
END;
$$ LANGUAGE plpgsql;

-- Function to check drug availability
CREATE OR REPLACE FUNCTION check_drug_availability(
    p_pharmacy_id UUID,
    p_drug_id UUID,
    p_quantity INTEGER
)
RETURNS TABLE(
    is_available BOOLEAN,
    available_quantity INTEGER,
    nearest_expiry DATE,
    batches JSONB
) AS $$
BEGIN
    RETURN QUERY
    SELECT
        SUM(mb.current_quantity) >= p_quantity,
        COALESCE(SUM(mb.current_quantity)::INTEGER, 0),
        MIN(mb.expiry_date),
        jsonb_agg(jsonb_build_object(
            'batch_number', mb.batch_number,
            'quantity', mb.current_quantity,
            'expiry_date', mb.expiry_date
        ) ORDER BY mb.expiry_date)
    FROM medicine_batches mb
    WHERE mb.pharmacy_id = p_pharmacy_id
    AND mb.drug_id = p_drug_id
    AND mb.current_quantity > 0
    AND mb.expiry_date > CURRENT_DATE
    AND mb.is_active = TRUE;

```

```

END;
$$ LANGUAGE plpgsql;

-- Function to get next available appointment slot
CREATE OR REPLACE FUNCTION get_next_available_slot(
    p_doctor_id UUID,
    p_branch_id UUID,
    p_from_date DATE DEFAULT CURRENT_DATE,
    p_days_ahead INTEGER DEFAULT 7
)
RETURNS TABLE(
    slot_date DATE,
    slot_time TIME,
    is_available BOOLEAN
) AS $$
DECLARE
    v_date DATE;
    v_day_of_week INTEGER;
    v_schedule RECORD;
    v_slot_time TIME;
    v_slot_duration INTEGER;
BEGIN
    -- Get consultation duration
    SELECT consultation_duration INTO v_slot_duration
    FROM doctors WHERE id = p_doctor_id;

    v_slot_duration := COALESCE(v_slot_duration, 15);

    FOR v_date IN SELECT generate_series(p_from_date, p_from_date + p_days_ahead,
'1 day')::DATE
    LOOP
        v_day_of_week := EXTRACT(DOW FROM v_date);

        -- Check if doctor has schedule for this day
        FOR v_schedule IN
            SELECT session1_start, session1_end, session2_start, session2_end
            FROM doctor_schedule_templates
            WHERE doctor_id = p_doctor_id
            AND branch_id = p_branch_id
            AND day_of_week = v_day_of_week
            AND is_active = TRUE
            AND NOT EXISTS (
                SELECT 1 FROM doctor_schedule_overrides
                WHERE doctor_id = p_doctor_id
                AND override_date = v_date
                AND override_type = 'leave'
            )
        LOOP
            -- Session 1 slots
            IF v_schedule.session1_start IS NOT NULL THEN
                v_slot_time := v_schedule.session1_start;
                WHILE v_slot_time < v_schedule.session1_end LOOP
                    slot_date := v_date;
                    slot_time := v_slot_time;

```

```

        is_available := NOT EXISTS (
            SELECT 1 FROM appointments
            WHERE doctor_id = p_doctor_id
            AND appointment_date = v_date
            AND appointment_time = v_slot_time
            AND status NOT IN ('cancelled')
        );

        IF is_available THEN
            RETURN NEXT;
        END IF;

        v_slot_time := v_slot_time + (v_slot_duration || '
minutes')::INTERVAL;
    END LOOP;
END IF;

-- Session 2 slots
IF v_schedule.session2_start IS NOT NULL THEN
    v_slot_time := v_schedule.session2_start;
    WHILE v_slot_time < v_schedule.session2_end LOOP
        slot_date := v_date;
        slot_time := v_slot_time;
        is_available := NOT EXISTS (
            SELECT 1 FROM appointments
            WHERE doctor_id = p_doctor_id
            AND appointment_date = v_date
            AND appointment_time = v_slot_time
            AND status NOT IN ('cancelled')
        );

        IF is_available THEN
            RETURN NEXT;
        END IF;

        v_slot_time := v_slot_time + (v_slot_duration || '
minutes')::INTERVAL;
    END LOOP;
END IF;
END LOOP;
END LOOP;

RETURN;
END;
$$ LANGUAGE plpgsql;

-- Function to calculate LOS (Length of Stay)
CREATE OR REPLACE FUNCTION calculate_los(
    p_admission_date DATE,
    p_admission_time TIME,
    p_discharge_date DATE DEFAULT NULL,
    p_discharge_time TIME DEFAULT NULL
)
RETURNS TABLE(

```

```

        total_days INTEGER,
        billable_days INTEGER,
        hours INTEGER,
        minutes INTEGER
    ) AS $$
DECLARE
    v_admission TIMESTAMPTZ;
    v_discharge TIMESTAMPTZ;
    v_interval INTERVAL;
BEGIN
    v_admission := (p_admission_date || ' ' || COALESCE(p_admission_time,
'00:00:00'))::TIMESTAMPTZ;
    v_discharge := (COALESCE(p_discharge_date, CURRENT_DATE) || ' ' ||
COALESCE(p_discharge_time, CURRENT_TIME))::TIMESTAMPTZ;

    v_interval := v_discharge - v_admission;

    total_days := EXTRACT(DAY FROM v_interval);
    billable_days := GREATEST(CEIL(EXTRACT(EPOCH FROM v_interval) / 86400),
1)::INTEGER;
    hours := EXTRACT(HOUR FROM v_interval);
    minutes := EXTRACT(MINUTE FROM v_interval);

    RETURN NEXT;
END;
$$ LANGUAGE plpgsql;

-- =====
-- SECTION 46: NOTIFICATION TRIGGERS
-- =====

-- Trigger for critical lab results
CREATE OR REPLACE FUNCTION notify_critical_lab_result()
RETURNS TRIGGER AS $$
BEGIN
    IF NEW.is_critical = TRUE AND NEW.critical_notified = FALSE THEN
        -- Insert into notification queue
        INSERT INTO notification_queue (
            organization_id,
            channel,
            recipient_type,
            recipient_id,
            recipient_contact,
            subject,
            body,
            reference_type,
            reference_id,
            status
        )
        SELECT
            lo.organization_id,
            'push',
            'doctor',
            a.primary_doctor_id,

```

```

        u.email,
        'CRITICAL LAB RESULT',
        'Critical result for patient ' || p.full_name || ': ' || tc.name || '
= ' || NEW.result_value,
        'lab_result',
        NEW.id,
        'pending'
FROM lab_orders lo
JOIN patients p ON lo.patient_id = p.id
LEFT JOIN admissions a ON lo.admission_id = a.id
LEFT JOIN doctors d ON a.primary_doctor_id = d.id
LEFT JOIN users u ON d.user_id = u.id
JOIN test_catalog tc ON NEW.test_id = tc.id
WHERE lo.id = NEW.order_id;

-- Update notification flag
NEW.critical_notified := TRUE;
NEW.critical_notified_at := CURRENT_TIMESTAMP;
END IF;

RETURN NEW;
END;
$$ LANGUAGE plpgsql;

CREATE TRIGGER trg_critical_lab_notification
BEFORE INSERT OR UPDATE ON lab_results
FOR EACH ROW
WHEN (NEW.is_critical = TRUE)
EXECUTE FUNCTION notify_critical_lab_result();

-- Trigger for appointment reminders
CREATE OR REPLACE FUNCTION queue_appointment_reminder()
RETURNS TRIGGER AS $$
BEGIN
    -- Queue SMS reminder for next day appointments
    IF NEW.appointment_date = CURRENT_DATE + 1 THEN
        INSERT INTO notification_queue (
            organization_id,
            channel,
            recipient_type,
            recipient_id,
            recipient_contact,
            subject,
            body,
            reference_type,
            reference_id,
            scheduled_at
        )
        SELECT
            NEW.organization_id,
            'sms',
            'patient',
            NEW.patient_id,
            p.primary_phone,

```

```

        'Appointment Reminder',
        'Dear ' || p.first_name || ', you have an appointment tomorrow at ' ||
        TO_CHAR(NEW.appointment_time, 'HH:MI AM') || ' with Dr. ' ||
u.full_name,
        'appointment',
        NEW.id,
        CURRENT_TIMESTAMP
    FROM patients p
    JOIN doctors d ON NEW.doctor_id = d.id
    JOIN users u ON d.user_id = u.id
    WHERE p.id = NEW.patient_id
    AND p.consent_for_sms = TRUE;
END IF;

RETURN NEW;
END;
$$ LANGUAGE plpgsql;

-- =====
-- SECTION 47: REPORTING VIEWS
-- =====

-- Daily OPD Report View
CREATE OR REPLACE VIEW vw_daily_opd_report AS
SELECT
    a.organization_id,
    a.branch_id,
    a.appointment_date,
    dep.name as department,
    u.full_name as doctor_name,
    d.specialization,
    COUNT(*) as total_appointments,
    COUNT(CASE WHEN a.status = 'completed' THEN 1 END) as completed,
    COUNT(CASE WHEN a.status IN ('scheduled', 'checked_in') THEN 1 END) as
waiting,
    COUNT(CASE WHEN a.status = 'no_show' THEN 1 END) as no_shows,
    COUNT(CASE WHEN a.status = 'cancelled' THEN 1 END) as cancelled,
    COUNT(CASE WHEN a.appointment_type = 'new' THEN 1 END) as new_patients,
    COUNT(CASE WHEN a.appointment_type = 'followup' THEN 1 END) as followups,
    SUM(CASE WHEN a.status = 'completed' THEN a.consultation_fee ELSE 0 END) as
revenue
FROM appointments a
JOIN doctors d ON a.doctor_id = d.id
JOIN users u ON d.user_id = u.id
LEFT JOIN departments dep ON u.department_id = dep.id
GROUP BY a.organization_id, a.branch_id, a.appointment_date, dep.name,
u.full_name, d.specialization;

-- Daily IPD Census View
CREATE OR REPLACE VIEW vw_daily_ipd_census AS
SELECT
    w.organization_id,
    w.branch_id,
    w.id as ward_id,

```



```

w.name as ward_name,
w.ward_type,
CURRENT_DATE as census_date,
COUNT(b.id) FILTER (WHERE b.is_active) as total_beds,
COUNT(b.id) FILTER (WHERE b.status = 'occupied') as occupied,
COUNT(b.id) FILTER (WHERE b.status = 'available') as available,
ROUND(
    COUNT(b.id) FILTER (WHERE b.status = 'occupied')::NUMERIC /
    NULLIF(COUNT(b.id) FILTER (WHERE b.is_active), 0) * 100, 1
) as occupancy_percent,
(
    SELECT COUNT(*) FROM admissions
    WHERE admission_date = CURRENT_DATE
    AND branch_id = w.branch_id
) as today_admissions,
(
    SELECT COUNT(*) FROM admissions
    WHERE actual_discharge_date = CURRENT_DATE
    AND branch_id = w.branch_id
) as today_discharges
FROM wards w
LEFT JOIN beds b ON w.id = b.ward_id
WHERE w.is_active = TRUE
GROUP BY w.organization_id, w.branch_id, w.id, w.name, w.ward_type;

-- Revenue Summary View
CREATE OR REPLACE VIEW vw_revenue_summary AS
SELECT
    b.organization_id,
    b.branch_id,
    b.bill_date,
    b.bill_type,
    bi.category as service_category,
    COUNT(DISTINCT b.id) as bill_count,
    SUM(bi.net_amount) as gross_revenue,
    SUM(bi.discount_amount) as discount,
    SUM(CASE WHEN b.is_insurance THEN bi.net_amount ELSE 0 END) as
insurance_revenue,
    SUM(CASE WHEN b.is_corporate THEN bi.net_amount ELSE 0 END) as
corporate_revenue,
    SUM(CASE WHEN NOT b.is_insurance AND NOT b.is_corporate THEN bi.net_amount
ELSE 0 END) as cash_revenue
FROM bills b
JOIN bill_items bi ON b.id = bi.bill_id
WHERE b.is_cancelled = FALSE
GROUP BY b.organization_id, b.branch_id, b.bill_date, b.bill_type, bi.category;

-- Pending Tasks Dashboard View
CREATE OR REPLACE VIEW vw_pending_tasks AS
SELECT
    'pending_lab_orders' as task_type,
    lo.organization_id,
    lo.branch_id,
    COUNT(*) as count,

```

```
    MIN(lo.order_datetime) as oldest_since
FROM lab_orders lo
WHERE lo.status IN ('ordered', 'collected', 'received')
GROUP BY lo.organization_id, lo.branch_id

UNION ALL

SELECT
    'pending_radiology' as task_type,
    ro.organization_id,
    ro.branch_id,
    COUNT(*) as count,
    MIN(ro.order_datetime) as oldest_since
FROM radiology_orders ro
WHERE ro.status IN ('ordered', 'scheduled')
GROUP BY ro.organization_id, ro.branch_id

UNION ALL

SELECT
    'pending_doctor_orders' as task_type,
    do.organization_id,
    do.branch_id,
    COUNT(*) as count,
    MIN(do.order_datetime) as oldest_since
FROM doctor_orders do
WHERE do.status = 'pending'
GROUP BY do.organization_id, do.branch_id

UNION ALL

SELECT
    'beds_under_cleaning' as task_type,
    bct.organization_id,
    bct.branch_id,
    COUNT(*) as count,
    MIN(bct.created_at) as oldest_since
FROM bed_cleaning_tasks bct
WHERE bct.status IN ('pending', 'in_progress')
GROUP BY bct.organization_id, bct.branch_id

UNION ALL

SELECT
    'pending_discharges' as task_type,
    a.organization_id,
    a.branch_id,
    COUNT(*) as count,
    MIN(a.expected_discharge_date) as oldest_since
FROM admissions a
WHERE a.status = 'admitted'
AND a.expected_discharge_date <= CURRENT_DATE
GROUP BY a.organization_id, a.branch_id;
```

```

-- =====
-- SECTION 48: SAMPLE DATA GENERATORS
-- =====

-- Function to generate sample patients
CREATE OR REPLACE FUNCTION generate_sample_patients(
    p_organization_id UUID,
    p_branch_id UUID,
    p_count INTEGER DEFAULT 100
)
RETURNS INTEGER AS $$
DECLARE
    v_first_names TEXT[] := ARRAY['Abdul', 'Mohammad', 'Fatima', 'Ayesha',
    'Rahman', 'Karim', 'Nasreen', 'Sultana', 'Hassan', 'Hossain', 'Rahim', 'Khadija',
    'Ahmed', 'Begum', 'Ali'];
    v_last_names TEXT[] := ARRAY['Islam', 'Ahmed', 'Rahman', 'Hossain', 'Khan',
    'Chowdhury', 'Akter', 'Khatun', 'Miah', 'Sheikh', 'Siddiqui', 'Uddin', 'Bhuiyan',
    'Sarkar', 'Das'];
    v_genders gender_type[] := ARRAY['male', 'female']::gender_type[];
    v_blood_groups blood_group_type[] := ARRAY['A+', 'A-', 'B+', 'B-', 'AB+', 'AB-',
    'O+', 'O-']::blood_group_type[];
    v_counter INTEGER := 0;
    v_mrn_prefix VARCHAR;
BEGIN
    -- Get MRN prefix
    SELECT COALESCE(code, 'MRN') INTO v_mrn_prefix FROM branches WHERE id =
p_branch_id;

    FOR i IN 1..p_count LOOP
        INSERT INTO patients (
            organization_id,
            mrn,
            first_name,
            last_name,
            gender,
            date_of_birth,
            blood_group,
            primary_phone,
            present_address_line1,
            present_city,
            present_district,
            registered_branch_id
        ) VALUES (
            p_organization_id,
            v_mrn_prefix || '-' || TO_CHAR(CURRENT_DATE, 'YYYYMM') || '-' ||
LPAD((100000 + i)::TEXT, 6, '0'),
            v_first_names[1 + floor(random() * array_length(v_first_names, 1))],
            v_last_names[1 + floor(random() * array_length(v_last_names, 1))],
            v_genders[1 + floor(random() * 2)],
            CURRENT_DATE - (floor(random() * 25000 + 365)::INTEGER),
            v_blood_groups[1 + floor(random() * array_length(v_blood_groups, 1))],
            '01' || (3 + floor(random() * 7))::TEXT || (10000000 + floor(random()
* 90000000))::TEXT,
            'House ' || (1 + floor(random() * 500))::TEXT || ', Road ' || (1 +

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```

floor(random() * 50))::TEXT,
    'Dhaka',
    'Dhaka',
    p_branch_id
);

    v_counter := v_counter + 1;
END LOOP;

RETURN v_counter;
END;
$$ LANGUAGE plpgsql;

-- =====
-- SECTION 49: PERFORMANCE INDEXES
-- =====

-- Composite indexes for common query patterns
CREATE INDEX CONCURRENTLY IF NOT EXISTS idx_appointments_lookup
ON appointments(organization_id, doctor_id, appointment_date, status);

CREATE INDEX CONCURRENTLY IF NOT EXISTS idx_admissions_active
ON admissions(organization_id, branch_id, status)
WHERE status = 'admitted';

CREATE INDEX CONCURRENTLY IF NOT EXISTS idx_bills_unpaid
ON bills(organization_id, patient_id, payment_status)
WHERE payment_status IN ('pending', 'partial');

CREATE INDEX CONCURRENTLY IF NOT EXISTS idx_lab_orders_pending
ON lab_orders(organization_id, branch_id, status, order_datetime)
WHERE status IN ('ordered', 'collected', 'received', 'processing');

CREATE INDEX CONCURRENTLY IF NOT EXISTS idx_medicine_low_stock
ON medicine_inventory(pharmacy_id, drug_id)
WHERE quantity_available <= reorder_level;

CREATE INDEX CONCURRENTLY IF NOT EXISTS idx_maintenance_open
ON maintenance_tickets(organization_id, status, priority)
WHERE status IN ('open', 'assigned', 'in_progress');

-- Partial indexes for active records
CREATE INDEX CONCURRENTLY IF NOT EXISTS idx_patients_active
ON patients(organization_id, mrn)
WHERE is_active = TRUE;

CREATE INDEX CONCURRENTLY IF NOT EXISTS idx_users_active
ON users(organization_id, username)
WHERE is_active = TRUE;

CREATE INDEX CONCURRENTLY IF NOT EXISTS idx_beds_available
ON beds(ward_id, status)
WHERE status = 'available' AND is_active = TRUE;

```

```

-- =====
-- SECTION 50: FINAL CLEANUP & OPTIMIZATION
-- =====

-- Analyze all tables for query planner
ANALYZE;

-- Create statistics for frequently joined columns
CREATE STATISTICS IF NOT EXISTS stat_patient_org ON organization_id, mrn FROM
patients;
CREATE STATISTICS IF NOT EXISTS stat_admission_patient ON organization_id,
patient_id, status FROM admissions;
CREATE STATISTICS IF NOT EXISTS stat_appointment_doctor ON organization_id,
doctor_id, appointment_date FROM appointments;

-- Function to vacuum and analyze specific schema
CREATE OR REPLACE FUNCTION maintenance_vacuum_analyze()
RETURNS VOID AS $$
DECLARE
    r RECORD;
BEGIN
    FOR r IN SELECT tablename FROM pg_tables WHERE schemaname = 'public'
    LOOP
        EXECUTE 'VACUUM ANALYZE ' || quote_ident(r.tablename);
    END LOOP;
END;
$$ LANGUAGE plpgsql;

-- Schedule maintenance (requires pg_cron)
-- SELECT cron.schedule('nightly-vacuum', '0 2 * * *', 'SELECT
maintenance_vacuum_analyze()');

-- =====
-- SCHEMA COMPLETION SUMMARY
-- =====

DO $$
DECLARE
    v_tables INTEGER;
    v_views INTEGER;
    v_indexes INTEGER;
    v_functions INTEGER;
    v_triggers INTEGER;
    v_types INTEGER;
BEGIN
    SELECT COUNT(*) INTO v_tables
    FROM information_schema.tables
    WHERE table_schema = 'public' AND table_type = 'BASE TABLE';

    SELECT COUNT(*) INTO v_views
    FROM information_schema.views
    WHERE table_schema = 'public';

    SELECT COUNT(*) INTO v_indexes

```

```

FROM pg_indexes
WHERE schemaname = 'public';

SELECT COUNT(*) INTO v_functions
FROM information_schema.routines
WHERE routine_schema = 'public';

SELECT COUNT(*) INTO v_triggers
FROM information_schema.triggers
WHERE trigger_schema = 'public';

SELECT COUNT(*) INTO v_types
FROM pg_type t
JOIN pg_namespace n ON t.typnamespace = n.oid
WHERE n.nspname = 'public' AND t.typtype = 'e';

RAISE NOTICE '';
RAISE NOTICE
'||
RAISE NOTICE '||      HOSPITAL ERP POSTGRESQL SCHEMA - COMPLETE
'||;
RAISE NOTICE
'||
RAISE NOTICE '|| Tables Created:      %
'||, LPAD(v_tables::TEXT, 4);
RAISE NOTICE '|| Views Created:      %
'||, LPAD(v_views::TEXT, 4);
RAISE NOTICE '|| Indexes Created:      %
'||, LPAD(v_indexes::TEXT, 4);
RAISE NOTICE '|| Functions Created:      %
'||, LPAD(v_functions::TEXT, 4);
RAISE NOTICE '|| Triggers Created:      %
'||, LPAD(v_triggers::TEXT, 4);
RAISE NOTICE '|| Custom Types Created: %
'||, LPAD(v_types::TEXT, 4);
RAISE NOTICE
'||
RAISE NOTICE '|| MODULES COVERED:
'||;
RAISE NOTICE '||  ✓ Multi-Tenancy & Access Control
'||;
RAISE NOTICE '||  ✓ Patient Management
'||;
RAISE NOTICE '||  ✓ OPD/Appointments
'||;
RAISE NOTICE '||  ✓ IPD/Admissions
'||;
RAISE NOTICE '||  ✓ EMR/Clinical Documentation
'||;
RAISE NOTICE '||  ✓ Nurse Station
'||;
RAISE NOTICE '||  ✓ Bed Management
'||;
RAISE NOTICE '||  ✓ OT Management

```

```
||';
  RAISE NOTICE '||  ✓ Diagnostics (Lab & Radiology)
||';
  RAISE NOTICE '||  ✓ Pharmacy
||';
  RAISE NOTICE '||  ✓ Inventory & Supply Chain
||';
  RAISE NOTICE '||  ✓ Insurance & Claims
||';
  RAISE NOTICE '||  ✓ Billing & Payments
||';
  RAISE NOTICE '||  ✓ HR & Payroll
||';
  RAISE NOTICE '||  ✓ Housekeeping
||';
  RAISE NOTICE '||  ✓ Maintenance
||';
  RAISE NOTICE '||  ✓ Diet & Kitchen
||';
  RAISE NOTICE '||  ✓ Telemedicine
||';
  RAISE NOTICE '||  ✓ Complaints & Feedback
||';
  RAISE NOTICE '||  ✓ Consent Management
||';
  RAISE NOTICE '||  ✓ Ambulance
||';
  RAISE NOTICE '||  ✓ Blood Bank
||';
  RAISE NOTICE '||  ✓ Visitor Management
||';
  RAISE NOTICE '||  ✓ Waste Management
||';
  RAISE NOTICE '||  ✓ Reports & Dashboards
||';
  RAISE NOTICE '||  ✓ Notifications
||';
  RAISE NOTICE '||  ✓ Audit Logging
||';
  RAISE NOTICE
'||=====||';
  RAISE NOTICE '';
END;
$;$;
```

 COMPLETE MODULE CHECKLIST

#	Module	Tables	Status
1	Organizations & Branches	3	☒
2	Users & Access Control	8	☒

#	Module	Tables	Status
3	Patient Management	12	☒
4	Doctor Management	4	☒
5	Appointments	2	☒
6	OPD Visits	1	☒
7	Emergency Visits	1	☒
8	IPD Admissions	4	☒
9	Bed Management	8	☒
10	EMR/Vitals	3	☒
11	Clinical Notes	2	☒
12	Diagnoses & Procedures	2	☒
13	Prescriptions	5	☒
14	Nurse Station	12	☒
15	Doctor Orders	2	☒
16	OT Management	8	☒
17	Lab Management	8	☒
18	Radiology	4	☒
19	Pharmacy	12	☒
20	Inventory	18	☒
21	Insurance	10	☒
22	Billing	6	☒
23	Housekeeping	8	☒
24	HR & Payroll	15	☒
25	Maintenance	8	☒
26	Diet Management	5	☒
27	Telemedicine	1	☒
28	Complaints	5	☒
29	Consent Forms	3	☒
30	Ambulance	2	☒
31	Blood Bank	3	☒
32	Visitor Management	1	☒

#	Module	Tables	Status
33	Waste Management	2	☒
34	Canteen/Kitchen	3	☒
35	Reports & Dashboards	4	☒
36	Notifications	3	☒
37	Audit Logs	2	☒
38	Master Data	8	☒

DEPLOYMENT CHECKLIST

SQL

```
-- Pre-deployment checks
SELECT 'Extensions' as check_type,
      CASE WHEN COUNT(*) >= 4 THEN '☑ OK' ELSE '✗ Missing' END as status
FROM pg_extension
WHERE extname IN ('uuid-ossf', 'pgcrypto', 'pg_trgm', 'btree_gist');

-- Post-deployment verification
SELECT
  (SELECT COUNT(*) FROM information_schema.tables WHERE table_schema = 'public'
   AND table_type = 'BASE TABLE') as tables,
  (SELECT COUNT(*) FROM pg_indexes WHERE schemaname = 'public') as indexes,
  (SELECT COUNT(*) FROM information_schema.routines WHERE routine_schema =
   'public') as functions;
```

This completes the **FULL Hospital ERP PostgreSQL Schema** with:

- **190+ Tables**
- **300+ Indexes**
- **50+ Functions**
- **30+ Triggers**
- **20+ Custom Types**
- **10+ Views**
- **5+ Materialized Views**
- **Row Level Security**
- **Full-Text Search**
- **Table Partitioning**
- **Audit Logging**
- **API Helper Functions**
- **Validation Functions**
- **Business Logic Functions**
- **Sample Data Generators**

Complete Hospital ERP PostgreSQL Schema (Extended Components)

📁 PART 20: HEALTHCARE INTEGRATIONS & STANDARDS

SQL

```
-- =====
-- SECTION 51: HL7/FHIR INTEGRATION TABLES
-- =====

-- HL7 Message Log
CREATE TABLE hl7_message_log (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),

    message_type VARCHAR(10) NOT NULL, -- ADT, ORM, ORU, RDE, MDM, etc.
    message_event VARCHAR(10) NOT NULL, -- A01, A02, A03, O01, R01, etc.
    message_control_id VARCHAR(50) NOT NULL,

    direction VARCHAR(10) NOT NULL, -- inbound, outbound

    -- Endpoints
    sending_application VARCHAR(100),
    sending_facility VARCHAR(100),
    receiving_application VARCHAR(100),
    receiving_facility VARCHAR(100),

    -- Content
    raw_message TEXT NOT NULL,
    parsed_message JSONB,

    -- Processing
    processing_status VARCHAR(20) DEFAULT 'pending', -- pending, processed,
    failed, acknowledged
    processing_error TEXT,
    processed_at TIMESTAMPTZ,

    -- Acknowledgment
    ack_code VARCHAR(10), -- AA, AE, AR
    ack_message TEXT,
    ack_sent_at TIMESTAMPTZ,

    -- Reference
    reference_type VARCHAR(50), -- patient, admission, lab_order, etc.
    reference_id UUID,

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

    UNIQUE(organization_id, message_control_id)
```

```

);

-- FHIR Resource Cache
CREATE TABLE fhir_resources (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),

  resource_type VARCHAR(50) NOT NULL, -- Patient, Encounter, Observation, etc.
  resource_id VARCHAR(100) NOT NULL,
  version_id INTEGER DEFAULT 1,

  -- Content
  resource_json JSONB NOT NULL,

  -- Reference to internal record
  internal_table VARCHAR(100),
  internal_id UUID,

  -- Metadata
  last_updated TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

  is_deleted BOOLEAN DEFAULT FALSE,

  UNIQUE(organization_id, resource_type, resource_id)
);

-- External System Mappings
CREATE TABLE external_system_mappings (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),

  external_system VARCHAR(100) NOT NULL, -- lab_lis, pacs, pharmacy_system,
insurance_portal

  internal_table VARCHAR(100) NOT NULL,
  internal_id UUID NOT NULL,
  internal_code VARCHAR(100),

  external_id VARCHAR(255) NOT NULL,
  external_code VARCHAR(100),

  mapping_type VARCHAR(50), -- patient, test, drug, doctor, etc.

  is_active BOOLEAN DEFAULT TRUE,
  last_synced_at TIMESTAMPTZ,

  created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

  UNIQUE(organization_id, external_system, internal_table, internal_id)
);

-- Integration Endpoints Configuration
CREATE TABLE integration_endpoints (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),

```

```

organization_id UUID NOT NULL REFERENCES organizations(id),
branch_id UUID REFERENCES branches(id),

endpoint_name VARCHAR(100) NOT NULL,
endpoint_type VARCHAR(50) NOT NULL, -- hl7_mllp, hl7_http, fhir, rest_api,
soap, sftp

-- Connection
host VARCHAR(255) NOT NULL,
port INTEGER,
path VARCHAR(500),

-- Authentication
auth_type VARCHAR(50), -- none, basic, bearer, oauth2, certificate
auth_credentials JSONB, -- Encrypted credentials

-- SSL/TLS
use_ssl BOOLEAN DEFAULT TRUE,
ssl_certificate TEXT,

-- Message Configuration
message_types TEXT[], -- Which message types to send/receive

-- Retry Configuration
retry_count INTEGER DEFAULT 3,
retry_interval_seconds INTEGER DEFAULT 60,

-- Status
is_active BOOLEAN DEFAULT TRUE,
last_connection_at TIMESTAMPTZ,
last_connection_status VARCHAR(50),

created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

UNIQUE(organization_id, endpoint_name)
);

-- =====
-- SECTION 52: CLINICAL DECISION SUPPORT
-- =====

-- Clinical Rules Engine
CREATE TABLE clinical_rules (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID REFERENCES organizations(id), -- NULL for global rules

  rule_code VARCHAR(50) NOT NULL,
  rule_name VARCHAR(255) NOT NULL,
  description TEXT,

  rule_category VARCHAR(50), -- drug_interaction, allergy, dosing, lab_alert,
  clinical_pathway

```

```

-- Rule Definition
trigger_event VARCHAR(100), -- on_prescription, on_lab_result, on_admission,
etc.

condition_type VARCHAR(50), -- simple, complex, ml_model
condition_expression TEXT, -- SQL or JSON expression
condition_parameters JSONB,

-- Action
action_type VARCHAR(50), -- alert, block, suggest, auto_order, notify
action_severity VARCHAR(20), -- info, warning, critical
action_message TEXT,
action_message_bangla TEXT,
action_parameters JSONB,

-- Override
can_override BOOLEAN DEFAULT TRUE,
override_requires_reason BOOLEAN DEFAULT TRUE,
override_requires_approval BOOLEAN DEFAULT FALSE,

-- Validity
effective_from DATE,
effective_to DATE,

is_active BOOLEAN DEFAULT TRUE,
created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

UNIQUE(COALESCE(organization_id, '00000000-0000-0000-0000-0000-000000000000'::UUID), rule_code)
);

-- Clinical Rule Execution Log
CREATE TABLE clinical_rule_executions (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),

  rule_id UUID NOT NULL REFERENCES clinical_rules(id),

  -- Context
  patient_id UUID REFERENCES patients(id),
  user_id UUID REFERENCES users(id),

  trigger_event VARCHAR(100),
  trigger_data JSONB,

  -- Execution
  executed_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

  rule_matched BOOLEAN NOT NULL,
  action_taken VARCHAR(50),

  -- Override
  was_overridden BOOLEAN DEFAULT FALSE,

```

```

        override_reason TEXT,
        overridden_by UUID REFERENCES users(id),

        created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
    );

-- Drug Formulary
CREATE TABLE drug_formulary (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),

    drug_id UUID NOT NULL REFERENCES drug_master(id),

    formulary_status VARCHAR(20) NOT NULL, -- preferred, allowed, restricted,
non_formulary

    -- Restrictions
    restriction_type VARCHAR(50), -- none, prior_auth, specialist_only,
specific_indication
    restriction_criteria TEXT,

    -- Therapeutic Alternatives
    therapeutic_alternatives UUID[], -- Array of alternative drug IDs

    -- Cost
    acquisition_cost DECIMAL(10, 2),
    is_high_cost BOOLEAN DEFAULT FALSE,

    -- Monitoring
    requires_monitoring BOOLEAN DEFAULT FALSE,
    monitoring_parameters TEXT,

    effective_from DATE,
    effective_to DATE,

    approved_by UUID REFERENCES users(id),
    approved_at TIMESTAMPTZ,

    is_active BOOLEAN DEFAULT TRUE,
    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

    UNIQUE(organization_id, drug_id)
);

-- Clinical Protocols/Pathways
CREATE TABLE clinical_pathways (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),

    pathway_code VARCHAR(50) NOT NULL,
    name VARCHAR(255) NOT NULL,
    description TEXT,

    -- Applicability

```

```

diagnosis_codes VARCHAR(20)[], -- ICD codes
department_id UUID REFERENCES departments(id),

-- Pathway Definition
pathway_steps JSONB NOT NULL,
/* Structure:
[
  {
    "day": 1,
    "step": "Admission Assessment",
    "orders": [
      {"type": "lab", "tests": ["CBC", "RFT", "LFT"]},
      {"type": "medication", "drug": "...", "dose": "..."}
    ],
    "assessments": ["nursing_assessment", "nutrition_assessment"],
    "expected_outcomes": ["..."]
  }
]
*/

expected_los_days INTEGER,

-- Version Control
version VARCHAR(20),
previous_version_id UUID REFERENCES clinical_pathways(id),

-- Approval
status VARCHAR(20) DEFAULT 'draft', -- draft, review, approved, retired
approved_by UUID REFERENCES users(id),
approved_at TIMESTAMPTZ,

is_active BOOLEAN DEFAULT TRUE,
created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

UNIQUE(organization_id, pathway_code, version)
);

-- Patient Pathway Enrollment
CREATE TABLE patient_pathway_enrollments (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),

  pathway_id UUID NOT NULL REFERENCES clinical_pathways(id),
  patient_id UUID NOT NULL REFERENCES patients(id),
  admission_id UUID REFERENCES admissions(id),

  enrolled_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
  enrolled_by UUID REFERENCES users(id),

  current_day INTEGER DEFAULT 1,
  current_step INTEGER DEFAULT 1,

  -- Progress

```

```

progress_data JSONB, -- Tracks completion of each step

-- Variance
variances JSONB, -- Records deviations from pathway

-- Status
status VARCHAR(20) DEFAULT 'active', -- active, completed, discontinued,
deviated

completed_at TIMESTAMPTZ,
discontinued_at TIMESTAMPTZ,
discontinuation_reason TEXT,

created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- =====
-- SECTION 53: QUALITY INDICATORS & METRICS
-- =====

-- Quality Indicators Definition
CREATE TABLE quality_indicators (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID REFERENCES organizations(id), -- NULL for standard
indicators

    indicator_code VARCHAR(50) NOT NULL,
    name VARCHAR(255) NOT NULL,
    description TEXT,

    category VARCHAR(50), -- patient_safety, clinical_outcome, process,
patient_experience
    sub_category VARCHAR(100),

-- Measurement
    numerator_definition TEXT,
    denominator_definition TEXT,
    exclusion_criteria TEXT,

    data_source VARCHAR(100),
    calculation_query TEXT,

-- Target
    target_value DECIMAL(10, 4),
    target_direction VARCHAR(20), -- higher_better, lower_better, target_range
    target_min DECIMAL(10, 4),
    target_max DECIMAL(10, 4),

-- Unit
    unit VARCHAR(50), -- percentage, rate_per_1000, count, days

-- Frequency
    measurement_frequency VARCHAR(20), -- daily, weekly, monthly, quarterly

```



```

-- Standards
nabh_indicator BOOLEAN DEFAULT FALSE,
jci_indicator BOOLEAN DEFAULT FALSE,
who_indicator BOOLEAN DEFAULT FALSE,

is_active BOOLEAN DEFAULT TRUE,
created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

UNIQUE(COALESCE(organization_id, '00000000-0000-0000-0000-
000000000000'::UUID), indicator_code)
);

-- Quality Indicator Results
CREATE TABLE quality_indicator_results (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),
  branch_id UUID REFERENCES branches(id),

  indicator_id UUID NOT NULL REFERENCES quality_indicators(id),

  measurement_period_start DATE NOT NULL,
  measurement_period_end DATE NOT NULL,

  numerator_value DECIMAL(12, 2),
  denominator_value DECIMAL(12, 2),
  result_value DECIMAL(10, 4) NOT NULL,

  target_value DECIMAL(10, 4),

-- Status
meets_target BOOLEAN,
variance_from_target DECIMAL(10, 4),
trend VARCHAR(20), -- improving, stable, declining

-- Analysis
root_cause_analysis TEXT,
action_plan TEXT,

calculated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
calculated_by UUID REFERENCES users(id),

-- Review
reviewed_by UUID REFERENCES users(id),
reviewed_at TIMESTAMPTZ,
review_comments TEXT,

created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

UNIQUE(organization_id, branch_id, indicator_id, measurement_period_start)
);

-- Infection Control Surveillance
CREATE TABLE infection_surveillance (

```

```
id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
organization_id UUID NOT NULL REFERENCES organizations(id),
branch_id UUID NOT NULL REFERENCES branches(id),

surveillance_number VARCHAR(50) NOT NULL,

patient_id UUID NOT NULL REFERENCES patients(id),
admission_id UUID REFERENCES admissions(id),

-- Infection Details
infection_type VARCHAR(100) NOT NULL,
-- HAI, SSI, CAUTI, CLABSI, VAP, C_DIFF, MRSA, etc.

infection_site VARCHAR(100),

onset_date DATE NOT NULL,
onset_type VARCHAR(20), -- community, hospital_acquired

-- If hospital acquired
days_post_admission INTEGER,

-- Organism
organism_name VARCHAR(255),
is_mdros BOOLEAN DEFAULT FALSE, -- Multi-drug resistant organism

-- Source/Risk Factors
risk_factors JSONB,
-- {"central_line": true, "urinary_catheter": true, "ventilator": false}

device_related BOOLEAN DEFAULT FALSE,
device_type VARCHAR(100),
device_days INTEGER,

-- Surgery Related
surgery_id UUID REFERENCES surgeries(id),
surgery_class VARCHAR(50), -- clean, clean_contaminated, contaminated, dirty

-- Treatment
treatment_started DATE,
treatment_antibiotics JSONB,

-- Outcome
outcome VARCHAR(50), -- resolved, ongoing, death_related, death_unrelated
outcome_date DATE,

-- Investigation
investigated_by UUID REFERENCES users(id),
investigation_date DATE,
investigation_findings TEXT,

-- Prevention Analysis
preventable BOOLEAN,
prevention_measures TEXT,
```

```

-- Reporting
reported_to_authority BOOLEAN DEFAULT FALSE,
reporting_date DATE,

created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

UNIQUE(organization_id, surveillance_number)
);

-- Incident Reporting (Patient Safety)
CREATE TABLE incident_reports (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),
  branch_id UUID NOT NULL REFERENCES branches(id),

  incident_number VARCHAR(50) NOT NULL,

  -- When/Where
  incident_datetime TIMESTAMPTZ NOT NULL,
  location VARCHAR(255),
  department_id UUID REFERENCES departments(id),
  ward_id UUID REFERENCES wards(id),

  -- Who
  patient_id UUID REFERENCES patients(id),
  visitor_involved BOOLEAN DEFAULT FALSE,
  staff_involved BOOLEAN DEFAULT FALSE,
  involved_staff_ids UUID[],

  -- What
  incident_type VARCHAR(100) NOT NULL,
  -- medication_error, fall, pressure_injury, wrong_site, near_miss,
  -- equipment_failure, security, violence, other

  incident_category VARCHAR(50), -- clinical, non_clinical

  description TEXT NOT NULL,

  -- Severity
  severity_actual VARCHAR(20), -- none, mild, moderate, severe, death
  severity_potential VARCHAR(20),

  -- Harm
  harm_occurred BOOLEAN DEFAULT FALSE,
  harm_description TEXT,

  -- Immediate Actions
  immediate_actions TEXT,

  -- Reporter
  reported_by UUID NOT NULL REFERENCES users(id),
  reported_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
  reporter_role VARCHAR(100),

```

```

is_anonymous BOOLEAN DEFAULT FALSE,

-- Investigation
investigation_required BOOLEAN DEFAULT TRUE,
investigation_status VARCHAR(20) DEFAULT 'pending', -- pending, in_progress,
completed

assigned_investigator_id UUID REFERENCES users(id),
assigned_at TIMESTAMPTZ,

root_cause_analysis TEXT,
contributing_factors JSONB,

-- Actions
corrective_actions TEXT,
preventive_actions TEXT,
actions_due_date DATE,
actions_completed BOOLEAN DEFAULT FALSE,
actions_completed_date DATE,

-- Review
reviewed_by UUID REFERENCES users(id),
reviewed_at TIMESTAMPTZ,
review_comments TEXT,

-- Closure
status VARCHAR(20) DEFAULT 'open', -- open, investigating, action_pending,
closed
closed_at TIMESTAMPTZ,
closed_by UUID REFERENCES users(id),

-- Disclosure
patient_notified BOOLEAN DEFAULT FALSE,
notification_date DATE,

-- External Reporting
reportable_externally BOOLEAN DEFAULT FALSE,
external_report_date DATE,
external_report_reference VARCHAR(100),

attachments TEXT[],

created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

UNIQUE(organization_id, incident_number)
);

-- Mortality Review
CREATE TABLE mortality_reviews (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),
  branch_id UUID NOT NULL REFERENCES branches(id),

```

```

death_record_id UUID NOT NULL REFERENCES patient_death_records(id),
patient_id UUID NOT NULL REFERENCES patients(id),
admission_id UUID REFERENCES admissions(id),

-- Review Details
review_date DATE NOT NULL,
review_committee TEXT,

-- Classification
death_expected BOOLEAN,
death_category VARCHAR(50), -- natural, accidental, suicide, homicide,
undetermined

-- Care Analysis
care_appropriate BOOLEAN,
care_issues_identified BOOLEAN DEFAULT FALSE,
care_issues_description TEXT,

-- Preventability
potentially_preventable BOOLEAN DEFAULT FALSE,
preventability_score INTEGER, -- 1-6 scale

-- Contributing Factors
contributing_factors JSONB,
system_issues TEXT,

-- Recommendations
recommendations TEXT,
action_items JSONB,

-- Quality Metrics
early_warning_score_documented BOOLEAN,
escalation_appropriate BOOLEAN,
palliative_care_involved BOOLEAN,

-- Reviewers
primary_reviewer_id UUID REFERENCES users(id),
reviewed_by_committee BOOLEAN DEFAULT FALSE,
committee_date DATE,

-- Status
status VARCHAR(20) DEFAULT 'pending', -- pending, reviewed, closed

notes TEXT,

created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- =====
-- SECTION 54: DOCUMENT MANAGEMENT SYSTEM
-- =====

-- Document Categories

```

```

CREATE TABLE document_categories (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID REFERENCES organizations(id),

  category_code VARCHAR(50) NOT NULL,
  name VARCHAR(100) NOT NULL,

  parent_category_id UUID REFERENCES document_categories(id),

  -- Retention
  retention_period_years INTEGER,

  -- Access
  default_access_level VARCHAR(20), -- public, internal, confidential,
  restricted

  is_active BOOLEAN DEFAULT TRUE,

  UNIQUE(COALESCE(organization_id, '00000000-0000-0000-0000-0000-000000000000'::UUID), category_code)
);

-- Documents
CREATE TABLE documents (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),
  branch_id UUID REFERENCES branches(id),

  document_number VARCHAR(50) NOT NULL,

  category_id UUID REFERENCES document_categories(id),

  title VARCHAR(500) NOT NULL,
  description TEXT,

  -- File
  file_name VARCHAR(255) NOT NULL,
  file_path TEXT NOT NULL,
  file_size BIGINT,
  mime_type VARCHAR(100),
  file_hash VARCHAR(64), -- SHA-256 for integrity

  -- Version Control
  version INTEGER DEFAULT 1,
  parent_document_id UUID REFERENCES documents(id),
  is_latest_version BOOLEAN DEFAULT TRUE,

  -- Classification
  access_level VARCHAR(20) DEFAULT 'internal',

  -- Tags
  tags TEXT[],

  -- Dates

```

```

    document_date DATE,
    expiry_date DATE,

    -- Review
    requires_review BOOLEAN DEFAULT FALSE,
    next_review_date DATE,

    -- Status
    status VARCHAR(20) DEFAULT 'active', -- draft, active, archived, deleted

    -- Ownership
    owner_id UUID REFERENCES users(id),
    department_id UUID REFERENCES departments(id),

    -- Audit
    created_by UUID REFERENCES users(id),
    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
    updated_by UUID REFERENCES users(id),
    updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

    UNIQUE(organization_id, document_number)
);

-- Document Access Log
CREATE TABLE document_access_log (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    document_id UUID NOT NULL REFERENCES documents(id),

    accessed_by UUID NOT NULL REFERENCES users(id),
    accessed_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

    access_type VARCHAR(50), -- view, download, print, edit

    ip_address INET,
    user_agent TEXT
);

-- Document Workflows
CREATE TABLE document_workflows (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),

    workflow_code VARCHAR(50) NOT NULL,
    name VARCHAR(100) NOT NULL,

    -- Steps
    workflow_steps JSONB NOT NULL,
    /*
    [
        {"step": 1, "name": "Draft", "role": "creator", "action": "create"},
        {"step": 2, "name": "Review", "role": "reviewer", "action":
"approve_reject"},
        {"step": 3, "name": "Approve", "role": "approver", "action":
"approve_reject"},

```

```

        {"step": 4, "name": "Publish", "role": "publisher", "action": "publish"}
    ]
    */

    applicable_categories UUID[],

    is_active BOOLEAN DEFAULT TRUE,

    UNIQUE(organization_id, workflow_code)
);

-- Document Workflow Instances
CREATE TABLE document_workflow_instances (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    document_id UUID NOT NULL REFERENCES documents(id),
    workflow_id UUID NOT NULL REFERENCES document_workflows(id),

    current_step INTEGER DEFAULT 1,

    status VARCHAR(20) DEFAULT 'in_progress', -- in_progress, completed, rejected,
cancelled

    -- Step History
    step_history JSONB,

    started_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
    started_by UUID REFERENCES users(id),

    completed_at TIMESTAMPTZ,

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- =====
-- SECTION 55: ADVANCED REPORTING
-- =====

-- Report Schedules
CREATE TABLE report_schedules (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),

    schedule_name VARCHAR(255) NOT NULL,

    report_definition_id UUID REFERENCES report_definitions(id),

    -- Custom Query (if not using report_definition)
    custom_query TEXT,

    -- Parameters
    parameters JSONB,

    -- Schedule (cron expression)
    schedule_cron VARCHAR(100) NOT NULL, -- "0 8 * * *" for daily at 8 AM

```



```

    timezone VARCHAR(50) DEFAULT 'Asia/Dhaka',

    -- Output
    output_format VARCHAR(20) DEFAULT 'pdf', -- pdf, excel, csv

    -- Delivery
    delivery_method VARCHAR(50), -- email, sftp, storage

    email_recipients TEXT[],
    email_subject VARCHAR(500),
    email_body TEXT,

    sftp_endpoint_id UUID REFERENCES integration_endpoints(id),
    sftp_path VARCHAR(500),

    storage_path VARCHAR(500),

    -- Status
    is_active BOOLEAN DEFAULT TRUE,

    last_run_at TIMESTAMPTZ,
    last_run_status VARCHAR(20),
    last_run_error TEXT,

    next_run_at TIMESTAMPTZ,

    created_by UUID REFERENCES users(id),
    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Report Execution History
CREATE TABLE report_execution_history (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),

    report_definition_id UUID REFERENCES report_definitions(id),
    schedule_id UUID REFERENCES report_schedules(id),

    execution_type VARCHAR(20), -- scheduled, manual, api

    -- Parameters Used
    parameters JSONB,

    -- Execution
    started_at TIMESTAMPTZ NOT NULL,
    completed_at TIMESTAMPTZ,
    duration_seconds INTEGER,

    status VARCHAR(20), -- running, completed, failed
    error_message TEXT,

    -- Output
    output_format VARCHAR(20),

```

```

    output_file_path TEXT,
    output_file_size BIGINT,
    row_count INTEGER,

    -- Delivery
    delivery_status VARCHAR(20),
    delivered_at TIMESTAMPTZ,
    delivery_error TEXT,

    executed_by UUID REFERENCES users(id),

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Dashboard Configurations
CREATE TABLE dashboard_configurations (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),

    dashboard_code VARCHAR(50) NOT NULL,
    name VARCHAR(255) NOT NULL,
    description TEXT,

    -- Layout
    layout_config JSONB NOT NULL,
    /*
    {
        "columns": 12,
        "rows": 8,
        "widgets": [
            {
                "widget_id": "...",
                "position": {"x": 0, "y": 0, "w": 4, "h": 2},
                "config": {...}
            }
        ]
    }
    */

    -- Filters
    global_filters JSONB,

    -- Refresh
    auto_refresh BOOLEAN DEFAULT TRUE,
    refresh_interval_seconds INTEGER DEFAULT 300,

    -- Access
    is_public BOOLEAN DEFAULT FALSE,
    allowed_roles UUID[],

    is_default BOOLEAN DEFAULT FALSE,

    is_active BOOLEAN DEFAULT TRUE,

```

```

        created_by UUID REFERENCES users(id),
        created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
        updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

        UNIQUE(organization_id, dashboard_code)
    );

-- KPI Definitions
CREATE TABLE kpi_definitions (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID REFERENCES organizations(id),

    kpi_code VARCHAR(50) NOT NULL,
    name VARCHAR(255) NOT NULL,
    description TEXT,

    category VARCHAR(50), -- financial, operational, clinical, hr

    -- Calculation
    calculation_type VARCHAR(50), -- count, sum, average, percentage, ratio,
formula
    calculation_query TEXT,
    calculation_formula TEXT,

    -- Data Source
    source_tables TEXT[],

    -- Unit
    unit VARCHAR(50),
    decimal_places INTEGER DEFAULT 2,

    -- Target
    target_type VARCHAR(20), -- fixed, dynamic, benchmark
    target_value DECIMAL(15, 4),
    target_query TEXT, -- For dynamic targets

    -- Thresholds
    threshold_good DECIMAL(15, 4),
    threshold_warning DECIMAL(15, 4),
    threshold_critical DECIMAL(15, 4),

    -- Trend
    trend_direction VARCHAR(20), -- higher_better, lower_better

    -- Frequency
    calculation_frequency VARCHAR(20), -- realtime, hourly, daily, weekly, monthly

    is_active BOOLEAN DEFAULT TRUE,

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

    UNIQUE(COALESCE(organization_id, '00000000-0000-0000-0000-0000-000000000000'::UUID), kpi_code)
);

```

```

-- KPI Values Cache
CREATE TABLE kpi_values (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),
  branch_id UUID REFERENCES branches(id),

  kpi_id UUID NOT NULL REFERENCES kpi_definitions(id),

  period_type VARCHAR(20) NOT NULL, -- hour, day, week, month, quarter, year
  period_start TIMESTAMPTZ NOT NULL,
  period_end TIMESTAMPTZ NOT NULL,

  value DECIMAL(15, 4) NOT NULL,
  target_value DECIMAL(15, 4),

  previous_value DECIMAL(15, 4),
  change_percent DECIMAL(8, 2),

  status VARCHAR(20), -- good, warning, critical

  calculated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

  UNIQUE(organization_id, branch_id, kpi_id, period_type, period_start)
);

-- =====
-- SECTION 56: WORKFLOW ENGINE
-- =====

-- Workflow Definitions
CREATE TABLE workflow_definitions (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),

  workflow_code VARCHAR(50) NOT NULL,
  name VARCHAR(255) NOT NULL,
  description TEXT,

  -- Trigger
  trigger_type VARCHAR(50), -- manual, event, schedule
  trigger_event VARCHAR(100), -- on_admission, on_discharge, on_lab_result, etc.
  trigger_condition JSONB,

  -- States
  states JSONB NOT NULL,
  /*
  [
    {"state": "submitted", "type": "initial"},
    {"state": "under_review", "type": "intermediate"},
    {"state": "approved", "type": "final"},
    {"state": "rejected", "type": "final"}
  ]
  */

```

```

-- Transitions
transitions JSONB NOT NULL,
/*
[
  {
    "from": "submitted",
    "to": "under_review",
    "action": "review",
    "allowed_roles": ["reviewer"],
    "conditions": [],
    "actions": [{"type": "notify", "to": "approver"}]
  }
]
*/

-- Timeout/Escalation
sla_config JSONB,

version INTEGER DEFAULT 1,

is_active BOOLEAN DEFAULT TRUE,

created_by UUID REFERENCES users(id),
created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

UNIQUE(organization_id, workflow_code, version)
);

-- Workflow Instances
CREATE TABLE workflow_instances (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),

  workflow_id UUID NOT NULL REFERENCES workflow_definitions(id),

  instance_number VARCHAR(50) NOT NULL,

  -- Reference
  reference_type VARCHAR(100) NOT NULL, -- leave_application,
  purchase_requisition, etc.
  reference_id UUID NOT NULL,

  -- Current State
  current_state VARCHAR(100) NOT NULL,

  -- Data
  workflow_data JSONB,

  -- Status
  status VARCHAR(20) DEFAULT 'active', -- active, completed, cancelled,
  suspended

```

```

-- Timing
started_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
started_by UUID REFERENCES users(id),

completed_at TIMESTAMPTZ,

-- SLA
sla_due_at TIMESTAMPTZ,
sla_breached BOOLEAN DEFAULT FALSE,

created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

UNIQUE(organization_id, instance_number)
);

-- Workflow Tasks
CREATE TABLE workflow_tasks (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  workflow_instance_id UUID NOT NULL REFERENCES workflow_instances(id),

  task_number VARCHAR(50) NOT NULL,

  state VARCHAR(100) NOT NULL,
  action_required VARCHAR(100),

  -- Assignment
  assigned_to UUID REFERENCES users(id),
  assigned_role UUID REFERENCES roles(id),
  assigned_department_id UUID REFERENCES departments(id),

  -- Status
  status VARCHAR(20) DEFAULT 'pending', -- pending, in_progress, completed,
skipped

  -- Action Taken
  action_taken VARCHAR(100),
  action_by UUID REFERENCES users(id),
  action_at TIMESTAMPTZ,
  action_comments TEXT,

  -- Timing
  created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
  due_at TIMESTAMPTZ,

  completed_at TIMESTAMPTZ
);

-- Workflow History
CREATE TABLE workflow_history (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  workflow_instance_id UUID NOT NULL REFERENCES workflow_instances(id),

  from_state VARCHAR(100),

```

```

    to_state VARCHAR(100) NOT NULL,

    action VARCHAR(100),

    performed_by UUID REFERENCES users(id),
    performed_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

    comments TEXT,

    task_id UUID REFERENCES workflow_tasks(id),

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- =====
-- SECTION 57: LOCALIZATION & CONFIGURATION
-- =====

-- System Configuration
CREATE TABLE system_configuration (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID REFERENCES organizations(id), -- NULL for global config
    branch_id UUID REFERENCES branches(id), -- NULL for org-level config

    config_key VARCHAR(255) NOT NULL,
    config_value TEXT,
    config_value_json JSONB,

    data_type VARCHAR(50) DEFAULT 'string', -- string, integer, decimal, boolean,
json

    category VARCHAR(100),
    description TEXT,

    is_encrypted BOOLEAN DEFAULT FALSE,
    is_readonly BOOLEAN DEFAULT FALSE,

    default_value TEXT,

    updated_by UUID REFERENCES users(id),
    updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

    UNIQUE(COALESCE(organization_id, '00000000-0000-0000-0000-0000-000000000000'::UUID),
        COALESCE(branch_id, '00000000-0000-0000-0000-0000-000000000000'::UUID),
        config_key)
);

-- Translations
CREATE TABLE translations (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),

    language_code VARCHAR(10) NOT NULL, -- en, bn, ar, etc.

```

```

translation_key VARCHAR(500) NOT NULL,
translation_value TEXT NOT NULL,

context VARCHAR(100), -- ui, report, notification, clinical
module VARCHAR(100),

is_verified BOOLEAN DEFAULT FALSE,
verified_by UUID REFERENCES users(id),

created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

UNIQUE(language_code, translation_key)
);

-- Number Sequences
CREATE TABLE number_sequences (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),
  branch_id UUID REFERENCES branches(id),

  sequence_name VARCHAR(100) NOT NULL,

  prefix VARCHAR(20),
  suffix VARCHAR(20),

  -- Format
  number_format VARCHAR(100), -- {PREFIX}-{YYYY}{MM}-{#####}

  current_number BIGINT DEFAULT 0,

  -- Reset
  reset_frequency VARCHAR(20), -- never, daily, monthly, yearly
  last_reset_at TIMESTAMPTZ,

  -- Range
  min_value BIGINT DEFAULT 1,
  max_value BIGINT,

  padding_length INTEGER DEFAULT 5,

  is_active BOOLEAN DEFAULT TRUE,

  updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

  UNIQUE(organization_id, COALESCE(branch_id, '00000000-0000-0000-0000-000000000000'::UUID), sequence_name)
);

-- Function to get next sequence number
CREATE OR REPLACE FUNCTION get_next_sequence_number(
  p_organization_id UUID,
  p_branch_id UUID,
  p_sequence_name VARCHAR

```



```

)
RETURNS VARCHAR AS $$
DECLARE
    v_seq RECORD;
    v_next_number BIGINT;
    v_result VARCHAR;
BEGIN
    -- Lock and get sequence
    SELECT * INTO v_seq
    FROM number_sequences
    WHERE organization_id = p_organization_id
    AND COALESCE(branch_id, '00000000-0000-0000-0000-000000000000'::UUID) =
        COALESCE(p_branch_id, '00000000-0000-0000-0000-000000000000'::UUID)
    AND sequence_name = p_sequence_name
    AND is_active = TRUE
    FOR UPDATE;

    IF v_seq IS NULL THEN
        RAISE EXCEPTION 'Sequence not found: %', p_sequence_name;
    END IF;

    -- Check if reset needed
    IF v_seq.reset_frequency = 'daily' AND v_seq.last_reset_at::DATE <
CURRENT_DATE THEN
        v_next_number := v_seq.min_value;
    ELSIF v_seq.reset_frequency = 'monthly' AND DATE_TRUNC('month',
v_seq.last_reset_at) < DATE_TRUNC('month', CURRENT_TIMESTAMP) THEN
        v_next_number := v_seq.min_value;
    ELSIF v_seq.reset_frequency = 'yearly' AND DATE_TRUNC('year',
v_seq.last_reset_at) < DATE_TRUNC('year', CURRENT_TIMESTAMP) THEN
        v_next_number := v_seq.min_value;
    ELSE
        v_next_number := v_seq.current_number + 1;
    END IF;

    -- Check max value
    IF v_seq.max_value IS NOT NULL AND v_next_number > v_seq.max_value THEN
        RAISE EXCEPTION 'Sequence % has reached maximum value', p_sequence_name;
    END IF;

    -- Update sequence
    UPDATE number_sequences
    SET current_number = v_next_number,
        last_reset_at = CASE
            WHEN v_next_number = v_seq.min_value THEN CURRENT_TIMESTAMP
            ELSE last_reset_at
        END,
        updated_at = CURRENT_TIMESTAMP
    WHERE id = v_seq.id;

    -- Format result
    v_result := v_seq.number_format;
    v_result := REPLACE(v_result, '{PREFIX}', COALESCE(v_seq.prefix, ''));
    v_result := REPLACE(v_result, '{SUFFIX}', COALESCE(v_seq.suffix, ''));

```

```

    v_result := REPLACE(v_result, '{YYYY}', TO_CHAR(CURRENT_DATE, 'YYYY'));
    v_result := REPLACE(v_result, '{YY}', TO_CHAR(CURRENT_DATE, 'YY'));
    v_result := REPLACE(v_result, '{MM}', TO_CHAR(CURRENT_DATE, 'MM'));
    v_result := REPLACE(v_result, '{DD}', TO_CHAR(CURRENT_DATE, 'DD'));
    v_result := REPLACE(v_result, '{#####}', LPAD(v_next_number::TEXT,
v_seq.padding_length, '0'));
    v_result := REPLACE(v_result, '{#####}', LPAD(v_next_number::TEXT, 4, '0'));
    v_result := REPLACE(v_result, '{###}', LPAD(v_next_number::TEXT, 3, '0'));

    RETURN v_result;
END;
$$ LANGUAGE plpgsql;

-- =====
-- SECTION 58: ADVANCED SECURITY
-- =====

-- API Keys
CREATE TABLE api_keys (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),

    key_name VARCHAR(100) NOT NULL,
    api_key_hash VARCHAR(255) NOT NULL, -- Store hashed key
    api_key_prefix VARCHAR(10) NOT NULL, -- First 8 chars for identification

    -- Permissions
    permissions JSONB, -- Specific API permissions
    allowed_ips INET[],

    -- Rate Limiting
    rate_limit_per_minute INTEGER DEFAULT 60,
    rate_limit_per_day INTEGER DEFAULT 10000,

    -- Validity
    expires_at TIMESTAMPTZ,

    -- Status
    is_active BOOLEAN DEFAULT TRUE,
    last_used_at TIMESTAMPTZ,

    created_by UUID REFERENCES users(id),
    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

    UNIQUE(organization_id, key_name)
);

-- API Request Log
CREATE TABLE api_request_log (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),

    api_key_id UUID REFERENCES api_keys(id),
    user_id UUID REFERENCES users(id),

```

```

    request_timestamp TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

    method VARCHAR(10),
    endpoint VARCHAR(500),

    request_headers JSONB,
    request_body JSONB,

    response_status INTEGER,
    response_time_ms INTEGER,

    ip_address INET,
    user_agent TEXT,

    error_message TEXT
) PARTITION BY RANGE (request_timestamp);

-- Create partitions for API log
CREATE TABLE api_request_log_default PARTITION OF api_request_log DEFAULT;

-- Security Events
CREATE TABLE security_events (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID REFERENCES organizations(id),

    event_type VARCHAR(100) NOT NULL,
    -- login_success, login_failed, password_changed, role_changed,
    -- permission_denied, suspicious_activity, brute_force_detected

    severity VARCHAR(20), -- info, warning, critical

    user_id UUID REFERENCES users(id),
    username VARCHAR(100),

    ip_address INET,
    user_agent TEXT,
    location_info JSONB,

    details JSONB,

    -- Response
    blocked BOOLEAN DEFAULT FALSE,
    block_duration_minutes INTEGER,

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Password History
CREATE TABLE password_history (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    user_id UUID NOT NULL REFERENCES users(id),

    password_hash VARCHAR(255) NOT NULL,

```

```

        created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
    );

-- Function to check password history
CREATE OR REPLACE FUNCTION check_password_history(
    p_user_id UUID,
    p_new_password_hash VARCHAR,
    p_history_count INTEGER DEFAULT 5
)
RETURNS BOOLEAN AS $$
DECLARE
    v_exists BOOLEAN;
BEGIN
    SELECT EXISTS (
        SELECT 1 FROM password_history
        WHERE user_id = p_user_id
        AND password_hash = p_new_password_hash
        ORDER BY created_at DESC
        LIMIT p_history_count
    ) INTO v_exists;

    RETURN NOT v_exists;
END;
$$ LANGUAGE plpgsql;

-- Data Masking Rules
CREATE TABLE data_masking_rules (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID REFERENCES organizations(id),

    table_name VARCHAR(100) NOT NULL,
    column_name VARCHAR(100) NOT NULL,

    masking_type VARCHAR(50), -- full, partial, hash, custom

    -- Partial masking config
    visible_start INTEGER,
    visible_end INTEGER,
    mask_character CHAR(1) DEFAULT '*',

    -- Custom masking
    custom_function VARCHAR(100),

    -- Access
    exempt_roles UUID[],

    is_active BOOLEAN DEFAULT TRUE,

    UNIQUE(COALESCE(organization_id, '00000000-0000-0000-0000-0000-000000000000'::UUID), table_name, column_name)
);

-- =====
-- SECTION 59: INDEXES FOR NEW TABLES

```

```
-- =====

-- HL7/FHIR
CREATE INDEX idx_hl7_messages_org ON hl7_message_log(organization_id);
CREATE INDEX idx_hl7_messages_type ON hl7_message_log(message_type,
message_event);
CREATE INDEX idx_hl7_messages_status ON hl7_message_log(processing_status);
CREATE INDEX idx_hl7_messages_created ON hl7_message_log(created_at DESC);

CREATE INDEX idx_fhir_resources_type ON fhir_resources(resource_type,
resource_id);
CREATE INDEX idx_fhir_resources_internal ON fhir_resources(internal_table,
internal_id);

-- Clinical Decision Support
CREATE INDEX idx_clinical_rules_category ON clinical_rules(rule_category);
CREATE INDEX idx_clinical_rules_trigger ON clinical_rules(trigger_event);
CREATE INDEX idx_rule_executions_patient ON clinical_rule_executions(patient_id);
CREATE INDEX idx_rule_executions_rule ON clinical_rule_executions(rule_id);

-- Quality
CREATE INDEX idx_quality_results_indicator ON
quality_indicator_results(indicator_id);
CREATE INDEX idx_quality_results_period ON
quality_indicator_results(measurement_period_start);
CREATE INDEX idx_infection_surveillance_patient ON
infection_surveillance(patient_id);
CREATE INDEX idx_incident_reports_status ON incident_reports(status);
CREATE INDEX idx_incident_reports_type ON incident_reports(incident_type);

-- Documents
CREATE INDEX idx_documents_category ON documents(category_id);
CREATE INDEX idx_documents_owner ON documents(owner_id);
CREATE INDEX idx_documents_status ON documents(status);
CREATE INDEX idx_document_access_document ON document_access_log(document_id);

-- Workflow
CREATE INDEX idx_workflow_instances_status ON workflow_instances(status);
CREATE INDEX idx_workflow_instances_ref ON workflow_instances(reference_type,
reference_id);
CREATE INDEX idx_workflow_tasks_assigned ON workflow_tasks(assigned_to, status);
CREATE INDEX idx_workflow_tasks_instance ON workflow_tasks(workflow_instance_id);

-- KPI
CREATE INDEX idx_kpi_values_kpi ON kpi_values(kpi_id);
CREATE INDEX idx_kpi_values_period ON kpi_values(period_start, period_end);

-- Security
CREATE INDEX idx_security_events_type ON security_events(event_type);
CREATE INDEX idx_security_events_user ON security_events(user_id);
CREATE INDEX idx_security_events_ip ON security_events(ip_address);
CREATE INDEX idx_api_request_log_key ON api_request_log(api_key_id);
CREATE INDEX idx_api_request_log_endpoint ON api_request_log(endpoint);
```

```

-- =====
-- SECTION 60: FINAL STATISTICS & COMPLETION
-- =====

-- Comprehensive schema statistics
CREATE OR REPLACE FUNCTION get_schema_statistics()
RETURNS TABLE(
    category VARCHAR,
    count INTEGER
) AS $$
BEGIN
    RETURN QUERY
    SELECT 'Tables'::VARCHAR, COUNT(*)::INTEGER
    FROM information_schema.tables
    WHERE table_schema = 'public' AND table_type = 'BASE TABLE'

    UNION ALL

    SELECT 'Views'::VARCHAR, COUNT(*)::INTEGER
    FROM information_schema.views
    WHERE table_schema = 'public'

    UNION ALL

    SELECT 'Materialized Views'::VARCHAR, COUNT(*)::INTEGER
    FROM pg_matviews
    WHERE schemaname = 'public'

    UNION ALL

    SELECT 'Indexes'::VARCHAR, COUNT(*)::INTEGER
    FROM pg_indexes
    WHERE schemaname = 'public'

    UNION ALL

    SELECT 'Functions'::VARCHAR, COUNT(*)::INTEGER
    FROM information_schema.routines
    WHERE routine_schema = 'public' AND routine_type = 'FUNCTION'

    UNION ALL

    SELECT 'Procedures'::VARCHAR, COUNT(*)::INTEGER
    FROM information_schema.routines
    WHERE routine_schema = 'public' AND routine_type = 'PROCEDURE'

    UNION ALL

    SELECT 'Triggers'::VARCHAR, COUNT(*)::INTEGER
    FROM information_schema.triggers
    WHERE trigger_schema = 'public'

    UNION ALL

```

```
SELECT 'Custom Types (Enums) '::VARCHAR, COUNT(*)::INTEGER
FROM pg_type t
JOIN pg_namespace n ON t.typnamespace = n.oid
WHERE n.nspname = 'public' AND t.typtype = 'e'

UNION ALL

SELECT 'Foreign Keys '::VARCHAR, COUNT(*)::INTEGER
FROM information_schema.table_constraints
WHERE constraint_schema = 'public' AND constraint_type = 'FOREIGN KEY'

UNION ALL

SELECT 'Check Constraints '::VARCHAR, COUNT(*)::INTEGER
FROM information_schema.table_constraints
WHERE constraint_schema = 'public' AND constraint_type = 'CHECK';
END;
$$ LANGUAGE plpgsql;

-- Display final statistics
SELECT * FROM get_schema_statistics();
```

FINAL SCHEMA SUMMARY

Complete Module Coverage

Module Category	Modules Included	Table Count
Foundation	Multi-tenancy, Users, Roles, Permissions	15
Patient Management	Registration, History, Allergies, Documents	15
Clinical	OPD, IPD, Emergency, EMR, Vitals	25
Nursing	Station, Orders, MAR, Assessments	15
Diagnostics	Lab, Radiology, Microbiology, Pathology	20
Pharmacy	Inventory, Dispensing, Narcotics, Cold Chain	15
Surgery	OT, Scheduling, Anesthesia, Recovery	12
Billing	Services, Bills, Payments, Insurance	18
HR	Employees, Attendance, Payroll, Leave	18
Operations	Housekeeping, Diet, Maintenance, Ambulance	20
Quality	Indicators, Incidents, Infection Control	10
Integrations	HL7, FHIR, External Systems	8
Workflow	Definitions, Instances, Tasks	6

Module Category	Modules Included	Table Count
Reporting	Definitions, Schedules, Dashboards, KPIs	10
Security	API Keys, Audit, Password History	8
Configuration	Settings, Sequences, Translations	5

Total Statistics

text

HOSPITAL ERP POSTGRESQL SCHEMA - FINAL SUMMARY	
Total Tables:	220+
Total Views:	15+
Materialized Views:	6
Total Indexes:	400+
Functions:	60+
Stored Procedures:	10+
Triggers:	40+
Custom Enum Types:	25
Foreign Key Constraints:	300+
FEATURES IMPLEMENTED:	
✓ Multi-Tenancy with Row Level Security	
✓ Complete Audit Logging	
✓ Full-Text Search (English + Bangla)	
✓ Table Partitioning for Large Tables	
✓ HL7/FHIR Integration Support	
✓ Clinical Decision Support Engine	
✓ Quality Indicators & Metrics	
✓ Workflow Engine	
✓ Document Management System	
✓ Advanced Reporting & KPI	
✓ API Security & Rate Limiting	
✓ Localization Support	
✓ Number Sequence Generator	
✓ Data Masking Rules	

 DEPLOYMENT SCRIPT

SQL

-- Master deployment script
-- Run this to deploy the complete schema


```

-- Step 1: Create extensions
CREATE EXTENSION IF NOT EXISTS "uuid-ossf";
CREATE EXTENSION IF NOT EXISTS "pgcrypto";
CREATE EXTENSION IF NOT EXISTS "pg_trgm";
CREATE EXTENSION IF NOT EXISTS "btree_gist";

-- Step 2: Run all schema parts in order
-- \i part_01_foundation.sql
-- \i part_02_patient.sql
-- \i part_03_visits.sql
-- ... etc.

-- Step 3: Verify deployment
SELECT * FROM get_schema_statistics();

-- Step 4: Create initial admin user
-- (Run from application or separate secure script)

-- Step 5: Initialize sequences
INSERT INTO number_sequences (organization_id, sequence_name, prefix,
number_format, padding_length)
SELECT
    o.id,
    seq.name,
    seq.prefix,
    seq.format,
    seq.padding
FROM organizations o
CROSS JOIN (VALUES
    ('MRN', 'MRN', '{PREFIX}-{YYYY}{MM}-{#####}', 5),
    ('ADMISSION', 'ADM', '{PREFIX}-{YYYY}{MM}-{#####}', 4),
    ('BILL', 'BIL', '{PREFIX}-{YYYY}{MM}{DD}-{#####}', 4),
    ('PRESCRIPTION', 'RX', '{PREFIX}-{YYYY}{MM}{DD}-{#####}', 4),
    ('LAB_ORDER', 'LAB', '{PREFIX}-{YYYY}{MM}{DD}-{#####}', 4)
) AS seq(name, prefix, format, padding);

-- Step 6: Run analyze
ANALYZE;

RAISE NOTICE 'Hospital ERP Schema deployed successfully!';

```

Complete Hospital ERP PostgreSQL Schema (Enterprise Extensions)

PART 21: MOBILE & PORTAL SUPPORT

SQL

```

-- =====
-- SECTION 61: MOBILE APP SUPPORT
-- =====

-- Mobile Device Registration
CREATE TABLE mobile_devices (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    user_id UUID NOT NULL REFERENCES users(id),
    organization_id UUID NOT NULL REFERENCES organizations(id),

    device_id VARCHAR(255) NOT NULL, -- Unique device identifier
    device_token VARCHAR(500), -- FCM/APNs token

    device_type VARCHAR(20) NOT NULL, -- ios, android, web
    device_model VARCHAR(100),
    os_version VARCHAR(50),
    app_version VARCHAR(20),

    -- Push Notifications
    push_enabled BOOLEAN DEFAULT TRUE,
    push_token_updated_at TIMESTAMPTZ,

    -- Status
    is_active BOOLEAN DEFAULT TRUE,
    last_active_at TIMESTAMPTZ,

    -- Security
    is_trusted BOOLEAN DEFAULT FALSE,
    trusted_at TIMESTAMPTZ,
    trusted_by UUID REFERENCES users(id),

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

    UNIQUE(user_id, device_id)
);

-- Mobile App Versions
CREATE TABLE mobile_app_versions (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),

    platform VARCHAR(20) NOT NULL, -- ios, android
    version_code INTEGER NOT NULL,
    version_name VARCHAR(20) NOT NULL,

    release_date DATE,

    -- Requirements
    min_os_version VARCHAR(20),

    -- Update Policy
    is_mandatory BOOLEAN DEFAULT FALSE,
    force_update_message TEXT,

```

```

-- Features
release_notes TEXT,
new_features JSONB,

-- URLs
download_url TEXT,

is_active BOOLEAN DEFAULT TRUE,

created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

UNIQUE(platform, version_code)
);

-- Mobile Sync Queue
CREATE TABLE mobile_sync_queue (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  device_id UUID NOT NULL REFERENCES mobile_devices(id),
  user_id UUID NOT NULL REFERENCES users(id),

  sync_type VARCHAR(50) NOT NULL, -- full, incremental, selective

  -- Data to sync
  entity_type VARCHAR(100) NOT NULL, -- patients, appointments, prescriptions
  entity_id UUID,
  operation VARCHAR(20), -- create, update, delete

  payload JSONB,

  -- Status
  status VARCHAR(20) DEFAULT 'pending', -- pending, synced, failed, conflict

  -- Timestamps
  queued_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
  synced_at TIMESTAMPTZ,

  retry_count INTEGER DEFAULT 0,
  last_error TEXT,

  -- Conflict Resolution
  has_conflict BOOLEAN DEFAULT FALSE,
  conflict_data JSONB,
  resolved_at TIMESTAMPTZ,
  resolved_by UUID REFERENCES users(id)
);

-- Offline Data Cache
CREATE TABLE offline_data_cache (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  device_id UUID NOT NULL REFERENCES mobile_devices(id),
  user_id UUID NOT NULL REFERENCES users(id),

  cache_key VARCHAR(255) NOT NULL,

```

```

    entity_type VARCHAR(100) NOT NULL,

    data JSONB NOT NULL,
    data_hash VARCHAR(64), -- For change detection

    cached_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
    expires_at TIMESTAMPTZ,

    last_accessed_at TIMESTAMPTZ,

    UNIQUE(device_id, cache_key)
);

-- =====
-- SECTION 62: PATIENT PORTAL
-- =====

-- Patient Portal Users
CREATE TABLE patient_portal_users (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    patient_id UUID NOT NULL REFERENCES patients(id),
    organization_id UUID NOT NULL REFERENCES organizations(id),

    username VARCHAR(100) NOT NULL,
    email VARCHAR(255),
    phone VARCHAR(20),

    password_hash VARCHAR(255) NOT NULL,

    -- Verification
    email_verified BOOLEAN DEFAULT FALSE,
    email_verified_at TIMESTAMPTZ,
    phone_verified BOOLEAN DEFAULT FALSE,
    phone_verified_at TIMESTAMPTZ,

    -- Security
    two_factor_enabled BOOLEAN DEFAULT FALSE,
    two_factor_method VARCHAR(20), -- sms, email, authenticator
    two_factor_secret VARCHAR(255),

    -- Status
    is_active BOOLEAN DEFAULT TRUE,
    is_locked BOOLEAN DEFAULT FALSE,
    locked_until TIMESTAMPTZ,
    failed_login_attempts INTEGER DEFAULT 0,

    -- Activity
    last_login_at TIMESTAMPTZ,
    last_login_ip INET,

    -- Terms
    terms_accepted BOOLEAN DEFAULT FALSE,
    terms_accepted_at TIMESTAMPTZ,
    terms_version VARCHAR(20),

```

```

        created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
        updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

        UNIQUE(organization_id, username)
    );

-- Portal Sessions
CREATE TABLE patient_portal_sessions (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    portal_user_id UUID NOT NULL REFERENCES patient_portal_users(id),

    session_token VARCHAR(500) NOT NULL UNIQUE,
    refresh_token VARCHAR(500),

    device_info JSONB,
    ip_address INET,
    user_agent TEXT,

    is_active BOOLEAN DEFAULT TRUE,

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
    expires_at TIMESTAMPTZ NOT NULL,
    last_activity_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Patient Portal Preferences
CREATE TABLE patient_portal_preferences (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    portal_user_id UUID NOT NULL REFERENCES patient_portal_users(id),

    -- Notification Preferences
    notify_appointment_reminder BOOLEAN DEFAULT TRUE,
    notify_lab_results BOOLEAN DEFAULT TRUE,
    notify_prescription_ready BOOLEAN DEFAULT TRUE,
    notify_bill_generated BOOLEAN DEFAULT TRUE,
    notify_promotional BOOLEAN DEFAULT FALSE,

    -- Communication Channels
    preferred_channel VARCHAR(20) DEFAULT 'email', -- email, sms, push

    -- Language
    preferred_language VARCHAR(10) DEFAULT 'en',

    -- Display
    date_format VARCHAR(20) DEFAULT 'DD/MM/YYYY',
    time_format VARCHAR(10) DEFAULT '12h',

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

    UNIQUE(portal_user_id)
);

```

```

-- Patient Messages (Secure Messaging)
CREATE TABLE patient_messages (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),

  thread_id UUID, -- For conversation threading
  parent_message_id UUID REFERENCES patient_messages(id),

  -- Participants
  patient_id UUID NOT NULL REFERENCES patients(id),

  sender_type VARCHAR(20) NOT NULL, -- patient, staff
  sender_patient_id UUID REFERENCES patients(id),
  sender_user_id UUID REFERENCES users(id),

  recipient_type VARCHAR(20) NOT NULL,
  recipient_patient_id UUID REFERENCES patients(id),
  recipient_user_id UUID REFERENCES users(id),
  recipient_department_id UUID REFERENCES departments(id),

  -- Message
  subject VARCHAR(500),
  body TEXT NOT NULL,

  -- Attachments
  attachments JSONB, -- [{file_name, file_url, file_size, mime_type}]

  -- Category
  category VARCHAR(50), -- general, appointment, prescription, billing,
lab_result
  priority VARCHAR(20) DEFAULT 'normal',

  -- Status
  is_read BOOLEAN DEFAULT FALSE,
  read_at TIMESTAMPTZ,

  is_archived BOOLEAN DEFAULT FALSE,
  is_deleted BOOLEAN DEFAULT FALSE,

  -- Response
  requires_response BOOLEAN DEFAULT FALSE,
  response_due_by TIMESTAMPTZ,
  responded_at TIMESTAMPTZ,

  created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Online Appointment Requests
CREATE TABLE online_appointment_requests (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),
  branch_id UUID REFERENCES branches(id),

  request_number VARCHAR(50) NOT NULL,

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```

-- Patient
patient_id UUID REFERENCES patients(id),
portal_user_id UUID REFERENCES patient_portal_users(id),

-- For new patients
patient_name VARCHAR(255),
patient_phone VARCHAR(20),
patient_email VARCHAR(255),
date_of_birth DATE,
gender gender_type,

-- Request Details
preferred_doctor_id UUID REFERENCES doctors(id),
department_id UUID REFERENCES departments(id),
specialization VARCHAR(100),

preferred_date DATE,
preferred_time_slot VARCHAR(20), -- morning, afternoon, evening
alternate_date DATE,

consultation_type VARCHAR(50), -- opd, telemedicine

-- Reason
reason_for_visit TEXT,
is_followup BOOLEAN DEFAULT FALSE,
previous_visit_date DATE,

-- Insurance
has_insurance BOOLEAN DEFAULT FALSE,
insurance_company VARCHAR(255),
insurance_policy_number VARCHAR(100),

-- Status
status VARCHAR(20) DEFAULT 'pending', -- pending, confirmed, rejected,
cancelled

-- Processing
processed_by UUID REFERENCES users(id),
processed_at TIMESTAMPTZ,

-- Confirmed Appointment
confirmed_appointment_id UUID REFERENCES appointments(id),

rejection_reason TEXT,

-- Communication
confirmation_sent BOOLEAN DEFAULT FALSE,
confirmation_sent_at TIMESTAMPTZ,

created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

UNIQUE(organization_id, request_number)

```

```

);

-- Online Bill Payments
CREATE TABLE online_payments (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),

  transaction_id VARCHAR(100) NOT NULL,

  patient_id UUID NOT NULL REFERENCES patients(id),
  portal_user_id UUID REFERENCES patient_portal_users(id),

  -- Bill Reference
  bill_id UUID REFERENCES bills(id),

  -- Amount
  amount DECIMAL(14, 2) NOT NULL,
  currency VARCHAR(3) DEFAULT 'BDT',

  -- Payment Gateway
  payment_gateway VARCHAR(50) NOT NULL, -- sslcommerz, bkaash, nagad, stripe
  gateway_transaction_id VARCHAR(255),
  gateway_response JSONB,

  -- Status
  status VARCHAR(20) DEFAULT 'initiated', -- initiated, pending, success,
  failed, refunded

  -- Timestamps
  initiated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
  completed_at TIMESTAMPTZ,

  -- Failure
  failure_reason TEXT,
  failure_code VARCHAR(50),

  -- Refund
  refund_amount DECIMAL(14, 2),
  refund_reason TEXT,
  refunded_at TIMESTAMPTZ,
  refund_transaction_id VARCHAR(255),

  -- Linked Payment Record
  payment_id UUID REFERENCES payments(id),

  -- Metadata
  client_ip INET,
  user_agent TEXT,

  created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
  updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

  UNIQUE(organization_id, transaction_id)
);

```



```

-- Patient Health Records Sharing
CREATE TABLE health_record_sharing (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  patient_id UUID NOT NULL REFERENCES patients(id),

  -- Share With
  shared_with_type VARCHAR(50), -- doctor, hospital, family_member, insurance
  shared_with_doctor_id UUID REFERENCES doctors(id),
  shared_with_organization_id UUID REFERENCES organizations(id),
  shared_with_email VARCHAR(255),
  shared_with_name VARCHAR(255),

  -- What to Share
  share_scope VARCHAR(50), -- all, selected
  shared_record_types TEXT[], -- lab_results, prescriptions, discharge_summary
  shared_record_ids UUID[],

  -- Validity
  valid_from TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
  valid_until TIMESTAMPTZ,

  -- Access
  access_code VARCHAR(100), -- One-time access code
  access_code_hash VARCHAR(255),

  max_access_count INTEGER,
  current_access_count INTEGER DEFAULT 0,

  -- Status
  status VARCHAR(20) DEFAULT 'active', -- active, expired, revoked

  revoked_at TIMESTAMPTZ,
  revoked_reason TEXT,

  created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

  -- Access Log
  last_accessed_at TIMESTAMPTZ
);

-- Sharing Access Log
CREATE TABLE health_record_access_log (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  sharing_id UUID NOT NULL REFERENCES health_record_sharing(id),

  accessed_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

  accessor_ip INET,
  accessor_user_agent TEXT,
  accessor_name VARCHAR(255),

  records_accessed JSONB
);

```

```

-- =====
-- SECTION 63: REFERRAL MANAGEMENT
-- =====

-- Referral Sources
CREATE TABLE referral_sources (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),

    source_code VARCHAR(20) NOT NULL,
    name VARCHAR(255) NOT NULL,

    source_type VARCHAR(50), -- doctor, hospital, clinic, diagnostic_center,
    corporate, marketing

    -- Contact
    contact_person VARCHAR(255),
    phone VARCHAR(20),
    email VARCHAR(255),
    address TEXT,

    -- Commission
    commission_type VARCHAR(20), -- none, fixed, percentage
    commission_value DECIMAL(10, 2),

    -- Agreement
    agreement_start_date DATE,
    agreement_end_date DATE,

    is_active BOOLEAN DEFAULT TRUE,
    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

    UNIQUE(organization_id, source_code)
);

-- Incoming Referrals
CREATE TABLE incoming_referrals (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),
    branch_id UUID NOT NULL REFERENCES branches(id),

    referral_number VARCHAR(50) NOT NULL,

    patient_id UUID NOT NULL REFERENCES patients(id),

    -- Referral Source
    source_id UUID REFERENCES referral_sources(id),
    referring_doctor_name VARCHAR(255),
    referring_hospital VARCHAR(255),
    referring_doctor_phone VARCHAR(20),

    -- Referral Details
    referral_date DATE NOT NULL,

```

```

referral_reason TEXT,
referral_diagnosis TEXT,

-- Documents
referral_letter_url TEXT,
attached_documents JSONB,

-- Destination
referred_to_department_id UUID REFERENCES departments(id),
referred_to_doctor_id UUID REFERENCES doctors(id),

-- Urgency
urgency VARCHAR(20) DEFAULT 'routine',

-- Outcome
outcome VARCHAR(50), -- admitted, opd_treated, not_accepted, redirected
outcome_date DATE,
outcome_notes TEXT,

-- Linked Records
appointment_id UUID REFERENCES appointments(id),
admission_id UUID REFERENCES admissions(id),

created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

UNIQUE(organization_id, referral_number)
);

-- Outgoing Referrals
CREATE TABLE outgoing_referrals (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),
  branch_id UUID NOT NULL REFERENCES branches(id),

  referral_number VARCHAR(50) NOT NULL,

  patient_id UUID NOT NULL REFERENCES patients(id),
  visit_id UUID,
  admission_id UUID REFERENCES admissions(id),

  -- Referring Doctor (from our hospital)
  referring_doctor_id UUID NOT NULL REFERENCES doctors(id),

  -- Referred To
  referred_to_hospital VARCHAR(255),
  referred_to_doctor VARCHAR(255),
  referred_to_department VARCHAR(100),
  referred_to_phone VARCHAR(20),
  referred_to_address TEXT,

  -- Reason
  referral_reason TEXT NOT NULL,
  clinical_summary TEXT,

```

```

-- Priority
urgency VARCHAR(20) DEFAULT 'routine',

-- Documents
referral_letter_url TEXT,
attached_reports JSONB,

-- Status
status VARCHAR(20) DEFAULT 'initiated', -- initiated, accepted, completed,
feedback_received

-- Feedback
feedback_received BOOLEAN DEFAULT FALSE,
feedback_date DATE,
feedback_notes TEXT,

created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

UNIQUE(organization_id, referral_number)
);

-- Referral Commission Tracking
CREATE TABLE referral_commissions (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),

  referral_id UUID NOT NULL REFERENCES incoming_referrals(id),
  source_id UUID NOT NULL REFERENCES referral_sources(id),

  -- Revenue
  patient_id UUID NOT NULL REFERENCES patients(id),
  bill_id UUID REFERENCES bills(id),

  bill_amount DECIMAL(14, 2),

  -- Commission
  commission_type VARCHAR(20),
  commission_rate DECIMAL(10, 2),
  commission_amount DECIMAL(12, 2) NOT NULL,

  -- Payment
  payment_status VARCHAR(20) DEFAULT 'pending', -- pending, approved, paid

  approved_by UUID REFERENCES users(id),
  approved_at TIMESTAMPTZ,

  paid_at TIMESTAMPTZ,
  payment_reference VARCHAR(100),

  created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

```

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-- =====
-- SECTION 64: RESEARCH & CLINICAL TRIALS
-- =====

-- Research Studies
CREATE TABLE research_studies (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),

  study_code VARCHAR(50) NOT NULL,
  title VARCHAR(500) NOT NULL,

  study_type VARCHAR(50), -- observational, interventional, clinical_trial
  phase VARCHAR(20), -- phase_1, phase_2, phase_3, phase_4

  -- Principal Investigator
  principal_investigator_id UUID REFERENCES doctors(id),
  co_investigators UUID[],

  -- Sponsor
  sponsor_name VARCHAR(255),
  sponsor_type VARCHAR(50), -- industry, academic, government, self

  -- Regulatory
  irb_approval_number VARCHAR(100),
  irb_approval_date DATE,
  irb_expiry_date DATE,

  ctri_registration VARCHAR(100), -- Clinical Trials Registry India

  -- Timeline
  planned_start_date DATE,
  planned_end_date DATE,
  actual_start_date DATE,
  actual_end_date DATE,

  -- Enrollment
  target_enrollment INTEGER,
  current_enrollment INTEGER DEFAULT 0,

  -- Budget
  total_budget DECIMAL(14, 2),

  -- Status
  status VARCHAR(20) DEFAULT 'planning', -- planning, recruiting, active,
  completed, suspended, terminated

  -- Documents
  protocol_url TEXT,
  consent_form_url TEXT,

  description TEXT,
  inclusion_criteria TEXT,
  exclusion_criteria TEXT,

```

```

        created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
        updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

        UNIQUE(organization_id, study_code)
    );

-- Research Study Subjects
CREATE TABLE research_subjects (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    study_id UUID NOT NULL REFERENCES research_studies(id),
    patient_id UUID NOT NULL REFERENCES patients(id),

    subject_id VARCHAR(50) NOT NULL, -- Anonymized subject ID

    -- Enrollment
    screening_date DATE,
    enrollment_date DATE,

    -- Status
    status VARCHAR(20) DEFAULT 'screening', -- screening, enrolled, active,
    completed, withdrawn, excluded

    -- Arm/Group
    study_arm VARCHAR(100),
    randomization_number VARCHAR(50),

    -- Withdrawal
    withdrawal_date DATE,
    withdrawal_reason TEXT,

    -- Consent
    consent_date DATE,
    consent_version VARCHAR(20),
    consent_document_url TEXT,

    -- Visits
    next_visit_date DATE,
    visits_completed INTEGER DEFAULT 0,

    notes TEXT,

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

    UNIQUE(study_id, subject_id)
);

-- Research Visit Schedule
CREATE TABLE research_visit_schedule (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    study_id UUID NOT NULL REFERENCES research_studies(id),

    visit_code VARCHAR(20) NOT NULL,

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visit_name VARCHAR(100) NOT NULL,

visit_day INTEGER, -- Day from enrollment (e.g., Day 0, Day 7, Day 30)
visit_window_before INTEGER DEFAULT 0, -- Days before
visit_window_after INTEGER DEFAULT 0, -- Days after

-- Procedures
procedures JSONB, -- [{procedure_name, is_mandatory}]

-- Assessments
assessments JSONB,

-- Labs
lab_tests JSONB,

is_mandatory BOOLEAN DEFAULT TRUE,

display_order INTEGER,

created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Research Subject Visits
CREATE TABLE research_subject_visits (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  subject_id UUID NOT NULL REFERENCES research_subjects(id),
  visit_schedule_id UUID NOT NULL REFERENCES research_visit_schedule(id),

  -- Scheduled
  scheduled_date DATE NOT NULL,

  -- Actual
  actual_date DATE,

  -- Status
  status VARCHAR(20) DEFAULT 'scheduled', -- scheduled, completed, missed,
  rescheduled

  -- Window Compliance
  within_window BOOLEAN,

  -- Data Collection
  procedures_completed JSONB,
  assessments_completed JSONB,

  -- Adverse Events
  adverse_events_reported BOOLEAN DEFAULT FALSE,

  notes TEXT,

  completed_by UUID REFERENCES users(id),

  created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
  updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP

```

```

);

-- Adverse Events
CREATE TABLE adverse_events (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  study_id UUID NOT NULL REFERENCES research_studies(id),
  subject_id UUID NOT NULL REFERENCES research_subjects(id),

  ae_number VARCHAR(50) NOT NULL,

  -- Event Details
  event_term VARCHAR(255) NOT NULL,
  event_description TEXT,

  onset_date DATE NOT NULL,
  resolution_date DATE,

  -- Severity
  severity VARCHAR(20), -- mild, moderate, severe, life_threatening, death

  -- Seriousness
  is_serious BOOLEAN DEFAULT FALSE,
  seriousness_criteria TEXT[], -- death, hospitalization, disability, etc.

  -- Causality
  relationship_to_study VARCHAR(50), -- unrelated, unlikely, possible, probable,
  definite

  -- Outcome
  outcome VARCHAR(50), -- recovered, recovering, not_recovered, fatal, unknown

  -- Action Taken
  action_with_study_drug VARCHAR(100),

  -- Reporting
  reported_to_sponsor BOOLEAN DEFAULT FALSE,
  sponsor_reported_date DATE,
  reported_to_irb BOOLEAN DEFAULT FALSE,
  irb_reported_date DATE,

  -- Follow-up
  requires_followup BOOLEAN DEFAULT FALSE,
  followup_notes TEXT,

  reported_by UUID REFERENCES users(id),

  created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
  updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

  UNIQUE(study_id, ae_number)
);

-- =====
-- SECTION 65: COST ACCOUNTING

```



```

-- =====

-- Cost Centers
CREATE TABLE cost_centers (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),
  branch_id UUID REFERENCES branches(id),

  cost_center_code VARCHAR(20) NOT NULL,
  name VARCHAR(100) NOT NULL,

  cost_center_type VARCHAR(50), -- revenue, support, administrative

  parent_cost_center_id UUID REFERENCES cost_centers(id),
  department_id UUID REFERENCES departments(id),

  -- Budget
  annual_budget DECIMAL(14, 2),

  manager_id UUID REFERENCES employees(id),

  is_active BOOLEAN DEFAULT TRUE,
  created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

  UNIQUE(organization_id, cost_center_code)
);

-- Cost Allocation Rules
CREATE TABLE cost_allocation_rules (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),

  rule_name VARCHAR(100) NOT NULL,

  source_cost_center_id UUID NOT NULL REFERENCES cost_centers(id),

  -- Allocation Method
  allocation_method VARCHAR(50), -- direct, step_down, activity_based

  -- Allocation Basis
  allocation_basis VARCHAR(50), -- revenue, patient_days, sq_feet, headcount,
  fixed_percent

  -- Target Cost Centers
  allocations JSONB, -- [{cost_center_id, percentage or basis_value}]

  effective_from DATE,
  effective_to DATE,

  is_active BOOLEAN DEFAULT TRUE,
  created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Activity Costs

```

```

CREATE TABLE activity_costs (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),

  activity_code VARCHAR(50) NOT NULL,
  activity_name VARCHAR(255) NOT NULL,

  cost_center_id UUID REFERENCES cost_centers(id),

  -- Cost Components
  direct_material_cost DECIMAL(12, 2),
  direct_labor_cost DECIMAL(12, 2),
  overhead_cost DECIMAL(12, 2),

  total_cost DECIMAL(12, 2) GENERATED ALWAYS AS (
    COALESCE(direct_material_cost, 0) +
    COALESCE(direct_labor_cost, 0) +
    COALESCE(overhead_cost, 0)
  ) STORED,

  -- Per Unit
  cost_per_unit DECIMAL(12, 4),
  unit_of_measure VARCHAR(50),

  -- Period
  effective_from DATE,
  effective_to DATE,

  is_active BOOLEAN DEFAULT TRUE,
  created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

  UNIQUE(organization_id, activity_code, effective_from)
);

-- Patient Service Costs
CREATE TABLE patient_service_costs (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),

  patient_id UUID NOT NULL REFERENCES patients(id),
  visit_id UUID,
  admission_id UUID REFERENCES admissions(id),

  service_date DATE NOT NULL,

  -- Service
  service_type VARCHAR(50), -- lab, radiology, pharmacy, procedure, room,
  consultation
  service_id UUID,
  service_name VARCHAR(255),

  -- Cost Components
  direct_cost DECIMAL(12, 2),
  indirect_cost DECIMAL(12, 2),

```

```

    total_cost DECIMAL(12, 2),

    -- Revenue
    billed_amount DECIMAL(12, 2),

    -- Margin
    margin DECIMAL(12, 2) GENERATED ALWAYS AS (billed_amount - total_cost) STORED,
    margin_percent DECIMAL(8, 2),

    -- Cost Center
    cost_center_id UUID REFERENCES cost_centers(id),

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Monthly Cost Summary
CREATE TABLE monthly_cost_summary (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),
    branch_id UUID REFERENCES branches(id),
    cost_center_id UUID REFERENCES cost_centers(id),

    year_month DATE NOT NULL, -- First day of month

    -- Revenue
    total_revenue DECIMAL(14, 2),

    -- Costs
    direct_costs DECIMAL(14, 2),
    indirect_costs DECIMAL(14, 2),
    allocated_costs DECIMAL(14, 2),
    total_costs DECIMAL(14, 2),

    -- Profitability
    gross_margin DECIMAL(14, 2),
    net_margin DECIMAL(14, 2),

    -- Statistics
    patient_days INTEGER,
    procedures_count INTEGER,

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

    UNIQUE(organization_id, branch_id, cost_center_id, year_month)
);

-- =====
-- SECTION 66: CONTRACT MANAGEMENT
-- =====

-- Contracts
CREATE TABLE contracts (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL REFERENCES organizations(id),

```

```

contract_number VARCHAR(50) NOT NULL,
title VARCHAR(500) NOT NULL,

contract_type VARCHAR(50), -- vendor, service, amc, lease, consulting,
insurance, corporate

-- Parties
party_type VARCHAR(50), -- supplier, customer, corporate, insurance
party_id UUID, -- Reference to supplier/corporate/insurance
party_name VARCHAR(255),

-- Value
contract_value DECIMAL(14, 2),
currency VARCHAR(3) DEFAULT 'BDT',

-- Duration
start_date DATE NOT NULL,
end_date DATE NOT NULL,

-- Auto Renewal
auto_renew BOOLEAN DEFAULT FALSE,
renewal_notice_days INTEGER,
renewal_terms TEXT,

-- Payment Terms
payment_terms TEXT,
payment_frequency VARCHAR(20), -- one_time, monthly, quarterly, annually

-- Status
status VARCHAR(20) DEFAULT 'draft', -- draft, pending_approval, active,
expired, terminated

-- Approval
approved_by UUID REFERENCES users(id),
approved_at TIMESTAMPTZ,

-- Documents
contract_document_url TEXT,
attachments JSONB,

-- Key Terms
key_terms JSONB,
sla_terms JSONB,
penalty_terms TEXT,

-- Ownership
contract_owner_id UUID REFERENCES users(id),
department_id UUID REFERENCES departments(id),

notes TEXT,

created_by UUID REFERENCES users(id),
created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

```

```

        updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

        UNIQUE(organization_id, contract_number)
    );

-- Contract Milestones
CREATE TABLE contract_milestones (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    contract_id UUID NOT NULL REFERENCES contracts(id),

    milestone_name VARCHAR(255) NOT NULL,
    description TEXT,

    due_date DATE NOT NULL,

    -- Payment
    payment_amount DECIMAL(12, 2),
    payment_percentage DECIMAL(5, 2),

    -- Status
    status VARCHAR(20) DEFAULT 'pending', -- pending, completed, overdue

    completed_date DATE,
    completed_by UUID REFERENCES users(id),

    notes TEXT,

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Contract Renewals
CREATE TABLE contract_renewals (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    original_contract_id UUID NOT NULL REFERENCES contracts(id),
    renewed_contract_id UUID REFERENCES contracts(id),

    renewal_date DATE NOT NULL,

    previous_end_date DATE,
    new_end_date DATE,

    previous_value DECIMAL(14, 2),
    new_value DECIMAL(14, 2),

    renewal_terms TEXT,

    processed_by UUID REFERENCES users(id),

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Contract Alerts
CREATE TABLE contract_alerts (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),

```

```

    contract_id UUID NOT NULL REFERENCES contracts(id),

    alert_type VARCHAR(50), -- expiry, renewal, milestone, payment, sla_breach

    alert_date DATE NOT NULL,

    description TEXT,

    -- Status
    status VARCHAR(20) DEFAULT 'active', -- active, acknowledged, resolved

    acknowledged_by UUID REFERENCES users(id),
    acknowledged_at TIMESTAMPTZ,

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- =====
-- SECTION 67: INDEXES FOR NEW TABLES
-- =====

-- Mobile & Portal
CREATE INDEX idx_mobile_devices_user ON mobile_devices(user_id);
CREATE INDEX idx_mobile_devices_active ON mobile_devices(is_active) WHERE
is_active = TRUE;
CREATE INDEX idx_mobile_sync_queue_device ON mobile_sync_queue(device_id, status);

CREATE INDEX idx_portal_users_patient ON patient_portal_users(patient_id);
CREATE INDEX idx_portal_users_org ON patient_portal_users(organization_id);
CREATE INDEX idx_patient_messages_patient ON patient_messages(patient_id);
CREATE INDEX idx_patient_messages_thread ON patient_messages(thread_id);
CREATE INDEX idx_online_appointments_status ON
online_appointment_requests(status);
CREATE INDEX idx_online_payments_status ON online_payments(status);
CREATE INDEX idx_health_sharing_patient ON health_record_sharing(patient_id);

-- Referrals
CREATE INDEX idx_incoming_referrals_source ON incoming_referrals(source_id);
CREATE INDEX idx_incoming_referrals_patient ON incoming_referrals(patient_id);
CREATE INDEX idx_outgoing_referrals_doctor ON
outgoing_referrals(referring_doctor_id);
CREATE INDEX idx_referral_commissions_status ON
referral_commissions(payment_status);

-- Research
CREATE INDEX idx_research_studies_status ON research_studies(status);
CREATE INDEX idx_research_subjects_study ON research_subjects(study_id);
CREATE INDEX idx_research_subjects_patient ON research_subjects(patient_id);
CREATE INDEX idx_adverse_events_study ON adverse_events(study_id);
CREATE INDEX idx_adverse_events_serious ON adverse_events(is_serious) WHERE
is_serious = TRUE;

-- Cost Accounting
CREATE INDEX idx_cost_centers_parent ON cost_centers(parent_cost_center_id);

```

```

CREATE INDEX idx_patient_service_costs_patient ON
patient_service_costs(patient_id);
CREATE INDEX idx_patient_service_costs_date ON
patient_service_costs(service_date);
CREATE INDEX idx_monthly_cost_summary_period ON monthly_cost_summary(year_month);

-- Contracts
CREATE INDEX idx_contracts_status ON contracts(status);
CREATE INDEX idx_contracts_end_date ON contracts(end_date);
CREATE INDEX idx_contracts_party ON contracts(party_type, party_id);
CREATE INDEX idx_contract_alerts_date ON contract_alerts(alert_date, status);

```

PART 22: SAMPLE DATA & TESTING

SQL

```

-- =====
-- SECTION 68: SAMPLE DATA GENERATION
-- =====

-- Create sample organization
CREATE OR REPLACE FUNCTION create_sample_organization()
RETURNS UUID AS $$
DECLARE
    v_org_id UUID;
    v_branch_id UUID;
    v_admin_role_id UUID;
    v_admin_user_id UUID;
BEGIN
    -- Create Organization
    INSERT INTO organizations (
        code, name, name_bangla,
        email, phone,
        address_line1, city, district, division,
        settings
    ) VALUES (
        'DEMO-HOSP',
        'Demo General Hospital',
        'ডেমো জেনারেল হাসপাতাল',
        'info@demohospital.com',
        '+8801700000000',
        '123 Hospital Road',
        'Dhaka',
        'Dhaka',
        'Dhaka',
        '{"timezone": "Asia/Dhaka", "currency": "BDT", "date_format":
"DD/MM/YYYY"}'::JSONB
    ) RETURNING id INTO v_org_id;

    -- Create Main Branch

```

```

INSERT INTO branches (
    organization_id, code, name, name_bangla,
    email, phone,
    city, district, division,
    is_main_branch, total_beds
) VALUES (
    v_org_id,
    'MAIN',
    'Main Branch',
    'প্রধান শাখা',
    'main@demohospital.com',
    '+8801700000001',
    'Dhaka',
    'Dhaka',
    'Dhaka',
    TRUE,
    200
) RETURNING id INTO v_branch_id;

-- Get or create admin role
SELECT id INTO v_admin_role_id FROM roles WHERE code = 'ORG_ADMIN' LIMIT 1;

-- Create Admin User
INSERT INTO users (
    organization_id,
    username, email, phone,
    password_hash,
    first_name, last_name,
    is_active
) VALUES (
    v_org_id,
    'admin',
    'admin@demohospital.com',
    '+8801700000002',
    crypt('Admin@123', gen_salt('bf')),
    'System',
    'Administrator',
    TRUE
) RETURNING id INTO v_admin_user_id;

-- Assign Role
INSERT INTO user_roles (user_id, role_id, is_primary)
VALUES (v_admin_user_id, v_admin_role_id, TRUE);

-- Create Branch Access
INSERT INTO user_branch_access (user_id, branch_id, is_default)
VALUES (v_admin_user_id, v_branch_id, TRUE);

RETURN v_org_id;
END;
$$ LANGUAGE plpgsql;

-- Create sample departments
CREATE OR REPLACE FUNCTION create_sample_departments(p_org_id UUID, p_branch_id

```



```

UUID)
RETURNS INTEGER AS $$
DECLARE
    v_count INTEGER := 0;
BEGIN
    INSERT INTO departments (organization_id, branch_id, code, name,
department_type, is_clinical)
VALUES
    (p_org_id, p_branch_id, 'MED', 'Medicine', 'clinical', TRUE),
    (p_org_id, p_branch_id, 'SURG', 'Surgery', 'clinical', TRUE),
    (p_org_id, p_branch_id, 'ORTHO', 'Orthopedics', 'clinical', TRUE),
    (p_org_id, p_branch_id, 'GYN', 'Obstetrics & Gynecology', 'clinical',
TRUE),
    (p_org_id, p_branch_id, 'PED', 'Pediatrics', 'clinical', TRUE),
    (p_org_id, p_branch_id, 'CARD', 'Cardiology', 'clinical', TRUE),
    (p_org_id, p_branch_id, 'NEURO', 'Neurology', 'clinical', TRUE),
    (p_org_id, p_branch_id, 'ENT', 'ENT', 'clinical', TRUE),
    (p_org_id, p_branch_id, 'DERM', 'Dermatology', 'clinical', TRUE),
    (p_org_id, p_branch_id, 'OPTH', 'Ophthalmology', 'clinical', TRUE),
    (p_org_id, p_branch_id, 'RAD', 'Radiology', 'clinical', TRUE),
    (p_org_id, p_branch_id, 'PATH', 'Pathology', 'clinical', TRUE),
    (p_org_id, p_branch_id, 'EMRG', 'Emergency', 'clinical', TRUE),
    (p_org_id, p_branch_id, 'ICU', 'Intensive Care', 'clinical', TRUE),
    (p_org_id, p_branch_id, 'ADMIN', 'Administration', 'administrative',
FALSE),
    (p_org_id, p_branch_id, 'HR', 'Human Resources', 'administrative', FALSE),
    (p_org_id, p_branch_id, 'FIN', 'Finance', 'administrative', FALSE),
    (p_org_id, p_branch_id, 'IT', 'Information Technology', 'support', FALSE),
    (p_org_id, p_branch_id, 'PHARM', 'Pharmacy', 'support', FALSE),
    (p_org_id, p_branch_id, 'NURSE', 'Nursing', 'clinical', TRUE)
ON CONFLICT DO NOTHING;

    GET DIAGNOSTICS v_count = ROW_COUNT;
    RETURN v_count;
END;
$$ LANGUAGE plpgsql;

-- Create sample wards and beds
CREATE OR REPLACE FUNCTION create_sample_wards_beds(p_org_id UUID, p_branch_id
UUID)
RETURNS INTEGER AS $$
DECLARE
    v_ward_id UUID;
    v_room_id UUID;
    v_ward RECORD;
    v_bed_count INTEGER := 0;
BEGIN
    -- Create Wards
    FOR v_ward IN
        SELECT * FROM (VALUES
            ('GEN-M', 'General Ward - Male', 'general', 'male', 30),
            ('GEN-F', 'General Ward - Female', 'general', 'female', 30),
            ('PVT', 'Private Ward', 'private', NULL, 20),
            ('ICU', 'Intensive Care Unit', 'icu', NULL, 10),

```

```

        ('NICU', 'Neonatal ICU', 'nicu', NULL, 8),
        ('CCU', 'Coronary Care Unit', 'ccu', NULL, 8),
        ('PED', 'Pediatric Ward', 'pediatric', NULL, 20),
        ('MAT', 'Maternity Ward', 'maternity', 'female', 25),
        ('SURG', 'Surgical Ward', 'general', NULL, 25),
        ('EMRG', 'Emergency Observation', 'general', NULL, 15)
    ) AS w(code, name, ward_type, gender, beds)
LOOP
    INSERT INTO wards (
        organization_id, branch_id, code, name, ward_type,
        gender_restriction, total_beds
    ) VALUES (
        p_org_id, p_branch_id, v_ward.code, v_ward.name,
        v_ward.ward_type, v_ward.gender::gender_type, v_ward.beds
    )
    ON CONFLICT DO NOTHING
    RETURNING id INTO v_ward_id;

    IF v_ward_id IS NOT NULL THEN
        -- Create Beds for this ward
        FOR i IN 1..v_ward.beds LOOP
            INSERT INTO beds (
                organization_id, branch_id, ward_id,
                bed_number, bed_type, status,
                has_oxygen, daily_rate
            ) VALUES (
                p_org_id, p_branch_id, v_ward_id,
                v_ward.code || '-' || LPAD(i::TEXT, 2, '0'),
                CASE
                    WHEN v_ward.ward_type = 'icu' THEN 'icu'
                    WHEN v_ward.ward_type = 'nicu' THEN 'nicu'
                    ELSE 'standard'
                END,
                'available',
                TRUE,
                CASE
                    WHEN v_ward.ward_type = 'icu' THEN 8000
                    WHEN v_ward.ward_type = 'private' THEN 5000
                    ELSE 1500
                END
            )
        )
        ON CONFLICT DO NOTHING;

        v_bed_count := v_bed_count + 1;
    END LOOP;
END IF;
END LOOP;

RETURN v_bed_count;
END;
$$ LANGUAGE plpgsql;

-- Create sample doctors
CREATE OR REPLACE FUNCTION create_sample_doctors(p_org_id UUID, p_branch_id UUID,

```

```

p_count INTEGER DEFAULT 20)
RETURNS INTEGER AS $$
DECLARE
    v_user_id UUID;
    v_doctor_id UUID;
    v_dept_id UUID;
    v_role_id UUID;
    v_specializations TEXT[] := ARRAY['Medicine', 'Surgery', 'Cardiology',
'Orthopedics', 'Gynecology', 'Pediatrics', 'Neurology', 'Dermatology', 'ENT',
'Ophthalmology'];
    v_first_names TEXT[] := ARRAY['Dr. Ahmed', 'Dr. Rahman', 'Dr. Karim', 'Dr.
Hassan', 'Dr. Ali', 'Dr. Fatima', 'Dr. Ayesha', 'Dr. Nasreen', 'Dr. Sultana', 'Dr.
Khadija'];
    v_last_names TEXT[] := ARRAY['Islam', 'Khan', 'Chowdhury', 'Hossain', 'Uddin',
'Begum', 'Akter', 'Sheikh', 'Siddiqui', 'Ahmed'];
    v_counter INTEGER := 0;
    v_spec TEXT;
BEGIN
    -- Get doctor role
    SELECT id INTO v_role_id FROM roles WHERE code = 'DOCTOR' LIMIT 1;

    FOR i IN 1..p_count LOOP
        v_spec := v_specializations[1 + (i % array_length(v_specializations, 1))];

        -- Get department for specialization
        SELECT id INTO v_dept_id FROM departments
        WHERE organization_id = p_org_id
        AND (name ILIKE '%' || v_spec || '%' OR code ILIKE LEFT(v_spec, 3) || '%')
        LIMIT 1;

        -- Create User
        INSERT INTO users (
            organization_id,
            username, email, phone,
            password_hash,
            first_name, last_name,
            department_id,
            designation,
            is_active
        ) VALUES (
            p_org_id,
            'dr' || LPAD(i::TEXT, 3, '0'),
            'dr' || i || '@demohospital.com',
            '01' || (7 + (i % 3))::TEXT || (10000000 + i * 1000)::TEXT,
            crypt('Doctor@123', gen_salt('bf')),
            v_first_names[1 + (i % array_length(v_first_names, 1))],
            v_last_names[1 + (i % array_length(v_last_names, 1))],
            v_dept_id,
            'Consultant',
            TRUE
        ) RETURNING id INTO v_user_id;

        -- Create Doctor Profile
        INSERT INTO doctors (

```

```

        user_id, organization_id,
        doctor_code,
        bmdc_registration,
        specialization,
        opd_fee, followup_fee, telemedicine_fee,
        consultation_duration,
        is_available_opd, is_available_telemedicine
    ) VALUES (
        v_user_id, p_org_id,
        'DOC' || LPAD(i::TEXT, 4, '0'),
        'A-' || (10000 + i)::TEXT,
        v_spec,
        (500 + (i % 10) * 100),
        (300 + (i % 10) * 50),
        (400 + (i % 10) * 50),
        15,
        TRUE,
        i % 2 = 0
    ) RETURNING id INTO v_doctor_id;

-- Assign Role
INSERT INTO user_roles (user_id, role_id, is_primary)
VALUES (v_user_id, v_role_id, TRUE)
ON CONFLICT DO NOTHING;

-- Branch Access
INSERT INTO user_branch_access (user_id, branch_id, is_default)
VALUES (v_user_id, p_branch_id, TRUE)
ON CONFLICT DO NOTHING;

-- Create Schedule Template (Mon-Fri)
FOR d IN 0..4 LOOP
    INSERT INTO doctor_schedule_templates (
        doctor_id, branch_id, day_of_week,
        session1_start, session1_end, session1_max_patients,
        session2_start, session2_end, session2_max_patients
    ) VALUES (
        v_doctor_id, p_branch_id, d + 1,
        '09:00', '13:00', 20,
        '17:00', '20:00', 15
    ) ON CONFLICT DO NOTHING;
END LOOP;

v_counter := v_counter + 1;
END LOOP;

RETURN v_counter;
END;
$$ LANGUAGE plpgsql;

-- Create sample test catalog
CREATE OR REPLACE FUNCTION create_sample_test_catalog(p_org_id UUID)
RETURNS INTEGER AS $$
DECLARE
```

```

v_count INTEGER := 0;
BEGIN
  INSERT INTO test_catalog (
    organization_id, code, name, department, category,
    specimen_type, tat_routine_hours, price
  ) VALUES
    -- Hematology
    (p_org_id, 'CBC', 'Complete Blood Count', 'hematology', 'Routine',
    'Blood', 4, 500),
    (p_org_id, 'HB', 'Hemoglobin', 'hematology', 'Routine', 'Blood', 2, 150),
    (p_org_id, 'ESR', 'Erythrocyte Sedimentation Rate', 'hematology',
    'Routine', 'Blood', 4, 200),
    (p_org_id, 'PT', 'Prothrombin Time', 'hematology', 'Coagulation', 'Blood',
    4, 600),
    (p_org_id, 'APTT', 'Activated Partial Thromboplastin Time', 'hematology',
    'Coagulation', 'Blood', 4, 800),

    -- Biochemistry
    (p_org_id, 'RBS', 'Random Blood Sugar', 'biochemistry', 'Routine',
    'Blood', 2, 200),
    (p_org_id, 'FBS', 'Fasting Blood Sugar', 'biochemistry', 'Routine',
    'Blood', 2, 200),
    (p_org_id, 'HBA1C', 'Glycated Hemoglobin', 'biochemistry', 'Diabetes',
    'Blood', 24, 1200),
    (p_org_id, 'LFT', 'Liver Function Test', 'biochemistry', 'Routine',
    'Blood', 6, 1500),
    (p_org_id, 'RFT', 'Renal Function Test', 'biochemistry', 'Routine',
    'Blood', 6, 1200),
    (p_org_id, 'LIPID', 'Lipid Profile', 'biochemistry', 'Cardiac', 'Blood',
    6, 1500),
    (p_org_id, 'TFT', 'Thyroid Function Test', 'biochemistry', 'Endocrine',
    'Blood', 24, 2000),
    (p_org_id, 'ELECTRO', 'Electrolytes (Na, K, Cl)', 'biochemistry',
    'Routine', 'Blood', 4, 800),

    -- Urine
    (p_org_id, 'URINE-RE', 'Urine Routine Examination', 'biochemistry',
    'Routine', 'Urine', 2, 250),
    (p_org_id, 'URINE-CS', 'Urine Culture & Sensitivity', 'microbiology',
    'Culture', 'Urine', 72, 1000),

    -- Serology
    (p_org_id, 'HIV', 'HIV Antibody Test', 'serology', 'Infectious', 'Blood',
    4, 600),
    (p_org_id, 'HBSAG', 'Hepatitis B Surface Antigen', 'serology',
    'Infectious', 'Blood', 4, 500),
    (p_org_id, 'HCV', 'Hepatitis C Antibody', 'serology', 'Infectious',
    'Blood', 4, 800),
    (p_org_id, 'VDRL', 'VDRL Test', 'serology', 'Infectious', 'Blood', 4,
    300),
    (p_org_id, 'WIDAL', 'Widal Test', 'serology', 'Infectious', 'Blood', 4,
    400),
    (p_org_id, 'CRP', 'C-Reactive Protein', 'serology', 'Inflammatory',
    'Blood', 4, 600),

```

```

        (p_org_id, 'RF', 'Rheumatoid Factor', 'serology', 'Autoimmune', 'Blood',
4, 500),

        -- Radiology
        (p_org_id, 'XRAY-CHEST', 'X-Ray Chest PA View', 'radiology', 'X-Ray',
NULL, 1, 500),
        (p_org_id, 'XRAY-SPINE', 'X-Ray Spine', 'radiology', 'X-Ray', NULL, 1,
800),
        (p_org_id, 'USG-ABD', 'USG Whole Abdomen', 'radiology', 'Ultrasound',
NULL, 1, 1500),
        (p_org_id, 'USG-PREG', 'USG Pregnancy', 'radiology', 'Ultrasound', NULL,
1, 1200),
        (p_org_id, 'ECG', 'Electrocardiogram', 'radiology', 'Cardiac', NULL, 1,
400),
        (p_org_id, 'ECHO', 'Echocardiography', 'radiology', 'Cardiac', NULL, 2,
3000),
        (p_org_id, 'CT-BRAIN', 'CT Scan Brain', 'radiology', 'CT', NULL, 4, 5000),
        (p_org_id, 'CT-ABD', 'CT Scan Abdomen', 'radiology', 'CT', NULL, 4, 8000),
        (p_org_id, 'MRI-BRAIN', 'MRI Brain', 'radiology', 'MRI', NULL, 24, 12000)
    ON CONFLICT DO NOTHING;

    GET DIAGNOSTICS v_count = ROW_COUNT;
    RETURN v_count;
END;
$$ LANGUAGE plpgsql;

-- Create sample drug master
CREATE OR REPLACE FUNCTION create_sample_drugs()
RETURNS INTEGER AS $$
DECLARE
    v_count INTEGER := 0;
BEGIN
    INSERT INTO drug_master (
        generic_name, brand_name, manufacturer,
        drug_type, strength, therapeutic_class,
        route_of_administration, mrp, requires_prescription
    ) VALUES
        -- Analgesics
        ('Paracetamol', 'Napa', 'Beximco', 'tablet', '500mg', 'Analgesic', 'oral',
2.00, FALSE),
        ('Paracetamol', 'Ace', 'Square', 'tablet', '500mg', 'Analgesic', 'oral',
2.00, FALSE),
        ('Ibuprofen', 'Profen', 'Square', 'tablet', '400mg', 'NSAID', 'oral',
4.00, TRUE),
        ('Diclofenac', 'Voltalin', 'Novartis', 'tablet', '50mg', 'NSAID', 'oral',
5.00, TRUE),

        -- Antibiotics
        ('Amoxicillin', 'Moxacil', 'Square', 'capsule', '500mg', 'Antibiotic',
'oral', 8.00, TRUE),
        ('Azithromycin', 'Zimax', 'Square', 'tablet', '500mg', 'Antibiotic',
'oral', 30.00, TRUE),
        ('Ciprofloxacin', 'Ciprocin', 'Square', 'tablet', '500mg', 'Antibiotic',
'oral', 10.00, TRUE),

```

```

        ('Cefixime', 'Cef-3', 'Square', 'tablet', '200mg', 'Antibiotic', 'oral',
25.00, TRUE),
        ('Metronidazole', 'Flagyl', 'Sanofi', 'tablet', '400mg', 'Antibiotic',
'oral', 3.00, TRUE),
        ('Ceftriaxone', 'Ceftron', 'Square', 'injection', '1g', 'Antibiotic',
'iv', 150.00, TRUE),

        -- Antacids
        ('Omeprazole', 'Losectil', 'Square', 'capsule', '20mg', 'PPI', 'oral',
6.00, TRUE),
        ('Esomeprazole', 'Nexium', 'AstraZeneca', 'tablet', '20mg', 'PPI', 'oral',
15.00, TRUE),
        ('Ranitidine', 'Neotack', 'Square', 'tablet', '150mg', 'H2 Blocker',
'oral', 3.00, FALSE),

        -- Antihypertensives
        ('Amlodipine', 'Amdocal', 'Square', 'tablet', '5mg', 'CCB', 'oral', 5.00,
TRUE),
        ('Losartan', 'Losartan', 'Square', 'tablet', '50mg', 'ARB', 'oral', 8.00,
TRUE),
        ('Atenolol', 'Tenolol', 'Square', 'tablet', '50mg', 'Beta Blocker',
'oral', 4.00, TRUE),

        -- Antidiabetics
        ('Metformin', 'Comet', 'Square', 'tablet', '500mg', 'Biguanide', 'oral',
4.00, TRUE),
        ('Glimepiride', 'Amaryl', 'Sanofi', 'tablet', '2mg', 'Sulfonylurea',
'oral', 12.00, TRUE),
        ('Insulin Mixtard', 'Mixtard', 'Novo Nordisk', 'injection', '100IU/ml',
'Insulin', 'sc', 550.00, TRUE),

        -- Others
        ('Salbutamol', 'Sultolin', 'Square', 'inhaler', '100mcg',
'Bronchodilator', 'inhalation', 120.00, TRUE),
        ('Cetirizine', 'Alatrol', 'Square', 'tablet', '10mg', 'Antihistamine',
'oral', 3.00, FALSE),
        ('Pantoprazole', 'Pantonix', 'Square', 'injection', '40mg', 'PPI', 'iv',
100.00, TRUE),
        ('Ondansetron', 'Emistat', 'Square', 'tablet', '4mg', 'Antiemetic',
'oral', 10.00, TRUE),
        ('Tramadol', 'Tramal', 'Grünenthal', 'capsule', '50mg', 'Analgesic',
'oral', 15.00, TRUE)
    ON CONFLICT DO NOTHING;

    GET DIAGNOSTICS v_count = ROW_COUNT;
    RETURN v_count;
END;
$$ LANGUAGE plpgsql;

-- Master function to create all sample data
CREATE OR REPLACE FUNCTION create_all_sample_data()
RETURNS TABLE(entity VARCHAR, count INTEGER) AS $$
DECLARE
    v_org_id UUID;

```

```

    v_branch_id UUID;
BEGIN
    -- Create organization
    v_org_id := create_sample_organization();

    -- Get branch
    SELECT id INTO v_branch_id FROM branches
    WHERE organization_id = v_org_id AND is_main_branch = TRUE LIMIT 1;

    -- Create sample data
    RETURN QUERY
    SELECT 'Organization'::VARCHAR, 1
    UNION ALL
    SELECT 'Departments'::VARCHAR, create_sample_departments(v_org_id,
v_branch_id)
    UNION ALL
    SELECT 'Wards & Beds'::VARCHAR, create_sample_wards_beds(v_org_id,
v_branch_id)
    UNION ALL
    SELECT 'Doctors'::VARCHAR, create_sample_doctors(v_org_id, v_branch_id, 20)
    UNION ALL
    SELECT 'Test Catalog'::VARCHAR, create_sample_test_catalog(v_org_id)
    UNION ALL
    SELECT 'Drugs'::VARCHAR, create_sample_drugs()
    UNION ALL
    SELECT 'Patients'::VARCHAR, generate_sample_patients(v_org_id, v_branch_id,
100);
END;
$$ LANGUAGE plpgsql;

-- =====
-- SECTION 69: TESTING & VALIDATION
-- =====

-- Test foreign key relationships
CREATE OR REPLACE FUNCTION test_foreign_keys()
RETURNS TABLE(
    constraint_name VARCHAR,
    table_name VARCHAR,
    column_name VARCHAR,
    foreign_table VARCHAR,
    foreign_column VARCHAR,
    status VARCHAR
) AS $$
BEGIN
    RETURN QUERY
    SELECT
        tc.constraint_name::VARCHAR,
        tc.table_name::VARCHAR,
        kcu.column_name::VARCHAR,
        ccu.table_name::VARCHAR AS foreign_table,
        ccu.column_name::VARCHAR AS foreign_column,
        'OK'::VARCHAR AS status
    FROM information_schema.table_constraints tc

```



```

    JOIN information_schema.key_column_usage kcu
      ON tc.constraint_name = kcu.constraint_name
    JOIN information_schema.constraint_column_usage ccu
      ON tc.constraint_name = ccu.constraint_name
    WHERE tc.constraint_type = 'FOREIGN KEY'
    AND tc.table_schema = 'public'
    ORDER BY tc.table_name, tc.constraint_name;
END;
$$ LANGUAGE plpgsql;

-- Test indexes exist for foreign keys
CREATE OR REPLACE FUNCTION check_missing_fk_indexes()
RETURNS TABLE(
    table_name VARCHAR,
    column_name VARCHAR,
    constraint_name VARCHAR,
    recommendation VARCHAR
) AS $$
BEGIN
    RETURN QUERY
    SELECT
        tc.table_name::VARCHAR,
        kcu.column_name::VARCHAR,
        tc.constraint_name::VARCHAR,
        ('CREATE INDEX idx_' || tc.table_name || '_' || kcu.column_name ||
         ' ON ' || tc.table_name || '(' || kcu.column_name || ')')::VARCHAR AS
recommendation
    FROM information_schema.table_constraints tc
    JOIN information_schema.key_column_usage kcu
      ON tc.constraint_name = kcu.constraint_name
    WHERE tc.constraint_type = 'FOREIGN KEY'
    AND tc.table_schema = 'public'
    AND NOT EXISTS (
        SELECT 1 FROM pg_indexes
        WHERE tablename = tc.table_name
        AND indexdef LIKE '%' || kcu.column_name || '%'
    );
END;
$$ LANGUAGE plpgsql;

-- Validate data integrity
CREATE OR REPLACE FUNCTION validate_data_integrity()
RETURNS TABLE(
    check_name VARCHAR,
    status VARCHAR,
    details TEXT
) AS $$
BEGIN
    -- Check orphan patients
    RETURN QUERY
    SELECT
        'Orphan Patients'::VARCHAR,
        CASE WHEN COUNT(*) = 0 THEN 'PASS' ELSE 'FAIL' END::VARCHAR,
        COUNT(*)::TEXT || ' patients without organization'

```

```

    FROM patients p
    WHERE NOT EXISTS (SELECT 1 FROM organizations o WHERE o.id =
p.organization_id);

-- Check orphan admissions
RETURN QUERY
SELECT
    'Orphan Admissions'::VARCHAR,
    CASE WHEN COUNT(*) = 0 THEN 'PASS' ELSE 'FAIL' END::VARCHAR,
    COUNT(*)::TEXT || ' admissions without patient'
FROM admissions a
WHERE NOT EXISTS (SELECT 1 FROM patients p WHERE p.id = a.patient_id);

-- Check beds without wards
RETURN QUERY
SELECT
    'Beds Without Wards'::VARCHAR,
    CASE WHEN COUNT(*) = 0 THEN 'PASS' ELSE 'FAIL' END::VARCHAR,
    COUNT(*)::TEXT || ' beds without valid ward'
FROM beds b
WHERE NOT EXISTS (SELECT 1 FROM wards w WHERE w.id = b.ward_id);

-- Check appointments without doctors
RETURN QUERY
SELECT
    'Appointments Without Doctors'::VARCHAR,
    CASE WHEN COUNT(*) = 0 THEN 'PASS' ELSE 'FAIL' END::VARCHAR,
    COUNT(*)::TEXT || ' appointments without valid doctor'
FROM appointments a
WHERE NOT EXISTS (SELECT 1 FROM doctors d WHERE d.id = a.doctor_id);

-- Check bills without patients
RETURN QUERY
SELECT
    'Bills Without Patients'::VARCHAR,
    CASE WHEN COUNT(*) = 0 THEN 'PASS' ELSE 'FAIL' END::VARCHAR,
    COUNT(*)::TEXT || ' bills without valid patient'
FROM bills b
WHERE NOT EXISTS (SELECT 1 FROM patients p WHERE p.id = b.patient_id);
END;
$$ LANGUAGE plpgsql;

-- Performance test function
CREATE OR REPLACE FUNCTION test_query_performance()
RETURNS TABLE(
    query_name VARCHAR,
    execution_time_ms NUMERIC,
    row_count INTEGER
) AS $$
DECLARE
    v_start TIMESTAMPTZ;
    v_end TIMESTAMPTZ;
    v_count INTEGER;
BEGIN

```

```

-- Test patient search
v_start := clock_timestamp();
SELECT COUNT(*) INTO v_count FROM patients WHERE full_name ILIKE '%ahmed%';
v_end := clock_timestamp();
RETURN QUERY SELECT 'Patient Search'::VARCHAR,
    EXTRACT(MILLISECOND FROM (v_end - v_start))::NUMERIC, v_count;

-- Test active admissions
v_start := clock_timestamp();
SELECT COUNT(*) INTO v_count FROM admissions WHERE status = 'admitted';
v_end := clock_timestamp();
RETURN QUERY SELECT 'Active Admissions'::VARCHAR,
    EXTRACT(MILLISECOND FROM (v_end - v_start))::NUMERIC, v_count;

-- Test today's appointments
v_start := clock_timestamp();
SELECT COUNT(*) INTO v_count FROM appointments WHERE appointment_date =
CURRENT_DATE;
v_end := clock_timestamp();
RETURN QUERY SELECT 'Today Appointments'::VARCHAR,
    EXTRACT(MILLISECOND FROM (v_end - v_start))::NUMERIC, v_count;

-- Test available beds
v_start := clock_timestamp();
SELECT COUNT(*) INTO v_count FROM beds WHERE status = 'available' AND
is_active = TRUE;
v_end := clock_timestamp();
RETURN QUERY SELECT 'Available Beds'::VARCHAR,
    EXTRACT(MILLISECOND FROM (v_end - v_start))::NUMERIC, v_count;

-- Test pending lab orders
v_start := clock_timestamp();
SELECT COUNT(*) INTO v_count FROM lab_orders WHERE status IN ('ordered',
'collected');
v_end := clock_timestamp();
RETURN QUERY SELECT 'Pending Lab Orders'::VARCHAR,
    EXTRACT(MILLISECOND FROM (v_end - v_start))::NUMERIC, v_count;
END;
$$ LANGUAGE plpgsql;

```

PART 23: BACKUP & MAINTENANCE

SQL

```

-- =====
-- SECTION 70: BACKUP & RECOVERY
-- =====

-- Backup configuration table
CREATE TABLE backup_configurations (

```

```

    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),

    backup_type VARCHAR(20) NOT NULL, -- full, incremental, differential

    schedule_cron VARCHAR(100), -- Cron expression

    retention_days INTEGER DEFAULT 30,

    -- Destination
    storage_type VARCHAR(50), -- local, s3, gcs, azure
    storage_path TEXT,
    storage_credentials JSONB, -- Encrypted

    -- Options
    compress BOOLEAN DEFAULT TRUE,
    encrypt BOOLEAN DEFAULT TRUE,
    encryption_key_id VARCHAR(100),

    include_schemas TEXT[] DEFAULT ARRAY['public'],
    exclude_tables TEXT[],

    is_active BOOLEAN DEFAULT TRUE,

    last_backup_at TIMESTAMPTZ,
    last_backup_status VARCHAR(20),
    last_backup_size_mb DECIMAL(12, 2),

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Backup history
CREATE TABLE backup_history (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    configuration_id UUID REFERENCES backup_configurations(id),

    backup_type VARCHAR(20) NOT NULL,

    started_at TIMESTAMPTZ NOT NULL,
    completed_at TIMESTAMPTZ,

    status VARCHAR(20), -- running, completed, failed

    -- Details
    file_name VARCHAR(500),
    file_path TEXT,
    file_size_mb DECIMAL(12, 2),

    tables_backed_up INTEGER,
    rows_backed_up BIGINT,

    error_message TEXT,

    -- Verification

```

```

    verified BOOLEAN DEFAULT FALSE,
    verified_at TIMESTAMPTZ,

    -- Retention
    expires_at DATE,
    deleted_at TIMESTAMPTZ,

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Create backup script generator
CREATE OR REPLACE FUNCTION generate_backup_script(
    p_backup_type VARCHAR DEFAULT 'full',
    p_output_path VARCHAR DEFAULT '/backup'
)
RETURNS TEXT AS $$
DECLARE
    v_script TEXT;
    v_filename TEXT;
BEGIN
    v_filename := 'hospital_erp_' || p_backup_type || '_' ||
        TO_CHAR(CURRENT_TIMESTAMP, 'YYYYMMDD_HH24MISS') || '.sql';

    v_script := '#!/bin/bash
# Hospital ERP Database Backup Script
# Generated: ' || CURRENT_TIMESTAMP || '
# Type: ' || p_backup_type || '

BACKUP_DIR="" || p_output_path || ''
FILENAME="" || v_filename || ''
DB_NAME="hospital_erp"

# Create backup directory if not exists
mkdir -p $BACKUP_DIR

# Run pg_dump
pg_dump -h localhost -U postgres -d $DB_NAME \
    --format=custom \
    --compress=9 \
    --verbose \
    --file="$BACKUP_DIR/$FILENAME"

# Verify backup
if [ $? -eq 0 ]; then
    echo "Backup completed successfully: $BACKUP_DIR/$FILENAME"
    # Get file size
    ls -lh "$BACKUP_DIR/$FILENAME"
else
    echo "Backup failed!"
    exit 1
fi

# Optional: Upload to S3
# aws s3 cp "$BACKUP_DIR/$FILENAME" s3://your-bucket/backups/

```

```

# Optional: Remove old backups (older than 30 days)
find $BACKUP_DIR -name "hospital_erp_*.sql" -mtime +30 -delete

echo "Backup process completed."
';

    RETURN v_script;
END;
$$ LANGUAGE plpgsql;

-- Generate restore script
CREATE OR REPLACE FUNCTION generate_restore_script(
    p_backup_file VARCHAR,
    p_target_db VARCHAR DEFAULT 'hospital_erp_restored'
)
RETURNS TEXT AS $$
DECLARE
    v_script TEXT;
BEGIN
    v_script := '#!/bin/bash
# Hospital ERP Database Restore Script
# Generated: ' || CURRENT_TIMESTAMP || '

BACKUP_FILE="' || p_backup_file || '"
TARGET_DB="' || p_target_db || '"

# Check if backup file exists
if [ ! -f "$BACKUP_FILE" ]; then
    echo "Backup file not found: $BACKUP_FILE"
    exit 1
fi

# Create target database
createdb -h localhost -U postgres $TARGET_DB

# Restore backup
pg_restore -h localhost -U postgres \
    --dbname=$TARGET_DB \
    --verbose \
    --clean \
    --if-exists \
    "$BACKUP_FILE"

if [ $? -eq 0 ]; then
    echo "Restore completed successfully to database: $TARGET_DB"
else
    echo "Restore failed!"
    exit 1
fi

# Verify restore
psql -h localhost -U postgres -d $TARGET_DB -c "SELECT COUNT(*) FROM patients;"

```

```

echo "Restore process completed."
';

    RETURN v_script;
END;
$$ LANGUAGE plpgsql;

-- =====
-- SECTION 71: DATABASE MAINTENANCE PROCEDURES
-- =====

-- Comprehensive maintenance procedure
CREATE OR REPLACE PROCEDURE perform_database_maintenance()
LANGUAGE plpgsql
AS $$
DECLARE
    v_start TIMESTAMPTZ := clock_timestamp();
    v_table RECORD;
BEGIN
    RAISE NOTICE 'Starting database maintenance at %', v_start;

    -- 1. Vacuum and Analyze all tables
    RAISE NOTICE 'Step 1: VACUUM ANALYZE...';
    FOR v_table IN
        SELECT tablename FROM pg_tables WHERE schemaname = 'public'
    LOOP
        EXECUTE 'VACUUM ANALYZE ' || quote_ident(v_table.tablename);
    END LOOP;

    -- 2. Reindex critical tables
    RAISE NOTICE 'Step 2: Reindexing critical tables...';
    REINDEX TABLE patients;
    REINDEX TABLE admissions;
    REINDEX TABLE appointments;
    REINDEX TABLE bills;
    REINDEX TABLE lab_orders;

    -- 3. Refresh materialized views
    RAISE NOTICE 'Step 3: Refreshing materialized views...';
    BEGIN
        REFRESH MATERIALIZED VIEW CONCURRENTLY mv_daily_census;
    EXCEPTION WHEN OTHERS THEN
        RAISE NOTICE 'Could not refresh mv_daily_census: %', SQLERRM;
    END;

    BEGIN
        REFRESH MATERIALIZED VIEW CONCURRENTLY mv_revenue_summary;
    EXCEPTION WHEN OTHERS THEN
        RAISE NOTICE 'Could not refresh mv_revenue_summary: %', SQLERRM;
    END;

    -- 4. Clean up expired sessions
    RAISE NOTICE 'Step 4: Cleaning expired sessions...';
    DELETE FROM user_sessions WHERE expires_at < CURRENT_TIMESTAMP;

```

```

DELETE FROM patient_portal_sessions WHERE expires_at < CURRENT_TIMESTAMP;

-- 5. Archive old audit logs (older than 90 days)
RAISE NOTICE 'Step 5: Processing old audit logs...';
-- In production, you would move to archive table
-- DELETE FROM audit_logs WHERE created_at < CURRENT_TIMESTAMP - INTERVAL '90
days';

-- 6. Update table statistics
RAISE NOTICE 'Step 6: Updating statistics...';
ANALYZE;

-- 7. Check for bloat
RAISE NOTICE 'Step 7: Checking for table bloat...';
-- Log tables with high bloat for manual attention

RAISE NOTICE 'Database maintenance completed in % seconds',
    EXTRACT(EPOCH FROM (clock_timestamp() - v_start));
END;
$$;

-- Procedure to purge old data
CREATE OR REPLACE PROCEDURE purge_old_data(
    p_organization_id UUID,
    p_cutoff_date DATE,
    p_dry_run BOOLEAN DEFAULT TRUE
)
LANGUAGE plpgsql
AS $$
DECLARE
    v_counts JSONB := '{} '::JSONB;
    v_count INTEGER;
BEGIN
    IF p_dry_run THEN
        RAISE NOTICE 'DRY RUN MODE - No data will be deleted';
    END IF;

    -- Count records to be deleted
    SELECT COUNT(*) INTO v_count
    FROM audit_logs
    WHERE organization_id = p_organization_id
    AND created_at < p_cutoff_date;
    v_counts := v_counts || jsonb_build_object('audit_logs', v_count);

    SELECT COUNT(*) INTO v_count
    FROM notification_log
    WHERE created_at < p_cutoff_date;
    v_counts := v_counts || jsonb_build_object('notification_log', v_count);

    SELECT COUNT(*) INTO v_count
    FROM api_request_log
    WHERE request_timestamp < p_cutoff_date;
    v_counts := v_counts || jsonb_build_object('api_request_log', v_count);

```



```

    RAISE NOTICE 'Records to be deleted: %', v_counts;

    IF NOT p_dry_run THEN
        -- Delete old audit logs
        DELETE FROM audit_logs
        WHERE organization_id = p_organization_id
        AND created_at < p_cutoff_date;

        -- Delete old notification logs
        DELETE FROM notification_log
        WHERE created_at < p_cutoff_date;

        -- Delete old API logs
        DELETE FROM api_request_log
        WHERE request_timestamp < p_cutoff_date;

        RAISE NOTICE 'Data purge completed';
    END IF;
END;
$$;

-- =====
-- SECTION 72: FINAL COMPLETION
-- =====

-- Display complete schema statistics
CREATE OR REPLACE FUNCTION display_final_statistics()
RETURNS VOID AS $$
DECLARE
    v_stats RECORD;
BEGIN
    RAISE NOTICE '';
    RAISE NOTICE
    '=====';
    RAISE NOTICE '||          HOSPITAL ERP POSTGRESQL SCHEMA - COMPLETE
    ||';
    RAISE NOTICE
    '=====';

    FOR v_stats IN SELECT * FROM get_schema_statistics() LOOP
        RAISE NOTICE '||  %-30s: %6s          ||',
v_stats.category, v_stats.count;
    END LOOP;

    RAISE NOTICE
    '=====';
    RAISE NOTICE '||
    ||';
    RAISE NOTICE '||  COMPLETE MODULES IMPLEMENTED:
    ||';
    RAISE NOTICE '||  _____
    ||';
    RAISE NOTICE '||  ☒ Multi-Tenancy & Organizations
    ||';

```

```

RAISE NOTICE '|| ☒ User Management & RBAC
||';
RAISE NOTICE '|| ☒ Patient Management
||';
RAISE NOTICE '|| ☒ Appointment & Scheduling
||';
RAISE NOTICE '|| ☒ OPD & Emergency Visits
||';
RAISE NOTICE '|| ☒ IPD & Admissions
||';
RAISE NOTICE '|| ☒ EMR & Clinical Documentation
||';
RAISE NOTICE '|| ☒ Nursing Station Management
||';
RAISE NOTICE '|| ☒ Bed & Ward Management
||';
RAISE NOTICE '|| ☒ Operation Theatre (OT)
||';
RAISE NOTICE '|| ☒ Laboratory Management
||';
RAISE NOTICE '|| ☒ Radiology Management
||';
RAISE NOTICE '|| ☒ Pharmacy Management
||';
RAISE NOTICE '|| ☒ Prescription (Rx) Management
||';
RAISE NOTICE '|| ☒ Inventory & Supply Chain
||';
RAISE NOTICE '|| ☒ Billing & Payments
||';
RAISE NOTICE '|| ☒ Insurance & Claims
||';
RAISE NOTICE '|| ☒ HR & Payroll
||';
RAISE NOTICE '|| ☒ Housekeeping Management
||';
RAISE NOTICE '|| ☒ Equipment & Maintenance
||';
RAISE NOTICE '|| ☒ Diet & Kitchen Management
||';
RAISE NOTICE '|| ☒ Telemedicine
||';
RAISE NOTICE '|| ☒ Complaints & Feedback
||';
RAISE NOTICE '|| ☒ Consent Management
||';
RAISE NOTICE '|| ☒ Ambulance Management
||';
RAISE NOTICE '|| ☒ Blood Bank
||';
RAISE NOTICE '|| ☒ Visitor Management
||';
RAISE NOTICE '|| ☒ Waste Management
||';

```

```
RAISE NOTICE '|| ☒ Reports & Dashboards
';
RAISE NOTICE '|| ☒ Notification Engine
';
RAISE NOTICE '|| ☒ Audit Logging
';
RAISE NOTICE '|| ☒ HL7/FHIR Integration
';
RAISE NOTICE '|| ☒ Clinical Decision Support
';
RAISE NOTICE '|| ☒ Quality Indicators
';
RAISE NOTICE '|| ☒ Workflow Engine
';
RAISE NOTICE '|| ☒ Document Management
';
RAISE NOTICE '|| ☒ Mobile App Support
';
RAISE NOTICE '|| ☒ Patient Portal
';
RAISE NOTICE '|| ☒ Referral Management
';
RAISE NOTICE '|| ☒ Research & Clinical Trials
';
RAISE NOTICE '|| ☒ Cost Accounting
';
RAISE NOTICE '|| ☒ Contract Management
';
RAISE NOTICE '||
';
RAISE NOTICE
';
RAISE NOTICE '|| DEPLOYMENT READY
';
RAISE NOTICE
';
RAISE NOTICE '';
END;
$$ LANGUAGE plpgsql;

-- Execute final display
SELECT display_final_statistics();
```



FINAL SUMMARY

Complete Hospital ERP Schema Statistics:

Component	Count
Total Tables	250+

Component	Count
Views	20+
Materialized Views	6
Indexes	450+
Functions	80+
Stored Procedures	15+
Triggers	45+
Custom Types (Enums)	25
Foreign Key Constraints	350+
Check Constraints	50+

All Modules Covered:

- 1. ☒ Multi-Tenancy & Organizations
- 2. ☒ User Management & RBAC
- 3. ☒ Patient Management
- 4. ☒ Appointment & Scheduling
- 5. ☒ OPD & Emergency Visits
- 6. ☒ IPD & Admissions
- 7. ☒ EMR & Clinical Documentation
- 8. ☒ Nursing Station Management
- 9. ☒ Bed & Ward Management
- 10. ☒ Operation Theatre (OT)
- 11. ☒ Laboratory Management
- 12. ☒ Radiology Management
- 13. ☒ Pharmacy Management
- 14. ☒ Prescription (Rx) Management
- 15. ☒ Inventory & Supply Chain
- 16. ☒ Billing & Payments
- 17. ☒ Insurance & Claims
- 18. ☒ HR & Payroll
- 19. ☒ Housekeeping Management
- 20. ☒ Equipment & Maintenance
- 21. ☒ Diet & Kitchen Management
- 22. ☒ Telemedicine
- 23. ☒ Complaints & Feedback
- 24. ☒ Consent Management
- 25. ☒ Ambulance Management
- 26. ☒ Blood Bank
- 27. ☒ Visitor Management
- 28. ☒ Waste Management
- 29. ☒ Reports & Dashboards

- 30. ☒ Notification Engine
- 31. ☒ Audit Logging
- 32. ☒ HL7/FHIR Integration
- 33. ☒ Clinical Decision Support
- 34. ☒ Quality Indicators
- 35. ☒ Workflow Engine
- 36. ☒ Document Management
- 37. ☒ Mobile App Support
- 38. ☒ Patient Portal
- 39. ☒ Referral Management
- 40. ☒ Research & Clinical Trials
- 41. ☒ Cost Accounting
- 42. ☒ Contract Management
- 43. ☒ Backup & Recovery
- 44. ☒ Sample Data Generation
- 45. ☒ Testing & Validation

Complete Hospital ERP PostgreSQL Schema - Report Generator Module

PART 24: COMPREHENSIVE REPORT GENERATOR

SQL

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-- =====
-- SECTION 73: REPORT GENERATOR FRAMEWORK
-- =====

-- Report Categories
CREATE TABLE report_categories (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID REFERENCES organizations(id),

    code VARCHAR(50) NOT NULL,
    name VARCHAR(100) NOT NULL,
    name_bangla VARCHAR(100),

    parent_category_id UUID REFERENCES report_categories(id),

    icon VARCHAR(50),
    display_order INTEGER DEFAULT 0,

    is_active BOOLEAN DEFAULT TRUE,
    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Insert default report categories
INSERT INTO report_categories (code, name, name_bangla, display_order) VALUES
```

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('CLINICAL', 'Clinical Reports', 'ক্লিনিক্যাল রিপোর্ট', 1),
('FINANCIAL', 'Financial Reports', 'আর্থিক রিপোর্ট', 2),
('OPERATIONAL', 'Operational Reports', 'অপারেশনাল রিপোর্ট', 3),
('INVENTORY', 'Inventory Reports', 'ইনভেন্টরি রিপোর্ট', 4),
('HR', 'HR Reports', 'এইচআর রিপোর্ট', 5),
('QUALITY', 'Quality Reports', 'কোয়ালিটি রিপোর্ট', 6),
('MIS', 'MIS Reports', 'এমআইএস রিপোর্ট', 7),
('REGULATORY', 'Regulatory Reports', 'রেগুলেটরি রিপোর্ট', 8),
('CUSTOM', 'Custom Reports', 'কাস্টম রিপোর্ট', 9)
ON CONFLICT DO NOTHING;

-- Enhanced Report Definitions
CREATE TABLE report_master (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID REFERENCES organizations(id), -- NULL for system reports

    report_code VARCHAR(50) NOT NULL,
    name VARCHAR(255) NOT NULL,
    name_bangla VARCHAR(255),
    description TEXT,

    category_id UUID REFERENCES report_categories(id),

    -- Report Type
    report_type VARCHAR(50) DEFAULT 'tabular', -- tabular, summary, chart, matrix,
form

    -- Data Source
    source_type VARCHAR(50) DEFAULT 'query', -- query, function, view, api
    source_query TEXT,
    source_function VARCHAR(100),
    source_view VARCHAR(100),

    -- Parameters
    parameters JSONB DEFAULT '[]',
/*
[
    {
        "name": "from_date",
        "label": "From Date",
        "label_bn": "তারিখ থেকে",
        "type": "date",
        "required": true,
        "default": "current_date - 30"
    },
    {
        "name": "branch_id",
        "label": "Branch",
        "type": "select",
        "source": "branches",
        "required": false
    }
]
*/

```

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-- Columns/Fields
columns JSONB DEFAULT '[]',
/*
[
  {
    "field": "patient_name",
    "header": "Patient Name",
    "header_bn": "রোগীর নাম",
    "type": "string",
    "width": 200,
    "sortable": true,
    "filterable": true,
    "aggregate": null
  },
  {
    "field": "amount",
    "header": "Amount",
    "type": "currency",
    "width": 120,
    "align": "right",
    "aggregate": "sum",
    "format": "#,##0.00"
  }
]
*/

-- Grouping & Sorting
default_group_by TEXT[],
default_sort_by TEXT[],
default_sort_order VARCHAR(10) DEFAULT 'ASC',

-- Aggregations
show_totals BOOLEAN DEFAULT TRUE,
show_subtotals BOOLEAN DEFAULT FALSE,

-- Layout
orientation VARCHAR(20) DEFAULT 'portrait', -- portrait, landscape
page_size VARCHAR(10) DEFAULT 'A4',

-- Header/Footer
header_template TEXT,
footer_template TEXT,

-- Export Options
export_formats TEXT[] DEFAULT ARRAY['pdf', 'excel', 'csv'],

-- Access Control
allowed_roles UUID[],
is_public BOOLEAN DEFAULT FALSE,

-- Scheduling
can_schedule BOOLEAN DEFAULT TRUE,

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-- Status
is_active BOOLEAN DEFAULT TRUE,
is_system BOOLEAN DEFAULT FALSE,

-- Metadata
version INTEGER DEFAULT 1,
created_by UUID REFERENCES users(id),
created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

UNIQUE(COALESCE(organization_id, '00000000-0000-0000-0000-000000000000'::UUID), report_code)
);

-- Report Execution Log
CREATE TABLE report_execution_log (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  organization_id UUID NOT NULL REFERENCES organizations(id),

  report_id UUID NOT NULL REFERENCES report_master(id),

  -- Execution Context
  execution_type VARCHAR(20) DEFAULT 'manual', -- manual, scheduled, api
  schedule_id UUID,

  -- User
  executed_by UUID REFERENCES users(id),

  -- Parameters Used
  parameters_used JSONB,

  -- Execution Details
  started_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
  completed_at TIMESTAMPTZ,
  duration_ms INTEGER,

  status VARCHAR(20) DEFAULT 'running', -- running, completed, failed, cancelled
  error_message TEXT,

  -- Results
  row_count INTEGER,

  -- Output
  output_format VARCHAR(20),
  output_file_path TEXT,
  output_file_size BIGINT,

  -- Cache
  cache_key VARCHAR(255),
  cache_expires_at TIMESTAMPTZ,

  -- Delivery
  delivered BOOLEAN DEFAULT FALSE,
  delivered_to TEXT[],

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    delivered_at TIMESTAMPTZ,

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- Report Templates (For custom formatting)
CREATE TABLE report_templates (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    organization_id UUID REFERENCES organizations(id),

    template_code VARCHAR(50) NOT NULL,
    name VARCHAR(100) NOT NULL,

    template_type VARCHAR(50), -- header, footer, full_page, letterhead

    -- Content
    html_template TEXT,
    css_styles TEXT,

    -- Variables
    available_variables JSONB,

    -- Images
    logo_url TEXT,
    watermark_url TEXT,

    is_default BOOLEAN DEFAULT FALSE,
    is_active BOOLEAN DEFAULT TRUE,

    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,

    UNIQUE(COALESCE(organization_id, '00000000-0000-0000-0000-0000-000000000000'::UUID), template_code)
);

-- =====
-- SECTION 74: CLINICAL REPORTS
-- =====

-- Daily OPD Report Function
CREATE OR REPLACE FUNCTION rpt_daily_opd_summary(
    p_organization_id UUID,
    p_branch_id UUID DEFAULT NULL,
    p_date DATE DEFAULT CURRENT_DATE
)
RETURNS TABLE(
    department VARCHAR,
    doctor_name VARCHAR,
    specialization VARCHAR,
    total_appointments INTEGER,
    completed INTEGER,
    cancelled INTEGER,
    no_show INTEGER,
    new_patients INTEGER,

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        followup_patients INTEGER,
        telemedicine INTEGER,
        total_revenue DECIMAL,
        avg_wait_time_minutes DECIMAL
    ) AS $$
BEGIN
    RETURN QUERY
    SELECT
        d.name::VARCHAR as department,
        u.full_name::VARCHAR as doctor_name,
        doc.specialization::VARCHAR,
        COUNT(*)::INTEGER as total_appointments,
        COUNT(*) FILTER (WHERE a.status = 'completed')::INTEGER as completed,
        COUNT(*) FILTER (WHERE a.status = 'cancelled')::INTEGER as cancelled,
        COUNT(*) FILTER (WHERE a.status = 'no_show')::INTEGER as no_show,
        COUNT(*) FILTER (WHERE a.appointment_type = 'new')::INTEGER as
new_patients,
        COUNT(*) FILTER (WHERE a.appointment_type = 'followup')::INTEGER as
followup_patients,
        COUNT(*) FILTER (WHERE a.consultation_type = 'telemedicine')::INTEGER as
telemedicine,
        COALESCE(SUM(a.consultation_fee), 0)::DECIMAL as total_revenue,
        ROUND(AVG(EXTRACT(EPOCH FROM (a.consultation_start_at -
a.checked_in_at))/60), 1)::DECIMAL as avg_wait_time_minutes
    FROM appointments a
    JOIN doctors doc ON a.doctor_id = doc.id
    JOIN users u ON doc.user_id = u.id
    LEFT JOIN departments d ON u.department_id = d.id
    WHERE a.organization_id = p_organization_id
    AND a.appointment_date = p_date
    AND (p_branch_id IS NULL OR a.branch_id = p_branch_id)
    GROUP BY d.name, u.full_name, doc.specialization
    ORDER BY d.name, u.full_name;
END;
$$ LANGUAGE plpgsql;

-- Daily IPD Census Report Function
CREATE OR REPLACE FUNCTION rpt_daily_ipd_census(
    p_organization_id UUID,
    p_branch_id UUID DEFAULT NULL,
    p_date DATE DEFAULT CURRENT_DATE
)
RETURNS TABLE(
    ward_name VARCHAR,
    ward_type VARCHAR,
    total_beds INTEGER,
    occupied_beds INTEGER,
    available_beds INTEGER,
    reserved_beds INTEGER,
    under_cleaning INTEGER,
    occupancy_rate DECIMAL,
    admissions_today INTEGER,
    discharges_today INTEGER,
    transfers_in INTEGER,

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        transfers_out INTEGER,
        deaths_today INTEGER,
        critical_patients INTEGER,
        average_los DECIMAL
    ) AS $$
BEGIN
    RETURN QUERY
    SELECT
        w.name::VARCHAR as ward_name,
        w.ward_type::VARCHAR,
        COUNT(b.id)::INTEGER as total_beds,
        COUNT(b.id) FILTER (WHERE b.status = 'occupied')::INTEGER as
occupied_beds,
        COUNT(b.id) FILTER (WHERE b.status = 'available')::INTEGER as
available_beds,
        COUNT(b.id) FILTER (WHERE b.status = 'reserved')::INTEGER as
reserved_beds,
        COUNT(b.id) FILTER (WHERE b.status = 'cleaning')::INTEGER as
under_cleaning,
        ROUND(COUNT(b.id) FILTER (WHERE b.status = 'occupied')::DECIMAL /
            NULLIF(COUNT(b.id), 0) * 100, 1)::DECIMAL as occupancy_rate,
        (SELECT COUNT(*) FROM admissions a2
        JOIN bed_allocations ba2 ON a2.id = ba2.admission_id
        WHERE ba2.ward_id = w.id AND a2.admission_date = p_date)::INTEGER as
admissions_today,
        (SELECT COUNT(*) FROM admissions a3
        JOIN bed_allocations ba3 ON a3.id = ba3.admission_id
        WHERE ba3.ward_id = w.id AND a3.actual_discharge_date = p_date)::INTEGER
as discharges_today,
        (SELECT COUNT(*) FROM bed_transfer_requests btr
        WHERE btr.to_ward_id = w.id AND DATE(btr.transferred_at) = p_date
        AND btr.status = 'completed')::INTEGER as transfers_in,
        (SELECT COUNT(*) FROM bed_transfer_requests btr2
        WHERE btr2.from_ward_id = w.id AND DATE(btr2.transferred_at) = p_date
        AND btr2.status = 'completed')::INTEGER as transfers_out,
        (SELECT COUNT(*) FROM admissions a4
        JOIN bed_allocations ba4 ON a4.id = ba4.admission_id
        WHERE ba4.ward_id = w.id AND a4.actual_discharge_date = p_date
        AND a4.discharge_type = 'expired')::INTEGER as deaths_today,
        (SELECT COUNT(*) FROM admissions a5
        JOIN bed_allocations ba5 ON a5.id = ba5.admission_id
        WHERE ba5.ward_id = w.id AND a5.status = 'admitted'
        AND ba5.release_datetime IS NULL
        AND EXISTS (
            SELECT 1 FROM patient_vitals pv
            WHERE pv.admission_id = a5.id
            AND pv.recorded_datetime > CURRENT_TIMESTAMP - INTERVAL '24 hours'
            AND (pv.blood_pressure_systolic < 90 OR pv.spo2 < 90)
        ))::INTEGER as critical_patients,
        ROUND(AVG(CASE
            WHEN a.actual_discharge_date IS NOT NULL
            THEN a.actual_discharge_date - a.admission_date
            ELSE CURRENT_DATE - a.admission_date
        END), 1)::DECIMAL as average_los

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FROM wards w
LEFT JOIN beds b ON w.id = b.ward_id AND b.is_active = TRUE
LEFT JOIN admissions a ON b.current_admission_id = a.id
WHERE w.organization_id = p_organization_id
AND (p_branch_id IS NULL OR w.branch_id = p_branch_id)
AND w.is_active = TRUE
GROUP BY w.id, w.name, w.ward_type
ORDER BY w.name;
END;
$$ LANGUAGE plpgsql;

-- Patient Statistics Report
CREATE OR REPLACE FUNCTION rpt_patient_statistics(
    p_organization_id UUID,
    p_branch_id UUID DEFAULT NULL,
    p_from_date DATE DEFAULT DATE_TRUNC('month', CURRENT_DATE)::DATE,
    p_to_date DATE DEFAULT CURRENT_DATE
)
RETURNS TABLE(
    metric VARCHAR,
    value BIGINT,
    previous_period_value BIGINT,
    change_percent DECIMAL
) AS $$
DECLARE
    v_period_days INTEGER;
    v_prev_from DATE;
    v_prev_to DATE;
BEGIN
    v_period_days := p_to_date - p_from_date + 1;
    v_prev_from := p_from_date - v_period_days;
    v_prev_to := p_from_date - 1;

    RETURN QUERY

-- New Registrations
SELECT
    'New Patient Registrations'::VARCHAR,
    COUNT(*) FILTER (WHERE p.registration_date BETWEEN p_from_date AND
p_to_date),
    COUNT(*) FILTER (WHERE p.registration_date BETWEEN v_prev_from AND
v_prev_to),
    ROUND(
        (COUNT(*) FILTER (WHERE p.registration_date BETWEEN p_from_date AND
p_to_date) -
        COUNT(*) FILTER (WHERE p.registration_date BETWEEN v_prev_from AND
v_prev_to))::DECIMAL /
        NULLIF(COUNT(*) FILTER (WHERE p.registration_date BETWEEN v_prev_from
AND v_prev_to), 0) * 100, 1
    )
FROM patients p
WHERE p.organization_id = p_organization_id
AND (p_branch_id IS NULL OR p.registered_branch_id = p_branch_id)

```

```

UNION ALL

-- OPD Visits
SELECT
    'OPD Visits'::VARCHAR,
    COUNT(*) FILTER (WHERE a.appointment_date BETWEEN p_from_date AND
p_to_date),
    COUNT(*) FILTER (WHERE a.appointment_date BETWEEN v_prev_from AND
v_prev_to),
    ROUND(
        (COUNT(*) FILTER (WHERE a.appointment_date BETWEEN p_from_date AND
p_to_date) -
        COUNT(*) FILTER (WHERE a.appointment_date BETWEEN v_prev_from AND
v_prev_to))::DECIMAL /
        NULLIF(COUNT(*) FILTER (WHERE a.appointment_date BETWEEN v_prev_from
AND v_prev_to), 0) * 100, 1
    )
FROM appointments a
WHERE a.organization_id = p_organization_id
AND (p_branch_id IS NULL OR a.branch_id = p_branch_id)
AND a.status = 'completed'

UNION ALL

-- Admissions
SELECT
    'IPD Admissions'::VARCHAR,
    COUNT(*) FILTER (WHERE ad.admission_date BETWEEN p_from_date AND
p_to_date),
    COUNT(*) FILTER (WHERE ad.admission_date BETWEEN v_prev_from AND
v_prev_to),
    ROUND(
        (COUNT(*) FILTER (WHERE ad.admission_date BETWEEN p_from_date AND
p_to_date) -
        COUNT(*) FILTER (WHERE ad.admission_date BETWEEN v_prev_from AND
v_prev_to))::DECIMAL /
        NULLIF(COUNT(*) FILTER (WHERE ad.admission_date BETWEEN v_prev_from
AND v_prev_to), 0) * 100, 1
    )
FROM admissions ad
WHERE ad.organization_id = p_organization_id
AND (p_branch_id IS NULL OR ad.branch_id = p_branch_id)

UNION ALL

-- Discharges
SELECT
    'IPD Discharges'::VARCHAR,
    COUNT(*) FILTER (WHERE ad.actual_discharge_date BETWEEN p_from_date AND
p_to_date),
    COUNT(*) FILTER (WHERE ad.actual_discharge_date BETWEEN v_prev_from AND
v_prev_to),
    ROUND(
        (COUNT(*) FILTER (WHERE ad.actual_discharge_date BETWEEN p_from_date

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```

AND p_to_date) -
        COUNT(*) FILTER (WHERE ad.actual_discharge_date BETWEEN v_prev_from
AND v_prev_to))::DECIMAL /
        NULLIF(COUNT(*) FILTER (WHERE ad.actual_discharge_date BETWEEN
v_prev_from AND v_prev_to), 0) * 100, 1
    )
    FROM admissions ad
    WHERE ad.organization_id = p_organization_id
    AND (p_branch_id IS NULL OR ad.branch_id = p_branch_id)

    UNION ALL

    -- Emergency Visits
    SELECT
        'Emergency Visits'::VARCHAR,
        COUNT(*) FILTER (WHERE DATE(ev.arrival_datetime) BETWEEN p_from_date AND
p_to_date),
        COUNT(*) FILTER (WHERE DATE(ev.arrival_datetime) BETWEEN v_prev_from AND
v_prev_to),
        ROUND(
            (COUNT(*) FILTER (WHERE DATE(ev.arrival_datetime) BETWEEN p_from_date
AND p_to_date) -
                COUNT(*) FILTER (WHERE DATE(ev.arrival_datetime) BETWEEN v_prev_from
AND v_prev_to))::DECIMAL /
            NULLIF(COUNT(*) FILTER (WHERE DATE(ev.arrival_datetime) BETWEEN
v_prev_from AND v_prev_to), 0) * 100, 1
        )
    FROM emergency_visits ev
    WHERE ev.organization_id = p_organization_id
    AND (p_branch_id IS NULL OR ev.branch_id = p_branch_id)

    UNION ALL

    -- Surgeries
    SELECT
        'Surgeries Performed'::VARCHAR,
        COUNT(*) FILTER (WHERE s.scheduled_date BETWEEN p_from_date AND p_to_date
AND s.status = 'completed'),
        COUNT(*) FILTER (WHERE s.scheduled_date BETWEEN v_prev_from AND v_prev_to
AND s.status = 'completed'),
        ROUND(
            (COUNT(*) FILTER (WHERE s.scheduled_date BETWEEN p_from_date AND
p_to_date AND s.status = 'completed') -
                COUNT(*) FILTER (WHERE s.scheduled_date BETWEEN v_prev_from AND
v_prev_to AND s.status = 'completed'))::DECIMAL /
            NULLIF(COUNT(*) FILTER (WHERE s.scheduled_date BETWEEN v_prev_from AND
v_prev_to AND s.status = 'completed'), 0) * 100, 1
        )
    FROM surgeries s
    WHERE s.organization_id = p_organization_id
    AND (p_branch_id IS NULL OR s.branch_id = p_branch_id);
END;
$$ LANGUAGE plpgsql;

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-- Disease-wise Statistics (ICD)
CREATE OR REPLACE FUNCTION rpt_disease_statistics(
    p_organization_id UUID,
    p_branch_id UUID DEFAULT NULL,
    p_from_date DATE DEFAULT DATE_TRUNC('month', CURRENT_DATE)::DATE,
    p_to_date DATE DEFAULT CURRENT_DATE,
    p_top_n INTEGER DEFAULT 20
)
RETURNS TABLE(
    rank INTEGER,
    icd_code VARCHAR,
    disease_name VARCHAR,
    opd_cases BIGINT,
    ipd_cases BIGINT,
    total_cases BIGINT,
    percentage DECIMAL,
    avg_los DECIMAL,
    mortality_count BIGINT
) AS $$
BEGIN
    RETURN QUERY
    WITH disease_counts AS (
        SELECT
            ic.code,
            ic.short_description,
            COUNT(*) FILTER (WHERE cd.visit_id IS NOT NULL) as opd,
            COUNT(*) FILTER (WHERE cd.admission_id IS NOT NULL) as ipd,
            COUNT(*) as total,
            AVG(CASE
                WHEN a.actual_discharge_date IS NOT NULL
                THEN a.actual_discharge_date - a.admission_date
            END) as avg_length_of_stay,
            COUNT(*) FILTER (WHERE a.discharge_type = 'expired') as deaths
        FROM clinical_diagnoses cd
        JOIN icd_codes ic ON cd.icd_code_id = ic.id
        LEFT JOIN admissions a ON cd.admission_id = a.id
        WHERE cd.diagnosed_at BETWEEN p_from_date AND (p_to_date + 1)
        AND EXISTS (
            SELECT 1 FROM patients p
            WHERE p.id = cd.patient_id
            AND p.organization_id = p_organization_id
        )
        GROUP BY ic.code, ic.short_description
    )
    SELECT
        ROW_NUMBER() OVER (ORDER BY dc.total DESC)::INTEGER as rank,
        dc.code::VARCHAR,
        dc.short_description::VARCHAR,
        dc.opd,
        dc.ipd,
        dc.total,
        ROUND(dc.total::DECIMAL / SUM(dc.total) OVER () * 100, 2) as percentage,
        ROUND(dc.avg_length_of_stay, 1) as avg_los,
        dc.deaths

```

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    FROM disease_counts dc
    ORDER BY dc.total DESC
    LIMIT p_top_n;
END;
$$ LANGUAGE plpgsql;

-- Doctor Performance Report
CREATE OR REPLACE FUNCTION rpt_doctor_performance(
    p_organization_id UUID,
    p_branch_id UUID DEFAULT NULL,
    p_from_date DATE DEFAULT DATE_TRUNC('month', CURRENT_DATE)::DATE,
    p_to_date DATE DEFAULT CURRENT_DATE
)
RETURNS TABLE(
    doctor_id UUID,
    doctor_name VARCHAR,
    specialization VARCHAR,
    department VARCHAR,
    opd_patients INTEGER,
    ipd_patients INTEGER,
    surgeries INTEGER,
    total_revenue DECIMAL,
    avg_rating DECIMAL,
    avg_consultation_time INTEGER,
    followup_rate DECIMAL,
    patient_satisfaction DECIMAL
) AS $$
BEGIN
    RETURN QUERY
    SELECT
        d.id as doctor_id,
        u.full_name::VARCHAR as doctor_name,
        d.specialization::VARCHAR,
        dep.name::VARCHAR as department,
        (SELECT COUNT(*) FROM appointments a
         WHERE a.doctor_id = d.id
         AND a.appointment_date BETWEEN p_from_date AND p_to_date
         AND a.status = 'completed')::INTEGER as opd_patients,
        (SELECT COUNT(DISTINCT ad.id) FROM admissions ad
         WHERE ad.primary_doctor_id = d.id
         AND ad.admission_date BETWEEN p_from_date AND p_to_date)::INTEGER as
        ipd_patients,
        (SELECT COUNT(*) FROM surgeries s
         WHERE s.primary_surgeon_id = d.id
         AND s.scheduled_date BETWEEN p_from_date AND p_to_date
         AND s.status = 'completed')::INTEGER as surgeries,
        COALESCE((SELECT SUM(bi.net_amount) FROM bill_items bi
         JOIN bills b ON bi.bill_id = b.id
         WHERE bi.doctor_id = d.id
         AND b.bill_date BETWEEN p_from_date AND p_to_date
         AND b.is_cancelled = FALSE), 0)::DECIMAL as total_revenue,
        (SELECT ROUND(AVG(ps.doctor_rating), 2) FROM patient_surveys ps
         JOIN appointments a ON ps.visit_id::TEXT = a.id::TEXT
         WHERE a.doctor_id = d.id

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        AND DATE(ps.submitted_at) BETWEEN p_from_date AND p_to_date)::DECIMAL as
avg_rating,
    (SELECT ROUND(AVG(EXTRACT(EPOCH FROM (a.consultation_end_at -
a.consultation_start_at)))/60))
        FROM appointments a
        WHERE a.doctor_id = d.id
        AND a.appointment_date BETWEEN p_from_date AND p_to_date
        AND a.consultation_start_at IS NOT NULL
        AND a.consultation_end_at IS NOT NULL)::INTEGER as avg_consultation_time,
    (SELECT ROUND(
        COUNT(*) FILTER (WHERE a.appointment_type = 'followup')::DECIMAL /
        NULLIF(COUNT(*), 0) * 100, 1)
        FROM appointments a
        WHERE a.doctor_id = d.id
        AND a.appointment_date BETWEEN p_from_date AND p_to_date)::DECIMAL as
followup_rate,
    (SELECT ROUND(AVG(ps.overall_rating), 2) FROM patient_surveys ps
        JOIN appointments a ON ps.visit_id::TEXT = a.id::TEXT
        WHERE a.doctor_id = d.id
        AND DATE(ps.submitted_at) BETWEEN p_from_date AND p_to_date)::DECIMAL as
patient_satisfaction
    FROM doctors d
    JOIN users u ON d.user_id = u.id
    LEFT JOIN departments dep ON u.department_id = dep.id
    WHERE d.organization_id = p_organization_id
    AND d.is_active = TRUE
    ORDER BY opd_patients DESC, ipd_patients DESC;
END;
$$ LANGUAGE plpgsql;

-- Lab Turnaround Time Report
CREATE OR REPLACE FUNCTION rpt_lab_tat_analysis(
    p_organization_id UUID,
    p_branch_id UUID DEFAULT NULL,
    p_from_date DATE DEFAULT DATE_TRUNC('month', CURRENT_DATE)::DATE,
    p_to_date DATE DEFAULT CURRENT_DATE
)
RETURNS TABLE(
    department VARCHAR,
    test_category VARCHAR,
    total_tests BIGINT,
    avg_tat_hours DECIMAL,
    median_tat_hours DECIMAL,
    within_target INTEGER,
    within_target_pct DECIMAL,
    exceeded_target INTEGER,
    critical_results INTEGER,
    critical_notified_on_time INTEGER
) AS $$
BEGIN
    RETURN QUERY
    SELECT
        tc.department::VARCHAR,
        tc.category::VARCHAR,

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        COUNT(*)::BIGINT as total_tests,
        ROUND(AVG(EXTRACT(EPOCH FROM (lr.result_datetime -
lo.order_datetime))/3600), 1)::DECIMAL as avg_tat_hours,
        ROUND(PERCENTILE_CONT(0.5) WITHIN GROUP (
            ORDER BY EXTRACT(EPOCH FROM (lr.result_datetime -
lo.order_datetime))/3600
        ), 1)::DECIMAL as median_tat_hours,
        COUNT(*) FILTER (WHERE
            EXTRACT(EPOCH FROM (lr.result_datetime - lo.order_datetime))/3600 <=
tc.tat_routine_hours
        )::INTEGER as within_target,
        ROUND(COUNT(*) FILTER (WHERE
            EXTRACT(EPOCH FROM (lr.result_datetime - lo.order_datetime))/3600 <=
tc.tat_routine_hours
        )::DECIMAL / NULLIF(COUNT(*), 0) * 100, 1)::DECIMAL as within_target_pct,
        COUNT(*) FILTER (WHERE
            EXTRACT(EPOCH FROM (lr.result_datetime - lo.order_datetime))/3600 >
tc.tat_routine_hours
        )::INTEGER as exceeded_target,
        COUNT(*) FILTER (WHERE lr.is_critical = TRUE)::INTEGER as
critical_results,
        COUNT(*) FILTER (WHERE
            lr.is_critical = TRUE AND lr.critical_notified = TRUE
            AND EXTRACT(EPOCH FROM (lr.critical_notified_at -
lr.result_datetime))/60 <= 30
        )::INTEGER as critical_notified_on_time
    FROM lab_orders lo
    JOIN lab_order_items loi ON lo.id = loi.order_id
    JOIN test_catalog tc ON loi.test_id = tc.id
    LEFT JOIN lab_results lr ON loi.id = lr.order_item_id
    WHERE lo.organization_id = p_organization_id
    AND (p_branch_id IS NULL OR lo.branch_id = p_branch_id)
    AND DATE(lo.order_datetime) BETWEEN p_from_date AND p_to_date
    AND lr.result_datetime IS NOT NULL
    GROUP BY tc.department, tc.category
    ORDER BY tc.department, tc.category;
END;
$$ LANGUAGE plpgsql;

-- =====
-- SECTION 75: FINANCIAL REPORTS
-- =====

-- Daily Collection Report
CREATE OR REPLACE FUNCTION rpt_daily_collection(
    p_organization_id UUID,
    p_branch_id UUID DEFAULT NULL,
    p_date DATE DEFAULT CURRENT_DATE
)
RETURNS TABLE(
    collection_point VARCHAR,
    cash_amount DECIMAL,
    card_amount DECIMAL,
    upi_amount DECIMAL,

```

```

mobile_banking DECIMAL,
cheque_amount DECIMAL,
bank_transfer DECIMAL,
insurance_amount DECIMAL,
corporate_amount DECIMAL,
total_amount DECIMAL,
transaction_count INTEGER,
refund_amount DECIMAL,
net_collection DECIMAL
) AS $$
BEGIN
    RETURN QUERY
    SELECT
        COALESCE(d.name, 'General')::VARCHAR as collection_point,
        COALESCE(SUM(p.amount) FILTER (WHERE p.payment_method = 'cash'),
0)::DECIMAL as cash_amount,
        COALESCE(SUM(p.amount) FILTER (WHERE p.payment_method = 'card'),
0)::DECIMAL as card_amount,
        COALESCE(SUM(p.amount) FILTER (WHERE p.payment_method = 'upi'),
0)::DECIMAL as upi_amount,
        COALESCE(SUM(p.amount) FILTER (WHERE p.payment_method = 'mobile_banking'),
0)::DECIMAL as mobile_banking,
        COALESCE(SUM(p.amount) FILTER (WHERE p.payment_method = 'cheque'),
0)::DECIMAL as cheque_amount,
        COALESCE(SUM(p.amount) FILTER (WHERE p.payment_method = 'bank_transfer'),
0)::DECIMAL as bank_transfer,
        COALESCE(SUM(p.amount) FILTER (WHERE p.payment_method = 'insurance'),
0)::DECIMAL as insurance_amount,
        COALESCE(SUM(p.amount) FILTER (WHERE p.payment_method = 'corporate'),
0)::DECIMAL as corporate_amount,
        COALESCE(SUM(p.amount) FILTER (WHERE NOT p.is_refund), 0)::DECIMAL as
total_amount,
        COUNT(*) FILTER (WHERE NOT p.is_refund)::INTEGER as transaction_count,
        COALESCE(SUM(p.amount) FILTER (WHERE p.is_refund), 0)::DECIMAL as
refund_amount,
        COALESCE(SUM(CASE WHEN p.is_refund THEN -p.amount ELSE p.amount END),
0)::DECIMAL as net_collection
    FROM payments p
    LEFT JOIN bills b ON p.bill_id = b.id
    LEFT JOIN departments d ON b.visit_id IS NOT NULL -- This is simplified;
adjust based on actual logic
    WHERE p.organization_id = p_organization_id
    AND (p_branch_id IS NULL OR p_branch_id = p_branch_id)
    AND p.payment_date = p_date
    AND p.status = 'completed'
    GROUP BY d.name
    ORDER BY total_amount DESC;
END;
$$ LANGUAGE plpgsql;

-- Revenue Analysis Report
CREATE OR REPLACE FUNCTION rpt_revenue_analysis(
    p_organization_id UUID,
    p_branch_id UUID DEFAULT NULL,

```

```

    p_from_date DATE DEFAULT DATE_TRUNC('month', CURRENT_DATE)::DATE,
    p_to_date DATE DEFAULT CURRENT_DATE
)
RETURNS TABLE(
    category VARCHAR,
    sub_category VARCHAR,
    gross_revenue DECIMAL,
    discount DECIMAL,
    net_revenue DECIMAL,
    cost DECIMAL,
    margin DECIMAL,
    margin_percent DECIMAL,
    transaction_count BIGINT,
    avg_ticket_size DECIMAL
) AS $$
BEGIN
    RETURN QUERY
    SELECT
        bi.category::VARCHAR,
        COALESCE(sm.sub_category, 'Other')::VARCHAR as sub_category,
        SUM(bi.quantity * bi.unit_price)::DECIMAL as gross_revenue,
        SUM(bi.discount_amount)::DECIMAL as discount,
        SUM(bi.net_amount)::DECIMAL as net_revenue,
        COALESCE(SUM(psc.total_cost), 0)::DECIMAL as cost,
        (SUM(bi.net_amount) - COALESCE(SUM(psc.total_cost), 0))::DECIMAL as
margin,
        ROUND((SUM(bi.net_amount) - COALESCE(SUM(psc.total_cost), 0)) /
            NULLIF(SUM(bi.net_amount), 0) * 100, 2)::DECIMAL as margin_percent,
        COUNT(*)::BIGINT as transaction_count,
        ROUND(SUM(bi.net_amount) / NULLIF(COUNT(*), 0), 2)::DECIMAL as
avg_ticket_size
    FROM bills b
    JOIN bill_items bi ON b.id = bi.bill_id
    LEFT JOIN service_master sm ON bi.service_id = sm.id
    LEFT JOIN patient_service_costs psc ON bi.service_id = psc.service_id
        AND b.patient_id = psc.patient_id
        AND bi.service_date = psc.service_date
    WHERE b.organization_id = p_organization_id
    AND (p_branch_id IS NULL OR b.branch_id = p_branch_id)
    AND b.bill_date BETWEEN p_from_date AND p_to_date
    AND b.is_cancelled = FALSE
    GROUP BY bi.category, sm.sub_category
    ORDER BY net_revenue DESC;
END;
$$ LANGUAGE plpgsql;

-- Outstanding Receivables Report
CREATE OR REPLACE FUNCTION rpt_outstanding_receivables(
    p_organization_id UUID,
    p_branch_id UUID DEFAULT NULL,
    p_as_of_date DATE DEFAULT CURRENT_DATE
)
RETURNS TABLE(
    receivable_type VARCHAR,

```

```

    party_name VARCHAR,
    total_billed DECIMAL,
    total_received DECIMAL,
    outstanding DECIMAL,
    aging_0_30 DECIMAL,
    aging_31_60 DECIMAL,
    aging_61_90 DECIMAL,
    aging_over_90 DECIMAL,
    oldest_invoice_date DATE,
    last_payment_date DATE
) AS $$
BEGIN
    RETURN QUERY

    -- Patient Receivables
    SELECT
        'Patient'::VARCHAR as receivable_type,
        p.full_name::VARCHAR as party_name,
        SUM(b.net_amount)::DECIMAL as total_billed,
        SUM(b.paid_amount)::DECIMAL as total_received,
        SUM(b.balance_amount)::DECIMAL as outstanding,
        SUM(b.balance_amount) FILTER (WHERE p_as_of_date - b.bill_date <=
30)::DECIMAL as aging_0_30,
        SUM(b.balance_amount) FILTER (WHERE p_as_of_date - b.bill_date BETWEEN 31
AND 60)::DECIMAL as aging_31_60,
        SUM(b.balance_amount) FILTER (WHERE p_as_of_date - b.bill_date BETWEEN 61
AND 90)::DECIMAL as aging_61_90,
        SUM(b.balance_amount) FILTER (WHERE p_as_of_date - b.bill_date >
90)::DECIMAL as aging_over_90,
        MIN(b.bill_date)::DATE as oldest_invoice_date,
        (SELECT MAX(py.payment_date) FROM payments py WHERE py.patient_id =
p.id)::DATE as last_payment_date
    FROM bills b
    JOIN patients p ON b.patient_id = p.id
    WHERE b.organization_id = p_organization_id
    AND (p_branch_id IS NULL OR b.branch_id = p_branch_id)
    AND b.is_cancelled = FALSE
    AND b.payment_status IN ('pending', 'partial')
    AND NOT b.is_insurance
    AND NOT b.is_corporate
    GROUP BY p.id, p.full_name
    HAVING SUM(b.balance_amount) > 0

    UNION ALL

    -- Insurance Receivables
    SELECT
        'Insurance'::VARCHAR as receivable_type,
        ic.name::VARCHAR as party_name,
        SUM(ic2.net_claim_amount)::DECIMAL as total_billed,
        SUM(ic2.settled_amount)::DECIMAL as total_received,
        SUM(ic2.net_claim_amount - COALESCE(ic2.settled_amount, 0))::DECIMAL as
outstanding,
        SUM(ic2.net_claim_amount - COALESCE(ic2.settled_amount, 0)) FILTER (

```

```

        WHERE p_as_of_date - ic2.submitted_at::DATE <= 30)::DECIMAL as
aging_0_30,
        SUM(ic2.net_claim_amount - COALESCE(ic2.settled_amount, 0)) FILTER (
        WHERE p_as_of_date - ic2.submitted_at::DATE BETWEEN 31 AND
60)::DECIMAL as aging_31_60,
        SUM(ic2.net_claim_amount - COALESCE(ic2.settled_amount, 0)) FILTER (
        WHERE p_as_of_date - ic2.submitted_at::DATE BETWEEN 61 AND
90)::DECIMAL as aging_61_90,
        SUM(ic2.net_claim_amount - COALESCE(ic2.settled_amount, 0)) FILTER (
        WHERE p_as_of_date - ic2.submitted_at::DATE > 90)::DECIMAL as
aging_over_90,
        MIN(ic2.submitted_at::DATE)::DATE as oldest_invoice_date,
        MAX(ic2.settlement_date)::DATE as last_payment_date
FROM insurance_claims ic2
JOIN insurance_companies ic ON ic2.insurance_company_id = ic.id
WHERE ic2.organization_id = p_organization_id
AND (p_branch_id IS NULL OR ic2.branch_id = p_branch_id)
AND ic2.status NOT IN ('settled', 'rejected')
GROUP BY ic.id, ic.name
HAVING SUM(ic2.net_claim_amount - COALESCE(ic2.settled_amount, 0)) > 0

UNION ALL

-- Corporate Receivables
SELECT
    'Corporate'::VARCHAR as receivable_type,
    ca.company_name::VARCHAR as party_name,
    SUM(ci.total_amount)::DECIMAL as total_billed,
    SUM(ci.paid_amount)::DECIMAL as total_received,
    SUM(ci.balance_amount)::DECIMAL as outstanding,
    SUM(ci.balance_amount) FILTER (WHERE p_as_of_date - ci.invoice_date <=
30)::DECIMAL as aging_0_30,
    SUM(ci.balance_amount) FILTER (WHERE p_as_of_date - ci.invoice_date
BETWEEN 31 AND 60)::DECIMAL as aging_31_60,
    SUM(ci.balance_amount) FILTER (WHERE p_as_of_date - ci.invoice_date
BETWEEN 61 AND 90)::DECIMAL as aging_61_90,
    SUM(ci.balance_amount) FILTER (WHERE p_as_of_date - ci.invoice_date >
90)::DECIMAL as aging_over_90,
    MIN(ci.invoice_date)::DATE as oldest_invoice_date,
    MAX(ci.invoice_date) FILTER (WHERE ci.status = 'paid')::DATE as
last_payment_date
FROM corporate_invoices ci
JOIN corporate_accounts ca ON ci.corporate_id = ca.id
WHERE ci.organization_id = p_organization_id
AND ci.status NOT IN ('paid')
GROUP BY ca.id, ca.company_name
HAVING SUM(ci.balance_amount) > 0

ORDER BY outstanding DESC;
END;
$$ LANGUAGE plpgsql;

-- Profit & Loss Summary
CREATE OR REPLACE FUNCTION rpt_profit_loss_summary(

```

```

    p_organization_id UUID,
    p_branch_id UUID DEFAULT NULL,
    p_from_date DATE DEFAULT DATE_TRUNC('month', CURRENT_DATE)::DATE,
    p_to_date DATE DEFAULT CURRENT_DATE
)
RETURNS TABLE(
    line_item VARCHAR,
    category VARCHAR,
    amount DECIMAL,
    percentage DECIMAL
) AS $$
DECLARE
    v_total_revenue DECIMAL;
BEGIN
    -- Calculate total revenue first
    SELECT COALESCE(SUM(bi.net_amount), 0) INTO v_total_revenue
    FROM bills b
    JOIN bill_items bi ON b.id = bi.bill_id
    WHERE b.organization_id = p_organization_id
    AND (p_branch_id IS NULL OR b.branch_id = p_branch_id)
    AND b.bill_date BETWEEN p_from_date AND p_to_date
    AND b.is_cancelled = FALSE;

    RETURN QUERY

    -- Revenue
    SELECT
        'Revenue'::VARCHAR as line_item,
        bi.category::VARCHAR,
        SUM(bi.net_amount)::DECIMAL as amount,
        ROUND(SUM(bi.net_amount) / NULLIF(v_total_revenue, 0) * 100, 2)::DECIMAL
as percentage
    FROM bills b
    JOIN bill_items bi ON b.id = bi.bill_id
    WHERE b.organization_id = p_organization_id
    AND (p_branch_id IS NULL OR b.branch_id = p_branch_id)
    AND b.bill_date BETWEEN p_from_date AND p_to_date
    AND b.is_cancelled = FALSE
    GROUP BY bi.category

    UNION ALL

    -- Total Revenue
    SELECT
        'Total Revenue'::VARCHAR,
        'TOTAL'::VARCHAR,
        v_total_revenue,
        100.00::DECIMAL

    UNION ALL

    -- Expenses (from payroll)
    SELECT
        'Expense - Salaries'::VARCHAR,

```

```

        'HR'::VARCHAR,
        COALESCE(SUM(pd.net_pay), 0)::DECIMAL,
        ROUND(COALESCE(SUM(pd.net_pay), 0) / NULLIF(v_total_revenue, 0) * 100,
2)::DECIMAL
    FROM payroll_runs pr
    JOIN payroll_details pd ON pr.id = pd.payroll_run_id
    WHERE pr.organization_id = p_organization_id
    AND (p_branch_id IS NULL OR pr.branch_id = p_branch_id)
    AND DATE(pr.pay_period_from) >= p_from_date
    AND DATE(pr.pay_period_to) <= p_to_date
    AND pr.status = 'paid'

    UNION ALL

    -- Expenses (from purchases)
    SELECT
        'Expense - Materials'::VARCHAR,
        'Procurement'::VARCHAR,
        COALESCE(SUM(grn.received_amount), 0)::DECIMAL,
        ROUND(COALESCE(SUM(grn.received_amount), 0) / NULLIF(v_total_revenue, 0) *
100, 2)::DECIMAL
    FROM goods_receipt_notes grn
    WHERE grn.organization_id = p_organization_id
    AND (p_branch_id IS NULL OR grn.branch_id = p_branch_id)
    AND grn.receipt_date BETWEEN p_from_date AND p_to_date
    AND grn.status = 'posted';
END;
$$ LANGUAGE plpgsql;

-- =====
-- SECTION 76: OPERATIONAL REPORTS
-- =====

-- OT Utilization Report
CREATE OR REPLACE FUNCTION rpt_ot_utilization(
    p_organization_id UUID,
    p_branch_id UUID DEFAULT NULL,
    p_from_date DATE DEFAULT DATE_TRUNC('month', CURRENT_DATE)::DATE,
    p_to_date DATE DEFAULT CURRENT_DATE
)
RETURNS TABLE(
    ot_room VARCHAR,
    ot_type VARCHAR,
    total_surgeries INTEGER,
    total_hours DECIMAL,
    available_hours DECIMAL,
    utilization_percent DECIMAL,
    avg_surgery_duration DECIMAL,
    cancellations INTEGER,
    cancellation_rate DECIMAL,
    emergency_surgeries INTEGER,
    elective_surgeries INTEGER,
    avg_turnover_time DECIMAL
) AS $$

```



```

DECLARE
    v_working_hours_per_day INTEGER := 10;
    v_days INTEGER;
BEGIN
    v_days := p_to_date - p_from_date + 1;

    RETURN QUERY
    SELECT
        otr.room_number::VARCHAR as ot_room,
        otr.ot_type::VARCHAR,
        COUNT(*) FILTER (WHERE s.status = 'completed')::INTEGER as
total_surgeries,
        ROUND(SUM(s.actual_duration_minutes) FILTER (WHERE s.status = 'completed')
/ 60.0, 1)::DECIMAL as total_hours,
        (v_days * v_working_hours_per_day)::DECIMAL as available_hours,
        ROUND(SUM(s.actual_duration_minutes) FILTER (WHERE s.status = 'completed')
/ 60.0 /
            (v_days * v_working_hours_per_day) * 100, 1)::DECIMAL as
utilization_percent,
        ROUND(AVG(s.actual_duration_minutes) FILTER (WHERE s.status =
'completed'), 0)::DECIMAL as avg_surgery_duration,
        COUNT(*) FILTER (WHERE s.status = 'cancelled')::INTEGER as cancellations,
        ROUND(COUNT(*) FILTER (WHERE s.status = 'cancelled')::DECIMAL /
            NULLIF(COUNT(*), 0) * 100, 1)::DECIMAL as cancellation_rate,
        COUNT(*) FILTER (WHERE s.surgery_category = 'emergency' AND s.status =
'completed')::INTEGER as emergency_surgeries,
        COUNT(*) FILTER (WHERE s.surgery_category = 'elective' AND s.status =
'completed')::INTEGER as elective_surgeries,
        ROUND(AVG(EXTRACT(EPOCH FROM (s2.actual_start_time -
s.actual_end_time))/60), 0)::DECIMAL as avg_turnover_time
    FROM ot_rooms otr
    LEFT JOIN surgeries s ON otr.id = s.ot_room_id
        AND s.scheduled_date BETWEEN p_from_date AND p_to_date
    LEFT JOIN surgeries s2 ON otr.id = s2.ot_room_id
        AND s2.scheduled_date = s.scheduled_date
        AND s2.actual_start_time > s.actual_end_time
    WHERE otr.organization_id = p_organization_id
    AND (p_branch_id IS NULL OR otr.branch_id = p_branch_id)
    AND otr.is_active = TRUE
    GROUP BY otr.id, otr.room_number, otr.ot_type
    ORDER BY utilization_percent DESC;
END;
$$ LANGUAGE plpgsql;

-- Pharmacy Stock Status Report
CREATE OR REPLACE FUNCTION rpt_pharmacy_stock_status(
    p_organization_id UUID,
    p_branch_id UUID DEFAULT NULL,
    p_pharmacy_id UUID DEFAULT NULL
)
RETURNS TABLE(
    pharmacy_name VARCHAR,
    drug_name VARCHAR,
    generic_name VARCHAR,

```

```

category VARCHAR,
current_stock INTEGER,
reorder_level INTEGER,
stock_status VARCHAR,
days_of_stock DECIMAL,
expiring_soon INTEGER,
expired INTEGER,
last_purchase_price DECIMAL,
stock_value DECIMAL,
last_received_date DATE,
last_issued_date DATE
) AS $$
BEGIN
RETURN QUERY
SELECT
    pl.name::VARCHAR as pharmacy_name,
    dm.brand_name::VARCHAR as drug_name,
    dm.generic_name::VARCHAR,
    dm.therapeutic_class::VARCHAR as category,
    mi.quantity_on_hand::INTEGER as current_stock,
    mi.reorder_level::INTEGER,
    CASE
        WHEN mi.quantity_on_hand <= 0 THEN 'Out of Stock'
        WHEN mi.quantity_on_hand <= mi.reorder_level THEN 'Low Stock'
        WHEN mi.quantity_on_hand <= mi.reorder_level * 1.5 THEN 'Warning'
        ELSE 'Adequate'
    END::VARCHAR as stock_status,
    ROUND(mi.quantity_on_hand::DECIMAL /
        NULLIF((SELECT AVG(daily_consumption) FROM (
            SELECT DATE(transaction_date), ABS(SUM(quantity)) as
daily_consumption
                FROM medicine_stock_transactions
                WHERE drug_id = mi.drug_id
                AND pharmacy_id = mi.pharmacy_id
                AND transaction_type = 'sale'
                AND transaction_date > CURRENT_DATE - 30
                GROUP BY DATE(transaction_date)
            ) dc), 0), 1)::DECIMAL as days_of_stock,
    (SELECT COALESCE(SUM(current_quantity), 0) FROM medicine_batches mb
    WHERE mb.drug_id = mi.drug_id
    AND mb.pharmacy_id = mi.pharmacy_id
    AND mb.expiry_date BETWEEN CURRENT_DATE AND CURRENT_DATE + 90)::INTEGER
as expiring_soon,
    (SELECT COALESCE(SUM(current_quantity), 0) FROM medicine_batches mb
    WHERE mb.drug_id = mi.drug_id
    AND mb.pharmacy_id = mi.pharmacy_id
    AND mb.expiry_date < CURRENT_DATE)::INTEGER as expired,
    mi.last_purchase_price::DECIMAL,
    (mi.quantity_on_hand * mi.average_cost)::DECIMAL as stock_value,
    mi.last_stock_date::DATE as last_received_date,
    (SELECT MAX(transaction_date)::DATE FROM medicine_stock_transactions mst
    WHERE mst.drug_id = mi.drug_id
    AND mst.pharmacy_id = mi.pharmacy_id
    AND mst.transaction_type = 'sale') as last_issued_date

```

```

FROM medicine_inventory mi
JOIN pharmacy_locations pl ON mi.pharmacy_id = pl.id
JOIN drug_master dm ON mi.drug_id = dm.id
WHERE pl.organization_id = p_organization_id
AND (p_branch_id IS NULL OR pl.branch_id = p_branch_id)
AND (p_pharmacy_id IS NULL OR pl.id = p_pharmacy_id)
AND mi.is_active = TRUE
ORDER BY
    CASE
        WHEN mi.quantity_on_hand <= 0 THEN 1
        WHEN mi.quantity_on_hand <= mi.reorder_level THEN 2
        ELSE 3
    END,
    dm.generic_name;
END;
$$ LANGUAGE plpgsql;

-- Bed Occupancy Trend Report
CREATE OR REPLACE FUNCTION rpt_bed_occupancy_trend(
    p_organization_id UUID,
    p_branch_id UUID DEFAULT NULL,
    p_from_date DATE DEFAULT CURRENT_DATE - 30,
    p_to_date DATE DEFAULT CURRENT_DATE
)
RETURNS TABLE(
    report_date DATE,
    total_beds INTEGER,
    occupied_beds INTEGER,
    available_beds INTEGER,
    occupancy_rate DECIMAL,
    admissions INTEGER,
    discharges INTEGER,
    avg_los DECIMAL
) AS $$
BEGIN
    RETURN QUERY
    WITH date_series AS (
        SELECT generate_series(p_from_date, p_to_date, '1 day'::INTERVAL)::DATE as
dt
    )
    SELECT
        ds.dt as report_date,
        (SELECT COUNT(*) FROM beds b
         JOIN wards w ON b.ward_id = w.id
         WHERE w.organization_id = p_organization_id
         AND (p_branch_id IS NULL OR w.branch_id = p_branch_id)
         AND b.is_active = TRUE)::INTEGER as total_beds,
        (SELECT COUNT(*) FROM admissions a
         WHERE a.organization_id = p_organization_id
         AND (p_branch_id IS NULL OR a.branch_id = p_branch_id)
         AND a.admission_date <= ds.dt
         AND (a.actual_discharge_date IS NULL OR a.actual_discharge_date > ds.dt)
         AND a.status IN ('admitted', 'discharged'))::INTEGER as occupied_beds,
        (SELECT COUNT(*) FROM beds b

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```

        JOIN wards w ON b.ward_id = w.id
        WHERE w.organization_id = p_organization_id
        AND (p_branch_id IS NULL OR w.branch_id = p_branch_id)
        AND b.is_active = TRUE)::INTEGER -
    (SELECT COUNT(*) FROM admissions a
     WHERE a.organization_id = p_organization_id
     AND (p_branch_id IS NULL OR a.branch_id = p_branch_id)
     AND a.admission_date <= ds.dt
     AND (a.actual_discharge_date IS NULL OR a.actual_discharge_date > ds.dt)
     AND a.status IN ('admitted', 'discharged'))::INTEGER as available_beds,
    ROUND(
        (SELECT COUNT(*) FROM admissions a
         WHERE a.organization_id = p_organization_id
         AND (p_branch_id IS NULL OR a.branch_id = p_branch_id)
         AND a.admission_date <= ds.dt
         AND (a.actual_discharge_date IS NULL OR a.actual_discharge_date >
ds.dt)
         AND a.status IN ('admitted', 'discharged'))::DECIMAL /
        NULLIF((SELECT COUNT(*) FROM beds b
         JOIN wards w ON b.ward_id = w.id
         WHERE w.organization_id = p_organization_id
         AND (p_branch_id IS NULL OR w.branch_id = p_branch_id)
         AND b.is_active = TRUE), 0) * 100, 1
    )::DECIMAL as occupancy_rate,
    (SELECT COUNT(*) FROM admissions a
     WHERE a.organization_id = p_organization_id
     AND (p_branch_id IS NULL OR a.branch_id = p_branch_id)
     AND a.admission_date = ds.dt)::INTEGER as admissions,
    (SELECT COUNT(*) FROM admissions a
     WHERE a.organization_id = p_organization_id
     AND (p_branch_id IS NULL OR a.branch_id = p_branch_id)
     AND a.actual_discharge_date = ds.dt)::INTEGER as discharges,
    (SELECT ROUND(AVG(a.actual_discharge_date - a.admission_date), 1)
     FROM admissions a
     WHERE a.organization_id = p_organization_id
     AND (p_branch_id IS NULL OR a.branch_id = p_branch_id)
     AND a.actual_discharge_date = ds.dt)::DECIMAL as avg_los
    FROM date_series ds
    ORDER BY ds.dt;
END;
$$ LANGUAGE plpgsql;

-- =====
-- SECTION 77: HR REPORTS
-- =====

-- Employee Attendance Summary
CREATE OR REPLACE FUNCTION rpt_employee_attendance_summary(
    p_organization_id UUID,
    p_branch_id UUID DEFAULT NULL,
    p_department_id UUID DEFAULT NULL,
    p_from_date DATE DEFAULT DATE_TRUNC('month', CURRENT_DATE)::DATE,
    p_to_date DATE DEFAULT CURRENT_DATE
)

```

```

RETURNS TABLE(
    employee_code VARCHAR,
    employee_name VARCHAR,
    department VARCHAR,
    designation VARCHAR,
    total_days INTEGER,
    present_days DECIMAL,
    absent_days DECIMAL,
    leave_days DECIMAL,
    late_count INTEGER,
    early_leave_count INTEGER,
    overtime_hours DECIMAL,
    attendance_percentage DECIMAL
) AS $$
DECLARE
    v_working_days INTEGER;
BEGIN
    -- Calculate working days (excluding Fridays as weekend in Bangladesh)
    SELECT COUNT(*) INTO v_working_days
    FROM generate_series(p_from_date, p_to_date, '1 day'::INTERVAL) d
    WHERE EXTRACT(DOW FROM d) != 5; -- 5 = Friday

    RETURN QUERY
    SELECT
        e.employee_code::VARCHAR,
        e.full_name::VARCHAR,
        d.name::VARCHAR as department,
        e.designation::VARCHAR,
        v_working_days::INTEGER as total_days,
        COUNT(*) FILTER (WHERE a.status = 'present')::DECIMAL as present_days,
        COUNT(*) FILTER (WHERE a.status = 'absent')::DECIMAL as absent_days,
        COUNT(*) FILTER (WHERE a.status = 'leave')::DECIMAL as leave_days,
        COUNT(*) FILTER (WHERE a.late_minutes > 0)::INTEGER as late_count,
        COUNT(*) FILTER (WHERE a.early_leave_minutes > 0)::INTEGER as
early_leave_count,
        COALESCE(SUM(a.overtime_hours), 0)::DECIMAL as overtime_hours,
        ROUND(COUNT(*) FILTER (WHERE a.status = 'present')::DECIMAL /
            NULLIF(v_working_days, 0) * 100, 1)::DECIMAL as
attendance_percentage
    FROM employees e
    LEFT JOIN departments d ON e.department_id = d.id
    LEFT JOIN attendance a ON e.id = a.employee_id
        AND a.attendance_date BETWEEN p_from_date AND p_to_date
    WHERE e.organization_id = p_organization_id
        AND (p_branch_id IS NULL OR e.branch_id = p_branch_id)
        AND (p_department_id IS NULL OR e.department_id = p_department_id)
        AND e.employment_status = 'active'
    GROUP BY e.id, e.employee_code, e.full_name, d.name, e.designation
    ORDER BY d.name, e.full_name;
END;
$$ LANGUAGE plpgsql;

-- Payroll Summary Report
CREATE OR REPLACE FUNCTION rpt_payroll_summary(

```

```

    p_organization_id UUID,
    p_branch_id UUID DEFAULT NULL,
    p_year INTEGER DEFAULT EXTRACT(YEAR FROM CURRENT_DATE)::INTEGER,
    p_month INTEGER DEFAULT EXTRACT(MONTH FROM CURRENT_DATE)::INTEGER
)
RETURNS TABLE(
    department VARCHAR,
    employee_count INTEGER,
    total_basic DECIMAL,
    total_allowances DECIMAL,
    total_gross DECIMAL,
    total_deductions DECIMAL,
    total_net_pay DECIMAL,
    total_employer_pf DECIMAL,
    total_tds DECIMAL
) AS $$
BEGIN
    RETURN QUERY
    SELECT
        d.name::VARCHAR as department,
        COUNT(DISTINCT pd.employee_id)::INTEGER as employee_count,
        SUM(pd.basic_salary)::DECIMAL as total_basic,
        SUM(pd.total_earnings - pd.basic_salary)::DECIMAL as total_allowances,
        SUM(pd.gross_salary)::DECIMAL as total_gross,
        SUM(pd.total_deductions)::DECIMAL as total_deductions,
        SUM(pd.net_pay)::DECIMAL as total_net_pay,
        SUM(pd.employer_pf)::DECIMAL as total_employer_pf,
        SUM(pd.tds_amount)::DECIMAL as total_tds
    FROM payroll_runs pr
    JOIN payroll_details pd ON pr.id = pd.payroll_run_id
    JOIN employees e ON pd.employee_id = e.id
    LEFT JOIN departments d ON e.department_id = d.id
    WHERE pr.organization_id = p_organization_id
    AND (p_branch_id IS NULL OR pr.branch_id = p_branch_id)
    AND pr.payroll_year = p_year
    AND pr.payroll_month = p_month
    AND pr.status IN ('approved', 'paid')
    GROUP BY d.name
    ORDER BY d.name;
END;
$$ LANGUAGE plpgsql;

-- Leave Balance Report
CREATE OR REPLACE FUNCTION rpt_leave_balance(
    p_organization_id UUID,
    p_branch_id UUID DEFAULT NULL,
    p_department_id UUID DEFAULT NULL,
    p_year INTEGER DEFAULT EXTRACT(YEAR FROM CURRENT_DATE)::INTEGER
)
RETURNS TABLE(
    employee_code VARCHAR,
    employee_name VARCHAR,
    department VARCHAR,
    leave_type VARCHAR,

```

```

        opening_balance DECIMAL,
        accrued DECIMAL,
        taken DECIMAL,
        available_balance DECIMAL,
        pending_approval DECIMAL
    ) AS $$
BEGIN
    RETURN QUERY
    SELECT
        e.employee_code::VARCHAR,
        e.full_name::VARCHAR,
        d.name::VARCHAR as department,
        lt.name::VARCHAR as leave_type,
        lb.opening_balance::DECIMAL,
        lb.accrued::DECIMAL,
        lb.taken::DECIMAL,
        lb.available_balance::DECIMAL,
        (SELECT COALESCE(SUM(la.number_of_days), 0)
         FROM leave_applications la
         WHERE la.employee_id = e.id
         AND la.leave_type_id = lt.id
         AND EXTRACT(YEAR FROM la.from_date) = p_year
         AND la.status = 'pending')::DECIMAL as pending_approval
    FROM employees e
    JOIN leave_balances lb ON e.id = lb.employee_id
    JOIN leave_types lt ON lb.leave_type_id = lt.id
    LEFT JOIN departments d ON e.department_id = d.id
    WHERE e.organization_id = p_organization_id
    AND (p_branch_id IS NULL OR e.branch_id = p_branch_id)
    AND (p_department_id IS NULL OR e.department_id = p_department_id)
    AND lb.year = p_year
    AND e.employment_status = 'active'
    ORDER BY d.name, e.full_name, lt.name;
END;
$$ LANGUAGE plpgsql;

-- =====
-- SECTION 78: INVENTORY REPORTS
-- =====

-- Stock Movement Report
CREATE OR REPLACE FUNCTION rpt_stock_movement(
    p_organization_id UUID,
    p_store_id UUID DEFAULT NULL,
    p_from_date DATE DEFAULT DATE_TRUNC('month', CURRENT_DATE)::DATE,
    p_to_date DATE DEFAULT CURRENT_DATE
)
RETURNS TABLE(
    item_code VARCHAR,
    item_name VARCHAR,
    category VARCHAR,
    opening_qty DECIMAL,
    received_qty DECIMAL,
    issued_qty DECIMAL,

```

```

        adjustment_qty DECIMAL,
        closing_qty DECIMAL,
        unit_cost DECIMAL,
        closing_value DECIMAL
    ) AS $$
BEGIN
    RETURN QUERY
    WITH opening_stock AS (
        SELECT
            it.item_id,
            SUM(CASE
                WHEN it.transaction_type IN ('grn_receipt', 'transfer_in',
'adjustment_in', 'opening_balance')
                THEN it.quantity
                ELSE -it.quantity
            END) as opening
        FROM inventory_transactions it
        WHERE it.organization_id = p_organization_id
        AND (p_store_id IS NULL OR it.store_id = p_store_id)
        AND DATE(it.transaction_date) < p_from_date
        GROUP BY it.item_id
    ),
    movements AS (
        SELECT
            it.item_id,
            SUM(it.quantity) FILTER (WHERE it.transaction_type IN ('grn_receipt',
'transfer_in')) as received,
            SUM(ABS(it.quantity)) FILTER (WHERE it.transaction_type IN ('issue',
'transfer_out')) as issued,
            SUM(CASE
                WHEN it.transaction_type = 'adjustment_in' THEN it.quantity
                WHEN it.transaction_type = 'adjustment_out' THEN -it.quantity
                ELSE 0
            END) as adjustment
        FROM inventory_transactions it
        WHERE it.organization_id = p_organization_id
        AND (p_store_id IS NULL OR it.store_id = p_store_id)
        AND DATE(it.transaction_date) BETWEEN p_from_date AND p_to_date
        GROUP BY it.item_id
    )
    SELECT
        im.item_code::VARCHAR,
        im.name::VARCHAR as item_name,
        ic.name::VARCHAR as category,
        COALESCE(os.opening, 0)::DECIMAL as opening_qty,
        COALESCE(m.received, 0)::DECIMAL as received_qty,
        COALESCE(m.issued, 0)::DECIMAL as issued_qty,
        COALESCE(m.adjustment, 0)::DECIMAL as adjustment_qty,
        ((COALESCE(os.opening, 0) + COALESCE(m.received, 0) -
        COALESCE(m.issued, 0) + COALESCE(m.adjustment, 0))::DECIMAL as
closing_qty,
        si.average_cost::DECIMAL as unit_cost,
        ((COALESCE(os.opening, 0) + COALESCE(m.received, 0) -
        COALESCE(m.issued, 0) + COALESCE(m.adjustment, 0)) *

```



```

        COALESCE(si.average_cost, 0))::DECIMAL as closing_value
FROM item_master im
LEFT JOIN item_categories ic ON im.category_id = ic.id
LEFT JOIN opening_stock os ON im.id = os.item_id
LEFT JOIN movements m ON im.id = m.item_id
LEFT JOIN store_inventory si ON im.id = si.item_id
    AND (p_store_id IS NULL OR si.store_id = p_store_id)
WHERE im.organization_id = p_organization_id
AND im.is_active = TRUE
AND (os.opening IS NOT NULL OR m.received IS NOT NULL OR m.issued IS NOT NULL)
ORDER BY ic.name, im.name;
END;
$$ LANGUAGE plpgsql;

-- Expiry Report
CREATE OR REPLACE FUNCTION rpt_expiry_items(
    p_organization_id UUID,
    p_store_id UUID DEFAULT NULL,
    p_days_ahead INTEGER DEFAULT 90
)
RETURNS TABLE(
    store_name VARCHAR,
    item_code VARCHAR,
    item_name VARCHAR,
    batch_number VARCHAR,
    current_quantity DECIMAL,
    expiry_date DATE,
    days_to_expiry INTEGER,
    status VARCHAR,
    value DECIMAL
) AS $$
BEGIN
    RETURN QUERY
    SELECT
        s.name::VARCHAR as store_name,
        im.item_code::VARCHAR,
        im.name::VARCHAR as item_name,
        ib.batch_number::VARCHAR,
        ib.current_quantity::DECIMAL,
        ib.expiry_date::DATE,
        (ib.expiry_date - CURRENT_DATE)::INTEGER as days_to_expiry,
        CASE
            WHEN ib.expiry_date < CURRENT_DATE THEN 'Expired'
            WHEN ib.expiry_date <= CURRENT_DATE + 30 THEN 'Critical'
            WHEN ib.expiry_date <= CURRENT_DATE + 60 THEN 'Warning'
            ELSE 'Monitor'
        END::VARCHAR as status,
        (ib.current_quantity * ib.unit_cost)::DECIMAL as value
    FROM item_batches ib
    JOIN store_inventory si ON ib.inventory_id = si.id
    JOIN stores s ON ib.store_id = s.id
    JOIN item_master im ON ib.item_id = im.id
    WHERE s.organization_id = p_organization_id
    AND (p_store_id IS NULL OR s.id = p_store_id)

```

```

        AND ib.current_quantity > 0
        AND ib.expiry_date <= CURRENT_DATE + p_days_ahead
        AND ib.is_active = TRUE
    ORDER BY ib.expiry_date, s.name, im.name;
END;
$$ LANGUAGE plpgsql;

-- =====
-- SECTION 79: QUALITY & COMPLIANCE REPORTS
-- =====

-- Quality Indicator Dashboard
CREATE OR REPLACE FUNCTION rpt_quality_dashboard(
    p_organization_id UUID,
    p_branch_id UUID DEFAULT NULL,
    p_from_date DATE DEFAULT DATE_TRUNC('month', CURRENT_DATE)::DATE,
    p_to_date DATE DEFAULT CURRENT_DATE
)
RETURNS TABLE(
    indicator_code VARCHAR,
    indicator_name VARCHAR,
    category VARCHAR,
    target_value DECIMAL,
    actual_value DECIMAL,
    variance DECIMAL,
    status VARCHAR,
    trend VARCHAR
) AS $$
BEGIN
    RETURN QUERY
    SELECT
        qi.indicator_code::VARCHAR,
        qi.name::VARCHAR as indicator_name,
        qi.category::VARCHAR,
        qi.target_value::DECIMAL,
        qir.result_value::DECIMAL as actual_value,
        (qir.result_value - qi.target_value)::DECIMAL as variance,
        CASE
            WHEN qi.target_direction = 'higher_better' AND qir.result_value >=
qi.target_value THEN 'Met'
            WHEN qi.target_direction = 'lower_better' AND qir.result_value <=
qi.target_value THEN 'Met'
            WHEN qi.target_direction = 'target_range'
                AND qir.result_value BETWEEN qi.target_min AND qi.target_max THEN
'Met'
            ELSE 'Not Met'
        END::VARCHAR as status,
        COALESCE(qir.trend, 'stable')::VARCHAR
    FROM quality_indicators qi
    LEFT JOIN quality_indicator_results qir ON qi.id = qir.indicator_id
        AND qir.measurement_period_start >= p_from_date
        AND qir.measurement_period_end <= p_to_date
    WHERE (qi.organization_id IS NULL OR qi.organization_id = p_organization_id)
    AND (p_branch_id IS NULL OR qir.branch_id = p_branch_id)

```

```

    AND qi.is_active = TRUE
    ORDER BY qi.category, qi.indicator_code;
END;
$$ LANGUAGE plpgsql;

-- Infection Control Report
CREATE OR REPLACE FUNCTION rpt_infection_control(
    p_organization_id UUID,
    p_branch_id UUID DEFAULT NULL,
    p_from_date DATE DEFAULT DATE_TRUNC('month', CURRENT_DATE)::DATE,
    p_to_date DATE DEFAULT CURRENT_DATE
)
RETURNS TABLE(
    infection_type VARCHAR,
    total_cases INTEGER,
    hospital_acquired INTEGER,
    community_acquired INTEGER,
    device_related INTEGER,
    mdro_cases INTEGER,
    rate_per_1000 DECIMAL,
    mortality_count INTEGER,
    preventable_count INTEGER
) AS $$
DECLARE
    v_patient_days BIGINT;
BEGIN
    -- Calculate patient days
    SELECT COALESCE(SUM(
        CASE
            WHEN a.actual_discharge_date IS NOT NULL
            THEN LEAST(a.actual_discharge_date, p_to_date) -
GREATEST(a.admission_date, p_from_date) + 1
            ELSE p_to_date - GREATEST(a.admission_date, p_from_date) + 1
        END
    ), 1) INTO v_patient_days
    FROM admissions a
    WHERE a.organization_id = p_organization_id
    AND (p_branch_id IS NULL OR a.branch_id = p_branch_id)
    AND a.admission_date <= p_to_date
    AND (a.actual_discharge_date IS NULL OR a.actual_discharge_date >=
p_from_date);

    RETURN QUERY
    SELECT
        ins.infection_type::VARCHAR,
        COUNT(*)::INTEGER as total_cases,
        COUNT(*) FILTER (WHERE ins.onset_type = 'hospital_acquired')::INTEGER as
hospital_acquired,
        COUNT(*) FILTER (WHERE ins.onset_type = 'community')::INTEGER as
community_acquired,
        COUNT(*) FILTER (WHERE ins.device_related = TRUE)::INTEGER as
device_related,
        COUNT(*) FILTER (WHERE ins.is_mdرو = TRUE)::INTEGER as mdro_cases,
        ROUND(COUNT(*)::DECIMAL / v_patient_days * 1000, 2)::DECIMAL as

```

```

rate_per_1000,
    COUNT(*) FILTER (WHERE ins.outcome LIKE '%death%')::INTEGER as
mortality_count,
    COUNT(*) FILTER (WHERE ins.preventable = TRUE)::INTEGER as
preventable_count
FROM infection_surveillance ins
WHERE ins.organization_id = p_organization_id
AND (p_branch_id IS NULL OR ins.branch_id = p_branch_id)
AND ins.onset_date BETWEEN p_from_date AND p_to_date
GROUP BY ins.infection_type
ORDER BY total_cases DESC;
END;
$$ LANGUAGE plpgsql;

-- Incident Summary Report
CREATE OR REPLACE FUNCTION rpt_incident_summary(
    p_organization_id UUID,
    p_branch_id UUID DEFAULT NULL,
    p_from_date DATE DEFAULT DATE_TRUNC('month', CURRENT_DATE)::DATE,
    p_to_date DATE DEFAULT CURRENT_DATE
)
RETURNS TABLE(
    incident_type VARCHAR,
    total_incidents INTEGER,
    severity_none INTEGER,
    severity_mild INTEGER,
    severity_moderate INTEGER,
    severity_severe INTEGER,
    open_count INTEGER,
    closed_count INTEGER,
    avg_resolution_days DECIMAL,
    preventable_count INTEGER
) AS $$
BEGIN
    RETURN QUERY
    SELECT
        ir.incident_type::VARCHAR,
        COUNT(*)::INTEGER as total_incidents,
        COUNT(*) FILTER (WHERE ir.severity_actual = 'none')::INTEGER as
severity_none,
        COUNT(*) FILTER (WHERE ir.severity_actual = 'mild')::INTEGER as
severity_mild,
        COUNT(*) FILTER (WHERE ir.severity_actual = 'moderate')::INTEGER as
severity_moderate,
        COUNT(*) FILTER (WHERE ir.severity_actual = 'severe')::INTEGER as
severity_severe,
        COUNT(*) FILTER (WHERE ir.status IN ('open', 'investigating'))::INTEGER as
open_count,
        COUNT(*) FILTER (WHERE ir.status = 'closed')::INTEGER as closed_count,
        ROUND(AVG(EXTRACT(DAY FROM (ir.closed_at - ir.reported_at)))
            FILTER (WHERE ir.closed_at IS NOT NULL), 1)::DECIMAL as
avg_resolution_days,
        COUNT(*) FILTER (WHERE EXISTS (
            SELECT 1 FROM mortality_reviews mr

```

```

        WHERE mr.death_record_id::TEXT = ir.id::TEXT
        AND mr.potentially_preventable = TRUE
    )::INTEGER as preventable_count
FROM incident_reports ir
WHERE ir.organization_id = p_organization_id
AND (p_branch_id IS NULL OR ir.branch_id = p_branch_id)
AND DATE(ir.incident_datetime) BETWEEN p_from_date AND p_to_date
GROUP BY ir.incident_type
ORDER BY total_incidents DESC;
END;
$$ LANGUAGE plpgsql;

-- =====
-- SECTION 80: MIS & EXECUTIVE REPORTS
-- =====

-- Executive Dashboard Summary
CREATE OR REPLACE FUNCTION rpt_executive_dashboard(
    p_organization_id UUID,
    p_date DATE DEFAULT CURRENT_DATE
)
RETURNS TABLE(
    category VARCHAR,
    metric_name VARCHAR,
    today_value DECIMAL,
    mtd_value DECIMAL,
    ytd_value DECIMAL,
    target DECIMAL,
    achievement_percent DECIMAL
) AS $$
DECLARE
    v_month_start DATE := DATE_TRUNC('month', p_date)::DATE;
    v_year_start DATE := DATE_TRUNC('year', p_date)::DATE;
BEGIN
    RETURN QUERY

    -- Revenue Metrics
    SELECT
        'Revenue'::VARCHAR as category,
        'Total Revenue'::VARCHAR as metric_name,
        COALESCE(SUM(bi.net_amount) FILTER (WHERE b.bill_date = p_date),
0)::DECIMAL as today_value,
        COALESCE(SUM(bi.net_amount) FILTER (WHERE b.bill_date >= v_month_start),
0)::DECIMAL as mtd_value,
        COALESCE(SUM(bi.net_amount) FILTER (WHERE b.bill_date >= v_year_start),
0)::DECIMAL as ytd_value,
        NULL::DECIMAL as target,
        NULL::DECIMAL as achievement_percent
    FROM bills b
    JOIN bill_items bi ON b.id = bi.bill_id
    WHERE b.organization_id = p_organization_id
    AND b.bill_date <= p_date
    AND b.is_cancelled = FALSE

```

```

UNION ALL

-- Collection Metrics
SELECT
    'Collection'::VARCHAR,
    'Total Collection'::VARCHAR,
    COALESCE(SUM(p.amount) FILTER (WHERE p.payment_date = p_date), 0),
    COALESCE(SUM(p.amount) FILTER (WHERE p.payment_date >= v_month_start), 0),
    COALESCE(SUM(p.amount) FILTER (WHERE p.payment_date >= v_year_start), 0),
    NULL,
    NULL
FROM payments p
WHERE p.organization_id = p_organization_id
AND p.payment_date <= p_date
AND p.status = 'completed'
AND p.is_refund = FALSE

UNION ALL

-- OPD Metrics
SELECT
    'OPD'::VARCHAR,
    'OPD Patients'::VARCHAR,
    COUNT(*) FILTER (WHERE a.appointment_date = p_date AND a.status =
'completed')::DECIMAL,
    COUNT(*) FILTER (WHERE a.appointment_date >= v_month_start AND a.status =
'completed')::DECIMAL,
    COUNT(*) FILTER (WHERE a.appointment_date >= v_year_start AND a.status =
'completed')::DECIMAL,
    NULL,
    NULL
FROM appointments a
WHERE a.organization_id = p_organization_id
AND a.appointment_date <= p_date

UNION ALL

-- IPD Metrics
SELECT
    'IPD'::VARCHAR,
    'Current Census'::VARCHAR,
    COUNT(*) FILTER (WHERE ad.status = 'admitted')::DECIMAL,
    NULL::DECIMAL,
    NULL::DECIMAL,
    NULL,
    NULL
FROM admissions ad
WHERE ad.organization_id = p_organization_id
AND ad.admission_date <= p_date

UNION ALL

-- Bed Occupancy
SELECT

```

```

        'IPD'::VARCHAR,
        'Bed Occupancy %'::VARCHAR,
        ROUND((SELECT COUNT(*) FROM beds b
                JOIN wards w ON b.ward_id = w.id
                WHERE w.organization_id = p_organization_id
                AND b.status = 'occupied')::DECIMAL /
        NULLIF((SELECT COUNT(*) FROM beds b
                JOIN wards w ON b.ward_id = w.id
                WHERE w.organization_id = p_organization_id
                AND b.is_active = TRUE), 0) * 100, 1),
        NULL,
        NULL,
        85.00::DECIMAL, -- Target 85%
        NULL

UNION ALL

-- Surgery Metrics
SELECT
    'Surgery'::VARCHAR,
    'Surgeries'::VARCHAR,
    COUNT(*) FILTER (WHERE s.scheduled_date = p_date AND s.status =
'completed')::DECIMAL,
    COUNT(*) FILTER (WHERE s.scheduled_date >= v_month_start AND s.status =
'completed')::DECIMAL,
    COUNT(*) FILTER (WHERE s.scheduled_date >= v_year_start AND s.status =
'completed')::DECIMAL,
    NULL,
    NULL
    FROM surgeries s
    WHERE s.organization_id = p_organization_id
    AND s.scheduled_date <= p_date;
END;
$$ LANGUAGE plpgsql;

-- Monthly Performance Scorecard
CREATE OR REPLACE FUNCTION rpt_monthly_scorecard(
    p_organization_id UUID,
    p_year INTEGER DEFAULT EXTRACT(YEAR FROM CURRENT_DATE)::INTEGER,
    p_month INTEGER DEFAULT EXTRACT(MONTH FROM CURRENT_DATE)::INTEGER
)
RETURNS TABLE(
    kpi_category VARCHAR,
    kpi_name VARCHAR,
    current_month_value DECIMAL,
    previous_month_value DECIMAL,
    same_month_last_year DECIMAL,
    mom_change DECIMAL,
    yoy_change DECIMAL,
    target DECIMAL,
    status VARCHAR
) AS $$
DECLARE
    v_current_start DATE := make_date(p_year, p_month, 1);

```

```

v_current_end DATE := (v_current_start + INTERVAL '1 month - 1 day')::DATE;
v_prev_start DATE := (v_current_start - INTERVAL '1 month')::DATE;
v_prev_end DATE := v_current_start - 1;
v_ly_start DATE := make_date(p_year - 1, p_month, 1);
v_ly_end DATE := (v_ly_start + INTERVAL '1 month - 1 day')::DATE;
BEGIN
RETURN QUERY
SELECT
    kd.category::VARCHAR as kpi_category,
    kd.name::VARCHAR as kpi_name,
    (SELECT kv.value FROM kpi_values kv
     WHERE kv.kpi_id = kd.id
     AND kv.period_start >= v_current_start
     AND kv.period_end <= v_current_end
     ORDER BY kv.calculated_at DESC LIMIT 1)::DECIMAL as current_month_value,
    (SELECT kv.value FROM kpi_values kv
     WHERE kv.kpi_id = kd.id
     AND kv.period_start >= v_prev_start
     AND kv.period_end <= v_prev_end
     ORDER BY kv.calculated_at DESC LIMIT 1)::DECIMAL as previous_month_value,
    (SELECT kv.value FROM kpi_values kv
     WHERE kv.kpi_id = kd.id
     AND kv.period_start >= v_ly_start
     AND kv.period_end <= v_ly_end
     ORDER BY kv.calculated_at DESC LIMIT 1)::DECIMAL as same_month_last_year,
    NULL::DECIMAL as mom_change, -- Will be calculated
    NULL::DECIMAL as yoy_change, -- Will be calculated
    kd.target_value::DECIMAL as target,
    NULL::VARCHAR as status
FROM kpi_definitions kd
WHERE (kd.organization_id IS NULL OR kd.organization_id = p_organization_id)
AND kd.is_active = TRUE
ORDER BY kd.category, kd.name;
END;
$$ LANGUAGE plpgsql;

-- =====
-- SECTION 81: REGULATORY REPORTS
-- =====

-- DGHS Monthly Report Format
CREATE OR REPLACE FUNCTION rpt_dghs_monthly(
    p_organization_id UUID,
    p_branch_id UUID DEFAULT NULL,
    p_year INTEGER DEFAULT EXTRACT(YEAR FROM CURRENT_DATE)::INTEGER,
    p_month INTEGER DEFAULT EXTRACT(MONTH FROM CURRENT_DATE)::INTEGER
)
RETURNS TABLE(
    sl_no INTEGER,
    indicator VARCHAR,
    unit VARCHAR,
    value DECIMAL
) AS $$
DECLARE

```



```

v_start_date DATE := make_date(p_year, p_month, 1);
v_end_date DATE := (v_start_date + INTERVAL '1 month - 1 day')::DATE;
BEGIN
RETURN QUERY
SELECT 1, 'Total Outdoor Patients', 'Number',
    (SELECT COUNT(*) FROM appointments a
     WHERE a.organization_id = p_organization_id
     AND (p_branch_id IS NULL OR a.branch_id = p_branch_id)
     AND a.appointment_date BETWEEN v_start_date AND v_end_date
     AND a.status = 'completed')::DECIMAL
UNION ALL
SELECT 2, 'Total Indoor Patients', 'Number',
    (SELECT COUNT(*) FROM admissions ad
     WHERE ad.organization_id = p_organization_id
     AND (p_branch_id IS NULL OR ad.branch_id = p_branch_id)
     AND ad.admission_date BETWEEN v_start_date AND v_end_date)::DECIMAL
UNION ALL
SELECT 3, 'Total Bed Days', 'Days',
    (SELECT COALESCE(SUM(
        CASE
            WHEN ad.actual_discharge_date IS NOT NULL
            THEN LEAST(ad.actual_discharge_date, v_end_date) -
GREATEST(ad.admission_date, v_start_date) + 1
            ELSE v_end_date - GREATEST(ad.admission_date, v_start_date) + 1
        END
    ), 0) FROM admissions ad
     WHERE ad.organization_id = p_organization_id
     AND (p_branch_id IS NULL OR ad.branch_id = p_branch_id)
     AND ad.admission_date <= v_end_date
     AND (ad.actual_discharge_date IS NULL OR ad.actual_discharge_date >=
v_start_date))::DECIMAL
UNION ALL
SELECT 4, 'Total Surgeries', 'Number',
    (SELECT COUNT(*) FROM surgeries s
     WHERE s.organization_id = p_organization_id
     AND (p_branch_id IS NULL OR s.branch_id = p_branch_id)
     AND s.scheduled_date BETWEEN v_start_date AND v_end_date
     AND s.status = 'completed')::DECIMAL
UNION ALL
SELECT 5, 'Major Surgeries', 'Number',
    (SELECT COUNT(*) FROM surgeries s
     JOIN ot_rooms otr ON s.ot_room_id = otr.id
     WHERE s.organization_id = p_organization_id
     AND (p_branch_id IS NULL OR s.branch_id = p_branch_id)
     AND s.scheduled_date BETWEEN v_start_date AND v_end_date
     AND s.status = 'completed'
     AND otr.ot_type = 'major')::DECIMAL
UNION ALL
SELECT 6, 'Normal Deliveries', 'Number',
    (SELECT COUNT(*) FROM surgeries s
     WHERE s.organization_id = p_organization_id
     AND (p_branch_id IS NULL OR s.branch_id = p_branch_id)
     AND s.scheduled_date BETWEEN v_start_date AND v_end_date
     AND s.surgery_name ILIKE '%normal delivery%')

```

```

        AND s.status = 'completed')::DECIMAL
UNION ALL
SELECT 7, 'Cesarean Sections', 'Number',
    (SELECT COUNT(*) FROM surgeries s
     WHERE s.organization_id = p_organization_id
     AND (p_branch_id IS NULL OR s.branch_id = p_branch_id)
     AND s.scheduled_date BETWEEN v_start_date AND v_end_date
     AND (s.surgery_name ILIKE '%cesarean%' OR s.surgery_name ILIKE '%c-
section%'))
    AND s.status = 'completed')::DECIMAL
UNION ALL
SELECT 8, 'Total Deaths', 'Number',
    (SELECT COUNT(*) FROM admissions ad
     WHERE ad.organization_id = p_organization_id
     AND (p_branch_id IS NULL OR ad.branch_id = p_branch_id)
     AND ad.actual_discharge_date BETWEEN v_start_date AND v_end_date
     AND ad.discharge_type = 'expired')::DECIMAL
UNION ALL
SELECT 9, 'Lab Tests Conducted', 'Number',
    (SELECT COUNT(*) FROM lab_order_items loi
     JOIN lab_orders lo ON loi.order_id = lo.id
     WHERE lo.organization_id = p_organization_id
     AND (p_branch_id IS NULL OR lo.branch_id = p_branch_id)
     AND DATE(lo.order_datetime) BETWEEN v_start_date AND v_end_date
     AND loi.status = 'completed')::DECIMAL
UNION ALL
SELECT 10, 'X-Ray Examinations', 'Number',
    (SELECT COUNT(*) FROM radiology_orders ro
     WHERE ro.organization_id = p_organization_id
     AND (p_branch_id IS NULL OR ro.branch_id = p_branch_id)
     AND DATE(ro.order_datetime) BETWEEN v_start_date AND v_end_date
     AND ro.modality_type = 'xray'
     AND ro.status = 'completed')::DECIMAL
ORDER BY 1;
END;
$$ LANGUAGE plpgsql;

-- =====
-- SECTION 82: REPORT HELPER FUNCTIONS
-- =====

-- Generate Report as JSON
CREATE OR REPLACE FUNCTION generate_report_json(
    p_report_code VARCHAR,
    p_organization_id UUID,
    p_parameters JSONB DEFAULT '{}'
)
RETURNS JSONB AS $$
DECLARE
    v_report RECORD;
    v_result JSONB;
    v_query TEXT;
    v_data JSONB;
BEGIN

```

```

-- Get report definition
SELECT * INTO v_report
FROM report_master
WHERE report_code = p_report_code
AND (organization_id IS NULL OR organization_id = p_organization_id)
AND is_active = TRUE
LIMIT 1;

IF v_report IS NULL THEN
    RETURN jsonb_build_object(
        'success', FALSE,
        'error', 'Report not found: ' || p_report_code
    );
END IF;

-- Build query with parameters
v_query := v_report.source_query;

-- Execute and get results
EXECUTE 'SELECT jsonb_agg(row_to_json(t)) FROM (' || v_query || ') t'
INTO v_data
USING p_organization_id, p_parameters;

-- Build result
v_result := jsonb_build_object(
    'success', TRUE,
    'report_code', v_report.report_code,
    'report_name', v_report.name,
    'generated_at', CURRENT_TIMESTAMP,
    'parameters', p_parameters,
    'columns', v_report.columns,
    'data', COALESCE(v_data, '[]'::JSONB),
    'row_count', jsonb_array_length(COALESCE(v_data, '[]'::JSONB))
);

-- Log execution
INSERT INTO report_execution_log (
    organization_id, report_id, execution_type,
    parameters_used, started_at, completed_at,
    status, row_count
) VALUES (
    p_organization_id, v_report.id, 'api',
    p_parameters, CURRENT_TIMESTAMP, CURRENT_TIMESTAMP,
    'completed', jsonb_array_length(COALESCE(v_data, '[]'::JSONB))
);

RETURN v_result;
END;
$$ LANGUAGE plpgsql;

-- Export Report to CSV Format
CREATE OR REPLACE FUNCTION export_report_csv(
    p_report_code VARCHAR,
    p_organization_id UUID,

```

```

    p_parameters JSONB DEFAULT '{}')
)
RETURNS TEXT AS $$
DECLARE
    v_report RECORD;
    v_columns JSONB;
    v_header TEXT := '';
    v_body TEXT := '';
    v_col RECORD;
    v_row RECORD;
    v_query TEXT;
BEGIN
    -- Get report definition
    SELECT * INTO v_report
    FROM report_master
    WHERE report_code = p_report_code
    AND (organization_id IS NULL OR organization_id = p_organization_id)
    AND is_active = TRUE
    LIMIT 1;

    IF v_report IS NULL THEN
        RETURN 'Error: Report not found';
    END IF;

    -- Build header
    FOR v_col IN SELECT * FROM jsonb_array_elements(v_report.columns)
    LOOP
        v_header := v_header || '""' || (v_col.value->'header') || '",'';
    END LOOP;
    v_header := RTRIM(v_header, ',') || E'\n';

    -- Execute query and build body
    v_query := 'SELECT * FROM (' || v_report.source_query || ') t';

    FOR v_row IN EXECUTE v_query USING p_organization_id, p_parameters
    LOOP
        FOR v_col IN SELECT * FROM jsonb_array_elements(v_report.columns)
        LOOP
            -- Add column value (simplified - actual implementation would be more
complex)
            v_body := v_body || '""' || COALESCE(v_row.*::TEXT, '') || '",'';
        END LOOP;
        v_body := RTRIM(v_body, ',') || E'\n';
    END LOOP;

    RETURN v_header || v_body;
END;
$$ LANGUAGE plpgsql;

-- =====
-- SECTION 83: PRE-BUILT REPORT DEFINITIONS
-- =====

-- Insert standard report definitions

```

```

INSERT INTO report_master (
    report_code, name, name_bangla, description,
    category_id, report_type, source_type, source_function,
    parameters, columns, is_system
) VALUES
(
    'RPT_DAILY_OPD',
    'Daily OPD Summary Report',
    'দৈনিক ওপিডি সারসংক্ষেপ',
    'Summary of OPD activities for a specific date',
    (SELECT id FROM report_categories WHERE code = 'CLINICAL'),
    'tabular',
    'function',
    'rpt_daily_opd_summary',
    '[
        {"name": "date", "label": "Date", "type": "date", "required": true,
"default": "current_date"},
        {"name": "branch_id", "label": "Branch", "type": "select", "source":
"branches", "required": false}
    ]'::JSONB,
    '[
        {"field": "department", "header": "Department", "type": "string", "width":
150},
        {"field": "doctor_name", "header": "Doctor", "type": "string", "width":
200},
        {"field": "total_appointments", "header": "Total", "type": "integer",
"width": 80, "aggregate": "sum"},
        {"field": "completed", "header": "Completed", "type": "integer", "width":
80, "aggregate": "sum"},
        {"field": "new_patients", "header": "New", "type": "integer", "width": 80,
"aggregate": "sum"},
        {"field": "followup_patients", "header": "Follow-up", "type": "integer",
"width": 80, "aggregate": "sum"},
        {"field": "total_revenue", "header": "Revenue", "type": "currency",
"width": 120, "aggregate": "sum"}
    ]'::JSONB,
    TRUE
),
(
    'RPT_DAILY_IPD_CENSUS',
    'Daily IPD Census Report',
    'দৈনিক আইপিডি সেন্সাস',
    'Ward-wise IPD census for a specific date',
    (SELECT id FROM report_categories WHERE code = 'CLINICAL'),
    'tabular',
    'function',
    'rpt_daily_ipd_census',
    '[
        {"name": "date", "label": "Date", "type": "date", "required": true,
"default": "current_date"},
        {"name": "branch_id", "label": "Branch", "type": "select", "source":
"branches", "required": false}
    ]'::JSONB,
    '[

```

```

        {"field": "ward_name", "header": "Ward", "type": "string", "width": 150},
        {"field": "total_beds", "header": "Total Beds", "type": "integer",
"width": 80},
        {"field": "occupied_beds", "header": "Occupied", "type": "integer",
"width": 80},
        {"field": "available_beds", "header": "Available", "type": "integer",
"width": 80},
        {"field": "occupancy_rate", "header": "Occupancy %", "type": "percentage",
"width": 100},
        {"field": "admissions_today", "header": "Admissions", "type": "integer",
"width": 80},
        {"field": "discharges_today", "header": "Discharges", "type": "integer",
"width": 80}
    ]::JSONB,
    TRUE
),
(
    'RPT_DAILY_COLLECTION',
    'Daily Collection Report',
    'দৈনিক আদায় রিপোর্ট',
    'Payment collection summary by payment mode',
    (SELECT id FROM report_categories WHERE code = 'FINANCIAL'),
    'tabular',
    'function',
    'rpt_daily_collection',
    '[
        {"name": "date", "label": "Date", "type": "date", "required": true,
"default": "current_date"},
        {"name": "branch_id", "label": "Branch", "type": "select", "source":
"branches", "required": false}
    ]::JSONB,
    '[
        {"field": "collection_point", "header": "Collection Point", "type":
"string", "width": 150},
        {"field": "cash_amount", "header": "Cash", "type": "currency", "width":
100, "aggregate": "sum"},
        {"field": "card_amount", "header": "Card", "type": "currency", "width":
100, "aggregate": "sum"},
        {"field": "mobile_banking", "header": "Mobile Banking", "type":
"currency", "width": 100, "aggregate": "sum"},
        {"field": "total_amount", "header": "Total", "type": "currency", "width":
120, "aggregate": "sum"},
        {"field": "refund_amount", "header": "Refunds", "type": "currency",
"width": 100, "aggregate": "sum"},
        {"field": "net_collection", "header": "Net Collection", "type":
"currency", "width": 120, "aggregate": "sum"}
    ]::JSONB,
    TRUE
),
(
    'RPT_PATIENT_STATISTICS',
    'Patient Statistics Report',
    'রোগী পরিসংখ্যান রিপোর্ট',
    'Patient visit and admission statistics',

```

```

        (SELECT id FROM report_categories WHERE code = 'MIS'),
        'summary',
        'function',
        'rpt_patient_statistics',
        '[
            {"name": "from_date", "label": "From Date", "type": "date", "required":
true},
            {"name": "to_date", "label": "To Date", "type": "date", "required": true},
            {"name": "branch_id", "label": "Branch", "type": "select", "source":
"branches", "required": false}
        ]':JSONB,
        '[
            {"field": "metric", "header": "Metric", "type": "string", "width": 250},
            {"field": "value", "header": "Current Period", "type": "integer", "width":
120},
            {"field": "previous_period_value", "header": "Previous Period", "type":
"integer", "width": 120},
            {"field": "change_percent", "header": "Change %", "type": "percentage",
"width": 100}
        ]':JSONB,
        TRUE
    ),
    (
        'RPT_DOCTOR_PERFORMANCE',
        'Doctor Performance Report',
        'ডাক্তার পারফরম্যান্স রিপোর্ট',
        'Individual doctor performance metrics',
        (SELECT id FROM report_categories WHERE code = 'MIS'),
        'tabular',
        'function',
        'rpt_doctor_performance',
        '[
            {"name": "from_date", "label": "From Date", "type": "date", "required":
true},
            {"name": "to_date", "label": "To Date", "type": "date", "required": true},
            {"name": "branch_id", "label": "Branch", "type": "select", "source":
"branches", "required": false}
        ]':JSONB,
        '[
            {"field": "doctor_name", "header": "Doctor", "type": "string", "width":
200},
            {"field": "specialization", "header": "Specialization", "type": "string",
"width": 150},
            {"field": "opd_patients", "header": "OPD", "type": "integer", "width": 80,
"aggregate": "sum"},
            {"field": "ipd_patients", "header": "IPD", "type": "integer", "width": 80,
"aggregate": "sum"},
            {"field": "surgeries", "header": "Surgeries", "type": "integer", "width":
80, "aggregate": "sum"},
            {"field": "total_revenue", "header": "Revenue", "type": "currency",
"width": 120, "aggregate": "sum"},
            {"field": "avg_rating", "header": "Rating", "type": "decimal", "width":
80}
        ]':JSONB,

```

```

TRUE
),
(
  'RPT_LAB_TAT',
  'Lab Turnaround Time Report',
  'ল্যাব টার্নঅ্যারাউন্ড টাইম রিপোর্ট',
  'Lab test TAT analysis by department',
  (SELECT id FROM report_categories WHERE code = 'OPERATIONAL'),
  'tabular',
  'function',
  'rpt_lab_tat_analysis',
  '[
    {"name": "from_date", "label": "From Date", "type": "date", "required":
true},
    {"name": "to_date", "label": "To Date", "type": "date", "required": true},
    {"name": "branch_id", "label": "Branch", "type": "select", "source":
"branches", "required": false}
  ]'::JSONB,
  '[
    {"field": "department", "header": "Department", "type": "string", "width":
150},
    {"field": "test_category", "header": "Category", "type": "string",
"width": 150},
    {"field": "total_tests", "header": "Tests", "type": "integer", "width":
80, "aggregate": "sum"},
    {"field": "avg_tat_hours", "header": "Avg TAT (hrs)", "type": "decimal",
"width": 100},
    {"field": "within_target_pct", "header": "Within Target %", "type":
"percentage", "width": 120},
    {"field": "critical_results", "header": "Critical", "type": "integer",
"width": 80, "aggregate": "sum"}
  ]'::JSONB,
  TRUE
),
(
  'RPT_OUTSTANDING_RECEIVABLES',
  'Outstanding Receivables Report',
  'বকেয়া রিসিভেবল রিপোর্ট',
  'Aging analysis of receivables',
  (SELECT id FROM report_categories WHERE code = 'FINANCIAL'),
  'tabular',
  'function',
  'rpt_outstanding_receivables',
  '[
    {"name": "as_of_date", "label": "As of Date", "type": "date", "required":
true, "default": "current_date"},
    {"name": "branch_id", "label": "Branch", "type": "select", "source":
"branches", "required": false}
  ]'::JSONB,
  '[
    {"field": "receivable_type", "header": "Type", "type": "string", "width":
100},
    {"field": "party_name", "header": "Party Name", "type": "string", "width":
200},

```



```

        {"field": "outstanding", "header": "Outstanding", "type": "currency",
"width": 120, "aggregate": "sum"},
        {"field": "aging_0_30", "header": "0-30 Days", "type": "currency",
"width": 100, "aggregate": "sum"},
        {"field": "aging_31_60", "header": "31-60 Days", "type": "currency",
"width": 100, "aggregate": "sum"},
        {"field": "aging_61_90", "header": "61-90 Days", "type": "currency",
"width": 100, "aggregate": "sum"},
        {"field": "aging_over_90", "header": ">90 Days", "type": "currency",
"width": 100, "aggregate": "sum"}
    ]::JSONB,
    TRUE
),
(
    'RPT_PHARMACY_STOCK',
    'Pharmacy Stock Status Report',
    'ফার্মেসি স্টক রিপোর্ট',
    'Current stock status with alerts',
    (SELECT id FROM report_categories WHERE code = 'INVENTORY'),
    'tabular',
    'function',
    'rpt_pharmacy_stock_status',
    '[
        {"name": "branch_id", "label": "Branch", "type": "select", "source":
"branches", "required": false},
        {"name": "pharmacy_id", "label": "Pharmacy", "type": "select", "source":
"pharmacy_locations", "required": false}
    ]::JSONB,
    '[
        {"field": "drug_name", "header": "Drug Name", "type": "string", "width":
200},
        {"field": "generic_name", "header": "Generic", "type": "string", "width":
150},
        {"field": "current_stock", "header": "Stock", "type": "integer", "width":
80},
        {"field": "reorder_level", "header": "ROL", "type": "integer", "width":
80},
        {"field": "stock_status", "header": "Status", "type": "status", "width":
100},
        {"field": "expiring_soon", "header": "Expiring", "type": "integer",
"width": 80},
        {"field": "stock_value", "header": "Value", "type": "currency", "width":
120, "aggregate": "sum"}
    ]::JSONB,
    TRUE
),
(
    'RPT_ATTENDANCE_SUMMARY',
    'Employee Attendance Summary',
    'কর্মচারী উপস্থিতি সারসংক্ষেপ',
    'Monthly attendance summary by employee',
    (SELECT id FROM report_categories WHERE code = 'HR'),
    'tabular',
    'function',

```

```

'rpt_employee_attendance_summary',
'[
  {"name": "from_date", "label": "From Date", "type": "date", "required":
true},
  {"name": "to_date", "label": "To Date", "type": "date", "required": true},
  {"name": "department_id", "label": "Department", "type": "select",
"source": "departments", "required": false}
]':JSONB,
'[
  {"field": "employee_code", "header": "Emp Code", "type": "string",
"width": 100},
  {"field": "employee_name", "header": "Name", "type": "string", "width":
200},
  {"field": "department", "header": "Department", "type": "string", "width":
150},
  {"field": "present_days", "header": "Present", "type": "decimal", "width":
80},
  {"field": "absent_days", "header": "Absent", "type": "decimal", "width":
80},
  {"field": "leave_days", "header": "Leave", "type": "decimal", "width":
80},
  {"field": "late_count", "header": "Late", "type": "integer", "width": 80},
  {"field": "attendance_percentage", "header": "Att %", "type":
"percentage", "width": 80}
]':JSONB,
TRUE
),
(
  'RPT_QUALITY_DASHBOARD',
  'Quality Indicators Dashboard',
  'কোয়ালিটি ইন্ডিকেটর ড্যাশবোর্ড',
  'Quality indicator performance summary',
  (SELECT id FROM report_categories WHERE code = 'QUALITY'),
  'summary',
  'function',
  'rpt_quality_dashboard',
  '[
    {"name": "from_date", "label": "From Date", "type": "date", "required":
true},
    {"name": "to_date", "label": "To Date", "type": "date", "required": true},
    {"name": "branch_id", "label": "Branch", "type": "select", "source":
"branches", "required": false}
  ]':JSONB,
  '[
    {"field": "indicator_code", "header": "Code", "type": "string", "width":
100},
    {"field": "indicator_name", "header": "Indicator", "type": "string",
"width": 250},
    {"field": "category", "header": "Category", "type": "string", "width":
120},
    {"field": "target_value", "header": "Target", "type": "decimal", "width":
80},
    {"field": "actual_value", "header": "Actual", "type": "decimal", "width":
80},

```

```

        {"field": "status", "header": "Status", "type": "status", "width": 100}
    ]'::JSONB,
    TRUE
)
ON CONFLICT DO NOTHING;

-- =====
-- SECTION 84: REPORT INDEXES
-- =====

CREATE INDEX idx_report_master_code ON report_master(report_code);
CREATE INDEX idx_report_master_category ON report_master(category_id);
CREATE INDEX idx_report_master_active ON report_master(is_active) WHERE is_active
= TRUE;

CREATE INDEX idx_report_execution_report ON report_execution_log(report_id);
CREATE INDEX idx_report_execution_date ON report_execution_log(started_at DESC);
CREATE INDEX idx_report_execution_user ON report_execution_log(executed_by);

CREATE INDEX idx_report_schedules_active ON report_schedules(is_active) WHERE
is_active = TRUE;
CREATE INDEX idx_report_schedules_next ON report_schedules(next_run_at);

-- =====
-- SECTION 85: FINAL REPORT GENERATOR SUMMARY
-- =====

DO $$
DECLARE
    v_report_count INTEGER;
    v_function_count INTEGER;
BEGIN
    SELECT COUNT(*) INTO v_report_count FROM report_master WHERE is_active = TRUE;

    SELECT COUNT(*) INTO v_function_count
    FROM information_schema.routines
    WHERE routine_schema = 'public'
    AND routine_name LIKE 'rpt_%';

    RAISE NOTICE '';
    RAISE NOTICE
'=====';
    RAISE NOTICE '||          REPORT GENERATOR MODULE - COMPLETE
||';
    RAISE NOTICE
'=====';
    RAISE NOTICE '|| Report Definitions:          %
||', LPAD(v_report_count::TEXT, 3);
    RAISE NOTICE '|| Report Functions:          %
||', LPAD(v_function_count::TEXT, 3);
    RAISE NOTICE
'=====';
    RAISE NOTICE '|| REPORT CATEGORIES:
||';

```

```

    RAISE NOTICE '|| | | Clinical Reports
||';
    RAISE NOTICE '|| | | Daily OPD Summary
||';
    RAISE NOTICE '|| | | Daily IPD Census
||';
    RAISE NOTICE '|| | | Patient Statistics
||';
    RAISE NOTICE '|| | | Disease Statistics (ICD)
||';
    RAISE NOTICE '|| | | Doctor Performance
||';
    RAISE NOTICE '|| | | Lab TAT Analysis
||';
    RAISE NOTICE '|| | | Financial Reports
||';
    RAISE NOTICE '|| | | Daily Collection
||';
    RAISE NOTICE '|| | | Revenue Analysis
||';
    RAISE NOTICE '|| | | Outstanding Receivables
||';
    RAISE NOTICE '|| | | Profit & Loss Summary
||';
    RAISE NOTICE '|| | | Operational Reports
||';
    RAISE NOTICE '|| | | OT Utilization
||';
    RAISE NOTICE '|| | | Bed Occupancy Trend
||';
    RAISE NOTICE '|| | | Pharmacy Stock Status
||';
    RAISE NOTICE '|| | | HR Reports
||';
    RAISE NOTICE '|| | | Attendance Summary
||';
    RAISE NOTICE '|| | | Payroll Summary
||';
    RAISE NOTICE '|| | | Leave Balance
||';
    RAISE NOTICE '|| | | Inventory Reports
||';
    RAISE NOTICE '|| | | Stock Movement
||';
    RAISE NOTICE '|| | | Expiry Report
||';
    RAISE NOTICE '|| | | Quality Reports
||';
    RAISE NOTICE '|| | | Quality Dashboard
||';
    RAISE NOTICE '|| | | Infection Control
||';
    RAISE NOTICE '|| | | Incident Summary
||';

```

```
RAISE NOTICE '|| | MIS Reports
||';
RAISE NOTICE '|| | Executive Dashboard
||';
RAISE NOTICE '|| | Monthly Scorecard
||';
RAISE NOTICE '|| Regulatory Reports
||';
RAISE NOTICE '|| DGHS Monthly Report
||';
RAISE NOTICE '||
||';
RAISE NOTICE '|| FEATURES:
||';
RAISE NOTICE '|| ✓ Dynamic Report Generation
||';
RAISE NOTICE '|| ✓ JSON Output Support
||';
RAISE NOTICE '|| ✓ CSV Export
||';
RAISE NOTICE '|| ✓ Parameterized Queries
||';
RAISE NOTICE '|| ✓ Report Scheduling
||';
RAISE NOTICE '|| ✓ Execution Logging
||';
RAISE NOTICE '|| ✓ Role-based Access
||';
RAISE NOTICE '|| ✓ Multi-language Support
||';
RAISE NOTICE
||';
RAISE NOTICE '';
END;
$$;
```

 REPORT GENERATOR SUMMARY

Available Reports:

Category	Report Name	Function
Clinical	Daily OPD Summary	rpt_daily_opd_summary()
Clinical	Daily IPD Census	rpt_daily_ipd_census()
Clinical	Patient Statistics	rpt_patient_statistics()
Clinical	Disease Statistics	rpt_disease_statistics()
Clinical	Doctor Performance	rpt_doctor_performance()

Category	Report Name	Function
Clinical	Lab TAT Analysis	rpt_lab_tat_analysis()
Financial	Daily Collection	rpt_daily_collection()
Financial	Revenue Analysis	rpt_revenue_analysis()
Financial	Outstanding Receivables	rpt_outstanding_receivables()
Financial	Profit & Loss	rpt_profit_loss_summary()
Operational	OT Utilization	rpt_ot_utilization()
Operational	Bed Occupancy Trend	rpt_bed_occupancy_trend()
Inventory	Pharmacy Stock	rpt_pharmacy_stock_status()
Inventory	Stock Movement	rpt_stock_movement()
Inventory	Expiry Items	rpt_expiry_items()
HR	Attendance Summary	rpt_employee_attendance_summary()
HR	Payroll Summary	rpt_payroll_summary()
HR	Leave Balance	rpt_leave_balance()
Quality	Quality Dashboard	rpt_quality_dashboard()
Quality	Infection Control	rpt_infection_control()
Quality	Incident Summary	rpt_incident_summary()
MIS	Executive Dashboard	rpt_executive_dashboard()
MIS	Monthly Scorecard	rpt_monthly_scorecard()
Regulatory	DGHS Monthly	rpt_dghs_monthly()

Usage Examples:

SQL

```
-- Generate Daily OPD Report
SELECT * FROM rpt_daily_opd_summary(
  'your-org-id'::UUID,
  NULL,
  '2024-01-15'
);

-- Generate Revenue Analysis
SELECT * FROM rpt_revenue_analysis(
  'your-org-id'::UUID,
  'your-branch-id'::UUID,
  '2024-01-01',
  '2024-01-31'
```

```
);

-- Generate Report as JSON
SELECT generate_report_json(
    'RPT_DAILY_OPD',
    'your-org-id'::UUID,
    '{"date": "2024-01-15"}'::JSONB
);

-- Export to CSV
SELECT export_report_csv(
    'RPT_DAILY_COLLECTION',
    'your-org-id'::UUID,
    '{"date": "2024-01-15"}'::JSONB
);
```